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AI IN CONSIDERATION OF HER: ACCOUNTING FOR GENDERED WORKPLACE DYNAMICS IN THE DESIGN AND EVALUATION OF HUMAN-CENTERED AI INTEGRATION IN EVERYDAY WORKPLACES

A Dissertation
Presented to
the Graduate School of
Clemson University

In Partial Fulfillment
of the Requirements for the Degree
Doctor of Philosophy
Human-Centered Computing

by
Kelsea Sierra Schulenberg
December 2025

Accepted by:
Dr. Guo Freeman, Committee Chair
Dr. Nathan J. McNeese
Dr. Bart Knijnenburg
Dr. Emma Dixon

Plain Language Abstract

As AI technology continues to advance, so too will its impacts on the people who use and work with it. This dissertation provides a comprehensive look into how the design and evaluation of AI technologies for use in everyday workplaces (i.e., office-type settings) may differently impact women workers in the years to come. While there has been plenty of popular and scholarly discourse drawing attention to issues such as gender bias in AI design and disparities in AI literacy in women workers, this dissertation is the first of its kind to empirically investigate how AI integration into workplaces might intermingle with workplace gender dynamics to differently impact the way that women workers are evaluated and attributed credit for the work they do in an AI-enabled workplace. To accomplish this goal, this dissertation presents three research studies. Study 1 examined how women in the AI development field perceive their experiences with workplace gender dynamics and how these experiences influence their approaches to designing AI through a qualitative interview study. Study 2 conducted an online experiment with 685 participants along with follow-up expert interviews to empirically test and explain how the evaluations of women and men workers might differ when workplace gender dynamics and AI functional design mix with each other. Lastly, Study 3 also utilized the substantial data set from the 685 evaluators and follow-up expert interviews to model and explain the impacts that AI integration will have on perceptions of human worker job security

when taking into consideration workplace gender dynamics and AI functional design. Through these three studies, this dissertation will enable the design, integration, and evaluation of AI technology to account for workplace gender dynamics and to improve the safety and security of *all* workers in AI-enabled workplaces.

Abstract

Rapid advancements in the technical capabilities and availability of generative Artificial Intelligence (AI) systems, such as OpenAI's ChatGPT, have drawn widespread attention to the opportunities and challenges associated with AI integration into everyday workplaces (i.e., office-type work). Following calls for organizations to consider the ethical and workplace-specific impacts of generative AI's use before integrating it into the workplace, this dissertation addresses three critical gaps in Human-Centered Computing (HCC) and AI workplace integration research. First, this dissertation unpacks the underdeveloped links between women's representation - or lack thereof - in AI-related fields and how their experiences with gendered workplace dynamics in said fields affects their decision-making when designing and evaluating AI workplace integration. Second, this dissertation directly addresses the critical need to clarify further how the introduction of AI into everyday workplaces will impact the ways through which other people within the everyday workplace will evaluate and attribute credit to the human workers who work with the AI, especially considering the differing impacts that workplace gender dynamics can have on women workers. Third, this dissertation ultimately aims to understand how the design and evaluation of human-AI collaborative integration into everyday workplaces might contribute to perceptions of human worker job security and whether workplace gender dynamics impact these understandings.

In doing so, Study 1 constitutes the “design” portion of this dissertation’s larger narrative and addresses the aforementioned gaps via eight in-depth, qualitative interviews with diverse women who work in AI design, development, and research. These interviews focused on understanding how these women perceive the role that their gender identity plays in their experiences within the male-dominated AI industry, and how said perceptions of their workplaces’ gender dynamics influence their approaches to designing and developing AI. Study 1’s findings provide one of the only empirical investigations linking the enduring underrepresentation of women in AI development to the reinforcement of gendered assumptions in the design and development of AI, and how these women subsequently challenge these assumptions in AI development by advocating for use-case specificity and intersectionality in development practices.

Study 2 constitutes the “evaluation” portion of this dissertation’s larger narrative, focusing on empirically investigating how evaluations and attributions of credit in the workplace are influenced by technological and gender dynamics within everyday workplaces. To do this, Study 2 conducted a large 2x4x2 online factorial survey experiment with 685 participants that explored how the gender presentation of a human worker (i.e., woman v. man), the gender or AI functional design of said worker’s collaborative partner (i.e., an AI tool v. an AI teammate v. a human female teammate v. a human male teammate), and the type of task completed by the partnership (i.e., meeting scheduling v. investment banking) all contribute to the evaluation of and attributions of credit to workers and their collaborative partners in AI-enabled workplaces. Study 2 then leveraged interviews with six experts in people management in AI-enabled workplaces to gain real-world insights on these results. Study 2 thus informs expert-crafted recommendations for how workplaces looking to integrate AI into their practices can best design AI to work collaboratively with humans as team-

mates. These insights strike a balance between gaining the benefits of AI use without inadvertently disadvantaging and disenfranchising human workers, especially women workers.

Finally, Study 3 offers a deeper examination of the pathways that lead from dyadic composition in AI-enabled workplaces to human worker job security, with the ultimate goal of providing workers, particularly women, with new insights into how AI workplace integration may impact their job security. Built upon data collected during Study 2, Study 3 conducted a more comprehensive analysis using structural equation modeling. Similar to Study 2, Study 3 then leveraged interviews with six experts in people management in AI-enabled workplaces to gain real-world insights. Study 3 thus provides comprehensive insights into how workplace gender dynamics, AI workplace technology, individual differences, partnership evaluations, and attributions of credit interact to influence the job security of future workers in AI-enabled workplaces. Such insights highlight how the revealed paths can inform expert-crafted recommendations for helping workplaces best integrate AI into their workflows while still accounting for the well-being and sense of security of their human workforce.

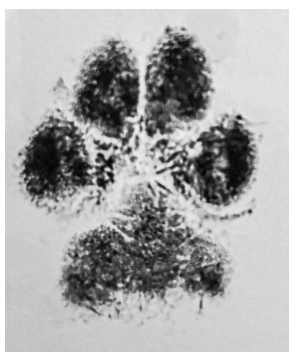
Overall, the three studies of this dissertation stand to make significant contributes towards research and ongoing global conversations in HCC and Human-Centered AI research about AI's impacts on the working lives of women as well as gendered aspects of AI design. It does so by explicitly considering how workplace gender and technological dynamics shape the design and evaluation of AI integration into everyday workplaces, how these dynamics intermingle with AI functional design to impact evaluations of and attributions of credit to women workers, and how these dynamics and evaluations lead to perceptions of human worker job security in AI-enabled workplaces to impact women workers specifically. As such, this dissertation provides much-needed insight into current and future efforts to integrate generative AI

into everyday workplaces in light of ongoing ambiguity and discourse surrounding the ethical and causal attributional challenges that AI workplace integration presents.

Dedication

This dissertation is dedicated to the love of my life, the wind betwixt her mother's wings, the greatest thing that has ever happened or will ever happen to her mother, the world's smallest polar bear (I checked), Miss Coco Puff Schulenberg. Miss Nut Butter, it is through your love that I know what true, unconditional, safe, and pure love feels like. It is also through my love for you that I have come to understand grief as just Love with her funeral blacks on. I know you would have stayed if I had asked you to, and I hope you know that letting you go was the ultimate testament to my love for you. Thank you for every late night spent waiting for me to finish my work and come to bed with you. Thank you for leading the way for me so confidently into our new life, with not even a glance back over your little shoulder. Thank you for showing me that, with your papa, I am safe because he made *you* feel safe. I'll always be taken care of, because he took care of *you* without hesitation. I'll be listened to, because he listened to *you* when I could not bear to hear yet. We both believe he has inherited your will, and he's been doing a great job since you left!

I love you, and my greatest wish in this life is still to squeeze you until we become one, but in the meantime, go run up and down those dog heaven stairs - I'll be okay. ♡



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I must also give a massive thank you to each of my committee members, without whom this dissertation, quite literally, would not be what it is today. To Dr. Bart Knijnenburg, thank you so much for all of the time and effort you put into helping me craft an experimental design for my last two studies that made this work so much more important and interesting than I had ever imagined it could be. I am especially grateful for your patience and guidance during data analysis, and for continuously taking the time to meet with me to go over (and over) the results to make sure I

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Chapter 1

Introduction and Motivation

1.1 Background and Problem Space

Amidst the craze of generative AI, many industry and technology reports [172, 338, 343, 444, 458] and scholarly research efforts [122, 141, 293, 349, 365, 457, 472] have drawn attention to the disproportionate impacts AI might have on everyday workers, **especially women**, in the coming years. While gender bias in algorithmic decision-making is nothing new [72, 209, 270], generative AI’s widespread accessibility and usability mean that harms, such as the perpetuation of gender stereotyping of STEM professionals as automatically male, are made to proliferate at a far grander scale [449, 453]. Another example of such harms is how generative AI’s ability to create content in a way that can be indistinguishable from human-made content (e.g., passing the Turing Test [128]) has led to the rise of “deepfakes,” i.e., AI-generated content designed to fool the consumer into believing that it is real documentation of events or people [202, 389, 422]. Women have been identified as being disproportionately harmed by this ease of deepfake creation, specifically by people using their likenesses to create convincing deepfake pornographic images of them, which can be incredibly

socially, emotionally, and professionally damaging to victims [126, 254, 341, 389]. The above examples illustrate how the mere existence and use of widely accessible generative AI systems create new and more severe avenues for the subjugation and victimization of women. This knowledge thus serves as a baseline justification for investigating further AI’s impacts on women’s everyday lives in more context-specific (i.e., everyday workplaces) applications, where these impacts are likely to intermingle with existing workplace gender dynamics, such as **sexism** in the workplace.

Indeed, the rise of these widely accessible generative AI systems is making various workplaces and industries think more about using AI to enhance their workforces, such as in healthcare [326, 378, 423, 489, 497], manufacturing [121, 275, 395, 485], and everyday (i.e., office-type) workplaces [88, 113, 134, 202], with the latter being the focus of this dissertation. Doing so requires companies to first understand what types of human-AI interaction could best benefit their workplaces [6, 231, 428]. The term *human-AI interaction* (HAI) within this dissertation generally refers to when humans and AI are made to work either with or in parallel to each other within the same working context and/or on a shared task [231]. It can be difficult for businesses to parse out what kinds of AI and human-AI interaction work best within their organization, as there are many nuances to how human-AI interaction operates based upon several factors, including how an AI is designed to function, how humans are trained to use or work with it, and the nature of the work itself [231]. This difficulty is further compounded by various ethical considerations of using generative AI in workplaces, such as how and whether AI should be used in decision-making and content generation [208].

Recognizing these nuances and challenges, researchers in industry and academia alike have put forth significant efforts to explicate (1) the opportunities and challenges of using generative AI in the everyday workplace (e.g., [60, 88, 113, 133, 134, 185,

237, 328, 490]); and (2) the design of human-AI interaction in the workplace to enhance such opportunities and address challenges via human-centered design principles (e.g., [10, 169, 231, 237, 351, 484]). However, while these efforts have provided necessary foundational knowledge, they provide limited insights into the potential for AI workplace integration to **disproportionately impact workers with marginalized identities** [69, 351]. In fact, they rarely go beyond acknowledging that the data used to train an AI system can have biases that can lead to discriminatory outcomes if not mitigated early in the development life cycle [351]. As such, this dissertation specifically argues for greater attention to be paid to how workplace gender dynamics, particularly **sexism in the workplace** [143, 272, 375, 381, 400], could intermingle with AI integration in ways that disproportionately hurt women workers if not properly considered.

Indeed, this lack of attention to marginalized individuals, such as women workers, in discussions on human-AI interaction design is especially alarming in light of an influx of reports detailing how AI is projected to disproportionately target and replace jobs whose workers are mostly women, such as customer service and administrative roles [134, 170, 300, 334]. More specifically, a comprehensive 2023 report from the McKinsey Institute projects women to be 1.5 times more likely to face occupational transitions than men due to generative AI automation [134]. This is reflective of women’s over-representation in office support and customer service roles, which are projected to shrink by 3.7 million and 2.0 million jobs each by 2030 as generative AI automates these positions [134]. This unequal impact is yet another example of unequal treatment of women workers within and outside of the US that has been a topic of discussion for decades, e.g., the gender pay gap [50], the devaluing of “feminine” labor at home and at work [204], and the “glass ceiling” preventing the promotion of women [81], to name a few.

Recognizing this long history, an extensive report entitled “*The Effects of AI on the Working Lives of Women*” was released in 2022 as a collaborative effort between the world organizations of UNESCO, Organisation for Economic Co-operation and Development (OECD), and Inter-American Development Bank (IDB) [449]. This report provides an overview of how AI systems impact women’s workplace treatment, status, and opportunities globally, given current knowledge at the time (2022) and predictions for the future, given patterns of AI and automation integration. In doing so, it highlights three major impact areas that heavily influence this dissertation’s motivations to explore this problem space:

(1) A lack of women in AI development [43, 163] and corresponding AI design that perpetuates gender stereotypes and reinforces women’s marginalization, often allowing sexist beliefs to manifest in the design and implementation of technology [41, 120, 294, 449];

(2) Gender disparities in tech literacy and STEM education impacting women’s ability to reskill and upskill in the event of AI job replacement, increasing the perceived vulnerability of some women to trends in technology [170, 300, 334, 449]; and

(3) Gender bias in algorithmic hiring and workplace monitoring devaluing women workers [24, 115, 281, 449]. It especially highlights how context-dependent these impacts are, varying by societal factors such as country and region, as well as the social, economic, and cultural position of an individual woman [449].

Given these issues, the aforementioned report advocates for programmatic efforts to (a) reskill and upskill women workers to meet the demand for STEM/AI professionals, (b) encourage women to pursue STEM careers early on to increase women’s involvement in AI design, (c) account for contextual and cultural complexity in AI design, (d) leverage multi-stakeholder approaches towards AI governance, and (e) to continue applied research to further understanding of AI’s impacts on women’s

working lives [449].

Although the aforementioned report provides valuable insights and recommendations for addressing disproportional harm to women workers in the era of AI, this dissertation is motivated to significantly extend upon those insights to better account for a fourth factor:

(4) The potential for workplace gender dynamics, such as sexism and technological biases, to intermingle with the functional design of an AI system that a woman works with in her workplace in a way that impacts how *other humans evaluate women workers*.

In doing so, this dissertation recognizes several critically under-addressed gaps in existing understandings of women’s perspectives on AI design and workplace integration, as well as their experiences with workplace gender dynamics and AI functional design. The following section thus delves deeper into these gaps, outlining how each of this dissertation’s three studies will address them.

1.1.1 Research Motivations

First, this dissertation is motivated to **Unpack Underdeveloped Links Between Women Representation in AI-Related Fields and Their Decision-Making in the Design and Evaluation of AI Workplace Integration**.

The solutions put forth by the collaborative UNESCO report place heavy emphasis on increasing women’s participation in STEM broadly and AI development specifically in order to reduce gender bias in AI design [449]. This makes sense, as there is a staggering lack of diversity in both AI workforces and AI design and development (see 2.2.2 for more details) [162, 450]. There is certainly merit in this approach, as the design, development, and data training of AI technologies often directly re-

flect the life experiences and prejudices of their creators [61, 337, 363]. However, this prevailing discourse relies on the assumption that increased representation inherently engenders *better* representation of women in AI design without interrogating how workplace dynamics, including sexism and technology biases, may complicate how women are able to operate in their AI workplaces. Indeed, there are likely many factors that would complicate this dynamic, such as an individual woman’s perceptions of her own ability [504] and responsibility for mitigating bias in AI [276]. Even more critically, existing within a patriarchal system where non-conforming women are often punished for exhibiting counter-stereotypical behavior can lead to the internalization of sexist beliefs within women [28, 106, 148, 193, 307] (see section 2.2.1 for more details), which could prevent women in AI development from feeling empowered to prevent bias when workplace gender dynamics uphold such beliefs.

Moreover, given the rapid evolution of accessible generative AI technologies, the ever-changing landscape and considerations involved in the design and evaluation of AI workplace integration might impact women’s abilities to participate in these integration steps or to prioritize women’s considerations in their choices. Ultimately, this dissertation recognizes that there is a critical need to empirically investigate how women’s experiences of their own gender identity within their AI-related workplaces might shape how they design and evaluate AI workplace integration, thereby motivating **Study 1**.

Second, this dissertation is motivated to **Further Clarify How Introducing AI into Workplaces Will Impact Human Evaluations of the Women Who Work With It**.

When it comes to AI and evaluations of human workers generally and of women workers specifically, current discourse almost exclusively centers on the use of algorithmic workplace monitoring (e.g, [99, 112, 433]) and the use of algorithms in human

resource management (e.g., [21, 25, 265]) (see 2.1.1 for more details). However, the increasing use of generative AI in workplaces creates a different kind of evaluation consideration, i.e., how credit and blame for generative AI outputs are attributed [127, 128, 169, 246, 364]. Indeed, scholars and industry leaders alike have warned that the use of generative AI in the workplace comes with new concerns regarding privacy, fair use, and a credit/blame asymmetry [127, 128, 134, 169, 246, 364]. Moreover, prior works in HCC-related domains, such as Human-Robot Interaction (HRI) (e.g., [140, 266, 289, 434, 456]) and autonomous vehicle research (e.g., [19, 166, 498]), have been exploring how autonomous agents are attributed blame when working with humans.

Despite this, comparatively little attention has been paid to how introducing generative AI systems into everyday workplaces will impact the ways in which *humans evaluate other humans*. More importantly to this dissertation, it is unclear how, if at all, women workers will be evaluated and attributed credit for their work differently than men workers when working with AI collaboratively (i.e., AI teammate) versus non-collaboratively (i.e., AI tool)(see section 2.1.2 for more details), especially when taking workplace gender dynamics into consideration, such as sexism and technology biases. This is a critical gap in our existing knowledge, as evaluations and attributions of credit could impact perceptions of a woman worker’s job security and potential for replacement in AI-enabled workplaces, thus motivating **Study 2**.

Moreover, fears surrounding job security in the age of generative AI continue to grow despite recent reports demonstrating no noticeable impacts of AI integration on the job market as of October 2025 [182, 184]. However, even the mere *perception* of job insecurity can have demonstrable impacts on worker performance [459, 503]. This thus creates the potential for a self-fulfilling prophecy for women workers if efforts are not made to explicitly consider workplace gender dynamics when investigating the

impacts of AI integration on job security. Taken together, then, there is an urgent need to understand how women who are made to work with AI will be evaluated and attributed credit when working with AI in AI-enabled workplaces, thus motivating **Study 3**.

Additionally, while there have been reports from institutions such as the National Science Foundation that demonstrate that working with an AI tool can result in worse evaluations of human workers [380], it is unclear how workers generally and women workers specifically will be evaluated and attributed less credit when working with AI depending upon the functional agency of said AI (i.e., AI tool v. AI teammate) [298, 299, 353], again motivating **Studies 2 and 3**.

Third, this dissertation is motivated to **Understand and Account for how Workplace Gender Dynamics, Including the Sexist Beliefs of Evaluators, Will Impact AI Workplace Integration, Contrasting the Current Overemphasis on Mitigation Strategies That Place the Burden on Individual Women Workers to Prevent Harm**.

Finally, much of the current discourse surrounding ways to mitigate the projected harms of AI on women’s workplace lives tends to mainly emphasize reskilling and upskilling women workers to make them less replaceable and better able to adapt to new positions that will be more secure [95, 147, 172, 449]. While these are certainly vital and important pursuits, it is also the case that they (1) place the burden on individual women to prevent harm and devaluation of their work, and (2) assume women are going to get replaced because of their lack of adaptability to new types of work when AI is involved.

However, what is less clear is whether there might still be disproportionate harms to women even if they are successfully able to adapt to AI workplace use. More specifically, this dissertation aims to explore whether workplace gender dynam-

ics that are beyond an individual woman’s control may still serve to subjugate women workers in AI development and AI-enabled workplaces. For example, existing literature on ambivalent sexism [27, 188] does demonstrate that endorsement of ambivalent sexist beliefs in the workplace leads to evaluations that devalue and discredit women workers when working alongside men [34, 203, 420]. Yet, to the best of this dissertation’s knowledge, there are no studies that account for this type of workplace gender dynamic in the design and evaluation of AI integration, within or outside the HCC and HCI domains, thus motivating **Studies 1, 2, and 3**.

1.1.2 Dissertation-Wide Research Questions

In light of the aforementioned research motivations, this dissertation endeavors to address the following dissertation-wide research questions (see Table 1.1):

D.W. RQ1: *How does the identity of “woman” shape the design and implementation of generative AI, especially within the workplace?* (Studies 1 & 3)

D.W. RQ2: *How will the introduction of generative AI be influenced by the gender dynamics, including sexist beliefs, of a workplace where AI is integrated?* (Studies 1 & 2)

D.W. RQ3: *How will the introduction of AI influence the work and evaluation of women workers, including their job security in AI-enabled workplaces?* (Studies 2 & 3)

D.W. RQ4: *How should the design and integration of generative AI into the workplace evolve to better account for women workers?* (Studies 1, 2, & 3)

Research Motivation	RQ#	Research Question
Unpack Underdeveloped Links Between Women Representation in AI-Related Fields and Their Decision-Making in the Designing, Planning, and/or Implementation of AI.	RQ1	<i>How does the identity of “woman” shape the design and implementation of AI, especially within the workplace?</i>
Understand and Account for how Workplace Gender-Dynamics, including the sexist beliefs of individuals, Will Impact AI Workplace Integration, Contrasting the Current Overemphasis on Mitigation Strategies That Place the Burden on Individual Women Workers to Prevent Harm.	RQ2	<i>How will the introduction of AI influence be influenced by the gender dynamics, including sexist beliefs, of a workplace where AI is integrated?</i>
	RQ4	<i>How should the design and integration of AI into the workplace evolve to better account for women workers?</i>
Further Clarify How the Introduction of AI into Workplaces Will Impact Human Evaluations of the Women Who Work With It.	RQ3	<i>How will the introduction of AI influence the work and evaluation of women workers, including their job security in AI-enabled workplaces?</i>

Table 1.1: Research Motivations & Corresponding Research Questions

1.2 Summary of Studies

Table 1.2 provides a summary of studies involved in this dissertation and how they collectively address the dissertation-wide research questions.

Study 1 of this dissertation addresses D.W. RQ1, RQ2, and RQ4 via eight in-depth, qualitative interviews with a diverse group of women AI designers, developers, and researchers. These interviews were designed to unpack how these women perceive the role their gender identity plays in their AI-related workplace (i.e., workplace gender dynamics), and how said perceptions influence their AI design and development approaches. Study 1’s findings provide firsthand empirical evidence from women’s perspectives on how the enduring gender gap in AI development and participation reinforces inaccurate and biased gendered perspectives and assumptions about how to design AI for and from the perspective of women. As a result, women in AI design and development demonstrate a firm resistance to these unnuanced gendered considerations and assumptions and instead, advocate for design perspectives that center on use-case specificity and intersectionality.

Study 2 of this dissertation addresses D.W. RQ2, RQ3, and RQ4 via a 2x2x4 online factorial experiment ($n = 685$) where participants were tasked with evaluating the performance of either a woman or man worker when working with an AI tool, an AI teammate, human female teammate, or a human male teammate to successfully complete either a meeting scheduling or investment banking task. The results of this study show that the evaluations of human workers are most strongly influenced by the dyadic partner they work with, with AI tools and AI teammates creating different impacts on evaluations and attributions of credit to human workers when compared to each other and other human dyadic partners. Results also show that the gender of a worker can create an interaction effect with the type of dyadic partner, resulting in

evaluations with specific dyadic partners differing between men and women workers.

Additionally, Study 2 explores the role of workplace gender dynamics and technological biases through the lens of ambivalent sexism and negative attitudes toward AI, highlighting that the influence of sexist beliefs is also highly influenced by the type of dyadic partner that humans find themselves working with. These results provide a clear contribution to understanding how the gender identity of a human worker and the gendered dynamics of a workplace are likely to not only impact the evaluation of human workers but also influence how people evaluate future AI technologies.

As the last component of Study 2, these results were reviewed by six (four women and two men) experts (four women and two men) in people management and AI workplace integration to gain deeper qualitative insights into how AI teammate integration into workplaces aligns with the industry trend towards evaluating workers' AI usage, and how they interpret results concerning sexist beliefs and evaluations of workers and partners in AI-enabled workplaces. Study 2 concludes with design recommendations that integrate Study 2's findings with expert insights to address the impacts of workplace gender and technical biases on the design and evaluation of AI workplace integration.

Finally, Study 3 of this dissertation addresses D.W.RQ1, D.W.RQ3, and D.W.RQ4 by leveraging the same dataset from Study 2 to create a first-of-its-kind model of how workplace gender dynamics and AI functional design intersect to impact perceptions of human worker job security. To accomplish this, Study 3 employs structural equation modeling (SEM) to conduct a comprehensive analysis of how workplace gender dynamics, AI attitudes, worker effectiveness, and credit interact to model the expected job security of human workers in AI-enabled workplaces. Based on the confirmatory factor analysis and SEM performed, a clear pathway toward job security was modeled.

This pathway has four highlights: (1) the gender of a human worker, type of dyadic partner, type of task, and gender of an evaluator directly impact and interact to impact the perceived mutual dependence of an AI-enabled collaboration; (2) the outcome of (1) in turn influences the perceived effectiveness of human workers and their collaborators; (3) the outcome of (2) then, in turn, influences the credit distributed between the worker and partner in the collaboration; and (4) all of these factors then either directly or indirectly feed into the perceived job security of a human worker. Based on this model, the same six experts were interviewed to deepen these results via qualitative insights that highlight how these results should inform the future of evaluation practices and AI design in the workplace. As such, Study 3 provides key design recommendations to ensure that the integration of AI technologies does not cause undue harm to the job security of human workers, particularly women.

Study #	Short Study Title	RQs Addressed
1	Understanding Women Developers' and Researchers' Perspectives on Gender in Approaching AI Design and Development	RQ1, RQ2, RQ4
2	Understanding the Influence of Workplace Gender and Technological Dynamics on the Evaluation of Women Workers in AI-Enabled Workplaces	RQ2, RQ3, RQ4
3	Understanding the Job Security of Women Workers in AI-Enabled Workplaces Based on Who or What they Work With	RQ1, RQ3, RQ4

Table 1.2: Studies that Address Each Research Question

1.3 Dissertation Positionality

Before proceeding to the next chapter of this dissertation, it is first important to acknowledge how this dissertation’s focus on women workers in AI-enabled workplaces is informed by the researcher’s position as a woman and the sociotechnical climate at the time of writing (late 2025). Indeed, being transparent about one’s positionality as a researcher is often encouraged within certain Human-Centered Computing (HCC) and Human-Computer Interaction (HCI) paradigms, such as feminist HCI, as a vital part of the ethical research process when focusing on sensitive or marginalized populations or topic areas to promote researcher accountability [26].

First, as this dissertation’s author and researcher, it is important for me to acknowledge that I am a woman in STEM, and that this aspect of my identity does play an intimate role in my motivations for investigating how AI integration might intermingle with workplace gender dynamics in ways that may disproportionately impact women workers. It is because of the first-hand knowledge that comes from my experiences in STEM and hearing other women’s stories throughout the years that I was able to recognize that there are critical gaps in empirical research concerning AI workplace integration and women workers, and that workplace gender dynamics must be an explicit consideration in AI integration research and practice. Moreover, my identity as a woman gave me a certain level of credibility and rapport with the women I interviewed in all three studies of this dissertation, which is a critical component of high-quality qualitative research [446].

However, it is also important to acknowledge that, as a cisgender, White, straight woman in the U.S., my experiences in workplaces, STEM, and society as a whole are shaped by the heteronormative values, racial climate, and gender politics of the U.S. that give preference to women like me, historically and present-day. This

positionality is evident in the dissertation’s use of U.S.-centric labor statistics, as well as its reliance on Western-leaning and heteronormative cultural understandings of workplace evaluations and gender dynamics. Yet, this positionality should not serve to discount the value and importance of other identities; I welcome continued work that challenges the findings and perspectives of this dissertation due to differences in identity.

Continued advancements will be especially needed in understanding the specific impacts that intersectional identity can have on women workers from diverse genders, ethnicities, sexual orientations, and cultural backgrounds. For example, the successful creation and use of a large experimental survey with almost 700 participants, which manipulates AI design and the gendered dynamics of a dyad being evaluated by others, in Studies 2 and 3, provides a first-of-its-kind model. These results and this model can be used by researchers and practitioners alike as a baseline model, which can and should be modified and tested by others to better account for intersectional factors such as ethnicity, gender identities beyond woman and man, and a variety of cultural and workplace contexts.

Chapter 2

Literature Review

This dissertation’s focus is built upon two strands of interrelated research: (1) Artificial Intelligence (AI) & workplace integration, including both opportunities and threats to (women) workers (**section 2.1**); and (2) women’s marginalization in AI design as a symptom of a broader, sexist problem in our modern society (**section 2.2**). This chapter ends with a summary of research gaps identified in the existing literature (**section 2.3**), which directly motivate the research questions presented at the beginning of this dissertation.

2.1 Artificial Intelligence (AI) & Workplace Integration

2.1.1 The Rise of Generative AI in the Everyday Workplace: Opportunities and Threats to (Women) Workers

Following the release of OpenAI’s ChatGPT in November 2022, the rise of widely available and usable generative Artificial Intelligence (AI) systems has ar-

guably ushered in a new era of economic and societal considerations that pose unique opportunities and challenges for workplace integration. At a high level, AI as a technology is characterized by its ability “*to perform cognitive functions that we associate with human minds*” ([373], p.iii) with varying levels of autonomy [161, 231, 301]. Generative AI is a type of AI that leverages Large-Scale Language Models (LLMs) to represent, predict, and subsequently produce human-like content such as text and art [128, 224, 364, 468].

The opportunities that generative AI use can provide workforces have received widespread attention in scholarly and public discourse. Researchers in a variety of disciplines point out that the ever-growing sophistication, autonomy, and pervasiveness of AI technologies create the potential for reshaping workplace environments to focus more on collaborative efforts between humans and AI [22, 231, 386]. The explosive potential of generative AI use in organizational enterprises such as marketing and sales, operations, IT/engineering, risk and legal, and research and development [88] has also been lauded by influential management consulting firms (e.g., McKinsey & Company [88, 134] and Deloitte [1, 4]) and private research organizations (e.g., National Bureau of Economic Research [60, 133]) alike. These institutions have collectively highlighted the tremendous potential for generative AI to increase efficiency and productivity in a variety of ways, such as enabling tech capabilities in non-technical workforces and freeing up human capital and creative potential when AI and humans work together [1, 4, 60, 88, 133, 134].

Alongside these opportunities exist challenges that can pose serious threats to the efficacy and ethicality of AI workforce integration if not properly considered. These challenges are often associated with the nature of the data used and given to an AI for algorithmic incorporation, including discrimination and exclusion stemming from algorithmic bias, data privacy and security concerns, and output assess-

ment uncertainties given the black-box nature of AI [5, 52, 128, 180, 231, 364, 417]. Generative AI specifically poses unique challenges for establishing and defending intellectual property rights over generated content as ongoing debates about who can be attributed credit for said content continue to wage [88, 128, 348, 364]. In light of these challenges, industry discourse on AI workplace integration most often centers on minimizing the occurrence of risky use and outputs, e.g., restricting what types of company documents can be shared with publicly accessible generative AI systems such as ChatGPT to minimize privacy and security risks (e.g., [88, 134]).

Moreover, there are widespread calls for organizations to consider ethical and workplace-specific environmental impacts when making decisions about whether and how to integrate AI into their business practices to ensure the well-being of their human workforce [40, 128, 231, 306]. **This dissertation thus directly responds to and significantly extends these calls by recognizing the unique cultural and organizational forces that will render women in the workplace especially vulnerable to the negative impacts of AI workforce integration if not adequately accounted for in AI design, evaluation and implementation of workplace integration.**

In particular, below we dive deeper into research concerning AI’s impacts on both (1) *worker well-being generally* and (2) *the well-being of **women workers** specifically*.

(1) Worker Anxieties Surrounding Algorithmic Decision-Making & Workplace Monitoring, Especially in Light of (Gender) Biases. There is a wealth of prior work examining worker well-being in light of the incorporation of algorithmic decision-making in the workplace, such as in the human resources management (HRM) domain (e.g., [21, 25, 265]). This is otherwise known as algorithmic management [21], or “*autonomous data-driven decision making in people’s manage-*

ment by the adoption of self-learning algorithms and artificial intelligence” [265]. Such works have collectively highlighted how the use of algorithms to make decisions related to sensitive and important workplace decisions, such as hiring decisions, makes employees and even third-party observers (e.g., public perception of a company) feel less respected [25], that these systems lack a sense of fairness and trustworthiness [277], and have a negative impact on feelings of job autonomy and worker well-being [265]. Although some argue that AI seems to provide benefits for the practice of HRM, such as increased efficiency and advanced data analytic capabilities, researchers have argued that a lack of human-centered considerations in the actual design of these systems stifles their potential, pointing out the need for transparency in its use in and design for HRM decision making [25, 277].

More importantly to this dissertation, algorithmic decision-making in HR/hiring decisions appears to uniquely target and disadvantage women workers [269, 308, 333, 342, 439], reflecting broader patterns of gender bias in algorithmic decision-making and the use of LLMs generally [425, 437]. This is often attributed to issues with the data that a hiring algorithm is trained on, such that there is a cycle wherein historical data preferences the characteristics of those who have historically had the most access to career opportunities and success within the organization, i.e., men [14, 372, 377, 402]. In other words, in domains where women have been historically hired at lower rates, the data used to train the HRM algorithmic decision-making system skews towards favoring male applicants because there are not as many ‘successful’ women candidates in the data pool to begin with. Moreover, said systems have been shown to inadvertently disadvantage women workers by reproducing male-dominated patterns of what it means to be competent and successful [100].

Indeed, when it comes to algorithmic monitoring and management of workers, algorithmic monitoring often prefers quantifiable measures of performance [245, 265],

whereas much of the valuable labor that women contribute to the workplace environment is often more qualitative and relational [449]. Women are also more likely to be unfairly assessed in remote work because they are more likely than men to have to simultaneously juggle unpaid domestic labor with remote work, especially after COVID-19 (e.g., [99, 112]), which also makes women more concerned with the privacy implications of algorithmic workplace monitoring [433].

Taken in combination, there are legitimate concerns about the use of algorithmic decision-making and monitoring in workplaces uniquely disadvantaging and devaluing women workers. While this dissertation is not specifically concerned with algorithmic decision-making in HRM or algorithmic management/monitoring, these issues do still permeate discussions on AI’s impacts on women workers worldwide (e.g., [449]), hence their acknowledgment here. In light of the aforementioned concerns, researchers and practitioners alike have repeatedly emphasized the need for continuous human involvement in the design of algorithmic and AI-based technology to prevent discomfort and disproportionate harm to workers (e.g., [25, 277]). Moreover, it is likely the case that humans will always be involved to some capacity in HRM practices due to the distrust that can be fostered without human decision-making on some level [281], but little is known about how the evaluation of women workers will be impacted by AI integration, even if it is other humans evaluating the dynamics.

(2) Fears Surrounding the Replaceability of (Women) Workers. Above all, generative AI’s ability to perform tasks and produce outputs in a human-like manner means that jobs requiring a level of human-level decision-making and creativity can now be performed by a comparatively faster, cheaper, and more efficient system than a team of human workers [113, 177]. Popular discourse has thus expressed widespread concerns about AI replacing jobs previously considered safe from automation [173]. These concerns are not unwarranted, as large tech and management corpo-

rations, such as Workday, have implemented proactive mass layoffs of human workers in preparation for AI-driven worker replacement [63, 218]. Furthermore, reports state that 80% of these jobs are mostly held by women worldwide (e.g., customer service and administrative roles) [444, 458]. Indeed, one study estimates that administrative positions will account for 60% of jobs lost in the US to AI automation, with 70% of the clerical and administrative workforce being women [170, 334].

This disproportionate impact is thought to result in part from societal expectations pushing women towards roles that value interpersonal 'helper' skills and away from STEM-related fields [167, 170, 232, 272, 381, 400]. These expectations also drive a lack of investment in STEM and tech literacy education for girls and women worldwide [3, 449], making it more difficult for women to meet the requirements for the jobs that will (theoretically) remain in an AI-integrated workforce (e.g., AI developers) [3, 449]. In sum, **concerns about AI's impacts on women's working lives assume that companies will be *incentivized to replace* human workers with AI and that this *replacement* will disproportionately target women workers.**

However, this assumption can feel incongruent with many of the recommendations by influential management consulting firms and scholarly research alike, which collectively focus on AI integration into existing workforces for worker augmentation or task redistribution (i.e., types of human-AI interaction) rather than outright replacement [42, 88, 113, 134]. It makes sense to employ a narrative focused on human-AI interaction for several reasons. For example, early users and adopters of generative AI systems like ChatGPT quickly found that the information it provides and generates is not always accurate or unbiased [127, 431], leading some companies to ban rather than embrace generative AI's use at even a basic level, much less replace human workers with it [169, 263]. Instead, when designed well, human-AI interaction

can reap the benefits of AI-assisted decision-making (e.g., exponentially faster analysis of massive amounts of data) while reducing inaccuracies and biases via human oversight [15, 146, 283, 351].

Additionally, while generative AI’s human-like qualities have expanded AI’s potential use, many duties and positions within a workforce require a deep understanding of the human experience and social practices that AI can ‘hallucinate’ but never truly know ([283, 316]. Again, well-designed human-AI interaction can foster better outcomes by delegating routine duties to AI, thereby freeing up human time and energy for more complex and creative tasks [177, 351]. In sum, **companies might be more incentivized to implement *human-AI interaction systems*, rather than AI replacement, demonstrating that the underlying assumption driving concerns about disproportionate harm against women workers is *not always met*.**

This dissertation wonders, though, if these top management consulting firms might be pushing a more human-AI interaction narrative as a strategic choice to **reduce potential economic fallout and backlash for themselves and their clients in the short term *without* ever having to formally commit to *never replacing* human workers in the long term.** On the issue of economic fallout, and given the relative novelty of generative AI use in the workplace, replacing human workers without first testing the long-term viability of AI integration into a workflow practice through prolonged use would be exceedingly risky.

Indeed, companies that engaged in the pre-preemptive firing of humans to make room for AI integration are starting to feel the economic fallout associated with that decision. For example, Spotify’s CEO acknowledges that the company’s performance and output have suffered greatly after mass layoffs, which were carried out in the name of “efficiency,” coinciding with Spotify’s shift towards AI [238, 258].

This is partly because they failed to properly understand the contributions made by those human workers and, therefore, implemented an AI system that was not designed with the workplace’s personnel needs in mind [197, 238, 258, 316, 505]. In contrast, the implementation of human-AI interaction can initially be a good way to explore how AI can benefit specific workplaces and their processes without incurring too much risk to the company in case the AI integration fails [366].

Regarding public perception, even signaling the intent to replace human worker roles with AI has been shown to foster massive public backlash, for example, IBM’s 2023 announcement that it plans to ‘hire’ AI for over a thousand roles originally slated for human workers [315, 382]. Studies have shown that this type of move has demonstrably negative impacts on perceptions of managers, workers’ sense of job security, and even on third-party observers’ (e.g., the public’s) perceptions of AI’s ethical use in the workplace, thus pushing people toward opinions that favor governmental regulation over private use freedom [277]. With these considerations in mind, we further explore existing efforts to incorporate Human-AI Interaction in the workplace in the next section.

2.1.2 Human-AI Interaction (HAI) & its Application in the Workplace

To better understand the impacts of AI integration on women in the workplace, it is essential to consider the factors that should be taken into account when planning the implementation of AI in the workplace in general. This dissertation argues that there are two foundational categories of AI workplace integration considerations: **(1) *technical design considerations***, and **(2) *integration design considerations***. For **(1)**, *technical design considerations* refers to how what a workplace needs and

desires from a human-AI interaction drives how an AI is conceptualized and built for said workplace. For **(2)**, *integration design considerations* can best be thought of as the policies crafted, implemented, and evaluated by key decision-makers within the specific workplace that seek to design how people interact with and use AI.

These two categories of considerations intermingle bidirectionally, such that the technical design of preexisting AI systems and models can inform integration design considerations, while integration design considerations can inform the customization of AI systems' technical design to fit specific workplace needs.

Technical Design Considerations for Human-AI Interaction in the Workplace

At a high level, the technical design of an AI system for workplace integration primarily depends on how the AI system is designed, built, and/or selected from pre-existing models in consideration of the workplace's specific needs. This dissertation identifies three primary technical design considerations that workplaces must consider when integrating AI: the level of autonomy an AI possesses, the embodied and behavioral presentation of an AI, and the type of AI model that best suits the contextual needs of the workplace in question.

Level of Autonomy. One of the most critical technical design considerations for how to design an AI for workplace integration is determining the level of autonomy (LOA) the AI needs to operate at, as these levels form the basis of how AI can operate in conjunction with humans in the workplace. At a high level, an artificial agent such as AI can be considered autonomous when it has either partial or a high degree of self-governance when it comes to making decisions and executing subsequent actions and has some level of adaptability and communication abilities [114, 325, 336, 353]. More specifically, this dissertation utilizes the guiding framework of LOA as detailed in O'Neill et al.'s [353] adaptation of Parasuraman et al.'s [356–358]

Levels of Automation (LOA) continuum, with the adaptation being the addition of Agent Autonomy Level categories. It is essential to note that the distinction between automation and autonomy can be somewhat murky and is arguably a moving target as AI continues to experience exponential growth [320, 357]. However, researchers do seem to agree that they can be distinguished from each other [110, 212], namely by the agent’s ability to govern itself (i.e., a necessary component of being perceived to have agency) [353].

Moving back to the LOA framework, it consists of 10 total Levels of Autonomy that can be clustered into groupings of manual control (Level 1), “automation (autonomy) as a tool” (Levels 2-4), partial autonomy (Levels 5-6), and high levels of autonomy (Levels 7-10) [136, 353, 356]. First, starting with the lowest levels of automation, Level 1 is when a computerized agent acts with no assistance to the human and must be manually controlled at every decision point, or is restricted in their capabilities to a narrow range of actions designed to help humans widen or narrow down their decisions (Levels 2-4). O’Neill et al. explain that these levels are unlikely to be perceived as agentic and thus autonomous because they lack the ability to make decisions independently or operate without human direction [353]. This matters to the classification of what type of HAI an organization is looking to incorporate, as humans are less likely to view an automated system as a teammate if it lacks agency [298, 299, 353]. In fact, in a later article revisiting the LOA model in light of the ever-expanding body of HAT research, O’Neill et al. recommend using a Human-Automation Interaction rather than a Human-AI/Autonomy Interaction perspective when conducting research with agents who fall at Level 4 or below in their capabilities because of the perceived lack of agency such systems have [353].

Second, partial autonomy occurs at Levels 5 and 6, wherein the agent does have the ability to generate and execute decisions, but is restricted by the need for human

approval prior to execution (Level 5) or can be vetoed by humans when the agent communicates its intentions (Level 6) [353, 357]. Here, there is a level of proactivity and decision-making that can lead humans to perceive the system as having some agency [325, 353, 481], but the agent falls short of high levels of autonomy because it still requires some human input to operate. It is also at this point in the continuum that O’Neill et al. recommend that researchers utilize a Human-Autonomy Interaction framework when conducting research with agents [347].

It is essential to note that there is a second factor, the agent’s role, which determines whether a research study should utilize more specific types of HAI-adjacent frameworks, such as Human-Autonomy Teaming [347]. This aspect of the continuum will be explored in greater depth later in this section when discussing integration design considerations, as role assignment goes beyond just the technical design of an AI system by encompassing both human *and* AI management and social dynamics.

Finally, high levels of autonomy (Levels 7 - 10) are characterized by agents that act proactively to make *and* execute decisions without the need for human input or intervention [353, 357]. Looking at the levels with more granularity, prior work has pointed out that there is a natural division between partial autonomy at Level 6 and high autonomy at Level 7 because agents at Level 7 execute actions prior to communicating with humans, while agents at Level 6 allow humans to veto a decision within a given period of time prior to executing the action [353, 356, 480]. Indeed, for agents at Levels 8 and above, agent communication with humans is minimal, from agents that only communicate with humans about their actions when prompted (Level 8) or when the agent chooses to (Level 9), to complete autonomy in their actions without ever communicating with humans at Level 10. It is arguably the case that excessively high levels of autonomy create a situation where agent use begins to stray away from HAI and its adjacent research paradigms (e.g., HAT research), as

there is little to no interaction between humans and agents. From a technical design perspective, however, an agent at any of these higher levels could still operate within an HAI application, depending on its role within that HAI system, which will be discussed in greater detail later on.

Overall, for the purposes of this dissertation, the technical design consideration of levels of autonomy serves as one of the most critical design considerations, as it is a key determinant of the types of interaction that can exist between a human and an AI. Prior works have also highlighted other important interaction considerations that are informed by an agent’s level of autonomy, such as the role that autonomy levels play in designing for safer AI teammates in contextually risky situations [220], and how autonomy levels and AI explainability interact to form team constructs (e.g., trust) [219].

Moreover, as AI development becomes increasingly more technically complex and advanced, researchers such as Hauptman, Schelble, and McNeese [221] have drawn attention to the idea of adaptive autonomy, or the notion that an AI should be designed to monitor a situation and change how autonomous it is in its decision-making and action execution based on contextual information. More specifically, they argue that adopting an adaptive autonomy approach to design AI teammates is critical for enacting ethical decision-making in human-AI teams given two notions: that people tend to attribute the ethicality of an AI’s decision-making to its level of human oversight/involvement [13], and that ethical codes within a team are formed through dynamic team interaction over time [157]. While this dissertation does not seek to study or make contributions to adaptive autonomy research, it is still important to acknowledge this growing body of work because those who seek to integrate AI into their workforces will likely need to consider both the level of autonomy they need an AI system to have and if that level needs to be static or dynamic across time and/or

contextual needs.

AI Embodiment. Another critical technical design consideration for AI workplace integration is how the AI system is designed to interface with the user, especially in terms of its embodied characteristics [227]. Within the realm of AI design, embodiment essentially refers to whether the AI system presents itself to the user in a way that has some kind of body characteristics attached to it, be it physical or digital [86, 285, 295, 324].

There is a wide range of possibilities when it comes to the embodied characteristics of an AI. On one end of the spectrum, there would be no embodiment whatsoever, e.g., algorithmic decision-making that human users interact with via a digital UI/UX [287]. On the other end of the spectrum is the fully physicalized and embodied, as seen in Human-Robot Interaction (HRI). HRI as a field of study has thrived for decades [109, 199, 419], pre-dating the relatively more recent study of HAI, although there are overlapping principles of interaction in each. What fundamentally distinguishes HRI from other types of HAI, though, is that robots necessarily have a physicalized presence and body that need to be accounted for when investigating how humans interact with them and their applied usage [36, 206, 414]. For example, Boston Dynamics’ Spot is a robotic system that has various contextual uses and considerations due to its embodied design and capabilities, such as supporting LiDAR scanning on active construction sites [473].

It is also important to note that much of the HRI literature focuses on systems that cannot be considered autonomous or AI, but rather constitute automated systems or machines [82, 262, 414]. However, when this dissertation discusses HRI literature, it refers to literature where robotic systems possess at least some level of autonomy. In between these two different examples is a wide variety of other types of embodiment, such as AI assistant voice embodiment (e.g., Siri, Cortona) [214, 239];

textual embodiment through tonality in writing, such as what can be seen in chatbots that utilize Natural Language Processing models [68, 241]; and digital embodiment, which can constitute a wide range of sophistication from simply giving an avatar to an AI assistant [493] to advanced embodied conversational agents (ECAs) [71, 409].

When it comes to workplace integration, the embodied design of an AI system must be carefully chosen based on the contextual needs of the workplace in question [451]. For example, research related to HRI and cobot integration into workplaces primarily focuses on contexts where there are explicit needs for embodied systems to perform physicalized tasks on behalf of or alongside human workers [471], such as manufacturing [78, 314, 486], customer service [46, 83, 427], and construction [57, 286, 473, 496]. However, in workplace contexts such as business management, prior works have investigated more digital forms of embodiment, including the use of AI-enabled chatbots in digitized workplaces [186, 260, 302, 462]. These examples illustrate how different levels of the embodiment of an AI are often driven by the needs of the workplace in question, such that a physicalized robotic system would likely be unnecessarily complex and expensive in a workplace where a digital AI would suffice, while a digital AI would not serve the physical labor needs of a manufacturing or food service role.

While this section primarily focuses on the practical considerations of embodiment, there is a wealth of research on other factors related to how an AI presents itself to users and its impacts on important user perceptions and behaviors, such as the role of emotions [18, 117, 185, 291, 304]. For example, Gkinko and Elbanna investigated employees' emotional experiences when working with an AI chatbot designed to be conversational and learn about the user, thereby establishing a social presence [185]. They found that users' evaluations of the chatbot were influenced by its form and functional design, and that these factors elicited a wide range of emotional responses,

engendering a sense of social connection with the chatbot. Conversely, embodiment can create complications with user acceptance when designed poorly or in a way that creeps into the Uncanny Valley territory, i.e., the creeped-out feeling humans get when a non-human entity tries to mimic being human but falls short [330].

The aforementioned examples offer a small glimpse into the broader context surrounding the embodiment of technology, AI, and non-human digital entities, which have been extensively discussed and studied in HCI/HCC works for decades. For this section of the dissertation, though, the most important takeaway here is that the embodied form of an AI is a technical design consideration that AI integrators must consider when thinking about how the design of an AI system should be informed by their workplace needs and worker perceptions.

Model Type. Finally, the last technical design consideration is perhaps the simplest but crucial component, i.e., determining what kind of AI model is most appropriate for the workplace context in question. Indeed, prior works have demonstrated that organizational readiness for successful AI integration is driven by an awareness of AI technology, how AI-based technologies function, and where they can be most effectively employed [237, 247]. Much like the embodiment consideration, different types of models serve different functions that are better suited for certain types of work and workplaces than others. For example, Natural Language Processing (NLP) models are frequently used in contexts where the AI needs to be conversational, such as customer service support chatbots [179, 274, 376, 494] and Personal Virtual Assistants (PVAs) [53, 67, 362, 369, 415]. However, if an organization’s needs call for, say, algorithmic detection of radiology scans [73, 215, 345], then an NLP model would be an inappropriate model type to choose as it is not designed for that type of image scanning and detection.

This dissertation’s focus is on generative AI, which leverages Large Language

Models (LLMs) to process, organize, and generate large amounts of information and content, making it best suited for types of work that need those skills, e.g., administrative work [45, 134, 383, 478, 491]. While this dissertation does not seek to dissect model types within workplace settings, it is still important to acknowledge the role that model type fit plays in the technical design consideration of AI integration into workplaces.

Integration Design Considerations for Human-AI Interaction in the Workplace

As a reminder, *integration design considerations* can be thought of as the policies crafted, implemented, and evaluated by key decision-makers within a specific workplace that aim to design how people interact with and use AI. More specifically, this dissertation argues that integration considerations can be broken down into three intertwined considerations: managing how workers will interact with the AI, how the AI will be designed to interact with humans, and how the human-AI unit will be classified and will operate (e.g., a tool versus a collaborator versus a teammate).

Managing Human Interaction with AI. One primary integration design consideration that must be considered is how the organization plans to manage human interaction with AI. More specifically, this dissertation argues that there are two primary subcategories of human management here: (1) ensuring that humans are well-equipped for AI integration by managing human expectations for AI and how integration is framed to workers (e.g., [156]), and (2) managing the ethical use of AI in the workplace, which involves questions of privacy and bias (e.g., [231]).

For (1), prior works on HAI and HAT have highlighted various factors that impact people’s perceptions of AI use that can shape worker AI adoption willingness if not properly accounted for [490], such as participants’ pre-existing attitudes towards AI and prior experiences with human collaboration [501] and participant un-

derstanding of the specific AI technology they would be working with (e.g., chatbots [186]). Additionally, multiple studies have highlighted how an imbalance in workload distribution responsibility that favors the AI over the human in a proposed workplace human-AI interaction scenario can induce feelings of AI identity threat, job insecurity, and adoption uncertainty in participants [156, 328]. As such, organizations are motivated to deliberately craft integration design with AI capability expectation setting, utility and use case justification, and human perception management in mind [156, 234, 371, 501]. Prior works have highlighted a variety of ways to accomplish these management objectives [6, 479, 488], such as giving human workers first-hand experience with the AI they will be working with prior to full HAI deployment via practice use [152] and/or training programs [145, 276, 479].

Even the way that an organization talks about and frames the AI’s role, regardless of its actual capabilities, can shape worker perception. Indeed, Flathmann et al.’s study on the purposeful presentation of AI teammates found that simply framing the AI as a “tool” (versus a “teammate”) can have small benefits to workers’ perceptions of job security and adoption, despite each AI condition being identical in workload responsibility and capabilities [156]. Overall, then, the management of human expectations for AI plays an important role in successful HAI workplace integration, although more research working in applied settings to uncover AI integrators’ real-world experiences and perceptions is needed to gain a holistic picture.

For (2), managing the ethical use of AI in the workplace is a critical yet complex component of integration design consideration, both as a means of preventing unintended negative consequences of AI use [373](see section 3.2 for more on attributing responsibility for AI outcomes) and to promote positive outcomes such as worker well-being [40, 128, 231, 306]. Indeed, it is difficult to overstate how critical discussions of ethical design and the use of AI are, prompting a wealth of research

exploring different theoretical perspectives for prescribing and enacting ethics within HAIs (e.g., [6, 12, 17, 231]) and HATs (e.g., [157, 296, 360, 406]) work. At a high level, though, these works collectively point to the need for organizations to establish guidelines and principles that lay out what is considered ethical behavior for both humans and AI, given the organization’s ethical values and contextual uses for HAI [16, 157, 211, 231, 374].

Of particular concern to the use of generative AI in workplaces is the ethics of data privacy [76, 142, 196], fair use [335, 393, 430], and content attribution in LLM usage [23, 178]. These concerns are deeply contextually driven and require a level of top-down policy creation and enforcement that is likely driven by the expertise of organizational leaders in privacy, security, and governance, such as AI governance consultants. Although this dissertation does not specifically aim to contribute to the creation of ethical frameworks for AI development and use, it is valuable to acknowledge the role organizations play in managing human interaction with AI, particularly in light of ethical concerns such as privacy and algorithmic bias.

Designing AI to Interact with Humans. While the technical design considerations covered earlier in this section referred to how the AI is developed to operate effectively within the workplace context, the integration design consideration of AI design focuses explicitly on how AI is designed to interact with humans in ways that make it easier for humans to accept AI integration into their work processes and capitalize on its benefits. There are, however, certain design functions that operate as both technical and integration design considerations, e.g., utilizing models that allow for conversational AI chatbots to learn over time the communication patterns of individual users in order to better align its responses to the needs of the individual in light of organizational values [186].

Focusing more on the HAI dynamic arguably requires a shift towards thinking

about how well-known concepts in human-human collaboration and teaming literature inform how AI is designed to work with humans [155]. In other words, when interacting in a collaborative capacity with AI, human mental models draw upon what we already have experienced and know about working with other humans, including what we desire and expect out of a teammate [154, 323, 501]. As such, research on perceptions of AI teammates in HATs have highlighted that humans expect a level of human-like behaviors, communication, and performance from their AI teammates [501], and that these expectations inform AI-designed communication style [499], preferences for levels of explanation in AI decision-making depending upon contextual and individual need for cognition [405, 407], and how individuals perceive and respond to the designed social etiquette of an AI [154].

However, research also warns that the specific nature of said design preferences is highly contextual to the situation and to individual differences in users themselves [154]. For example, Zhang et al.’s study found that the same level of AI-provided explanation that built trust in users after the AI made a mistake lowered trust when the AI lied to the user [500]. Overall, then, organizations and AI integrators must carefully understand the social dynamics and needs at the organizational *and* individual level in order to get a more usable picture of how to design AI to interact positively with human workers.

Classifying and Operationalizing the Human-AI Interaction Dynamic. Finally, the culmination of all of the prior technical and integration design considerations for HAI integration into workplaces forms the basis of the final integration consideration, i.e., classifying and operationalizing the HAI dynamic that best fits the needs of an organization. The remainder of this section thus breaks down this consideration into its two components of (1) classifying the type of HAI that is possible depending upon the interdependent nature of the collaborative work in

question in addition to the AI’s level of autonomy, and (2) operationalizing the HAI dynamic based upon its classification in combination with the social and functional role the human and AI fulfill within the dynamic.

For (1), classifying the type of interaction between AI and humans when working together comprises a dual evaluation of the level of autonomy the AI is designed to have (as previously mentioned earlier in this section) *and* the level of interdependence that exists between the human and AI when working towards a shared goal [347, 353]. In simple terms, interdependence in collaborative work is when each member of a team occupies a distinct and necessary role on a team [278, 279], with each member of the team needing to interact and coordinate with the others to varying degrees to perform their role and/or to achieve the shared outcome [347, 443]. Applied to computerized systems, seminal research from Nass et al. demonstrates that interdependence is a critical determinant of whether an agent can be perceived as a teammate by humans [339, 347].

Bringing interdependence and levels of autonomy together, it is helpful to remember that differing levels of autonomy correspond to different types of AI agents, classified based on their ability to operate independently of human intervention [353, 356, 357]. Agents whose capabilities fall under the Level 5 mark of the LOA continuum are best classified as tools because they lack the necessary proactivity and independent operation that is needed to be considered agentic. As such, O’Neill et al.’s framework for guiding future research on HATs recommends utilizing a Human-Automation Interaction perspective when conceptualizing interaction dynamics for such agents [347]. While there is some contention between scholars regarding if AI must always be classified as a tool, regardless of its LOA (e.g., [421]), this dissertation follows the prevailing narrative that there are distinct characteristics of AI design and interaction design that separate AI tools from AI teammates, and, furthering these perspectives,

this dissertation argues that these differences are perceivable to those who work with it.

Moving up the continuum, AI agents at Level 5 and above have the necessary components to constitute partial or full autonomy. Here is where interdependence plays a role in differentiating between generalized Human-Autonomy/AI Interaction (non-interdependent) and Human-AI Teaming (interdependent) [347, 353]. More specifically, in order for a human-AI interaction to be classified as a team, there must be at least one “autonomous agent” (Level 5 and above) that collaborates with humans *and* fulfill a distinguishable role on the team that has the agent(s) and human(s) work interdependently towards a shared goal [347]. Other types of human-AI interaction that are common to study in the context of workplace integration, such as Human-AI Collaboration/Cooperation (e.g., [131]), fall short of the classification of “teaming” because either there is a lack of clarity about what unique roles each member serves on the team [319], or the roles are distinct but the activities of each member do not interact with the actions of the other members (i.e., not interdependent) [353].

It is important to make these classification distinctions because the use of HATs in collaborative work has been shown to be particularly advantageous in creating better performance and overall outcomes than human-human teams [323], and is continuously projected to be the dominant human-AI interaction dynamic in the future of work [441]. The specific type of human-AI interaction should be driven by the contextual needs of the workplace in which it will be incorporated. Moreover, the nature of the dynamic is simultaneously shaped by and reinforced through its operationalization in the workplace, leading to the final component of this integration design consideration, which is discussed in the following passages.

Additionally, operationalization of the human-AI dynamic can best be thought of in terms of the roles that AI and humans are made to occupy within the dynamic,

and how these roles impact perceptions and outcomes of the dynamic, and can be social (e.g., relational norms as seen in [131], functional (e.g., tool versus teammate and level of autonomy in [347]), or workload distribution in [156], or a combination of the two (e.g., the framing of tool versus teammate in [156]) in nature. The contextual designation of roles in a human-AI interaction can drive important perceptions and outcomes. For example, perceptions of who is responsible for biased outcomes in human-AI collaborative decision-making are impacted by who (AI or human) occupies the role of a decision-generator versus a decision overseer [418].

Viewing integration design considerations through this human-AI dynamic lens is particularly beneficial for organizational integrators, as the ever-evolving nature of AI technologies makes it nearly impossible to craft policies that account for every type of current and future AI technology. For example, Earp et al. argue that viewing human-AI cooperative endeavors through a relational norm perspective could aid in the creation of AI use policy and governing by focusing on what types of relationships (e.g., romantic) between humans and AI should or should not be restricted rather than focusing on regulating the virtually endless list of types of AI technologies [131]. Overall, then, careful consideration of the operationalization of an HAI dynamic for workplace integration is especially necessary as AI continues its exponential evolution.

However, despite these efforts and promising approaches to design, integrate, and manage human-AI interaction in the workplace, women workers may face unexpected and more severe risks due to gender bias in AI design, which this dissertation argues is a symptom of a broader problem with sexism in our modern society. The following literature review section delves deeply into this dissertation’s approach to interrogating gender bias in AI design and development, with the understanding that pervasive sexist ideology both shapes and is shaped by workplace dynamics and technology design.

2.2 Women’s Marginalization in AI Design as Symptom of a Broader, Sexist Problem

To gain a clearer understanding of how women’s marginalization in AI design can be viewed as a symptom of broader societal issues related to sexism, this section will first outline the theoretical underpinnings of ambivalent sexism [188] and its operationalization in workplaces. Then we will explore how ambivalent sexism may manifest as challenges for women in the AI development workforce, as well as within gender bias in AI design, limiting how AI workplace integration can currently be designed to support rather than harm women workers.

2.2.1 Applying Ambivalent Sexism in Everyday Workplace Contexts to Unpack Women Workers’ Challenges

Sexism has been a widely recognized prejudice across cultures and societies since the 1970s and 80s (e.g., [37, 379, 429]), and was often attributed to inter-group conflict between the genders that gives men dominant sociopolitical power over women [27, 66, 424, 477]. In heteronormative societies, however, there is a high level of mutual dependency and cooperation between men and women to meet certain sociocultural needs [150, 244, 385, 477]. Recognizing this, the **Ambivalent Sexism Theory (AST)** and its associated Inventory (ASI) were developed by prominent social psychologists Peter Glick and Susan Fiske to account for both perspectives [188, 191].

At a high level, AST is made up of two complimentary components: *hostile sexism (HS)*, or the belief that women actively seek to steal power from men (i.e., competition), and *benevolent sexism (BS)*, or the belief that women and men are

fundamentally different in capabilities and in the roles they are meant to fulfill in society (i.e., cooperation), such that it reinforces women’s continued marginalization [188]. AST has since been solidified as a valid means for conceptualizing the beliefs that drive gender bias and discrimination across cultures, continents, and generations [27, 92], making it uniquely beneficial for contextualizing the marginalization that women face and are continuously exposed to in the workplace and through AI development and design

In order to fully understand the extent of AST’s use in research over the years and across domains, authors Orly Barket and Susan Fiske (i.e., one of AST’s founders) conducted a systematic literature review synthesizing 654 peer-reviewed articles focusing on AST and the use of its associated inventory from 1997 to 2021 [27]. Three major takeaways from this review inform the use of AST in this dissertation to theoretically inform the investigation of how women’s marginalization shapes the design, evaluation, and workplace integration of AI.

(1) AST’s *well-reasoned theoretical development* and ASI’s *persistent use as a valid measure across cultures and majority/minority populations* make it a suitable theoretical foundation to use to contextualize the marginalization that women face and are continuously exposed to through AI development and design. More specifically, the ASI continues to be the most widely used measure of sexist attitudes among gender and sexual majorities (i.e., men/women, heterosexual) and minorities (i.e., LGBTQ+; especially when modified to exclude heterosexual references [97]) to date [101]. For example, several studies have utilized AST and associated ASI to study sexist beliefs amongst gender/sexual minorities in comparison to gender/sexual majorities and have found that the former often (but not always) display less ambivalently sexist attitudes compared to the latter (e.g., [51, 119, 465]) [101].

(2) AST understands that *sexist attitudes can and do exist even within those who are most marginalized by its ideology, particularly women.*

This makes it a useful theory when dealing with gender dynamics in research because it understands that identifying as a woman does not inherently mean that one’s views would support the consideration and uplifting of other women. For example, benevolent sexism (BS) characterizes women in a subjectively positive light, “*warm, pure, and supportive, who therefore deserve men’s protection, provision, and admiration,*” [27]. It is therefore thought that women can hold and embody BS beliefs as a way to (un)consciously gain the material benefits [27, 148, 193] and, more critically, the psychological safety bestowed upon them by this protective paternalism in situations (e.g., endorsing BS beliefs as a response to perceived threats against women [106]).

Moreover, the positive nature of BS stereotypes makes it difficult for people to perceive these beliefs as sexist and thus problematic in the first place [11, 30, 307, 332]. Without the ability to recognize the problem, women who endorse these beliefs are thus more concerned with embodying them rather than challenging them [307, 332], in turn reinforcing gender stereotypes for themselves and others [187]. The nuance AST provides, then, is arguably more reflective of the reality of the current US environment, e.g., instances of conservative white women playing an active role in their own gender’s subjugation via support of far-right conservative values. In this sense, this dissertation’s use of AST allows for both scientific and real-world contributions.

(3) AST has been associated with a variety of factors related to *workplace discrimination, bias, and aspirations that affect how women are evaluated by others and how women evaluate themselves.* As such, AST provides theoretical insights to overlay existing understandings of women’s participation in AI development and design (Study 1), as well as women’s participation in AI

integration (Study 3). Moreover, combined with AST’s robust capacity for measuring sexist attitudes, AST is well-positioned for further exploration on how sexism may moderate the evaluations of women who work with AI (Study 2).

Now that the groundwork has been laid for what AST is and how it becomes internalized in those it marginalizes, it is time to turn toward understanding how it manifests in workplaces. Ambivalent sexism’s relationship to women in the workplace can best be broken down into categories of workplace harassment (HS), gender bias in evaluations of women workers (BS), and expectations of what constitutes women’s work (BS).

Before we begin, it is important to note that this dissertation’s use of the term **workplace** generally refers to **office-type work (e.g., banking, administrative workers, business management)**, which is in line with Bareket and Fiske’s [27] 2023 metareview’s operationalization of the term. This is important to reiterate because there are certain sectors of work, such as the healthcare industry, where general sexism intermingles with specific workplace considerations to create more nuanced issues for women workers (e.g., the lack of research on women’s health impacting women’s health outcomes [85, 236]). While future work should expand upon this dissertation’s work by looking at more sectors, such as the healthcare industry, this dissertation focuses on the dynamics of office-type workplaces and positions as a foundational starting point.

Hostile Sexism (HS) & Discrimination, Harassment in Everyday Workplaces.

At a high level, hostile sexist beliefs have been linked to a wide variety of discriminatory attitudes and behaviors related to women in the workplace. HS generally predicts negative attitudes towards women’s abilities to succeed within everyday workplaces [87, 396] and is associated with acceptance of workplace gender inequality

[93]. More specifically, Bareket and Fiske [27] explain that HS is associated with women being perceived as less competent [165, 198, 381] and hireable [248, 464] than their male counterparts, while HS is associated with positive perceptions of and preferences for men in the workplace [235, 280, 313, 401]. Overall, then, hostile sexism plays a clear role in discriminatory practices and organizational environments that systematically devalue women workers as less competent and therefore less capable of succeeding, especially in more 'masculine' working environments [396].

Moreover, although this dissertation does not specifically focus on harassment in the workplace, it is still important to acknowledge the connection between hostile sexist beliefs in the workplace and harassment. This is because it provides insights into the power dynamics present in mixed-gender workplaces, which are often driven by competition and the need to get ahead of one's coworkers on the climb to the top. Previous work on the impacts of ambivalent sexism in the workplace has found that hostile sexism plays a role in the tolerance and perpetration of explicit workplace discrimination and harassment against women [38, 118, 229, 354]. In cases of workplace sexual harassment, the existence of HS beliefs (more than BS) among both men *and* women predicts the blame of women victims (i.e. labeling their behavior inappropriate for the workplace [201]), and is associated with the trivialization or refusal to acknowledge the existence of such workplace harassment as it is happening [228, 259, 346, 412].

In this sense, HS beliefs in the workplace arguably serve to discredit and undermine women workers who dare to stand up for themselves in the hopes of weeding out said women. Indeed, experiencing hostile sexism in the workplace has been linked to decreased satisfaction in workplace engagement and organizational support for women workers [116, 329], which could encourage women workers to leave the workplace entirely. This, in turn, reduces the perceived threat that such women pose to

the patriarchal status quo, thus elevating men and conforming women in comparison, which could explain the existence of said beliefs in both men and women.

Indeed, while the occurrence of hostile sexist beliefs in women is less well understood, there is some thought that women who display hostile sexism when evaluating other women do so as a competitive maneuver to elevate themselves in comparison, especially within the workplace [124, 139]. This especially makes sense within a workplace context where there are limited seats at the metaphorical table for women in positions of power [102, 149, 492]. Hostile sexism within women arguably embodies the ideology that *other* women are conniving and duplicitous, using their sexuality as a weapon to get ahead of others, but not *me* - the “*I am not like other girls,*” mentality, or a level of psychological entitlement to better treatment [98]. There are arguably undercurrents of self-preservation within these beliefs as well, specifically in the way that it positions the believer as inherently better and therefore safer than the women they look down upon.

More specifically, it is theorized that HS beliefs in women do not necessarily reflect hostility towards *all* women, but rather against women who fail to conform to patriarchal norms, such as women who are sexually empowered [138] and feminists or career-focused women [33]. It is especially telling that hostile sexism in women is most often associated with judgments about one’s own sexualization (e.g., [164]) and other women’s sexual activity (e.g., [492]), victim blaming and disbelief in cases of sexual harassment against women in the workplace (e.g., [201, 475]), and endorsement of related beliefs such as rape myths (e.g., [164]). In this sense, HS arguably acts as a way to psychologically distance oneself from the possibility of violence and harm by giving the believer a (false) sense of control over her situation.

Benevolent Sexism (BS) & Gender Bias in Evaluations of Women in the Workplace.

Benevolent sexism is most often linked to less overt but still impactful forms of gender bias in everyday workplaces by subtly justifying and reinforcing workplace gender inequity in ways that are “*paternalistic yet conditional*” [27] towards the women who are already in the workplace. In simpler terms, BS does create more favorable outcomes for women in the workplace than HS does, but these outcomes actually serve to continuously marginalize women in the workplace. For example, male supervisors who hold BS beliefs are less likely to distribute challenging career development experiences to women workers to “protect” them from challenging work [264]. This implicit communication of BS beliefs at the organizational level can have significant negative effects on women workers’ self-esteem [463].

These issues are compounded in women who endorse BS beliefs because such women have been shown to be less likely to seek out managers for advancement in leadership positions [233]. Moreover, these “benefits” apply only to women who conform to the societal expectations set forth by BS. For example, women who conform to BS standards are evaluated by others as more agreeable and cooperative within the workplace [251], making them more endorsed for leadership positions [8] when evaluated by mixed-gendered teams [194]. Conversely, failing to uphold such standards within a workplace setting can result in negative evaluations of women who exhibit counter-stereotypical behavior such as assertiveness and confidence [317, 361, 397].

This dynamic is made more complicated when women and men work collaboratively because benevolent sexism also perpetuates expectations for men (the “stronger” gender) to protect and help women (the “weaker” gender) within the workplace [420]. For example, Becker et al. [34] found that women who accepted help from men in a workplace setting that was more competence-based (e.g., management consulting) were evaluated by study participants as being warm and caring (i.e., likable) but less competent at their jobs. In contrast, women who did not ac-

cept help from a male colleague were evaluated by participants as being cold and impersonal, and therefore not well-suited to more warmth-based occupations, such as daycare workers. Yet, the male colleague was evaluated equally favorably in both conditions [34].

This demonstrates the double bind that benevolent sexism puts women in when working with men: Workplace performance metrics directly linked to perceptions of likability (e.g., easy to work with, cooperative, team player) and competence (e.g., good at her job, knowledgeable, reliable) often cannot be satisfactorily met simultaneously for women workers when working in the same workplace as men. This can be an especially challenging dilemma at the organizational level, given the systemic boxing out of women in leadership positions [487]. For example, Gullen et al. found that self-confidence in high-performing women only led to organizational influence when these women also displayed a prosocial orientation, which is often associated with warmth dimensions, whereas high-performing men were able to achieve organizational influence without this prosocial orientation [203].

Benevolent Sexism (BS) & Gender Bias in Expectations for What Qualifies as Women’s Work.

An interesting exception to the prior section’s discussion on gender bias in evaluations is that women are often evaluated as more competent workers than their male counterparts when performing traditionally “feminine” work-related tasks (e.g., caregiving, administrative roles) [232]. This leads to a greater discussion about how BS beliefs work to undermine women even before they enter the workforce by setting expectations of what kinds of work women are best suited for, thereby influencing women’s career options and aspirations [167, 232]. More specifically, some studies suggest a link between BS beliefs in women and a lack of STEM participation, specifically that women temper their STEM-related aspirations because they view themselves as

less inclined towards STEM due to being women [272, 381, 400]. This message is arguably continuously espoused and reinforced by sources such as modern media trends (e.g., the TikTok trend of “girl math” [158, 399]) and implicit association biases in search queries, which assign STEM-related roles to men more often than to women (e.g., [365]).

There are even links between hostile sexism and STEM-related fields, such that men within STEM majors have often been shown to have more sexist attitudes than men in other types of majors [143, 272, 375, 400]. Taken in combination, BS and HS set women up to be continuously marginalized within STEM-related fields by preventing women from entering into said fields in the first place, either out of genuine but misguided belief in what work they are best suited for or as a protective measure against aggressive male presence in STEM fields.

This provides a smooth transition into discussing current understandings of women in the AI development workforce, specifically, gender bias in AI design, and how these concepts tie together with ambivalent sexism to motivate this dissertation.

2.2.2 How Ambivalent Sexism May Affect the Design, Evaluation, and Integration of AI in the Workplace

In this section, we further delve into how ambivalent sexism may affect the design, evaluation, and integration of AI in the workplace, especially in light of (1) women in the AI development workforce and (2) gender bias in AI design.

2.2.2.1 Women in the AI Development Workforce

The gender gap in Science, Technology, Engineering, and Mathematics (STEM) disciplines is a well-documented phenomenon [35], particularly in computing domains,

where women comprise only 25 percent of the computing workforce in the U.S. [168]. Extensive prior research has focused on investigating various factors that lead to the notable lack of women in computing and STEM-related professions, including gender-role associations from an early age discouraging young girls from associating STEM professions with women [252], thereby lowering feelings of computer self-efficacy in the long term [222].

Moreover, societal gender norms have been directly linked to being a driving factor for why women are, in numbers, less technologically oriented and less likely to pursue a STEM discipline [387]. Additional factors such as a disproportionate lack of STEM-readiness in high school [70], insufficient friend influence and peer exposure to STEM before secondary education [370], a lack of female role models [35], and gender-related stereotypes and biases [460] have been explicated in previous work to explain the STEM gender gap.

Even when women can overcome these barriers to entry during their primary and secondary years of education, women are more likely to leave the STEM profession because of a lack of advancement opportunities [442] and less family-friendly working hours and conditions [35]. However, how these issues of gender disparity in STEM, which is fundamentally driven by ambivalent sexist attitudes in women [272, 381, 400] and men [143, 272, 375, 400], may affect AI participation remains a question that motivates this dissertation investigation.

Unfortunately, the huge disparity between men and women in general STEM education is also significantly affecting the workforce of the emerging AI industry. Generous estimates suggest that women hold only 26 percent of data and AI positions in the workforce [162], and comprise a concerning 16 percent of tenure-track faculty globally whose focus centers on AI [450]. Additionally, as of 2019, only 22 percent of individuals pursuing AI and computer science Ph.D. degrees in North America were

women [113]. Furthermore, studies indicate that women comprise less than 12 percent of contributors to AI research conferences and only 10-15 percent of AI designers and developers at major technology firms [474].

Various reports examining the reasons for these gaps cite similar findings to previous research, such as a lack of role models and mentorship opportunities [113]. Here again, the ties of ambivalent sexism to reduced self-efficacy in women and girls regarding STEM proficiency are also a significant factor driving this divide [272, 381, 400]. A small body of work also explored how women in AI development experience marginalization. For instance, Strok uncovered a variety of issues with how women feel they are treated within the AI community: they perceive the AI community to possess a “boys club” atmosphere that isolates women and minimizes their opinions; they also consider this community more intolerant to women designers’ failures whereas men are given more leeway to try new things and innovate [435].

What is not known, however, is how said gender gap and women developers’ and workers’ perceptions of their marginalization affect their perceptions of gender bias and mitigation in AI design, which becomes a critically important research gap to fill when considering how the design of technology can become biased towards the perspectives of developers, and therefore often biased against women.

2.2.2.2 Gender Bias in AI Design

As the design and development of artificial intelligence increases in importance as the core driver of innovation in emerging technologies [297], so too have efforts within and outside of AI research communities to understand how gender biases are perpetuated by and integrated into AI in a way that creates large-scale social impacts. For example, facial recognition AI is a technological area that is often rife with gender bias [122], with many facial recognition AI systems being 99 percent

accurate when assessing white, Western males. At the same time, that percentage drops dramatically for women who are ethnic minorities [293]. Prates et al. [365] found that Google Translate defaults to using male pronouns more often than not when translating phrases such as “He/She is an engineer” from English to a gender-neutral language, especially in fields traditionally associated with unbalanced gender distributions or stereotypes, such as STEM.

Such instances of clear and systemic gender bias arguably obscure the professional accomplishments of women in STEM fields such as AI development, which may in turn hurt women’s self-development and self-worth [457]. Additionally, gender bias presents itself heavily in AI chatbot design (e.g., Siri, Alexa, and Cortana), such that the vast majority of chatbots are designed to speak, look, and act like women [141, 472]. This may lead to gender stereotyping of women as subservient due to the “commanding” ways in which we interact with these technologies [141, 391]. The aforementioned gender stereotypes in AI design are a one-to-one recreation of gender stereotypes associated with ambivalent sexism, particularly the dimension of benevolent sexism [188], although explicit acknowledgment of AST in HCC/HCI works is limited. For example, one of the few HCC studies to explicitly acknowledge benevolent sexism found that, in image search results, queries came back with more images of women that are associated with “warm” traits, such as “emotional”, and more agentic traits for men, like “rational” [349].

The increased awareness and recognition of gender bias in AI, evident in various versions of the technology, have thus prompted further investigation into the contributing factors leading to this bias and how to mitigate it. Much of this research has focused on the importance of diverse training datasets to avoid gender bias [305], as datasets can inherit the biases and prejudices of the humans who compile and use them [61]. Nadeem et al. [337] go further and classify several additional sources of

gender bias in AI, including a lack of diversity in data, developers, and programmers, as well as the amplification of existing gender bias in society by AI’s computational power.

This can be especially concerning in terms of AI tech that relies to some degree on gendered anthropomorphism for interaction (e.g., chatbots [141, 472]), as it means designers and developers are unconsciously building in their biases of what it means to exhibit and replicate human-like qualities [398]. Therefore, many researchers and practitioners alike argue that it is crucial for AI design and development teams to be diverse and possess a wide variety of backgrounds and capabilities [363].

Indeed, the lack of gender diversity in both AI workforces and AI design and development may lead to poor AI products that fail to cater to their users and instead exhibit harmful gender stereotypes [390]. This is because, generally speaking, gender is one of the most common ways to define user groups in the development of new technologies [455], and designers often have to guess who their users are and how they should design for them to save time during development [350], thus designers are inherently limited by their own experiences (i.e., with their gender identity) to inform their choices on how to design for and from the perspectives of women. As previously detailed in 2.1, AI development is overwhelmingly male-dominated. Therefore, even AI technologies that are assumed to be gender neutral generally favor the thought processes of the men who designed them [64], and are therefore often based on stereotypes and beliefs that men have about gender and women.

In this sense, it is crucial for AI design and development teams to be diverse, drawing from perspectives that are intimately tied to identity components such as gender. Yet, women are underrepresented in AI design and development in general, resulting in a lack of input from women regarding how AI technologies should be perceived, implemented, and approached. Consequently, there is also an inherent lack of

understanding of how women AI developers themselves consider the role their gender identity plays in their position within the AI industry and development process, and how such perceptions may affect their approaches to designing future AI technologies since most of the said tech does not reflect women’s perspectives and can therefore not be analyzed to make this connection. Answering these questions would be a vital step in understanding the role that gender identity plays - or, more crucially, should play - in AI design and development, which is essential to addressing the harmful effects of gender bias and assumptions in this rapidly growing technology area.

2.3 Summary of Research Gaps

In this section, we summarize existing research gaps to recognize the need to investigate women’s marginalization in AI design and development in light of societal ambivalent sexism and the push towards human-AI interaction in the workplace.

Overall, section 2.2’s extensive review of AST and its association with marginalization internalization in women, its impacts on women in the workplace, and its overlaps with gender bias in AI design and development clearly demonstrates AST’s internal and external validity for use in research on gender dynamics in the workplace in HCC/HCI. Yet, it is shockingly underutilized in HCC/HCI works focused on gender generally, but especially in relation to AI design. It is more often the case that HCC/HCI works related to social justice and feminist HCI design either do not explicitly acknowledge sexism as a central issue (e.g., [84]), or only do so in passing as a part of defining sexism (e.g., [413]).

HCC works that have utilized components of AST within HCC/HCI often do so when examining the occurrence and impacts of sexist beliefs within gaming spaces [54, 165], or when embodying sexualized virtual avatars [164]. When it comes to

HCC/HCI works on AI, though, there is an overall lack of explicit acknowledgment of ambivalent sexism and its underlying presence in gender bias in AI design and development.

When it comes to the study of women’s perspectives in development in HCC/HCI, Marsden et al.’s [307] interview study with developers on their perceptions of user personas for requirement analysis provides interesting insights into why utilizing AST specifically is important for accounting for sexism within women as well. Out of the 7 participants in the interview study, only one of the female participants mentioned wanting to highlight the needs of marginalized groups, while the other three women and three men focused on personas as a workflow tool based upon empirical data. In other words, Marsden et al.’s findings demonstrate that just because women are all a part of the same marginalized group of “women”, it does not mean that they equally prioritize or even recognize their own marginalization and how to prevent the marginalization of women through design [307].

The aforementioned study serves as a good example of why we cannot keep moving forward with the current trajectory of research into the impacts AI design will have on women workers without critically interrogating the core assumption that more women’s representation inherently equals better representation. This is especially important because the prevailing discourse surrounding AI’s projected impacts on women workers overemphasizes the need to upskill and reskill women workers, in part, to get more diversity into development to reduce gender bias. As such, the first research gap this dissertation seeks to close is as follows:

Research Gap #1: There are underdeveloped links between women’s representation in AI-related fields and their decision-making in the designing, planning, and/or implementation of AI.

Moreover, section 2.1’s extensive review of the concerns accompanying the rise of generative AI and the subsequent push towards human-AI interaction in the workplace demonstrates that popular discourse is already recognizing the various ways in which women workers might be uniquely negatively impacted by the integration of AI into the workplace. However, this discourse often places undue emphasis on women’s replaceability and the need to reskill/upskill women workers in response, without fully acknowledging the current trajectory of AI workplace integration, i.e., towards human-AI interaction rather than replacement.

As such, literature surrounding the projected impacts of AI on women workers often fails to bridge the gap between common concerns in generative AI integration about credit, with concerns about how women workers might be evaluated when working with an AI in a workplace human-AI interaction, in light of rampant gender bias in workplaces generally and AI systems specifically. Indeed, this dissertation argues that, if women workers are deemed replaceable with AI, it will only be after these interactions are evaluated and implemented, thus leading to the urgent need to close the following research gap:

Research Gap #2: There is currently a lack of clarity regarding how the introduction of AI into workplaces will impact human evaluations of the women who work with it.

Finally, section 2.1’s extensive review of the technical and integration design considerations involved in designing a human-AI interaction in the workplace provides a solid foundation for understanding the decision-making process for AI workplace integration. Overall, this body of literature emphasizes the highly contextual nature of this process, highlighting how the choices made by designers and implementers simultaneously shape and are shaped by organizational needs and practices.

Yet, it is unclear how, if at all, organizational gender dynamics are or should be taken into consideration when designing, planning, and integrating AI into the workplace. This is an especially glaring gap given the extensive literature demonstrating the disproportionate impacts that gender dynamics in the workplace have on women workers and on gender bias in AI design, as seen in section 2.2. In fact, this dissertation argues that addressing this gap is vital for uncovering the ways in which AI workplace integration can account for and preemptively mitigate harms to women workers, rather than continuously overemphasizing the reskilling/upskilling of individual women workers to mitigate these harms. As such, the final research gap that this dissertation seeks to fill is as follows:

Research Gap #3: There is an urgent need to understand and account for workplace gender dynamics to craft AI workplace integration, rather than overemphasizing mitigation strategies that place the burden on individual women workers to prevent harm.

Having identified these gaps, Chapter 3 will outline and explain how this dissertation utilizes an attributional lens as a theoretical framework to further unpack and subsequently address these gaps.

Chapter 3

Theoretical Framework: Who - or What - Will Get the Credit? An Attributional Approach to Understanding the Potential Impacts of AI on Women Workers

In reviewing how gender bias in AI design and development is underpinned by the theoretical foundation provided by Ambivalent Sexism Theory (AST), the value in applying theoretical lenses to this dissertation's investigations became clear. As such, this dissertation utilizes an attributional approach to address the proposed dissertation-wide research questions (see 1.1.2 for a detailed explanation), which serves as the theoretical framework to guide this dissertation's research. More specifically, this chapter will first start by providing a brief explanation of causal attributions and attribution theory (AT) to then unpack the effects of gender bias in evaluations of

attribution in groupwork when women work with men (**section 3.1**). The remainder of the chapter will then leverage attribution theory to unpack the effects of generative AI on women workers, particularly in terms of how various attributional considerations in HAI might interact with women’s existing marginalization and devaluation in group work (**section 3.2**).

3.1 Attribution Theory & Gender Bias in Group Work

The origins of Attribution Theory (AT) began with Heider’s [225] seminal work, which posits that people are inherently investigative in nature, leading us to seek out answers as to what might have caused success or failure to occur. Building upon these propositions, Kelley’s [256, 257] principles establish the role that observational information access plays in attributions, such that multiple observations lead to attributions that consider changes in situational and behavioral factors over time, whereas attributions based on singular observations have to rely more on the contextual information provided within that moment [311]. Of particular interest to this dissertation is Weiner’s [470] dimensions of attribution, which include locus of causality, stability, and controllability.

At a high level, *locus of causality* refers to whether the cause of success or failure is attributed to factors that are internal (e.g., amount of effort put into a task) or external (e.g., task difficulty) to the person; *stability* refers to how changeable or unstable (e.g. a worker’s day-to-day motivation levels) versus permanent or stable (e.g., worker innate ability or talent) the attribution is or is perceived to be over time; and *controllability* refers to the extent to which an observer of the sit-

uation perceives the cause of the outcome to be within the person’s control (e.g., the worker “earned it”) or outside of their control (e.g., the worker “got lucky”) [59, 217, 469, 470]. There are slight variations based on whether the dimensions are being investigated in terms of evaluating oneself (i.e., *self-attribution*) or evaluating others (i.e., *social attribution*), although previous work has demonstrated that concepts and measurements for *self-attribution* have an equivalent meaning in *social attribution* contexts [59, 217].

Each of these causal dimensions serves as a precursor to specific cognitive, emotional, and behavioral responses in the workplace [59], such as leader-member relationships between managers/subordinates (e.g., [65, 174]) and employee/organization (e.g., [230]). More specifically, this dissertation is concerned with how casual attributions of women workers relate to performance outcomes and evaluations [59, 217, 268, 436] and projected rewards/punishments for success/failure [217, 310].

For example, within a *social attribution* context of employers evaluating employees, an external attribution (e.g., task difficulty too high) for employees’ negative outcomes (e.g., poor performance) leads to lower intentions of punishment coming from employers [217, 436]. While this could seemingly be advantageous for women workers, research in hiring contexts suggests that evaluators are more likely to over-attribute the success of others to internal attributes while underestimating the impacts of external attributes [438]. As such, individuals (e.g., men) with societal and circumstantial privileges that lead to opportunities for success (e.g., prestigious education) are often misattributed as being more innately capable than their less privileged counterparts (e.g., women) [210]. This example thus demonstrates the need to critically examine how women are specifically impacted by causal attributions in the workplace.

Now that we have an understanding of causal attributions in workplaces gen-

erally, it is time to turn towards understanding how attribution theory can be used as a lens to investigate how women will be attributed credit depending on who - or what - they work with.

Interestingly, there is only a small body of work that explicitly investigates attributions and gender dynamics within group work. These works collectively highlight patterns of gender bias and stereotyping, such that women are generally attributed less credit for group work when working with men [48, 226, 404]. This is particularly the case in situations where there is ambiguity regarding the attribution of success in a mixed-gender group or dyad, as is often the case when evaluators have a limited amount of information to work from when making attributions, e.g., credit attributions for coauthored academic papers [404]. In situations of ambiguity or limited information, evaluators frequently default to biased rationalizations that devalue women as less influential, credible, and contributing unless countered with an abundance of explicit evidence to the contrary [226].

Unfortunately, gender biases in attributional assumptions can have widespread consequences for women in various sectors of the working world. For instance, Sarason et al. reviewed and compared tenure decisions for male and female academic economists to see whether gender might influence the allocation of credit when it comes to coauthored papers and whether these attributions could be linked to gender disparities in tenure awarding [404]. In doing so, they found that the number of co-authored papers on one's CV made no substantial difference in tenure decisions for men, regardless of who they co-authored with. For women, though, frequent co-authorship with men, but not other women, accounted for **40% of the gender gap** between tenure awarding to men and women without conditions and an astounding **65% of the tenure award gender gap** when controlling for average paper quality, citations, institution ranks, *and* field of study. Moreover, they found that women who

were rejected for tenure actually had *more* solo-authored papers (i.e., papers where attribution of credit is clear) than similarly rejected men, regardless of the number of co-authored papers [404]. The results of this study serve as a powerful demonstration of how gender bias devalues women and their efforts and how this bias serves as the default state when determining credit attributions in mixed-gender teams, leading to tangible harm to women’s careers regardless of the corresponding evidence of past success.

Furthermore, some limited prior work has shown that certain factors can moderate attributions of credit in mixed-gender groups, such as the racial composition of the team [48] and the gendered nature of the task [226]. First, regarding race, Biernat and Sesko found that men were given more credit for success and less blame for failure than women only when White women and White men were working together - the same did not apply to White men working with Black women or White women working with Black men [48]. These results are somewhat surprising given the wealth of studies demonstrating how Black women are uniquely targeted and marginalized within workplaces due to their compounding intersectional identities as Black and women (e.g., [416]). One thought is that Black women’s double marginalization makes them functionally invisible within the workplace due to their non-prototypical status [47, 48]. While this dissertation is not explicitly focused on race, it is essential to acknowledge that race will likely play a role in how women working with AI are evaluated, although the exact nature of this dynamic remains unclear at this time.

Second, the gendered nature of a task appears to play a complex role in the evaluations of women and men in the workplace. Sarsons et al. found that within the realm of masculine-stereotyped tasks (i.e., math quizzes), women were attributed less credit for joint efforts than their male counterparts [404]. These findings match what is often seen in the literature on ambivalent sexism in the workplace, i.e., that

women are often evaluated as being less inclined toward and capable of fulfilling more ‘masculine’ roles and tasks (see section 2.2.1 for more details) [27].

When it comes to ‘feminine’ tasks, though, the results here can be mixed. Sarsons et al. found that, for the joint feminine-stereotyped quiz (i.e., grammar exercises), women were attributed an equal amount of credit as their male partners, likely because evaluators viewed grammar exercises as gender-neutral [404]. However, women quiz-takers actually outperformed male quiz-takers, making the equal attribution evaluation inaccurate to the reality of the task credit distribution. Furthermore, showing evaluators the gender score distribution corrected the attribution bias for the feminine-stereotyped task, but did *not* correct the bias seen within the masculine-stereotyped task, suggesting that evaluators’ beliefs about what men and women are best suited for supersedes actual evidence of equal contribution in attributions of credit for ‘masculine’ tasks [404].

Taken in combination, the above-mentioned studies demonstrate that (white) women are often evaluated less favorably than their (white) male teammates, regardless of assumed or demonstrated skill and share of task success. However, this dissertation wonders **whether these evaluations and attributions of women workers will be more or less favorable towards women when they work with an artificial entity, AI, compared to when they work with men** (Study 2). Moreover, given the moderating effect that gender perceptions of tasks can have on said evaluations, this dissertation’s Study 2 aims to **account for participants’ perceptions of gendered associations with the task scenario**.

It is also important to acknowledge that attribution is made more complex by the power dynamics between the evaluator and the evaluatee, whether that relationship is peer-to-peer or more hierarchical in nature [311]. As such, this dissertation’s **Study 2** further focuses on **taking the power dynamics between evaluator and**

evaluatee into consideration when investigating how evaluators attribute credit to women. Additionally, although a limited amount of research has begun to recognize organizations as potential relational actors in causal attributions (e.g., [230]), it is still unclear how, if at all, organizations can shape how attributions of women workers play out depending upon their AI workplace integration design and evaluation, especially in light of the complexities associated with factors like workplace gender dynamics and worker job security, thus leading to this dissertation’s use of supplemental expert interviews with experts in AI-enabled workplaces in **Studies 2 and 3.**

3.2 Leveraging Attribution Theory to Unpack Impacts of Generative AI on Women Workers

3.2.1 Attribution of Responsibility in HCI/HCC and Causal Attributions

From a general HCI/HCC perspective, causal attribution has been investigated in a variety of contexts, such as a part of systems evaluations in usability testing [344] and within the context of consumer perceptions of technological failure [255]. Such studies employ causal attribution concepts to investigate whether the success or failure of a system’s operation can be attributed to elements within the system itself (i.e., external to the user), such as a faulty or laggy click-through rate, or to user knowledge/error (i.e., internal to the user). While these types of studies tend to focus on more “traditional” computing systems such as websites and software [255, 344, 426], the advent of autonomous vehicles and agents arguably represents an important shift in attributional research in HCI/HCC towards viewing technology as

a social actor capable of consequential decision-making.

Within the autonomous vehicle (AV) domain, attributional studies focus fairly exclusively on how blame is distributed between the autonomous vehicle and the human “driver“ of the vehicle when failure occurs, e.g., accidents with differing levels of severity [498] and failure to comply with complex traffic patterns [166, 213]. This makes sense, as causality and responsibility are central legal concerns when it comes to automotive accidents and injury [135]. The need to understand who is to blame in cases involving autonomous vehicles is especially heightened due to the often life-or-death consequences that result from vehicle failure.

Yet, study results are mixed on who - or what - is perceived to be primarily to blame in cases of autonomous vehicles, with some studies finding that the AV takes the blame over humans [166, 498] while others find that drivers are blamed more than their AVs if both the human and vehicle make mistakes [19]. Moreover, there is a third party that is often considered when assigning blame from a legal perspective, i.e., the companies that design and manufacture AVs [166].

In fact, there are ongoing debates within tech ethics (e.g., [90, 94, 183]) and legal (e.g., [123]) literature about who is actually morally and legally responsible for the detrimental impacts of autonomous decision-making in many areas of life, such as within patient care decision-making (e.g., [20]). These discussions have prompted researchers to investigate when and why people might attribute responsibility for harmful outcomes to autonomous systems, users of the systems, or the developers themselves [289]. In terms of AVs, people are more willing to attribute responsibility to the companies that own the autonomous system rather than the autonomous systems themselves in cases where there is no human collaborator to the failure that can share in some of the blame [7, 166]. From a robotics perspective, in cases where service robots fail, people are more likely to blame the service firm rather than the

robot itself. However, when less perceived controllability is attributed to the robot, there is less blame directed at the service firm [392]. In terms of AI specifically, Lima et al.’s large-scale study on the distribution of the blame for algorithmic harm amongst designers, users, and the systems themselves found that designers and users were both blamed to a large extent by the public, more so than the system itself [289].

While this dissertation does not seek to contribute to ongoing philosophical discussions on moral responsibility attributions, it is important to acknowledge the aforementioned discussions because they touch on **the uncertainty that surrounds the practical roles that real-world designers and builders of autonomous technology play in preventing harmful outcomes**. In other words, while academic and public discourse continues to explore the philosophical and perceptual attribution of responsibility to AI designers and developers, there is only a limited amount of empirical research that has sought to understand how AI designers and builders view their own practical role in mitigating bias in AI decision-making.

For example, Lancaster et al.’s interview study with AI designers and vision makers from industry and military contract backgrounds found that they often felt that it was not appropriate for them to judge what is and is not biased, or that they did not have the authority to make such decisions [276]. These results demonstrate the uncertainty surrounding the attribution of responsibility for bias mitigation, even among AI designers and developers [276]. Therefore, in recognition of the need to further explore the perspectives of key stakeholders in real-world AI design, evaluation, and workplace integration, this dissertation investigates the perspectives of women in AI design and development (**Study 1**), as well as women who are experts in AI workplace integration via supplemental expert interviews in **Studies 2 and 3**.

3.2.2 Attributing Human-Like Characteristics to Autonomous Agents, Levels of Autonomy and Agency, and Collaborative Roles in Causal Attributions in HAI/HRI

Perhaps the most fruitful area of research concerning attributions in HCI/HCC realms is within the Human-AI Interaction (HAI) and Human-Robot Interaction (HRI) domains. Before diving further into the HAI/HRI literature on attributions of failure and credit (i.e., causal attributions), it is essential to note that the term “attribution” is also often used when discussing how humans prescribe (or, “attribute”) different characteristics to an AI and/or robot (e.g., [108, 176, 445]).

This type of attribution is prevalent in research related to how humans anthropomorphize AI/robots, i.e., ascribe human-like qualities to inanimate objects [284]. Anthropomorphism is often discussed in terms of an artificial agent’s human-like designed qualities, such as pitch characteristics of a Virtual Assistant’s (VA) (e.g., Siri, Cortana) voice, leading to the attribution of gender to VAs (but not necessarily differing levels of trust in the VA as once speculated) [445]. However, attribution in this context also refers to how we attribute mental states and internal characteristics to artificial agents. For example, prior works have shown that expectation setting and the attribution of different personality characteristics (e.g., complementary personality types) can positively influence people’s evaluations of robots within HRI [108].

Additionally, there is evidence to suggest a potential correlation between humans attributing emotional intelligence and mental states to robots when they feel empathy towards the robot, which can be induced by robots that express concerns about their own mortality [176]. To conclude, characteristic attribution research helps to contextualize the tendency for humans to attribute autonomous agents with human-like qualities, such as mental states, thereby serving to foreground the remain-

der of this section’s discussion on HAI/HRI works that deal with causal attributions.

Building on the aforementioned discussion of attributions of mental states, causal attribution research in HAI/HRI and human perception of blame has highlighted how levels of agent autonomy drive perceptions of agency, which is a necessary precursor to causal attributions [298]. At a high level, agency refers to the ability to have mental processes that allow one to have self- and social awareness, and to think, plan, and act upon certain choices in response to a situation [107, 298, 299].

When considering causal dimensions, entities perceived as agentic can be more readily attributed credit or blame for an outcome, as agency implies a level of controllability and intentionality over one’s actions, suggesting that different choices could have been made. Prior works have shown that humans often assign agency to robots (e.g., [49]), likely as a result of a combination of perceptual processes such as the tendency to anthropomorphize robots [137] and as a way to make sense of outcomes involving robots by treating robots as capable of being subjected to causal attributions the way that humans are [107, 467].

Expanding upon the aforementioned works, Dawtry and Callan explored how agency and blame were attributed to robots when working collaboratively with humans as overseers, depending upon the robot’s level of autonomy (“autonomous robot” vs. “non-autonomous machine”) and technical sophistication (high/low) [107]. Their results from the three-part study showed that the more autonomous and sophisticated the robot was perceived to be, the more evaluators attributed agency, and thus blame, to the robot in cases of HRI-involved accidents across a variety of contexts. Indeed, other prior works in HRI have also found that the higher the computational sophistication of an AI system, the more cognitive capacities were attributed to the robot, thereby making the robot more culpable and blame-worthy in cases of failure [266].

However, Dawtry and Callan’s results also demonstrate that humans tasked with overseeing the robot as part of the HRI were blamed similarly, regardless of the robot’s level of autonomy or sophistication [107]. Generally speaking, causal attribution research in HAI/HRI has collectively demonstrated that human collaborators are attributed more blame for failure than the AI/robot they are working with when working collaboratively towards a shared goal [107, 240, 282, 418]. This is even the case when human collaborators are making attributions about themselves, such that people are often more inclined to attribute failure to internal self-attributes in HRI collaborative work and less blame on their autonomous and human group mates [240, 282].

Digging deeper into the nature of collaborative work, it becomes apparent that the distribution of collaborative roles and responsibilities between humans and autonomous agents has an impact on how blame is shared between the two. More specifically, Shank, DeSanti, and Maninger found that, in a collaborative decision-making task where one member of the dyad (human and autonomous agent) would make the decision recommendation and the other would oversee the execution of the decision, the participants evaluated the overseer autonomous agent as sharing the blame for a bad outcome to a greater extent than when the autonomous agent was the recommender [418].

In other words, the more control that an autonomous agent has over the final outcome of a task, the more it can be blamed in cases of failure, complementing other works that have found that perceptions of uncontrollability result in fewer attributions of failure to autonomous agents (e.g., [250]). Moreover, with more control and share in outcomes comes a higher position of power within a role dynamic, such that the higher position an AI is perceived to occupy in comparison to its human counterparts, the more blame it can share [418].

Taken in combination with Dawtry and Callen’s findings on agency and blame [107], it is arguably the case that, when working in a dyadic collaborative task, AI that is designed to be highly autonomous and to play a more active role in the final outcomes of a task would be more likely to share in causal attributions, **directly motivating Studies 2 and 3’s use of AI tool vs. teammate conditions.**

3.2.3 Ongoing Ambiguity Surrounding Attributions of Credit in Human-AI Interaction

Although existing literature has developed some understanding of how causal attributions for failure play out in human-AI interaction, it remains somewhat unclear how causal attributions unfold in cases of success, partly because there are comparatively fewer works in HAI/HRI domains that focus on the attribution of success and credit. This ambiguity is understandable from a research prioritization perspective, as the consequences of failure are more dire and require attribution for legal purposes, e.g., in cases of autonomous vehicle crashes.

However, the ever-growing prevalence of generative AI in various sectors of life has arguably shifted that prioritization, giving rise to more focused concerns on how credit can or should be attributed and to whom when generative AI is used (e.g., [128, 288, 364, 367]). These conversations often center on the ethical and practical implications of generative AI use in education, raising questions about whether students and academics who utilize generative AI to assist in or write academic papers on their behalf deserve credit for said work [128, 288]. From an organizational perspective, industry reports tend to focus on credit attribution as a legal and intellectual property protection issue, often advising companies to limit the access that a generative AI system has to proprietary or classified information [88, 113]. There

are also ethical and legal concerns about who rightfully owns the creative outcomes of generative AI, given that said outcomes rely on access to human-made examples and human-generated datasets [335, 393].

Overall, then, it is clear that credit attribution is a crucial consideration for the use of generative AI in everyday work contexts. Yet, as Porsdam et al. point out, there appears to be a credit/blame asymmetry when it comes to responsibility and attribution for success/failure, such that traditional AI responsibility and failure literature can be used to inform similar concepts in the use of generative AI but might not be equivalent in the case of credit/success attribution [364]. In other words, blame attributions for generative AI failure tend to mirror what can be seen with other types of automated systems, such as autonomous vehicles [212, 498], while credit attributions in generative AI use are rife with a level of ambiguity that makes Porsdam et al. wonder if other domains of knowledge might better serve as a lens for understanding credit [364]. They specifically propose examining the principles of technology use in human augmentation, particularly in the context of generative AI [105, 312]. This discussion is important because the lens used to investigate and understand credit attribution can shape how credit is defined, ethically and legally [335, 364, 393].

This dissertation argues, however, that a causal attributional lens is the most appropriate to use in light of what we know about the future of human-AI interaction in workplaces, i.e., that many workplaces will and already have begun to integrate human-AI teams, with AI acting as teammates that work interdependently with humans towards a shared goal rather than as a simple augmentation tool [321, 323]. The causal attribution lens better accounts for this level of human-AI interaction, as demonstrated by the prior section’s extensive discussion on blame attribution in HAI/HRI. As such, this dissertation is motivated to **use an attributional lens to**

explore how evaluators make and distribute attributions of credit and success when humans are made to work with an AI that possesses a level of autonomy that classifies as a group member, of which current understandings are limited to only a few studies. Hsiung et al.’s study of attributions in groups composed of two humans and an AI system found that people are more likely to attribute credit for task success to their AI and human teammate, more so than they are to themselves [240]. Additionally, Xin Lei and Pei-Luen Patrick Rau found in their study that women attributed less credit to themselves when working with an AI than men did when working with an AI [282].

While the aforementioned studies provide some insight into credit attribution in HAI, these insights are limited by their focus on how people make *self-attributions* in tandem with attributions of others, i.e. *social attributions*. This is an important distinction, as evaluations of attributions in groupwork function differently when the evaluator is a member of said group, such that people tend to give more credit to their group members and attribute more blame to themselves [432]. This consideration is not present when the evaluator is a third party to the scenario. Third-party evaluators instead use other considerations to drive perceptions, such as their preconceived notions of what types of work women are better suited towards when attributing credit for shared task success in a woman-man team [404].

Therefore, while the aforementioned studies provide necessary and valuable insight into attributions of credit in HAI group work, it is unclear how credit will be attributed for task success when the evaluator is a third party to the scenario, thus motivating **Studies 2 and 3**. Even more critically, though, it is unclear how, if at all, these evaluations of credit attribution in HAIs will interact with gender biases in credit attribution for groupwork as discussed in 3.1 to create disproportionate harms to women workers (**Study 2**), and how resulting evaluations of and attributions of

credit to women workers will influence perceptions of women workers' job security in AI-enabled workplaces (**Study 3**).

Chapter 4

Study 1: Understanding Women Developers’ and Researchers’ Perspectives on Gender in Approaching AI Design and Development

**Please note that this dissertation’s Study 1 has been published prior to the submission of this dissertation as [411], although some minor modifications have been made to the content to better articulate dissertation-wide contributions.*

Schulenberg, K., Hauptman, A.I., Schlesener, E.A., Watkins, H., & Freeman, G. (2023). “I Felt Like I Wasn’t Really Meant to be There”: Understanding Women’s Perceptions of Gender in Approaching AI Design & Development. The 2023 Hawaii International Conference on System Sciences (HICSS 2023). <https://hdl.handle.net/10125/102650> ***Best Paper Nomination Award***

4.1 Introduction

The design of an AI is a fundamental component of AI workplace integration, inherently shaped by the perspectives and priorities of the designers and integrators themselves. As such, this dissertation’s efforts to take women workers into explicit consideration when designing and evaluating AI workplace integration must necessarily account for the perspectives and priorities of women in AI design and development. Study 1 achieves this through a qualitative interview study that delves deeply into the perspectives of women who work in and research the design and development of AI technologies. Study 1 also examines workplace gender dynamics by investigating the role that women perceive their gender identity plays within their workplace and how these experiences may impact their design choices.

More specifically, the enduring underrepresentation of women in the area of AI design and development is increasingly noticeable in recent years. Studies show that women compose less than 12 percent of contributors to AI research conferences and only 10-15 percent of AI designers and developers at major technology firms [474]. Even the most generous estimates put women at 26 percent of the data and AI positions in the workforce [162], and only 16 percent of tenure-track faculty globally are focusing on AI research [450]. The gap widens even more substantially as women move up the corporate ladder [442].

This dissertation understands this large gender gap and the lack of gendered perceptions as a major concern for developing the next generation of AI agents, which are increasingly embedded in every aspect of people’s daily and working lives and seamlessly interacting with human beings. Indeed, technologies inherently reflect the bias of those who design them, such that a lack of diversity in AI development necessarily means a lack of diverse perspectives [363]. Therefore, Study 1 explores

the following two research questions about women, gender roles, and AI design and development:

S.1.RQ1 How do women in AI design and development perceive the current role their gender identity plays in their experiences within the field?

S.1.RQ2 How do these gender identity specific perceptions influence women’s approaches to designing and developing AI?

To answer these questions, eight in-depth semi-structured interviews were conducted with individuals who identify as women and have engaged in AI design and development in a variety of capacities. The richness of these findings contributes to HCI and the future of work in AI in two specific ways. First, Study 1 provides first-hand empirical evidence from women’s own perspectives on how the enduring gender gap in AI design is reinforcing harmful bias and inaccurate assumptions about how to design AI from and for the perspective of a woman. Second, Study 1 proposes initial design implications for AI agent systems based on the design practices of this study’s specific women participants, who repeatedly call for practices toward intersectionality and/or gender neutrality. Thus, Study 1 lays the groundwork for urgently needed future research to work to ensure that the long-standing gender disparities that have negatively impacted women in STEM do not continue to influence the design and development of AI agents.

4.2 Methods

Recruitment and Participants. This study’s participant recruitment and interview data collection took place in the fall of 2021. To recruit participants, recruitment messages were posted to specific forums for AI professionals on various

Participant ID	Age	Ethnicity	Occupation	Years of Experience
1	42	White	Data Scientist	6 Years
2	37	White	Business Owner	1 Year
3	33	White	University Professor (Military)	5 Years
4	27	Asian	Software Developer	2 Years
5	24	Latina	Analytic Support Officer (Military)	3 Years
6	67	White	Project Manager (Retired)	20 Years
7	24	Latina	Capability Developer (Military)	3 Years
8	28	Asian	Graduate Student	4 Years

Table 4.1: Participant Demographics in Study 1

social media platforms (e.g., Reddit, Facebook, and Twitter) to recruit potential interviewees who are over the age of eighteen, identify as women, and have professional work experience in AI design and development. Recruitment flyers were also sent to several email listservs for AI professionals. While the utilization of social media as a recruitment tool can bias a sample by suggesting algorithms include bias (e.g., users’ preferences for what is displayed in their feed), the highly specific population targeted (women in AI development) required a means of recruitment that was less randomized and more verifiable. It is interesting to note that the recruitment criteria received some push-back within co-ed social media forums. For example, in the subreddit “r/artificial”, a male-identifying user commented *“Not to be that guy but isn’t this just sexism. If someone made a men only AI event it probably would get ridiculed.”*. Such comments were dealt with appropriately and underscore the importance of prioritizing women’s voices in the face of opposition. Interviews were conducted with all participants who responded to the recruitment messages and agreed to participate voluntarily from September to November 2021. In total, eight professional women with current or previous positions in the field of AI design and development were interviewed. Participants range in age from 24 to 67 ($M=35.25$, $SD=14.32$) and originate from three different countries: China (1), England (1), and the United States (6). It is important to acknowledge that half of the participants are younger (i 30

years old) professionals with 1-5 years of professional experience living in the U.S. Table 4.1 shows the demographic information of Study 1’s participants.

Interviews. One-hour, one-on-one semi-structured interview sessions were held over the Zoom video conferencing platform in order to protect participants and researchers from the ongoing COVID-19 pandemic. Cameras and microphones were set to "on" for both the interviewer and the participant for the duration of the session. Participants first provided verbal informed consent and confirmed recording permissions.

The interview guide was constructed using a mixture of dialogic [466] and phenomenological [44] interviewing techniques, and included questions relating to participants’ lived experiences as women in AI development and their perceptions of designing AI in general and in regard to women users. In each interview, a researcher presented questions to the participant but allowed the participant to lead the conversation. The conversation was constructed to be open, but occasionally directed follow-up questions were necessary to ensure questions were answered in full. Some demographic questions were asked initially. Further questions were designed to gain insights about the environment, tasks, people, cultural influences, and systematic influences as they relate to women in AI. These consisted of a diverse set of open-ended questions related to thoughts and opinions about how women conceptualize their world and their role in designing human-like AI systems, what kinds of design decisions they make, and how they design for women users. The interview concluded with an opportunity for participant feedback and questions.

4.2.1 Data Analysis

Data analysis was conducted using in-depth qualitative thematic analysis [55, 56]. This qualitative approach does not focus on inter-rater reliability but endeavors to yield recurring concepts and themes of interest, find relationships among them, and formulate them into more complex groups and broader themes [318]. Following Braun and Clarke’s [55, 56] procedures for thematic analysis, four researchers first carefully read through the collected data line-by-line to familiarize themselves with the data, identifying and notating how these women participants experienced the role their gender identity plays in their development practices. Second, the research team began the iterative coding process by generating initial codes, where the research team individually assigned preliminary codes to the identified information, then came together to iterate on the codes to eliminate redundancy and make sure that each piece of data had only one code. Third, the four researchers categorized codes into thematic topics related to the research questions and developed sub-themes from participants’ descriptions of their development practices and experiences as women in AI design and development. Fourth, all researchers continued to discuss, integrate, and refine themes and sub-themes to streamline the data to better capture the data in relation to the research questions. Fifth, the research team collaborated to further refine these themes, considering all themes across the entirety of the data set to identify the ”essence” of each theme [55, 56]. Finally, the research team selected the most compelling quotes as examples to craft a narrative structure for the findings that flows logically and naturally [55, 56].

4.3 Findings

This section utilizes quotes from the participants' own accounts in order to present findings regarding S.1.RQ1 and S.1RQ2. Each section is divided into three main themes that emerged from the data.

4.3.1 How Women Perceive the Role of their Gender Identity in the AI Industry

In analyzing the interview data, Study 1 uncovered three main themes that relate to how women perceive their gender identity within the AI industry: the influence of gender imbalance in the workplace, expectations to change behaviors to fit organizational culture, and the existence of the “Bro Code” in a male-dominated AI industry.

The Influence of Gender Imbalance in the Workplace. First, there was a distinct division amongst participants' perceptions of gender roles and their gender identity in their workplace. This division in perception is due, in part, to each participant's background.

Some of the participants are used to being the only woman in the room and, as a result, do not generally notice the gender imbalance in their current AI workplace. For example, P3 (33, University Professor, White) had prior experience in a male-dominated field, the United States Army. When asked about her perception of gender imbalance in the workplace, she stated *“yes, [women are] definitely the minority, but not to the extent that we are in the army, right?...when I'm in environments with large percentages of females, I'm more surprised.”* P8 (28, Graduate Student, Asian), also echoed this sentiment, *“because my previous background was also in engineering, there were only like four girls in my previous like, undergraduate class. So it's like, I'm not*

totally sure what it is like, although I know it will be different.” For both participants, they are well aware of the status quo gender imbalance in their workplace and seem to even consider it as a “natural” state – they will be “surprised” if more women are present.

On the other hand, many of Study 1’s participants noticed less gender balance in the workplace and felt that they were treated as somewhat of a work “mom” because of their gender identity. P4 (27, Software Developer, Asian) described this phenomenon in the following way, *“the thing is, you work with a ton of men. And you tend to have - they tend to look at you to be more nurturing, caring, and maternal in the workplace. They want to come talk to you about their problems, they get confused if you are serious or maybe even stern.”* For P4, the influence of gender imbalance in her workplace is not only shown as women being outnumbered by men but also demonstrated by men’s expectations for traditional gender roles – such as women serving as *“more nurturing, caring, maternal.”* P4’s experience arguably speaks to the larger issue of benevolent sexism in workplaces generally [27] and STEM-related fields specifically [143, 272, 375, 400], demonstrating that the AI development industry is not immune to sexism’s influence.

Expectations to Change Behaviors to Fit Organizational Culture.

As a result of this significant gender divide in AI development, the majority of participants noted the need to adjust their normal behavior at work.

Multiple participants stated feelings of imposter syndrome in this male-dominated work environment. For example, when asked about her experience in the AI field, P2 (37, Business Owner, White) said: *“At first I felt like I wasn’t really meant to be there. And I’m kind of ashamed to say that as a feminist, and why shouldn’t I be? Why shouldn’t I be in a space?”* For P2, it seems to be necessary for her to make additional effort to prove why she belongs to this field and why she deserves to be in

this field.

Additionally, P6 (67, Retired Project Manager, White) described multiple accounts of male clients not listening to her ideas and, as a result, becoming more assertive in the workplace: *“I had a lot of instances where colleagues, all men, actually had a problem...[when they] get to the same level that you are.”* P1 (42, Data Scientists, White) echoed this sentiment, saying *“I had to be more aggressive, be forward, be a loud voice...and I actually had to dial it down.”* Both participants acknowledged that being assertive may not be their style in their personal lives. However, to make their voices heard and their opinions valued in their workplace, it seems necessary for them to adjust their behaviors *“to be more aggressive, be forward, be a loud voice.”* – in a sense, to be more masculine. This is particularly interesting in light of prior works on ambivalent sexism, which often find that women who adopt more “masculine” traits such as assertiveness run the risk of being labeled as too aggressive and unlikeable [313, 447]. Even here, though, P1 qualifies her statement by pointing out that she had to “dial it down” with the aggressive persona, further reinforcing the double bind that benevolent sexism in particular puts women in the workplace in, i.e., having to strike a delicate balance between appearing competent without stepping too far outside of gender role expectations.

The Existence of the ‘Bro Code’ in a Male-Dominant Industry Another interesting topic brought up within many of the interviews is the concept of the “bro code” within their workplace. Many participants mentioned their male colleagues backing each other up in group meetings, often at the expense of their female colleagues. This behavior evokes an unhealthy dynamic between employees, allowing for a *majority rules* mindset to dictate the workplace. For example, when recounting an incident where she came to the defense of another woman colleague’s idea in a meeting, P6 (67, Retired Project Manager, White) described: *“I think if she*

didn't have me there, I really think the guys would have just continued to say majority rules." P6's experience demonstrates how the uneven representation between men and women in AI workplaces inherently creates a dynamic in which men's voices are prioritized by other men, and how the burden of making their voices heard falls on all of the women in the room as a collective.

With respect to group work, participants also noted distinct differences in the ways their male and female colleagues behave in collaborative meetings, and how those differences affect whose voices are prioritized. Unfortunately, this is not necessarily surprising, as prior works on the attribution of success/failure in mixed-gender group work have shown that women are systematically under-credited and diminished when working with men [226, 404]. Study 1's participants' experiences add an additional consideration to this by detailing how different working styles might also affect how women's contributions are (de)valued. For instance, P1 (42, Data Scientist, White) stated that *"women are looking for a more collaborative approach; whereas, men are more inclined to steamroll the conversation. If women don't recognize that, their voice would often be dominated, because they didn't want to engage in confrontation."* For P1, such different working/collaborative styles unfortunately do not help women build a strong voice in the AI industry, as their working styles are not given the space they need in meetings to flourish. Indeed, even when women are able to have their voices heard, it is only in light of having a "back up option" to fall back on that makes the majority men feel comfortable, as P7 (24, Capability Developer, Latina) explains: *"If a female states an idea, they're like 'okay, we'll try it out but we'll have a Plan B."* Here it can be seen that, even if women are "allowed" by their male colleagues to have an idea, their contributions are undermined by their male colleagues' insular thinking of what is best.

Oftentimes, this *bro code* even exists outside of work hours, as many of the

participants mentioned their male colleagues bonding in activities outside of work and refusing to invite their female counterparts. For instance, P2 (37, Business Owner, White) discussed not being invited out for social outings with her male colleagues: *“at least in the Silicon Roundabout area in London, it was very much like the beards and the ping pong, and the beer, the microbrewery bar and all this kind of stuff.”* This additional bonding time further emphasizes the division between the majority and the minority members of the AI field, a stratification that is then re-codified in the meeting room. In this way, sexist divisions within and outside of the workplace combine with attributions of credit and value to disproportionately harm women workers in the AI industry.

4.3.2 Women’s Approaches to Design and Develop AI

The above perceptions of the role their gender identity as women plays in the AI workplace influenced participants’ design approaches in a variety of ways. In particular, they discussed three approaches to AI design that are important for reducing gender bias in the AI industry and developing more inclusive AI technologies: Design with Gender Neutrality in Mind, Focus on Reducing Gender Stereotypes, Incorporate More Specific and Intersectional Use Cases.

Design with Gender Neutrality in Mind. The majority of participants in Study 1 agreed that AI should have a gender-neutral design. In fact, most already design as gender-neutral as possible, seeing gender as a confounding variable rather than a defining one. For instance, P3 (33, University Professor, White) somewhat took offense to the question when it was posed: *“more than gender...I’m not going to design it pink, if you will, because it’s for females”*.

Therefore, many of Study 1’s participants choose not to develop with gender

in mind at all. For example, when asked about specific examples of an AI tool for women, P7 (24, Capability Developer, Latina), refuted the idea that you could group women users into a single category: *“I would focus on the process of how women users would be using, you know, whatever product... Women kind of approach it more from a problem solving perspective, whereas the guys have, like, a gaming perspective.”* P7 asserts that looking at an AI agent as a component to solve a problem, rather than something a specific gendered user operates, is the right approach. Moreover, P7 also seems to attribute a greater level of practicality to women designers’ approaches to AI development, arguably bucking benevolent sexist assumptions about women being better suited for more emotional and relational-type work [27].

Interestingly, one of the participants, P8 (28, Graduate Student, Asian), does not acknowledge AI as a gendered entity: *“I don’t necessarily put a gender on [Alexa], because I don’t know, it’s just a machine that you can tell it what to do but it does not need to be a girl or a boy...I feel like it’s more like a machine type of creature.”* For P8, even if the AI agent is an avatar with a defined gendered appearance, she still does not perceive gender as a necessary factor in AI design. From her perspective, gender is not something that can be ascribed to a synthetic entity and is rather based in characteristics associated with biological entities. P8’s view points to the complicated and evolving nature of gender in modern society – while the contemporary understanding of gender has gone beyond a binary dichotomy regarding one’s biological sex assigned at birth, the question becomes if the same understanding is or should be applied to AI. In this sense, the default design approach of gender-neutrality can serve as a mechanism to ensure that no one personal or societal understanding of the mechanisms that drive gender identification become ingrained within AI systems.

Focus on Reducing Gender Stereotypes. The vast majority of participants believe that current AI design is far too gendered. For instance, when asked

about the current state of AI design, P2 (37, Business Owner, White) discussed the gendered state of AI tools: *“I mean, I think people have been calling out for a little while, haven’t they? The fact that these tools seem to be too gendered.”* When asked the same question, P5 (24, Analytic Support Officer, Latina) stated *“I don’t ascribe to the idea of gender essentialism, where women are always this way and men are always this way.”* For both participants, how gender is approached in the current AI design and development process seems to simply reinforce traditional gender stereotypes, rather than incorporating the key values of inclusiveness and diversity embedded in a gendered perspective. Therefore, they consider such *“gender essentialism”* more harmful than beneficial to AI development. From a theory perspective, P2 and P5’s viewpoints can be viewed as direct rejections of ambivalent sexist ideals and assumptions, which are deeply rooted in binary (male/female) gender essentialism [27, 189].

When asked about what designs women would prefer, P8 (28, Graduate Student, Asian) points out exactly how gender essentialism can creep its way into the process of designing human-like AI systems: *“Males are different from us [women]. They probably would, you know, would like those [sexy female avatars].”* Here, P8 speculates on the effects that having predominantly male developers can have on design (i.e., *“[sexy female avatars]”*), which might not appeal to non-male users and thus alienate women (i.e., *“males are different from us”*). In essence, P8’s statement acts as a commentary on how design is often too focused on gendered attributes like sex appeal, when in fact, as she goes on to mention, AI is *“just a machine that you can tell it what to do but it does not need to be a girl or a boy.”*

In light of participants’ adamant resistance to gender stereotypes and gender essentialism, P8 goes on to highlight how designers can reduce gender stereotypes in AI design: *“There are different types of gender, different types of AI like different*

characteristics, but gender is not elicited to me not one thing on it is more like different behaviors of it. Our different knowledge it has, or even different algorithm it uses, but not gender.” For her, it is important not to associate certain users’ gender with their behavioral patterns. Rather, it may be promising to focus on characterizing behaviors and optimizing AI agents to accommodate different user behaviors rather than different user genders. Moreover, there is also a tacit acknowledgment here of the need for AI developers to carefully consider how their design choices could have biased consequences, tying into attributions of responsibility regarding gender bias in AI design.

Incorporate Use-Case Specificity and Intersectional Practices. Focusing on how to design AI for women users, participants collectively highlighted use-case-specific and intersectional AI design as promising approaches, especially in light of the aforementioned rejection of gender essentialism in AI development.

More specifically, many of Study 1’s participants mentioned the need for a use-case-specific approach to design AI for women. For example, P7 (24, Capability Developer, Latina) tried to reorient the conversation towards user-centric design when asked about designing a product for women users: *“I would focus on the process of how women users would be using, you know, whatever product.”* Rather than focusing on the user population’s gender identity as the primary design driver, P7’s reorientation focuses on how the product would be used as the primary driver with the user population’s gender identity as secondary, if considered at all. A few participants also provide a practical mechanism to help with this, i.e., developers should utilize social listening practices when designing for a particular audience rather than making assumptions about the intended user population. Social listening as described by Study 1’s participants refers to an investigatory process undertaken after securing a client with the purpose of establishing the problem and population area for a design

project. For instance, P4 (27, Software Developer, Asian) described how she would handle a client asking her to design for women users by redirecting the client toward a social listening approach, *“You might actually have a really big hole in the LGBT community, and you have no idea....or it could be a totally different income level than you realize...I try to like bring that perspective that is like, we come into something with our own biases, whether we realize it or not. So why don’t we take a step back and see what’s actually out there before we react?”*

Here P4 is pointing out that, while a client may believe that their intended user population is women, the reality of who the client should be targeting as their intended user *“could be a totally different”* population than the client realizes. Social listening, then, acts as a strategy for pushing back against the idea that AI designers can and should start from a single conception of a user (e.g., women). P4 additionally points out how making assumptions about a user and design based on the category of “women” can be problematic, *“Oftentimes, it’s pretty, like straight white woman, middle class...your audience could actually be totally wider than that...women is super broad...”* Here P4 touches on the assumptions that gender essentialism and sexism make about the cultural behavior of women regardless of their intersectional identities. Part of what makes the theoretical lens of ambivalent sexism a particularly powerful tool for understanding sexist attitudes is that it has been extensively tested and validated in a wide range of countries and cultures [27, 190], demonstrating that one of sexism’s greatest weapons is its tendency to reduce women to just their gender identity rather than acknowledging intersectional identity as P4 calls for.

Taken in combination with her previous quote, P4 emphasizes how social listening as a practice can provide a deeper understanding of factors like income and LGBTQ+ status that are crucial for designing at the intersection of multiple identity factors. That intersectionality is fundamentally lost when a user population is

reduced down to a singular broad category (e.g., women) because hegemonic views on what that category represents in the U.S. (*“pretty, straight white woman, middle class”*) excludes large swaths of potential users (e.g., women of color, women of different classes, LGBTQ+ individuals). This is not only problematic for the making of functional and effective AI designs that fulfill the needs of a client and their users, but is also socially problematic because it perpetuates social bias through the upholding of hegemonic social ideas of who is a “woman” in the U.S..

The majority of participants echoed P4’s emphasis on an intersectional approach to AI design. For example, when commenting on the importance of diverse data sets, P2 (37, Business Owner, White) mentioned *“...we’re living in this exponential age and this [fast] pace that comes with that...I guess it would be to kind of slow down and to ensure that the data sets that you’re using are ethical and have been considered from lots of different perspectives...”*. There is a recognition in P2’s comment that the AI design industry is moving and evolving rapidly, and that such a rapid pace has the potential to lead to the reductive reasoning previously mentioned (e.g., P4’s *“pretty, straight white...”*), which in turn can lead to ethical issues. Thus, the response to how to design AI for women needs to involve consideration from *“lots of different perspectives”* in order to truly fulfill the practical and ethical obligations of AI designers. Here again there is an implicit acknowledgment among participants that designers and developers should play an active role in taking responsibility for mitigating gender bias in AI design and development.

4.4 Discussion

The gender gap in technology at large [62] and within the field of AI development [474] prompted some limited past research on the experiences of women AI

developers [435]. However, existing literature on HCI and AI is lacking in theoretical conceptions and explorations of how the experiences associated with being a woman in AI development specifically impact the ways in which these women approach designing various AI systems. In this section, findings stemming from interviews with eight women in AI fields address these above-mentioned issues.

Overall, Study 1's findings show that the lack of representation of women in AI has led to a majority rules mentality and gendered approaches to AI development. These findings lend empirical credence to some of the recommendations organizations such as UNESCO have been making regarding how best to address the threat that gender bias in AI design pose to women's working lives globally, i.e., for more women representation in AI design and development [449]. More specifically, Study 1 is one of the few studies to this dissertation's knowledge to provide insight into how women's experiences with being a woman in the AI development industry and research field actually does play a role in the decisions they make in their development practices. However, Study 1's findings does provide a more nuanced picture that goes beyond the assumptions the aforementioned works make about women representation being inherently tied to less biased outcomes. Indeed, findings demonstrate that women in AI fields often find themselves altering their behavior in various ways to survive in such a culture, or do not inherently recognize their own marginalization. Prior works on ambivalent sexism in the workplace point out that the insidious nature of benevolent sexism lies in its subtle nature, which can make it difficult for women to recognize their own marginalization - a necessary precursor to identifying and resisting sexist ideology [11, 30, 307, 332].

Taken in isolation, then, these specific findings serve to remind the HCI and AI ethics community at large that a simplified view of diverse representation and its outcomes is not a silver bullet for fighting gender bias in AI design and development.

For example, Lancaster et al.'s interview study with AI developers and vision makers found that AI developers, regardless of gender identity, have deeply ambivalent views about their role, responsibility, and ability to address and mitigate bias in AI design [276]. By specifically interrogating the relationship between representation in AI development and actual practice of real-world AI designers and developers, Study 1 is able to make more nuanced contributions to these discussions. Indeed, Study 1's findings demonstrate that women developers' subsequent views on design display a resistance to the existing unnuanced and gendered approach of the man-dominated industry of AI development, which will be discussed in greater detail for the remainder of this discussion.

4.4.1 Women's Resistance to Unnuanced Gendered Ideas in AI Design

Study 1's findings provide a comprehensive understanding of women's perceptions of their gender identities within the field and the workplace, influencing - both implicitly and explicitly - how these women view AI design and development, particularly in terms of gender bias. First, despite the only other prominent study focusing on the perspectives of women in the AI industry being conducted nearly 30 years ago when the AI industry was in its infancy [435], interestingly - and sadly - Study 1's findings show that there still exists a similar sense of unequal treatment and barriers to entering the boy's club of AI today. However, despite the existence of other more recent works that include women's perspectives in AI development as a part of larger studies (e.g., [276, 390]), no other studies to this dissertation's knowledge have sought to link how these perspectives and experiences could be playing a role in how these women view and approach AI design from the perspective of a woman.

This gap in understanding was particularly concerning as technologies inherently reflect the experiences and biases of the people that design them, and this is especially apparent in the case of AI technologies such as Google Translate (e.g., [365]) and generative AI (e.g., [452]) where gendered stereotypes of women not belonging in STEM jobs are on display. This issue poses problems in AI development partly because developers inevitably define through their biases what it means for an AI system to replicate and exhibit human-like qualities [398], in turn perpetuating gender stereotypes such as women being in subservient positions as is arguably the case with chatbots that are designed with characteristics attributed to human women (e.g., feminine voice pitch characteristics as seen with Siri, Alexa, and Cortana) [141, 391].

Even more concerning is the fact that technologies act as key societal artifacts that can inadvertently reinforce and shape user identities, including within the realms of gender roles and stereotypes [350], further underscoring the danger inherent in excluding women’s perspectives from the design process. Prior to this study, HCI and HCC research has been missing how women’s perspectives could be manifesting themselves through the designs that are seen today, and how their viewpoints are shaping the future of AI design in various sectors of life. Therefore, Study 1’s findings fill this gap by going beyond simply describing women’s experiences and instead focusing on how their experiences shape their design choices and philosophies.

4.4.2 The Importance of Gender Considerations to Future AI Design

Study 1’s findings also provide crucial insights into as to why the representation of women and the diversity of viewpoints within the AI industry is important for design, in turn providing empirical backing to the calls made by global organi-

zations (e.g. UNESCO) for greater representation of women in AI development in light of projected harms to women in the workplace [449]. Prior research has shown that gender is one of the most common groupings for possible users when developing technology [455], leading to male AI developers making assumptions about women’s perspectives that in turn may lead to AI products that do not cater to their users and instead exhibit harmful gender stereotypes [390]. Study 1’s findings show that, despite this prevailing knowledge, women in the AI industry are fighting back against the very notion that using gender as a primary grouping for users is either appropriate or accurate. Instead, participants call for and enact various ways to resist this unnuanced approach to design by instead emphasizing the intersectional nature of women’s identities, and how the needs of the intended users are more important to focus on than the gender of the users.

Study 1’s participants also revealed specific design implications for AI development that are important to note. First, developers should engage in social listening and extensive user research to identify a client’s target users *regardless of the assumed characteristics (e.g., gender) of the users*. Second, if gender is still considered to be an important characteristic of the target users, it *still does not mean that the AI itself has to be designed in a gendered manner*. There are many circumstances in which an AI does not need to be explicitly gendered in appearance or voice in order to still provide use and value to the user, regardless of the user’s gender, and, according to participants, gender neutrality should be the standard to start from so as to avoid perpetuating harmful stereotypes. Third, human-like AI should be, to the greatest extent technically possible, customizable to the individual users’ wants and needs, because *women users are not a monolith*. The needs, desires, and backgrounds of individual women users, when applicable, can never fully be accounted for, even with careful user research and an intersectional approach, thus necessitating customization

to reach a more ideal state of design.

Finally, Study 1 provides vital foundational insight into women as designers and developers of AI technologies that serve to shape this dissertation's Studies 2 and 3. More specifically, Study 1's findings and recommendations to focus on the functionality and use-case specificity when approaching the design and development of AI form the basis of the factorial experiment used in both Studies 2 and 3, which focuses on exploring the potential impacts that differing functional designs of an AI (tool vs. teammate) might have on evaluations of women workers within a human-AI interaction (HAI). Despite often being silenced by the workplace gender dynamics characteristic of the AI industry, these studies give base consideration to women developers' views on AI design, ensuring that any resulting understanding of AI workplace integration gained from these studies is inherently considerate of women workers at the design level. Moreover, inspired by the richness of the data gleaned from interviews in Study 1, both Studies 2 and 3 incorporate interviews with a small but diverse group of experts in AI-enabled workplaces (4 women and 2 men in fields related to AI and/or people management) as a way to contextualize the results of the experiment in real-world perspectives. These interviews also draw upon the principles of intersectionality espoused by Study 1's participants through the recruitment of women, and men who work with women, with different types of expertise related to AI workplace integration, from different fields with different workplace gender dynamics, and who have different demographic and background characteristics from each other (see section 5.3.1 and Table 5.2 for specifics).

Chapter 5

Study 2: Understanding the Influence of Workplace Gender and Technological Dynamics on the Evaluation of Women Workers in AI-Enabled Workplaces

5.1 Introduction

Built upon the foundational insights gleaned from women in AI design and development in Study 1, Study 2 implements their functional design principles and investigates how evaluations of women workers will differ when working with different types of AI functional designs (i.e., tool vs. teammate). In doing so, Study 2 is one of the first empirical studies to date to tackle questions of evaluations and attributions of credit in the manifestation of human-AI interaction in AI-enabled workplaces at the

intersection of gendered dynamics in the workplace. Indeed, the rise of readily accessible AI systems in workplaces has stirred up much discourse surrounding how credit for AI-generated outputs can or should be attributed to the people and workplaces that utilize them to enhance their workplace productivity (e.g., [351, 364]). Yet, comparatively little research attention has been given to how various workplace dynamics and system designs can impact evaluations of human workers generally, such as in terms of effectiveness, ability, and future job security, despite the prevalence of such evaluations in everyday workplaces and other research domains that study everyday workplaces [144].

Simultaneously, concerns are growing about how women workers might be uniquely impacted by this rise, particularly in light of patriarchal gender dynamics in everyday workplaces that systemically disadvantage and devalue women workers [27]. This presents a unique challenge to AI integration into the everyday workplace, where complex workplace gender dynamics can also create differing impacts on evaluations of workers and credit in collaborative work. For example, women who work with men are often attributed less credit in a human-human teaming scenario than their male partners are, even when evaluators are presented with evidence that the woman worker contributed *more* than their male partner did [404].

In light of these gaps in understanding, this dissertation’s Study 2 explored the following research questions via a 2 (Human Worker Gender) x 4 (Dyadic Partner Type) x 2 (Task Type) online factorial experiment with 685 participants (Part I) alongside post-experiment supplemental expert interviews with experts in AI-enabled workplaces (4 women, 2 men)(Part II):

S.2. Overarching RQ: How are evaluations and attributions of credit to a human worker, their dyadic partner, and their

collaboration as a whole influenced by the gender dynamics of an AI-enabled workplace?

S.2.RQ1 How are these evaluations and attributions influenced by the gender identity of a human worker?

S.2.RQ2 How are these evaluations and attributions influenced by the AI identity or gender identity of a dyadic partner?

S.2.RQ3 How, if at all, do the influences of S.2.RQ1 and S.2.RQ2 vary based on the type of task completed in a workplace?

S.2.RQ4 How, if at all, do the discriminatory beliefs (i.e., sexism and AI attitudes) of an evaluator interact with gender dynamics to influence these evaluations and attributions?

S.2.RQ5 How do real-world experts in AI-enabled workplaces conceptualize AI workplace integration's impacts on evaluations of human workers?

In addressing these research questions, Study 2 endeavors to make significant contributions toward understanding the intersection of evaluations and attributions of credit in light of workplace gender dynamics in groupwork [48, 226, 404], and within Human-AI Interaction (HAI) and Human-AI/Autonomy Teaming (HAT) research paradigms [323, 347, 353]. The remainder of this introduction dives deeper into this dissertation's Study 2 motivations for exploring the aforementioned research questions.

First, this dissertation's Study 2 is motivated to investigate human worker gender identity as a vital component of evaluations and attributions of credit to human workers in AI-enabled workplaces, primarily because of the influx of reports identifying women workers as particularly vulnerable to replacement as a result of

AI workplace integration [91]. However, there is comparatively little attention given to how women workers might be evaluated and attributed credit differently in AI-enabled environments [273, 280], especially when in comparison to existing gender dynamics in workplaces that systematically disadvantage women workers. Indeed, women who work with men in a dyadic collaboration are most often given less credit and perceived less favorably than their male counterparts, even when contributing an equal or greater amount than the male partner [404]. This imbalance tends not to be seen in partnerships where women work with other women or when men work with other men [404].

Yet, no studies to the best of this dissertation’s knowledge have endeavored to empirically test if this same imbalance in evaluations and attributions applies when human workers work with AI. In other words, it is unclear if evaluations and attributions of credit to human workers generally and especially women workers specifically might be different depending upon whether the partner they work with is human or an AI. Indeed, when it comes to working with AI, there is some early evidence to suggest that evaluators are actually more suspicious and critical of *male* workers using generative AI for writing assistance than they are of female workers [440], which would suggest that the gender dynamics that play out in human-AI dyads in AI-enabled workplaces might differ from those seen in human-human dyads. However, this limited work did not empirically compare evaluations of women versus men workers working with an AI versus a human, thus motivating this dissertation’s Study 2 **S.2.RQ1** (*How are these evaluations influenced by the gender identity of a human worker?*).

Second, the need to compare how evaluations and attributions of success are shaped by the identity of a Dyadic Partner as an AI versus a human with different genders similarly motivates this dissertation’s Study 2 **S.2.RQ2** (*How are these*

evaluations influenced by the AI identity or gender identity of a dyadic partner?). However, there is also a need to break down the AI identity component of S.2.RQ2, namely via the potential differences in evaluations and attributions of credit that can manifest from AI that is designed to act as a tool versus as a teammate. More specifically, as AI advances in its technical capabilities and abilities to be interdependent with humans, we are closer to more widespread workplace integration of AI systems that can act as teammates rather than just simple tools [321, 323].

Yet, most works in HAI-related fields such as autonomous vehicle research (e.g., [166]) and Human-Robot Interaction (HRI) (e.g., [418]) tend to focus on how agents can share in the attribution of blame for joint human-technology efforts rather than credit. Even fewer have looked at this with the gender of the human teammate in mind. The most notable is Xin Lei and Pei-Luen Patrick Rau’s study [282], which demonstrates that women who were working in a human-AI group were more likely than men to give the AI entity more credit than themselves [282].

However, it is unclear how working with an AI system in an AI-enabled workplace context towards the successful completion of a task will impact an *evaluator’s* perceptions of and credit attributions to workers generally, and women workers specifically, when making *social attributions*, i.e., when making attributions about other people rather than oneself [59, 217]. Therefore, the inclusion of **S.2.RQ2** serves the three-pronged purpose of empirically comparing human-human gender dynamics and human-AI dynamics, as well as comparing AI functional designs and their impacts.

Third, given the wide range of workplaces that are working towards AI integration (e.g., in educational contexts or healthcare [80, 495]), it is necessary to acknowledge that the nature of the tasks involved in the workplace context in which a partnership is working can impact evaluator perceptions in different ways because different workplace contexts have different needs and values. Moreover, different

workplace contexts can also have different implications in terms of gender dynamics and evaluations of and attributions to human workers in ways that can favor women over men and vice versa, underscoring the intricate nature of gender dynamics in the workplace.

In particular, prior works on ambivalent sexism [188] and mixed-gendered partnerships [48] have shown that evaluations of women and men workers can differ depending on how closely associated a task is with gendered expectations, such that women are sometimes perceived to be more competent than their man counterparts when performing tasks typically associated as “feminine”, e.g., child-rearing [384]. However, the reverse happens in tasks or workplace environments that are traditionally considered to be “masculine”, e.g., STEM-related fields, where women are systematically perceived to be less accomplished and effective than their male counterparts, even when evaluators are explicitly told that the women accomplish more [132, 331]. Therefore, this dissertation’s Study 2 explicitly acknowledges these impacts by including **S.2.RQ3** (*How, if at all, do the influences of S.2.RQ1 and S.2.RQ2 vary based on the type of task completed in a workplace?*).

Fourth, there are inherent links between gender dynamics in the workplace and discriminatory beliefs within evaluators, thus motivating this dissertation’s Study 2 to explicitly acknowledge the unique organizational and cultural factors that devalue women’s competence and overall value in the workplace, as is discussed in literature on ambivalent sexism (e.g., [27]). Indeed, experiences with sexism in the workplace have demonstrably negative impacts on women workers’ self-efficacy and performance [249], yet this goes underaddressed when intersecting with AI workplace integration. As such, it is unclear how various evaluations and attributions of credit will interact with evaluators’ preexisting sexist beliefs to impact human workers generally and women workers specifically.

Moreover, negative attitudes towards AI’s use in society have also been linked to worsening perceptions of the use of AI in such contexts as education [340], and are particularly prominent in research concerning HAI and credit attribution [223] and trust [125]. Given all of this, this dissertation’s Study 2 is motivated to investigate how evaluations and attributions of credit are impacted by evaluators’ pre-existing discriminatory beliefs via **S.2.RQ4** (*How, if at all, do the discriminatory beliefs [i.e., sexism and AI attitudes] of an evaluator interact with gender dynamics to influence these evaluations?*).

Ultimately, this dissertation acknowledges the crucial need to bridge the gap between research outcomes and industry practices in the context of studying AI integration in everyday workplaces. Indeed, the rapid advancement of AI technologies and their use in workplaces means that what could be considered the “norm” for AI use can change from month to month, and even week to week. Therefore, this dissertation’s Study 2 is motivated to investigate how experts in AI workplace integration conceptualize and understand how workers are evaluated in AI-enabled workplaces to bridge the gap between this study’s research outcomes and real-world practice, thus motivating **S.2.RQ5** (*How do real-world experts in AI-enabled workplaces conceptualize AI workplace integration’s impacts on evaluations of human workers?*).

5.2 Methods Part I: Factorial Experiment

5.2.1 Factorial Survey Design

To address this study’s research questions, Study 2 consisted of an online 2 (Human Worker [HW] Gender Presentation) x 4 (Dyadic Partner [DP] Type) X 2 (Task Type) between-subjects factorial experiment where 685 participants were

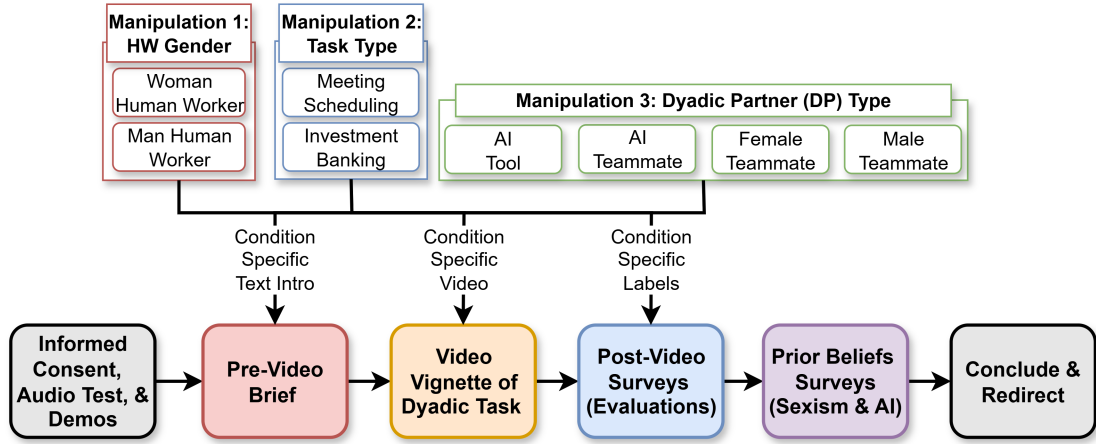


Figure 5.1: Between-Subject Manipulations Chart & Factorial Experiment Survey Flow in Study 2.

tasked with evaluating the performance of either a woman or a man human worker (HW) when working with a dyadic partner (DP) that was either an AI tool, an AI teammate, a human female teammate, or a human male teammate on either a meeting scheduling task or an investment banking task. This study design is driven by three motivations related to perceptions of women in group work [48, 226, 404], attributions of credit in human-AI interactions [107, 282, 418], and real-world concerns about how women’s working lives will be impacted by AI workplace integration [171, 449]. This subsection will describe in detail the three manipulations in this study, which can be seen in Figure 6.1, and how they were operationalized in the factorial survey to answer the research questions outlined in this study, as well as the dissertation-level research questions.

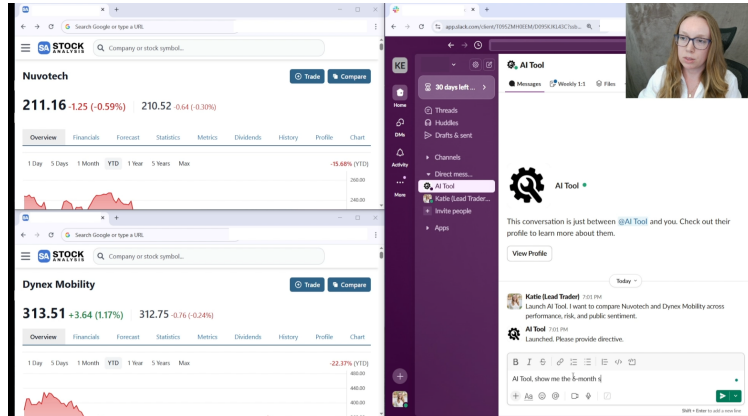


Figure 5.2: Screenshot taken from the video vignette for the between-subjects condition of the **Woman Human Worker (HW)** working with the **AI Tool Dyadic Partner (DP)** on the **Investment Banking Task**.

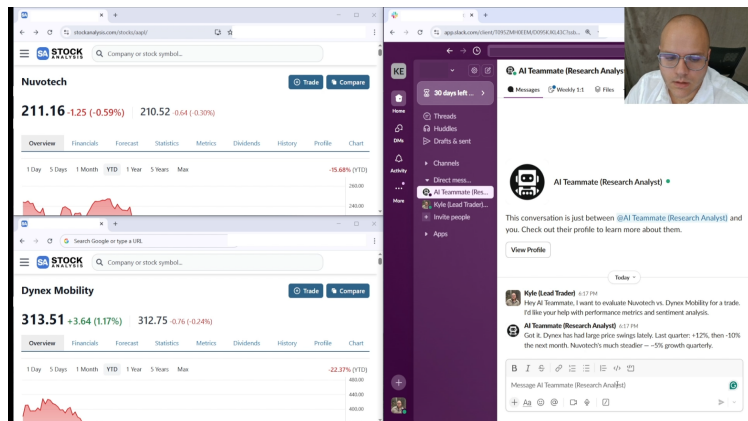


Figure 5.3: Screenshot taken from the video vignette for the between-subjects condition of the **Man Human Worker (HW)** working with the **AI Teammate (DP)** on the **Investment Banking Task**.

5.2.1.1 Manipulation #1: Human Worker (HW) Gender Presentation

Manipulation 1 in this study is Human Worker (HW) Gender Presentation, and consists of two conditions: a Woman Human Worker (HW) and a Man Human Worker (HW). This manipulation was included to better understand how working with different types of dyadic partners in an AI-enabled workplace might impact evaluator perceptions of women and men workers differently. This manipulation was operationalized in the factorial experimental survey design via (1) textual, (2) visual, and (3) auditory design.

Textual Design of Human Worker (HW) Gender. For (1) textual, the text contained within the parts of the survey that Figure 6.1 identifies as “Condition Specific” (i.e., Pre-Video Brief, Video Vignette, and Post-Video Surveys) was written to emphasize the gender presentation of the HWs by using conventionally gendered naming and pronouns, i.e., “Katie” and she/her for the Woman HW, “Kyle” and he/him for the Man HW. Examples of what this looked like in the Pre-Video Briefs can be seen in Figure 5.7 (Woman HW) and Figure 5.8 (Man HW). The Post-Video Survey questions were also worded in such a way that the HW name was repeated throughout the questions for clarity and continuing emphasis.

Visual Design of Human Worker (HW) Gender. For (2) visual, the condition-specific video contained in the Video Vignette of Dyadic Task block in Figure 6.1 communicated the gender presentation of the HW in two ways. First, the videos were filmed as if the human worker was sharing their screen with the evaluator, meaning that a human actor was live narrating the steps they were taking during the dyadic task. This set up meant that the HW actor appeared in a small screen in the top right corner of the video, as would be the case in live conferencing platforms like Zoom when sharing one’s screen.

These two human actors - one woman (see Figure 5.2) and one man (see Figure 5.4) - were chosen such that their basic presentation characteristics were as similar as possible in the hopes of reducing the impact that other differences besides gender presentation might have on evaluations. For example, both actors were White, wore glasses, had light-colored hair, wore white button-down shirts, and were instructed to make similar facial movements and expressions. Second, the video vignette featured a chat log that had profile pictures for the HW and the dyadic partner (DP). The profile picture featured next to the HW messages was a real picture that the two actors use as head-shots in professional settings, which can be seen in the example screenshots in Figure 5.2 (Woman HW) and Figure 5.4 (Man HW).

Before moving forward, it is important to note that the decision to use White actors for this study's operationalization of human workers was done based on knowledge gained from prior works about perceptions in group work (e.g., [47]). Such works have found that significant differences in participant evaluation of mixed-gender groups only manifests when White women work with White men, but not when White women work with Black men or when Black women work with White men [47]. Interestingly, it is theorized that the aforementioned results might be a reflection of how Black women's intersectional marginalization might render them invisible in the workplace [47, 48], and future work should bear this in mind. For this dissertation, though, the use of White actors was chosen because they would be the most likely to lead to significant differences between evaluations of women and men, if there would be any at all. This same motivation applies to the pictures used to represent the Female Teammate Dyadic Partner (DP) and Male Teammate Dyadic Partner (DP), both of which also appear White.

Auditory Design of Human Worker (HW) Gender. Finally, for (3) auditory, the two actors in the video vignettes narrated out loud all of the messages

sent between the HW and DP while working on the Dyadic Task. The pitch and tone of one’s voice is often used by other people as a perceptual cue to make assumptions about one’s gender [359], a topic this dissertation’s author has extensively explored in contexts of online social environments such as how women users may be identified and targeted for harassment in social Virtual Reality (VR) based on their voices [410]. Similarly, this dissertation’s exploration of differences in evaluations of women and men workers used audio narration to further emphasize the gender presentation of the HWs through pitch and tone.

More specifically, the actor for the Man HW condition had a lower pitch and tone more commonly associated with men/masculine individuals, while the actor for the Woman HW condition had a higher pitch and tone that is more commonly associated with women/feminine individuals. The use of audio also served the secondary purpose of ensuring that the participants had more than one way to understand what was happening during the Dyadic Task in case they had trouble keeping up visually with the chat in the video.

5.2.1.2 Manipulation #2: Dyadic Partner (DP) Type

Manipulation 2 of this study is Dyadic Partner (DP) Type, and consists of four conditions: an AI Tool Dyadic Partner (DP), an AI Teammate Dyadic Partner (DP), a Female Teammate Dyadic Partner (DP), and a Male Teammate Dyadic Partner (DP). By including both AI and Human DP conditions, this study is uniquely equipped to provide much needed insights on how AI-enabled workplaces might experience differing challenges in workplace evaluations when humans work with AI in comparison to when humans work with other humans, taking inspiration from prior works on human-AI teaming that utilize human-human teaming constructs to conceptualize human-AI teaming constructs [157, 405]. This manipulation contains

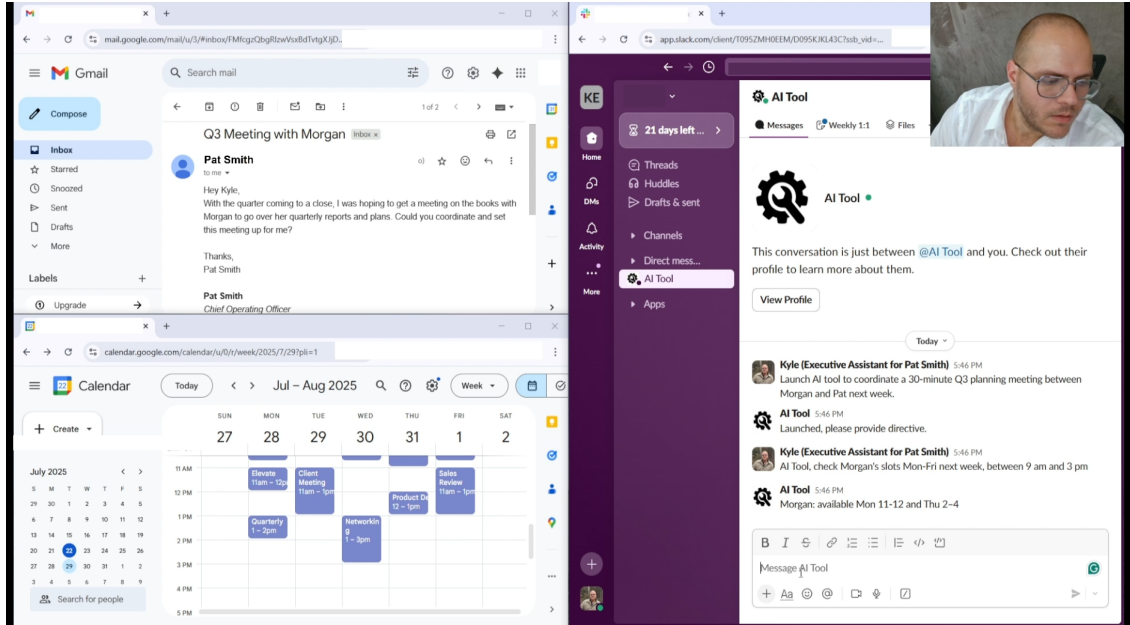


Figure 5.4: Screenshot taken from the video vignette for the between-subjects condition of the **Man Human Worker (HW)** working with the **AI Tool Dyadic Partner (DP)** on the **Meeting Scheduling Task**.

two different functional designs of an AI (Tool vs. Teammate) in order to better understand how working with an AI that has differing levels of argentic qualities might impact evaluator perceptions of human workers differently when working in an AI-enabled workplace.

This manipulation additionally contains two human partner conditions (Female vs. Male) to better understand how working with AI might impact evaluator perceptions differently than when human workers work with other humans, especially when considering workplace gender dynamics in group work. For example, women who work with men are often attributed less credit in a human-human teaming scenario than their male partners are, even when evaluators are presented with evidence that the woman worker contributed more than their male partner did [403]. This gender-based credit imbalance tends not to affect same-gendered partnerships to the same extent [403, 404], suggesting that there are perceptual disadvantages that are

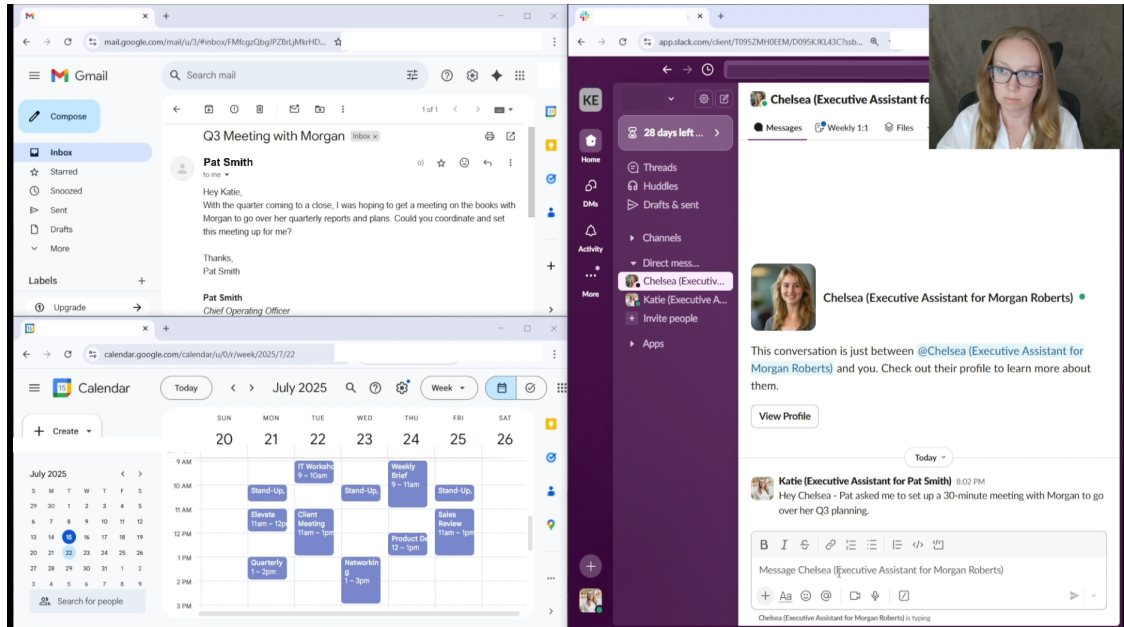


Figure 5.5: Screenshot taken from the video vignette for the between-subjects condition of the **Woman Human Worker (HW)** working with the **Female Teammate (DP)** on the **Meeting Scheduling Task**.

unique to women workers working with male workers versus other women workers. Given this, the Female Teammate DP condition essentially acts as the more favorable partnership condition for women workers being evaluated others, while the Male Teammate DP condition acts as the less favorable condition for women workers in workplaces.

Generally speaking, the DP manipulation was operationalized in the factorial experimental survey design via (2) functional design, (2) textual, and (3) visual cues, all of which are detailed below.

Functional Design of the Dyadic Partner (DP) Types. At a high level, the (1) functional design of the DPs differed between the AI Tool and all Teammate DP conditions, such that the AI Tool displayed behaviors that were less agentic and interdependent than the other conditions. For the AI Tool DP condition, the functional role of the AI was designed for it to work in a way that is intelligent and

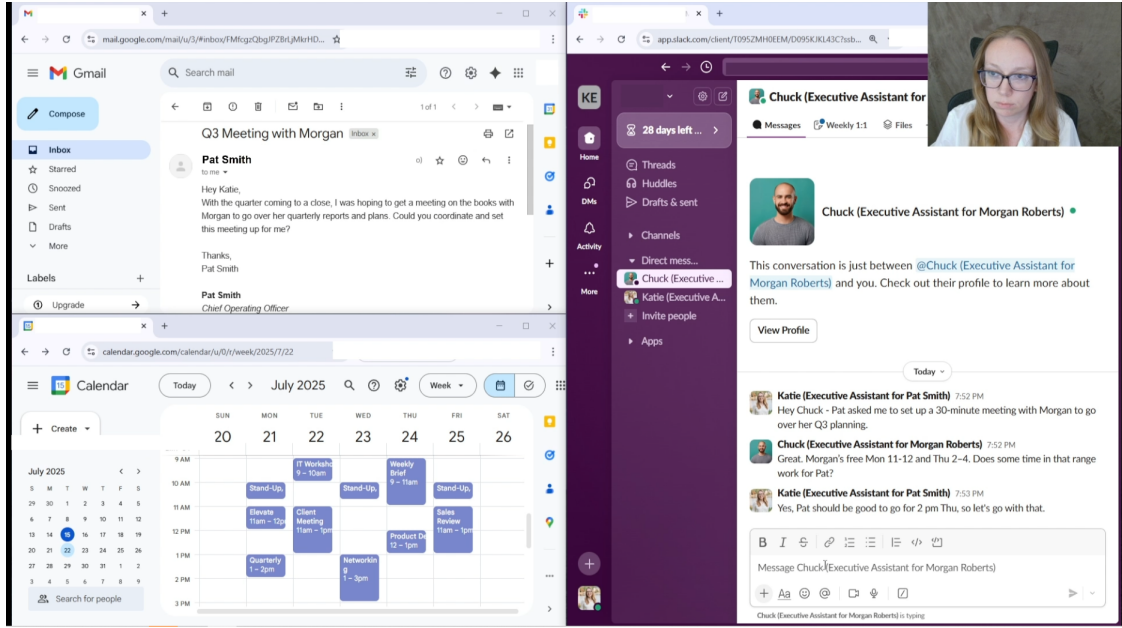


Figure 5.6: Screenshot taken from the video vignette for the between-subjects condition of the **Woman Human Worker (HW)** working with the **Male Teammate (DP)** on the **Meeting Scheduling Task**.

generative but lacks the ability to act autonomously from the human worker, such that the AI can assist or enhance the work of the HW but cannot be considered agentic.

More specifically, this study utilizes O’Neil et al.’s Levels of Autonomy continuum [347, 353] to conceptualize the functional design of the AI Tool DP as a Level 4 entity that can make suggestions and alternatives when prompted by the HW but lacks the necessary levels of autonomy to be considered agentic. This is an important distinction because being perceived as having agency is the necessary precursor to being able to make attributions of credit to the entity in question [298, 299]. This is accomplished within the conversational script used to craft the AI Tool’s responses during the course of the task, which has the AI Tool give responses to the HW that meet the HW’s request without providing proactive action or suggestions. It also lacks the interdependence that characterizes AI Teammates according to the LOA

continuum [347, 353], in that it does not proactively offer to share in task responsibilities with the HW. The script for the AI Tool DP conditions for the Meeting Scheduling task and the Investment Banking task can both be found in Appendix B.

By contrast, the AI Teammate DP condition was designed using O’Neil et al.’s [347] conceptualization, such that the AI Teammate was designed within the conversational script to work autonomously and interdependently with the HW assistant (Level 7), as opposed to nonautonomously and dependently as seen in the AI Tool DP condition. In doing so, while the HW still had to initiate the exchange with the AI Teammate DP via chat, the AI Teammate was designed to proactively anticipate what information the human worker would need to accomplish the task, and autonomously took on parts of the shared task (e.g., making a prior meeting summary for the upcoming meeting in the Meeting Scheduling Task). As such, the AI Teammate DP was designed to have a level of autonomy that prior works have demonstrated that human evaluators will perceive the AI entity as being agentic [298, 299, 353] and, therefore, able to attribute credit to.

Furthermore, the AI Teammate DP was designed to occupy a role within the dyadic partnership that will demonstrate its ability to be proactive and interdependent with the HW by autonomously taking on parts of the shared task, such as making a prior meeting summary for the upcoming meeting in the Meeting Scheduling Task, as seen in Appendix B. The inclusion of interdependence in its role design is critical because interdependence is a necessary condition for a human-AI/autonomy interaction to be classified as a human-AI/autonomy team [347].

Finally, when it comes to the way the conversational script was written, the functional design for the Female Teammate DP and Male Teammate DP conditions was identical to that of the AI Teammate DP condition. This was done in order to allow for direct comparisons to be made between how evaluators perceive DPs and

HWs that work with DPs on a collaborative tasks based on whether the teammate was being labeled and visually presented as an AI versus as a human despite having identical functional designs and roles.

However, there *is* a difference between the AI DP conditions (Tool and Teammate) and the human DP conditions in how quickly the DP responded to the HW in the chat conversation featured in the videos. For both AI conditions, the DP responds within 1-2 seconds after the HW finishes typing and sends the message. This was done as a way to further reinforce the appearance that these DPs were AI that were capable of processing and generating information at a speed far greater than a human could. Meanwhile, the response time between the HW sending the message and the DP responding was roughly 15 seconds for the human DP conditions. This was done as a way to further reinforce the feeling that the evaluator was watching an actual interaction between two humans, where the DP would need more time than an AI would to read, process, and generate information. In this sense, the functional design difference between the AI DP conditions and the human DP conditions was still delineated by functional design despite the identical scripting in the three Teammate conditions.

Textual Design of the Dyadic Partner (DP) Types. For (2), the textual design of the DP conditions was similar to the HW Manipulation in that briefings featured the names and pronouns (when applicable) of the DPs, and also included references within the Post-Video Surveys. More specifically, for the AI Tool DP condition, the condition-specific text featured in the Pre-Video Brief and the Video Vignette for explicitly told participants that they were about to see a human “*using*” an AI Tool to accomplish their task. This study’s use of explicit DP naming and framing takes inspiration from Flathmann et al.’s study on the purposeful presentation of AI teammates, which found that labeling a prospective workplace AI as a

tool versus a teammate changed participant perceptions, such as job security, despite having no functional design difference between the two [156]. This study expands upon this work by changing the actual functional design of the AI to correspond to the Tool versus Teammate label per the LOA continuum [353].

Furthermore, the specific phrasing of “*using*” was designed to reinforce the Tool aspect, much like one would “*use*” the other digital tools featured in the videos within an office-type situation (e.g., chat messaging). Additionally, the condition-specific text featured in the Pre-Video Brief and the Video Vignette for the AI Teammate DP condition explicitly told participants that they were about to see a human “*working with*” an AI Teammate. This phrasing was used to further emphasize the interdependent nature of the AI Teammate, contrasting the utilitarian phrasing of “*using*” an AI Tool.

Similarly to the functional design of the AI Teammate, the Female Teammate DP and Male Teammate DP conditions also used the phrasing of “*working with*” the HW. The primary difference here, though, is that the textual briefings for the human DP conditions explicitly referred to the DP by a gendered associated name - “Chelsea” for the Female Teammate DP, “Chuck” for the Male Teammate DP. Moreover, the partner was introduced as “*a woman named Chelsea,*” and “*a man named Chuck*” as a way to reinforce the intended gender presentation of the human DPs.

Visual Design of the Dyadic Partner (DP) Types. Finally, for (3), all DP conditions were operationalized visually via chat communication profile pictures and chat screen names. The pictures utilized for the AI DP conditions were intentionally designed to be plain black symbols that evoked imagery of a “tool” for the AI Tool DP, as shown in Figure 5.4, and that of an “artificial entity” for the AI Teammate DP, as shown in 5.3. These plain symbols were chosen because they provide visual cues for what kind of DP the HW was working with without creating a complex

anthropomorphic design, which would directly violate the recommendations made by this dissertation’s Study 1 women participants who advocated for avoiding gendered designs of AI. For the two human DP conditions, the profile pictures were designed to have similar characteristics as the HW actors, namely that they appear to be White, late 20s to early 30’s professionals with similar hair styles to the HWs. These DPs can be seen in Figures 5.5 (Female Teammate DP) and 5.6 (Male Teammate DP), respectfully.

It is also worth noting here that this study chose to operationalize the DP manipulation via a chat system for several reasons. First, most office-type workplaces that would be technically up-to-pace enough to have AI-enabled work also are likely to have collaborative chat systems such as Slack as a means of communicating and collaborating with other employees, thus making this an ecologically valid way of operationalizing an office-type collaborative effort. Second, this design allowed for the different identities of the DPs to be visible via clearly identified pictures and naming without introducing additional variables that could serve as confounding factors.

More specifically, this dissertation’s Study 1 continuously emphasizes the complications that come with “gendering” and anthropomorphizing AI, which is why this study’s design intentionally did not replicate the HW video design for the DPs so that the AI conditions could focus on functional design rather than anthropomorphizing AI unnecessarily. Finally, DP’s operationalization via chat, rather than video, serves to create visual separation between the HW and the DP. This makes sense, as the visual presentation of the HW needed to be as visible as possible to make sure that evaluators had enough information about the HW to form a more robust evaluation of them - especially because more of the Post-Video Survey measures are specific to the HW (i.e., HW job security) than the DP.

5.2.1.3 Manipulation # 3: Task Type

Manipulation 3 of this study is Task Type, and consists of two conditions: a Meeting Scheduling Task, which is a task associated with administrative assistant roles, and an Investment Banking Task, which is a task that is typically associated with higher-positioned roles. The type of dyadic task is important because the workplace context in which a partnership is working in can impact evaluator perceptions in ways that can favor women over men and vice versa, underscoring the intricate nature of gender dynamics in the workplace.

In particular, prior works on ambivalent sexism [188] and mixed-gendered partnerships [48] have shown that evaluations of women and men workers can differ depending on how closely associated a task is with gendered expectations. For example, women are sometimes perceived to be more competent than their man counterparts when performing tasks typically associated as “feminine”, e.g., child-rearing [129, 130]. However, the reverse happens in tasks or workplace environments that are traditionally considered to be “masculine”, e.g. STEM-related fields, where women are systematically perceived to be less accomplished and effective than their male counterparts even when evaluators are explicitly told that the women accomplish more [200]. As such, this study was motivated to pick two tasks that would engender “feminine” and “masculine” workplace context perceptions.

More specifically, Study 2 is motivated to used tasks with administrative-like duties as a test bed for the because prevailing discourse on the projected impacts of AI workforce integration on women workers has identified administrative assistant roles as particularly vulnerable to AI replacement and are majority held by women worldwide [3, 202, 449]. Indeed, the kinds of tasks associated with such a role, such as writing meeting notes and data gathering, are well-suited to the capabilities of

AI systems acting in the capacity of a highly skilled tool or as a teammate in an AI-integrated workplace (e.g., ChatGPT). Finally, the administrative assistant role embodied by the Meeting Scheduling Task is an interesting test bed because the role is currently and historically seen as “feminine”, or mostly associated with women workers. As such, there is a chance that perceptions of the woman HW might actually be *more* favorable than perceptions of the man HW when working in said task, as some limited prior works on ambivalent sexism in the workplace have found that women were rated more favorably than their male counterparts when the task was perceived to be “feminine” in nature [404].

Given this, there is the chance that Study 2’s results could serve to either reinforce or weaken AI integration concerns regarding evaluations of women workers in administrative assistant roles if participants perceive the task to be more “feminine” in nature. In contrast, there are also arguments that administrative roles within banking enterprises might be even more at risk for replacement as AI capabilities reach the level of human-like decision making most often seen in managerial roles that require more complex problem solving [449]. Given this, the Investment Banking Task evokes the images of a more male-dominated workplace and industry [2, 242] while also still falling under the umbrella of administrative-like work that has been flagged by some institutes as potentially vulnerable to worker replacement [449].

Similar to the other two manipulations, the Task Type manipulation was operationalized via (1) task functional design and (2) visual design. For (1), it can be seen how the two different tasks were operationalized and framed to participants in the Pre-Video Brief in Figures 5.8 (Investment Banking Task) and 5.9 (Meeting Scheduling Task). At a high level, for the Investment Banking Task, the partnership is working to help the HW decide which company to invest in by gathering information about each company and the DP’s recommendations. For the Meeting

Scheduling Task, the HW is tasked with scheduling a meeting for the person they are an Executive Assistant for, and coordinates with the other Executive Assistant who is the DP to get the meeting scheduled.

For (2), the visual operationalization of these Tasks can be seen in Figures 5.4 (Meeting Scheduling) and 5.3 (Investment Banking). At a high level, the screen recording for the Video Vignette shows a three-way screen split, with the left side of the screen showing two investment sites for the two companies that the HW will use to initiate the investment once they work with the DP on the task. The video in this condition cuts directly after the HW clicks the “*Invest*” button on the page, which quickly flashes a new page that confirms the investment before the video ends. For the Meeting Scheduling Task, there is also a three-way screen split, with the left side showing the email that the HW received from their boss alongside their boss’s Google calendar. The video cuts right after the HW clicks on the appropriate time slot that they negotiated with the DP during the collaborative task to schedule the meeting.

Both of these tasks were created to resemble platforms that are frequently utilized in everyday workplaces, such as Gmail and Google Calendar, in an effort to increase the ecological validity of the study. These sites were designed to contain enough visual information to simulate these environments while reducing potential confounding variables by keeping any titles, schedules, and names of companies vague. Similar logic and practice was applied to designing the chat visuals.

5.2.2 Factorial Survey Structure

The factorial survey was developed using Qualtrics, which is a commonly used professional survey platform in HCC research [156, 500]. It was distributed via an integrated link in Prolific, which is a professional online data collection platform that

offers high-quality and representative anonymous survey data for researchers [153]. Each participant in Prolific has a unique ID linked to their account to keep track of the studies that they participate in for payment, hence the inclusion of the Prolific ID fill-in in this study.

5.2.2.1 Informed Consent & Audio Test

Upon opening the link to the survey, participants were required to read and complete an informed consent form in accordance with Clemson University’s IRB requirements. If the Prolific user chose the option indicating that they did not consent to participation, Qualtrics was set up to automatically redirect the user back to Prolific without showing them any more of the factorial survey. For those who consented, they would be allowed to proceed to the rest of the factorial survey. Upon agreement, participants were asked to insert their Prolific IDs to continue, although the Qualtrics survey was designed to automatically input their Prolific IDs as a precautionary measure.

Then, participants were directed to test to make sure that their audio was working, as the video vignette had audio that participants needed to listen to. In an effort to ensure that all participants were, in fact, using audio, the audio test asked participants to listen to a four-second clip of a person reading a set of numbers. They were then required to input the set of numbers into a text field. The Qualtrics question was designed to allow the participants to listen to the audio more than once, but would not allow them to move forward if the participant could not enter the correct sequence of numbers.

5.2.2.2 Participant Demographics & Background

After passing the audio test, the factorial survey began with questions regarding participant demographics and background (see Appendix A.1), including their level of education (Figure 5), country of origin (Figure 7) and what U.S. state they currently reside in (Figure 7), and their position level within their current job role (e.g., student, self-employed, entry-level, manager, supervisor, executive leadership)(Figure 6). Questions regarding participant demographics were also asked, including age (Figure 1), ethnicity/race (Figure 4), sexual identity (e.g., straight, gay, bisexual, asexual, pansexual) (Figure 3), and gender identity (e.g., woman, man, agender, third gender/non-binary, gender fluid) (Figure 2). Participants were given the option not to disclose any of this information, or to specify if none of the options described them. Qualtrics screenshots displaying the structure of each of these questions can be found in Appendix A.1.1.

Importantly, gender identity was used to split participants evenly between the between-subjects conditions based on their response, i.e., 50% of women, 50% of men, and 50% of other participants (e.g., third gender, agender, non-binary, those who preferred not to answer) in each condition. There are two reasons for this style of split.

First, dividing participants evenly into each condition ensures that, should there be any individual gender differences in evaluations between women and non-women participants, these differences would be more readily accounted for in the study design. However, it is essential to note here that prior works on attributions of credit and workplace gender dynamics have more often found that there are no significant differences between women and men evaluators' perceptions [47]. As such, participant gender was treated as a covariate rather than a manipulation in the anal-

ysis. Second, this dissertation recognizes that gender identity is not a binary of just women and men, but rather there are many different types of gender identities that one can have, such as being agender, non-binary, and third gender, hence the intentional inclusion of a split to account for these more nuanced gender identities should there have been enough participants that identified in this category to include in data analysis.

5.2.2.3 Video Vignette Briefing

In the video on the next page, you will observe a woman named Katie, who is the Executive Assistant for Pat Smith at KE Enterprises, receive an email from Pat asking her to schedule a meeting with another KE Enterprises executive, Morgan Smith. To help with this task, Katie reaches out to Morgan's Executive Assistant, a woman named Chelsea, to get more information about Morgan's schedule and agenda expectations. Chelsea serves as an Executive Assistant to Morgan and can coordinate Morgan's schedule, provide information on prior meetings between Morgan and Pat, and help with making a notes tracker for the upcoming meeting.

Katie then uses the information she gained from working with Chelsea to create a coordinated meeting invitation on Pat's behalf that includes a Zoom link, agenda and summary of prior meetings, and a meeting notes tracker.

Based on your viewing of this video, your job is to evaluate Katie's performance, Chelsea's performance, and the performance of the partnership as a whole.

Figure 5.7: Qualtrics Screenshot of the Pre-Video Brief for the Between-Subjects Condition of the **Woman Human Worker (HW)** working with the **Female Teammate Dyadic Partner (DP)** on the **Meeting Scheduling Task**.

After completing the demographic and background questions, participants were met with a page explaining they would be moving to the next part of the study, which would require them to watch a video and evaluate the partnership they see in the video. This page also explained that clicking on the next page would bring them to a brief that would explain the scenario they were about to see and their role as the evaluator.

At a high level, the brief they were given explained that *“a man/woman named*

In the video on the next page, you will observe a man named Kyle, the Lead Investment Trader at KE Enterprises, deciding whether to invest in Nuvotech or Dynex Mobility stock. To help with this task, Kyle reaches out to KE Enterprise's Research Analyst, a man named Chuck, to get more information about these companies and their projections. Chuck serves as a research analyst for KE Enterprises and can provide real-time analytics on potential companies that traders are interested in.

Kyle then uses the information he gained from working with Chuck to make his final investment decision.

Based on your viewing of this video, your job is to evaluate Kyle's performance, Chuck's performance, and the performance of the partnership as a whole.

The following video also has sound, so please ensure that the sound on your device is turned up so that you can hear.

Figure 5.8: Qualtrics Screenshot of the Pre-Video Brief for the Between-Subjects Condition of the **Man Human Worker (HW)** working with the **Male Teammate Dyadic Partner (DP)** on the **Investment Banking Task**.

Katie/Kyle has a particular role at their company, *“KE Enterprises”*. For participants who were randomly assigned the Meeting Scheduling task, the job role was described as *“the Executive Assistant for Pat Smith”*, and that Pat had asked the HW to coordinate a meeting with a different executive at KE Enterprises. For participants who were randomly assigned the Investment Banking task, they were told that the human worker was *“the Lead Investor at KE Enterprises”*, and that they needed to pick between two companies to invest in. The brief then explained that the HW reaches out to the dyadic partner to help with the respective task. The descriptions for these partners were designed to emphasize their nature as either as a tool that the human worker uses (e.g., *“Katie/Kyle uses KE Enterprises’ AI Tool... The AI Tool Kyle uses can...”*) or a teammate that the human worker was working with (e.g., *“Katie/Kyle reaches out to Morgan’s Executive Assistant, a woman named Chelsea, to...”*). Then the brief explained the different capabilities and responsibilities of the dyadic partner, and how the human worker used the information they gained from working with the dyadic partner to successfully complete the task on their end.

In the video on the next page, you will observe a man named Kyle, who is the Executive Assistant for Pat Smith at KE Enterprises, receive an email from Pat asking him to schedule a meeting with another KE Enterprises executive, Morgan Smith. To help with this task, Kyle uses KE Enterprises' AI Tool to get more information about Morgan's schedule and agenda expectations. The AI Tool Kyle uses can coordinate Morgan's schedule, provide information on prior meetings between Morgan and Pat, and help with making a notes tracker for the upcoming meeting.

Kyle then uses the information he gained from using the AI Tool to create a coordinated meeting invitation on Pat's behalf that includes a Zoom link, agenda and summary of prior meetings, and a meeting notes tracker.

Based on your viewing of this video, your job is to evaluate Kyle's performance, the AI Tool's performance, and the performance of the partnership as a whole.

Figure 5.9: Qualtrics Screenshot of the Pre-Video Brief for the Between-Subjects Condition of the **Man Human Worker (HW)** working with the **AI Tool Dyadic Partner (DP)** on the **Meeting Scheduling Task**.

Importantly, after receiving the description of the scenario, the brief explicitly explained that the participant's job was to evaluate the performance of the HW, the DP, and the partnership based on their viewing of the video they were about to see. They were also reminded that the video would have sound, so they should ensure that their sound was on before going to the next page. Examples for what these briefs looked like can be found in Figures 5.7, 5.9, and 5.8. More specifically, an example of what the brief looked like for participants in the between-subjects condition of the Woman HW working with the Female DP on the Meeting Scheduling task can be found in Figure 5.7. Figure 5.9 also shows the Meeting Scheduling Task, but instead shows the Man HW working with the AI Tool DP to demonstrate how the wording changes when the condition is a tool rather than a teammate and a man HW rather than a woman HW. Finally, Figure 5.8 also shows the man HW, but instead shows how the brief looked in the Investment Banking task conditions, as well as the wording for the Male Teammate DP conditions.

5.2.2.4 Video Vignette Presentation

Upon clicking to the next page, participants were instructed to please watch the video contained on the page fully. They were also provided with a reminder of the scenario that they were going to be watching. This reminder broke down the briefing described in the prior subsection to highlight and emphasize the names of the HW and DP, the roles that each fulfilled, and that their job was to evaluate the performance of the HW, DP, and partnership.

This design was utilized for two reasons. First, it was suggested by pilot participants that having the brief on the same page as the video again would make it easier to remember what participants were looking for, as they could have a readily available reference to look back on if need be. Second, this reemphasizing was done as an extra measure to make the manipulations as explicit as possible through labeling and framing. This was further reinforced by the re-framing of the audio reminder on this page, which was phrased to ask participants to turn their sound on so that, *“you can hear Kyle/Katie’s narration of the messages being sent between Kyle/Katie and the AI Tool/AI Teammate/Chelsea/Chuck as they are working on the meeting scheduling/investment baking task.”*

Below the aforementioned descriptions, the Qualtrics page contained an embedded YouTube video corresponding to the between-subjects condition the participant was randomly assigned. This page was embedded with a Qualtrics feature that would not show the “Next” button to participants until two minutes after the participant had clicked the video. This was done as a way to decrease the possibility of participants moving through the survey without having watched the video first. While there is no way to guarantee that the participants truly watched the video because the participants were not being directly observed by the researcher, this hidden

embedded timer did help ensure that participants could not skip to the next screen without having spent time with the video playing. The two-minute mark was chosen because pilot testing revealed that this time frame struck the best balance between making sure the time period was long enough to last the majority of the video time for all conditions without being longer than the shortest of the between-subjects videos. There were instructions at the bottom of the page to let participants know to click the arrow that would appear once they were done watching the video.

5.2.2.5 Manipulation Checks, Vignette Measures, & Post-Survey Measures

Immediately after clicking to the next page, there were three questions to act as manipulation checks, asking participants to indicate (1) what they believed the HW's gender identity was; (2) what they believed the DP's gender identity was (if they had a gender at all, to account for the AI conditions); and (3) whether they believed the partnership successfully completed the task. These were used to ensure that the manipulations were interpreted as intended, which was found to be the case in this study's sample.

After answering these manipulation questions, participants were presented with the first of several vignette measures. These were measures designed to elicit participants' evaluations of HW and DP effectiveness, credit attributions to the HW and DP, comparative credit to HWs, HW job security, power perceptions of the HW, and partnership mutual dependence. Details of these measures can be found in 6.3.3 Measurements of this study. These measures were all included as a part of the evaluation component of this study's purpose within the dissertation.

After answering the vignette measures, participants were instructed that they would be entering into the final part of the survey. This final part consisted of post-

survey measures designed to measure participants’ preexisting beliefs. More specifically, there were measures for participant endorsement of hostile sexism, benevolent sexism, and negative attitudes towards AI. Details of these measures can also be found in 6.3.3 Measurements of this study. These measures were included to act as a way to account for participants’ pre-existing beliefs that could impact perceptions of the HWs, depending upon gender identity, as well as the DPs, depending on gender identity and feelings towards AI usage.

After answering all of the above-mentioned questions, participants were met with a screen thanking them for their participation. The participants were then instructed to click to the next page, where they were redirected back to Prolific via an embedded link in Qualtrics.

5.2.3 Measurements

This study’s factorial survey contained two sets of measurements: *vignette measures* and *post-survey measures*.

5.2.3.1 Vignette Measures

The *vignette measures* consist of the survey questions that participants were presented with directly after watching the video vignette. These include questions regarding human worker/dyadic partner effectiveness and credit, comparative credit, and partnership power and mutual dependence. Participants were instructed to answer the measures keeping in mind the partnership they had just seen.

Human Worker (HW)/Dyadic Partner (DP) Effectiveness. In order to understand how evaluators perceive human workers differently in terms of their effectiveness as a worker when working with different types of dyadic partners (DPs)

in light of DP effectiveness perceptions, participants were asked to rate the HW and DP on three attributes on a 1 - 9 numerical scale: incompetent (1) - competent (9); unproductive (1) - productive (9); and ineffective (1) - effective (9). This measure was derived from prior works on attributions of credit and employee evaluations in mixed-gendered groupwork, specifically [226] and [48].

Moreover, Study 2 took further inspiration from Biernat and Sesko [48] by including a single-item numerical rating measure for perceived warmth - Not Warm (1) - Warm (9). Warmth is a critical component of benevolent sexism ideology and gender stereotyping and may actually compensate for effectiveness/competence perceptions [48, 151]. Furthermore, there is early evidence to suggest that perceptions of warmth attributed to AI can also improve perceptions of AI [181], making it an interesting measure for this study to include on both fronts. The full survey measure for perceived competence and warmth can be found in Appendix A.2.1 (Table 2).

Credit Attributions to Human Workers (HWs) & Dyadic Partners (DPs). In order to understand how evaluators attribute credit to both human workers (HWs) and dyadic partners (DPs) depending on the gender presentation of the HW and the type of DP, participants were instructed to allocate a total of 100 points between four potential causes for task success: (1) ability (“(1) [INSERT HW/DP NAME] has above-average ability.”); (2) effort (“(2) [INSERT HW/DP NAME] exerted a great deal of effort.”); (3) task difficulty (“(3) This task was very easy.”); (4) luck (“(5) Good luck, chance, or some other irrational factor is important in this case.”). This method of measuring attributions is well-established in attribution research [9, 216, 253, 426, 470] and has been successfully utilized in HCI/HCC-related fields such as information systems research, e.g., [426].

Using the points allocated to each of these factors, an attribution score variable (ATTRIB) is formed by adding the two scores for the internal factors (i.e., ability and

effort) together, the two scores for the external factors (i.e., task difficulty and luck) together, and subtracting the combined internal score from the combined external score. The resulting score is used to determine if evaluators attribute the cause of the successful task completion to factors that are internal to the HW/DP, with +100 being a perfect internal score, or to factors that are external to the HW/DP, with -100 being a perfect external score. This formula is also well-established in attribution research [426, 470], and can be seen more clearly below:

$$\text{ATTRIB} = (\text{Ability} + \text{Effort}) - (\text{Task Difficulty} + \text{Luck})$$

Comparative Credit to Human Workers (HWs). In addition to the credit attribution measure, participants were also asked to indicate who they believed deserved the most amount of credit for the successful completion of the task. This was done via a one-item question on a scale from (1) “*Definitely [INSERT HW NAME]*” to (3) “*They Both Deserve An Equal Amount of Credit*” to (5) “*Definitely [INSERT DP NAME]*”. For analysis purposes, a lower value indicates that the participant believed the human worker (HW) deserves more credit for the successful completion of the task in comparison to their dyadic partner (DP), and vice versa.

Human Worker (HW) Job Security. Additionally, in order to understand how evaluations of Human Worker (HW) job security might be impacted by the gender presentation of the HW and the type of dyadic partner (DP) they work with, participants were instructed to evaluate the cognitive job security for the HW using Bazzoli and Probst’s [32] shortened and rescaled Job Security Index (JSI). The JSI consists of 9 items (7 point Likert scale, 1 = Strongly disagree, 7 = Strongly agree) designed to elicit cognitive perceptions of future employment, which were modified to fit this study’s social attribution perspective (e.g., “*After watching Katie work with the AI Tool, I believe Katie’s future employment is... **Unpredictable.***”). This

measure was included because job security is a dimension that other studies have used to compare worker perceptions of future impacts of workplace integration of AI tools versus AI teammates [156]. The full survey measures for job security can be found in A.2.3 (Table 3).

Partnership Power & Mutual Dependence. Finally, in order to understand how participants perceived the nature of the dyadic relationship between the HW and the DPs, two measures modified from the Human-Machine-Interaction-Interdependence-Questionnaire (HMII) by Woide et al. [476] were used. These two measures were modified versions of the 6-item, 5-point Likert scale (1 = Do not agree at all, 5 = Fully agree) Mutual Dependence dimension and the 4-item, 5-point Likert scale (1 = Definitely the [DP], 5 = Definitely the [HW]) Power dimension. The Mutual Dependence dimension is concerned with understanding how dependent the dyad was on each other to reach the outcome. The power dimension is concerned with the level of power evaluators felt that the Human Worker (HW) had in comparison to the dyadic partner. This has been included because limited prior works have pointed out that the position of power that each member of a human-AI interaction has can influence credit attribution [205, 418] and could reasonably be tied to relational attributions [131].

These two measures were only asked once because participants were instructed to answer these questions based on their perceptions of the dyad as a whole. The full survey measure can be found in A.2.4 (Table 4).

5.2.3.2 Post-Survey Measures

The *post-survey measures* were given to participants after completing all of the video vignette measures, and consisted of questions to measure participants' prior beliefs concerning hostile and benevolent sexism [192] and Negative Attitudes Towards

AI [408]. It is important to note that, while Study 2’s analysis will focus on the main effects of Human Worker (HW) gender presentation and Dyadic Partner (DP) type regardless of participants’ prior beliefs, these post-survey measures will still be detailed here as they are a part of the collection process for the data used in this study.

Perceived Task Femininity vs. Masculinity. The first post-survey measure was a one-item measure that asked participants if they felt that the type of administrative task that they saw in the vignette - i.e., the meeting scheduling task or investment banking task - would be considered more “feminine” or more “masculine” in nature. Responses were recorded on a numerical scale of “*feminine*” (1) to “*masculine*” (9). This measure was included as a way to account for the possibility that the perceived femininity/masculinity of the task would have an impact on perceptions of human workers or dyadic partners, as has been shown in prior research work on perceptions of mixed-gendered teams [27, 404].

Endorsement of Hostile & Benevolent Sexist Beliefs. In order to account for the possibility that participants’ pre-existing sexist beliefs might impact their evaluations of human workers (HWs) and dyadic partners (DPs), the factorial survey included a modified 16-item, 6-point Likert scale (0 = disagree strongly, 5 = agree strongly) Ambivalent Sexism Inventory (ASI) [188]. It is important to note that the ASI’s Likert scale design does not allow for a neutral answer, so participants had to either agree or disagree with the statements to some extent.

As a reminder, ambivalent sexism is a well-established construct that is comprised of two interrelated components: hostile sexism, or the belief that women are competitors to men who use their sexuality and/or feminism to gain power over men; and benevolent sexism, or the belief that women and men fill different roles in society such that women are more suited towards warmth and caregiving, and are thus

worthy of protection when complicit [27, 189].

The ASI was developed by the theory’s originators, Peter Glick and Susan Fiske, as a way to measure people’s levels of endorsement of ambivalent sexist beliefs, with 11 questions to measure endorsement of hostile sexism and 11 for benevolent sexism [188]. However, Study 2 modified the original 22-item ASI to exclude four benevolent sexism measures that are concerned with heterosexual intimacy questions. This is because this study was designed to accommodate participants of a variety of sexualities, which would make it difficult for all participants to be able to reliably answer the four items in question. Moreover, this study is not concerned with perceptions of sexual intimacy, so there were no practical measurement issues with excluding these items. Four hostile sexism measures were also removed to exclude three items related to heterosexual intimacy and one to reduce redundancy. The full survey measure for ambivalent sexism can be found in Appendix A.3.1 (Table 5).

Negative Attitudes Towards AI. Finally, in order to account for the possibility that participants’ pre-existing negative attitudes toward AI might impact their evaluations of AI DP conditions, the post-survey included a modified version of the General Attitudes towards Artificial Intelligence Scale (GAAIS) developed by Schepman and Rodway [408]. This is a 20-item, 5-point Likert scale (1 - Strongly disagree, 5- Strongly Agree) asking participants to rate their level of agreement with various statements about AI use in society in accordance with their personal views. To reduce redundancy and the length of the factorial survey, all positive-coded items were removed, leaving 10 negative-coded items. The full survey measure can be found in Appendix A.1.2 (Table 1).

5.2.4 Recruitment and Data Collection

The factorial survey was pilot tested and reviewed by this dissertation’s author and her professional contacts prior to Prolific deployment. The Prolific study was set to require respondents to be proficient in English, at least 18 years of age, and currently residing in the United States. This study was limited to the U.S. because it concerns cultural understandings of sexism that, while certainly present in other cultures [103], have been primarily observed and studied in the United States [75]. Additionally, requiring all respondents to be from the same country serves to reduce the chance of major sociocultural differences in societal expectations for women and men based on country of residence. Finally, the Prolific study parameters were set so that the platform would intentionally recruit an equal number of Prolific research participants whose accounts list their sex assigned at birth as male and an equal number whose sex assigned at birth as female. This was done to ensure that the study would recruit an equal number of people who are likely to identify as a woman and people who are likely to identify as a man. It is important to note that this study does not report on respondents’ sex assigned at birth, but rather only reports respondents’ answers to the demographic question on gender identity included in the factorial survey.

The Prolific study was deployed in July of 2025. First, 40 responses were collected and analyzed to ensure that the factorial survey was working properly. After confirming the quality of the data, the Prolific study participant number was increased on the same day to ultimately recruit a total of 703 participants. Of these 703 participants, 6 were rejected and not compensated for failing the attention checks per Prolific policy, and the remaining 697 participants were approved and compensated.

Variables	Total (N=685)	Percentage (%)
Gender Identity		
Woman	344	50.22
Man	341	49.78
Sexuality		
Asexual	6	0.88
Bisexual	57	8.32
Gay	19	2.77
Queer	3	0.44
Lesbian	12	1.75
Pansexual	8	1.17
Straight	576	84.09
Not Provided	4	0.58
Race/Ethnicity		
Asian	44	6.42
Black or African American	118	17.23
Hispanic or Latinx	21	3.07
Mixed	46	6.72
Native American or Alaska Native	4	0.58
Native Hawaiian or Pacific Islander	1	0.15
White	451	65.84
Age		
Average Age:	42.47 years old	
Age SD:	13.17 years	

Variables	Total (N=685)	Percentage (%)
Highest Level of Education		
2-year college degree (e.g., Associate's degree)	70	10.22
4-year college degree (e.g., Bachelor's degree)	278	40.58
Doctorate (e.g., JD, PhD, MD)	27	3.94
High school graduate or GED	93	13.58
Less than high school	5	0.73
Master's degree or equivalent graduate degree	104	15.18
Other/Prefer Not to Say	3	0.44
Some college	105	15.33
Job Position Level		
Department supervisor	49	7.15
Entry-level position	170	24.82
Executive leadership	15	2.19
Managerial position	225	32.85
Other/Prefer Not to Say	40	5.84
Retired	29	4.23
Self-employed	92	13.43
Student	17	2.48
Unemployed	48	7.01
Job Field		
Academic (e.g., student, professor, teacher)	46	6.72
Manufacturing industry (e.g., factory worker)	47	6.86
Medical profession (e.g., doctor, nurse, CNA)	52	7.59
Office-type work (e.g., administration, accounting, banking)	276	40.29
Other/Prefer Not to Say	168	24.53
Service industry (e.g., restaurant work, retail, daycare worker)	96	14.01

Table 5.1: Demographic information of 685 valid respondents.

5.2.5 Participants

In total, 697 respondents were approved and compensated in Prolific. Of those 697, data for 12 participants were removed from the data set post-approval and compensation because they identified outside of the gender binary of Woman or Man (e.g., agender, non-binary) or chose not to share their gender identity. This was done because part of this study’s focus is on perceptions of women vs men workers in the cultural context of the U.S., where the majority of the population identifies within the gender binary of women and men. Indeed, the low number of participants recruited who did not explicitly identify in this binary is reflective of the general U.S. population. These participants were thus dropped from the sample to allow for the statistical analysis of potential differences in evaluator perceptions based on a participant’s gender identities, which would have been statistically unfeasible if kept as a separate group and misleading if instead absorbed into the binary categories. On

average, respondents spent 19 minutes completing the factorial survey. Respondents who completed the survey and passed the attention check criteria were compensated at a rate of USD \$10.81 per hour through Prolific.

Table 5.1 summarizes the demographic information of the 685 valid respondents used in the data analysis for this study. As can be seen in said table, the sample includes a nearly equal number of women (50.22%) and men (49.78%) participants, indicating that Prolific recruitment criteria and respondent distribution performed successfully. The sample is fairly representative of the U.S. population, with the majority of the respondents identifying as straight (84.09%) or bisexual (8.32%), with other sexualities comprising less than 2% each of the sample. Additionally, the majority of respondents identify as White (65.84%), Black or African American (17.23%), Mixed (6.72%), Asian (6.42%), or Hispanic or Latinx (3.07%). The average age of the respondents is 42.47 ($SD = 13.17$), indicating that respondent perspectives represent a wide range of age groups.

Finally, the majority of the respondents hold a college degree of some kind (69.92%) or have completed some college education (15.33%). Interestingly, when it comes to respondents' job positions within their organizations, 42% of respondents indicated being in positions where they manage (32.85%), supervise (7.15%) or lead (2.19%) others. This indicates that a substantial number of respondents have prior experience evaluating employees in some form or another, making their perspectives especially relevant to their role as evaluators in this study. Furthermore, the job field that the largest proportional number of respondents indicated as representing their field is that of office-type work (e.g., administration, accounting, banking) (40.29%). This is important to note because this study uses an administrative assistant-type task (i.e., meeting scheduling) and a banking-related task (i.e., investment banking) as contexts for evaluation. As such, a large portion of the respondents have direct

insights into the nature of these tasks, making their perspectives especially valuable.

5.3 Methods Part II: Supplemental Expert Interviews

In HCC research, expert evaluations are often considered an important part of “formative evaluations,” which are typically characterized as evaluations that happen after a system is designed in order to work out the kinks of a system before the implementation of said system [160, 394]. As opposed to relying on observations of user behavior to inform system design, expert evaluations are considered uniquely effective in the design process because experts possess a level of domain knowledge that makes their insights particularly targeted to the needs of the environment that a system will be implemented into [79, 159, 243].

Taking inspiration from these works, Study 2 also conducted post-experiment expert interviews with 6 experts in AI-enabled workplaces (4 women, 2 men who work with women) to supplement the quantitative efforts detailed in Methods Part I. It is important to note that, while expert evaluations in HCC typically focus on the usability or interface design of a built system, expert evaluations of Study 2’s results is an especially valuable method in this dissertation because these experts are in positions to inform or craft the design and implementation of AI workplace integration now and into the future. Therefore, their insights are uniquely informative for conceptualizing future design recommendations for AI workplace integration that are considerate of the impacts that workplace gender dynamics and AI functional design can have if not properly considered.

More specifically, there are two primary motivations driving Study 2’s use

of these supplemental expert interviews. First, while the quantitative results can provide insights into *whether* and *how* evaluations of and attributions of credit to workers might differ depending on factors related to workplace gender dynamics and AI functional design, qualitative insights are still needed to understand *why* these differences might occur. Therefore, in order to understand why evaluations and credit differed when they did in light of what is already happening in real-world AI-enabled workplaces, these supplementary expert interviews served to explore how real-world experts in people management and AI workplace integration react to and interpret Study 2's quantitative results.

Second, this dissertation argues that there needs to be a greater emphasis on how companies and institutions that plan to or already are crafting AI workplace integration strategies should be responsible for preventing disproportionate harm to their women workers, rather than placing the burden on women workers themselves. Therefore, these supplementary expert interviews served to uncover how experts in AI-enabled workplaces understand and characterize the integration of AI within and outside of their own organizations, as well as their recommendations for organizations going forward for how to take women workers into consideration in their integration practices.

More importantly, to keep women's experiences with workplace gender dynamics continuously at the center of these conversations, it is necessary to understand how women and men who work with women in positions that allow them to craft AI use and integration policy can use insights gleaned from this dissertation's efforts to shed light on these little-understood impacts to inform future AI workplace integration that is considerate of *all* workers, thus motivating **Study 2's use of expert interviews**.

In light of the aforementioned motivations, and given this dissertation's Study 2's focus on unpacking the impacts that workplace gender dynamics and AI functional

design will have on women workers in AI-enabled workplaces, the supplemental expert interviews for Study 2 address the following research question:

S.2.RQ5 How do real-world experts in AI-enabled workplaces conceptualize AI workplace integration's impacts on evaluations of human workers?

5.3.1 Participants & Data Collection

As previously mentioned, to support the quantitative efforts performed by this study, supplemental interviews were conducted to explore how real-world experts in people management and AI workplace integration react to interpret the results of this work. In total, six experts in AI-enabled workplaces were interviewed, consisting of 4 women and 2 men who work with women. Their ages ranged from 28 to 59, and their professional roles ranged from workplace conflict management to AI research and development, and even being the CEO of a startup accelerator. A full list of participant demographics and work experience background can be seen in Table 5.2.

Approval by the university's Institutional Review Board was obtained prior to conducting these interviews. Interviews were conducted in October 2025 over Zoom and in person per the individual participants' preference. Participants were recruited via targeted recruitment, pulling from this dissertation's author's extensive connections in AI research and entrepreneurship. Participants were provided with a consent form upon starting the interview, and were asked if they gave consent to participate and consent to recording. All participants consented to interview participation, and all but P6 consented to being recorded for transcription purposes.

The structure of the interviews were inspired by expert interview qualitative methods, which comprises of qualitative semi-structured interviewing that encourages participants to share their insights in relation to their own experiences as experts in

their field [327]. During these interviews, experts were asked about their general opinions toward AI workplace integration, their own use of AI technologies in their workplace, and their predictions for the results of this study. After this section, the interview was split into two parts, one corresponding to Study 2 results and one corresponding to Study 3 results. The first of these sections was for Study 2, where participants were shown select results of Study 2 via a PowerPoint presentation, asked about their general reactions to these results, and asked how these results might impact their own workplaces or the way that they advise others on AI workplace integration. The full semi-structured interview guide used is available in Appendix B.5, and the review slides shown to participants during the interviews are presented in Appendix B.6.

PID	Age	Gender Identity	Ethnic Identity	Current Role	Exp. in Current Role	Exp. in Mngmt./AI
1	46	Man	White	Conflict Arbiter	> 1 Year	8 Years
2	29	Woman	White	AI Research & Development	> 1 Year	5 Years
3	34	Woman	Asian	Research Professor Human-Centered AI	3 Years	8 Years
4	59	Woman	Asian	CEO, Science/Tech Start-Up Accelerator	2 Years	20 Years
5	32	Man	White	Dir. of Business Intelligence	3 Years	5 Years
6	28	Woman	Asian	AI Governance Expert/Advisor	1 Year	6 Years

Table 5.2: Participant Demographics & Professional Backgrounds for Supplemental Expert Interviews for Studies 2 and 3.

5.3.2 Qualitative Data Analysis

Similarly to Study 1’s qualitative data analysis, data analysis for Study 2’s supplemental expert interviews was conducted using in-depth qualitative reflexive thematic analysis [55, 56]. This qualitative approach encourages researcher reflexivity rather than strict reliability as the researcher endeavors to yield recurring concepts and themes of interest, find relationships among them, and formulate them into more complex groups and broader themes [318]. Most importantly, the “reflexive” element of this type of thematic analysis conceptualizes the researcher’s subjectivity as a resource in the process of research, such that coding and interpreting data is recognized as an *“inherently and inescapably subjective practice”* [56].

Following Braun and Clarke’s [55, 56] procedures for reflexive thematic analysis, this dissertation’s author first carefully read through the collected data line-by-line to familiarize herself with the data, identifying and notating how these experts in AI-enabled workplaces interpreted Study 2’s quantitative findings and formulated their own views on AI workplace integration that is considerate of workplace gender dynamics. Next, the iterative coding process began with generating initial codes through assigning codes to individual data and then iterating on the codes to eliminate redundancy. Next, codes were categorized into thematic topics related to the research questions, which developed an initial set of themes related to experts’ insights on workplace gender dynamics and AI functional design on evaluations and attributions of credit to workers. Themes were further refined to better capture the data in relation to the research questions, and then again to identify the “essence” of each theme [55, 56]. Finally, the most compelling quotes were selected as examples to craft a narrative structure for the findings that flows logically and naturally [55, 56].

5.3.3 Researcher Positionality

While section 1.3 Dissertation Positionality in the Introduction and Motivations of this dissertation broadly details and establishes my positionality in terms of dissertation-wide foci and approaches, it is valuable here to outline specifics about how my positionality as a researcher impacted how I approached the supplemental expert interviews conducted in both Studies 2 and 3. Indeed, transparency about one’s positionality as a researcher is a vital component of reflexive approaches to thematic analysis, as it helps contextualize the researcher’s subjectivity [56]. With this in mind, there are two components of my positionality that impacted the way these interviews were conducted.

First, it is necessary to share my identity in relation to my participants, as my experiences with workplace gender dynamics within and outside of STEM related fields likely are rooted in my personal identity as a White, American woman in academia. This could be perceived as a limitation of this work, as three out of four of the women participants are Asian and might therefore experience or interpret workplace gender dynamics differently because of their intersectional identities. However, our shared identities as women in academic and industry fields related to AI and tech does allow me to be aware of the unique challenges that women face in U.S. markets, which all participants operate within.

Second, it is important to acknowledge that all 6 participants are individuals that I knew prior to conducting these interviews through my involvement in both academic AI research circles and local women in science and tech organizations. There is therefore a possibility that their insights might have been influenced by their pre-existing perceptions of me and my work. However, their insights as experts are still highly valuable given their well-established years of experience and specific exper-

tise in people management and AI workplace integration (see Table 5.2). Moreover, the pre-existing relationships that I had with these individuals gave me the unique benefit of already having established rapport and shared understanding with my interviewees, which can be seen as highly beneficial for eliciting rich qualitative data when conducting interviews [446].

5.4 Results Part I - Quantitative Results for Factorial Experiment

The following results outline the marginal effects of Study 2’s experimental manipulations on the perceived evaluation of performance, causal impacts, job security, and mutual relationships for the workers and dyadic partners depicted in the video vignettes. Unless otherwise specified, all of the following metrics were analyzed via ANCOVAs, where each dependent variable tested the significance of manipulations of Human Worker (HW) Gender, Dyadic Partner (DP) Type, and Task Type and their intervention effects. Significant results are supported by graphs and tables of estimated marginal means.

Study 2 is focused on understanding how evaluators perceive the performance of a human worker working with different types of AI or human dyadic partners in an AI-integrated everyday workplace. More specifically, Study 2 is interested in uncovering how, if at all, evaluator perceptions differ depending upon the gender presentation of the human worker, how they differ depending upon the type of dyadic partner the human worker is partnered with, and how these two factors might interact to create differing impacts on women vs. men workers in an everyday workplace. Given the interests of this study, this four-part results section looks at the manipulations

of Human Worker (HW) Gender Presentation (see 6.4.1) and Dyadic Partner (DP) Type (see 6.4.2) in isolation from each other first, such that any robust effects will be readily identifiable. Then the results will turn towards how these two manipulations interact to impact evaluator perceptions (see 6.4.3). Each of these three subsections is further divided by the main effects of the manipulations per dependent variable (Tables 5.3, 5.6, and 5.17) and the manipulations' interaction effects with Task Type (Tables 5.4, 5.14, and 5.19). It is worth noting that to better highlight the most novel and unique insights from this study, we primarily focus on reporting and analyzing results that are statistically significant, rather than those that do not reach statistical significance.

Before moving forward, it is helpful to further explain why Study 2 is treating Task Type as a secondary manipulation rather than a primary manipulation. Although Task Type was manipulated in the experimental design, its purpose as a manipulation is to act as a secondary factor that could impact how evaluators perceive the Human Worker (HW), the Dyadic Partner (DP), and/or the partnership differently depending upon the nature of the everyday workplace task being performed. In other words, Study 2 is most interested in how the gender presentation of the Human Worker (HW) and the type of Dyadic Partner (DP) they work with impact evaluator perceptions regardless of the type of task they are doing.

However, this study also recognizes that the type of task being performed can have differing impacts on how human workers are perceived. In particular, prior works on ambivalent sexism and mixed-gendered partnerships have shown that evaluations of women and men workers can differ depending on how closely associated a task is with gendered expectations, such that women are sometimes perceived to be more competent than their man counterparts when performing tasks typically associated as “feminine”, e.g., child-rearing [129, 130]. It was therefore still important to the

purpose of this study to account for this relationship between task type and gendered associations. As such, the final subsection of this results section unpacks the surprising ways that task-type gendered associations depend not just on the type of task being performed, but also the gender presentation of the Human Worker (HW) and type of Dyadic Partner (DP) (see 6.4.4).

5.4.1 Effects of Human Worker (HW) Gender Presentation on Evaluator Perceptions

Evaluation Perception	<i>F</i> (1,666)	<i>p-value</i>	η_p^2	Significance
Effectiveness of Human Worker (HW)	1.73	.19	—	—
Effectiveness of Dyadic Partner (DP)	3.6	.058	.01	Borderline
Attrib. of Credit to HW - Internal v. External	.9	.344	—	—
Attrib. of Credit to HW - Ability v. Effort	.79	.376	—	—
Attrib. of Credit to DP - Internal v. External	1.00	.318	—	—
Attrib. of Credit to DP - Ability v. Effort	.43	.514	—	—
Comparative Credit	.04	.837	—	—
Human Worker (HW) Job Security	.00	.991	—	—
Human Worker (HW) Expectations of Power	.17	.68	—	—
Partnership Mutual Dependence	.32	.571	—	—

Table 5.3: Summary of Results for Main Effects of Human Worker (HW) Gender Presentation on Evaluation Perceptions

5.4.1.1 Main Effects for Human Worker (HW) Gender Presentation on Evaluator Perceptions

Overall, results of the ANCOVAs demonstrated no significant main effects of Human Worker (HW) Gender Presentation for all evaluation perceptions, as seen in Table 5.3, although evaluation perceptions of effectiveness of the DP were borderline significant. This means that the data analysis did not reveal any significant differences in how human workers or dyadic partners were perceived as effective or

attributed credit solely based on whether the human worker was a woman or a man. Furthermore, data analysis did not reveal any significant differences in whether the human worker was attributed more comparative credit than their dyadic partners, solely based on whether the human worker was a woman or a man. Data analysis also did not reveal any significant differences in evaluators' perceptions of the human worker's job security and expectations of power solely based on whether the human worker was a woman or a man. Finally, data analysis also did not reveal any significant differences in evaluators' perceptions of the partnership's mutual dependence based solely on whether the human worker was a woman or a man. Overall, then, Study 2's results can be interpreted to mean that evaluators' perceptions cannot solely be attributed to whether the human worker in a dyadic partnership within an AI-integrated workforce is a woman or a man.

Table 5.3 provides the statistical results of the ANCOVAs and indications of their (non) significance, while a detailed description of these (non)significant results can be found below.

Perceived Effectiveness of the Dyadic Partner (DP). There was a **borderline significant** effect of Human Worker (HW) gender presentation on the perceived effectiveness of the dyadic partner ($F(1,666) = 3.6, p = .058, \eta_p^2 = .01$), although it did not reach significance and will therefore not be further reported.

5.4.1.2 Interactions Between Human Worker (HW) Gender Presentation & Task Type on Evaluator Perceptions

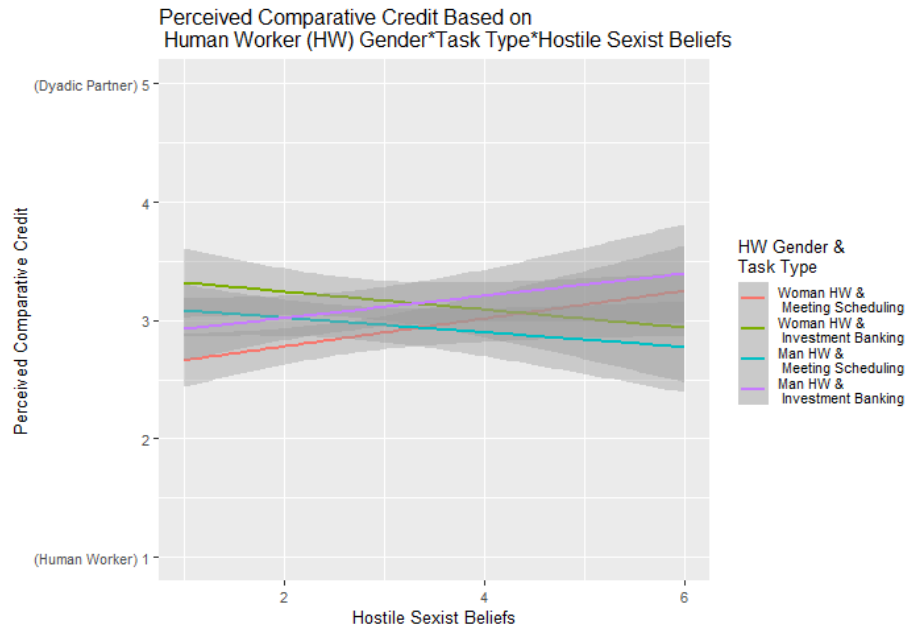
As can be seen in Table 5.4, the results of the ANCOVA performed for evaluators' perceptions of Human Worker (HW) Expectations of Power did reveal a significant interaction effect, such that there is a significant difference between how much power evaluators believed the Human Worker (HW) thought they had based

Evaluation Perception	$F(1,666)$	p -value	η_p^2	Significance
Effectiveness of Human Worker (HW)	.01	.936	—	—
Effectiveness of Dyadic Partner (DP)	.07	.791	—	—
Attrib. of Credit to HW - Internal v. External	.59	.444	—	—
Attrib. of Credit to HW - Ability v. Effort	1.31	.252	—	—
Attrib. of Credit to DP - Internal v. External	.06	.812	—	—
Attrib. of Credit to DP - Ability v. Effort	.34	.559	—	—
Comparative Credit	.04	.837	—	—
Human Worker (HW) Job Security	1.73	.189	—	—
Human Worker (HW) Expectations of Power	4.76	.03*	.01	*Significant
Partnership Mutual Dependence	.01	.905	—	—

Table 5.4: Summary of Results for Interaction Effects Between Human Worker (HW) Gender Presentation & Task Type on Evaluation Perceptions

on whether the human worker was a woman or a man working on a meeting scheduling task or an investment banking task.

The remainder of the results of the ANCOVAs demonstrated no significant interaction effects between Human Worker (HW) Gender Presentation and Task Type for all of the other evaluation perceptions, as seen in Table 5.4. This means that the data analysis did not reveal any significant differences in how human workers or dyadic partners were perceived and attributed credit based on whether the human worker was a woman or a man working on a meeting-scheduling task or an investment-banking task. Furthermore, data analysis did not reveal any significant differences in whether the human worker was attributed more comparative credit than their dyadic partners based on whether the human worker was a woman or a man working on a meeting-scheduling task or an investment-banking task. Data analysis also did not reveal any significant differences in evaluators' perceptions of the human worker's job security based on whether the human worker was a woman or a man working on a meeting-scheduling task or an investment-banking task. Finally, data analysis also did not reveal any significant differences in evaluators' perceptions of the partnership's mutual



Task Type	Worker Gender	Trend	SE	df	lower CL	upper CL
Meeting Scheduling	Woman	0.0628	0.0614	657	-0.0578	0.1835
	Man	-0.0826	0.0593	657	-0.1990	0.0338
Investment Banking	Woman	-0.0791	0.0628	657	-0.2024	0.0442
	Man	0.0815	0.0545	657	-0.0255	0.1885

Figure 5.10: Perceived Comparative Credit to Human Workers (HWs) Based on the Interaction Between Human Worker (HW) Gender, Task Type, and Evaluator Endorsement of Hostile Sexist Beliefs.

dependence based on whether the human worker was a woman or a man working on a meeting-scheduling task or an investment-banking task.

Overall, then, Study 2's results can be interpreted to mean that evaluators' perceptions cannot be attributed to whether the human worker in a dyadic partnership within an AI-integrated workforce is a woman or a man working on a meeting scheduling task or an investment banking task. Table 5.3 provides the statistical results of the ANCOVAs and indications of their (non) significance, while the technical breakdown of these (non)significant results can be found below.

Evaluation of Comparative Credit to Human Workers (HWs). There

was **no significant** interaction effect between human worker (HW) gender presentation and task type on the comparative credit to human workers (HWs) ($F(1,666) = .04$, $p = .837$). Interestingly, though, there *is* a **significant** three way interaction between Human Worker (HW) Gender, Task Type, and evaluators' endorsement of hostile sexist beliefs. Looking towards Figure 5.10, there are several trends worth noting. First, it can be seen that perceived comparative credit trends towards more credit being given to the dyadic partner as endorsement of hostile sexist beliefs increases in two conditions, i.e. when the **Woman HW** is working with a dyadic partner on a **Meeting Scheduling Task**, and when the **Man HW** is working with a dyadic partner on a **Investment Banking Task**. Conversely, Figure 5.10 shows that perceived comparative credit trends downward and towards the human worker in the two other conditions, i.e., when the **Woman HW** is working with a dyadic partner on an **Investment Banking Task**, and when the **Man HW** is working with a dyadic partner on a **Meeting Scheduling Task**.

These results are intriguing considering hostile sexism's close ties with concepts that degrade women and paint them in an unfavorable light (e.g., women are "manipulative") [29]. Given this, there is the sense that the Woman HW should have been evaluated less favorably as hostile sexism increases across both task types, which is not the case here. Furthermore, it can at first appear surprising that higher endorsements of hostile sexist beliefs would be associated with less credit to the Man HW in the Investment Banking Task. However, the results seen in Figure 5.10 arguably demonstrate the complex nature of hostile sexism as a measure of how people view gender roles in society in ways that can harm both women *and* men [29]. For example, it could be the case that individuals who endorse hostile sexist beliefs to a greater extent might have felt that the HWs deserved less credit when working with a dyadic partner on a task that matches patriarchal expectations of gendered work,

because that type of work should have been easy for them. In other words, it could be the case that these evaluators felt that the HWs were being lazy or unnecessarily offloading work that they should otherwise be well-suited for because of rigid perceptions of gender roles (i.e., women in admin roles, and men in investment banking roles).

In contrast, in conditions where the gender presentation of the HW did not align with hegemonic patriarchal view points of gendered work (i.e., women in investment banking, men in admin work), it could have been the case that individuals with greater levels of hostile sexist beliefs were more surprised or impressed that the HW could successfully accomplish a task that they would deem as less suitable given gender role norms. Alternatively, it could be the case that such evaluators simply did not apply the same level of punishment as they would in the gender role expectations conditions, thereby creating a somewhat false impression that individuals with more hostile sexist beliefs are more favorable to giving credit to HWs than individuals with less hostile sexist beliefs.

HW Gender	Task Type	Marginal Mean	95% CI for Mean Difference		SE
			Lower	Upper	
Woman	Meeting Scheduling	4.241	4.136	4.345	0.053
Man		4.379	4.274	4.485	0.054
Woman	Investment Banking	4.207	4.103	4.312	0.053
Man		4.113	4.008	4.217	0.053

Table 5.5: Marginal Means - Perceived Comparative Power Based on Interaction Between Presentation of Human Worker Gender and Task Type

Evaluations of Power Perceptions. There was a **significant** interaction effect between human worker (HW) gender presentation and task type on the evaluations of power within the partnership ($F(1,666) = 4.76, p = .03, \eta_p^2 = .01$) (Figure 5.11). An analysis of simple main effects was conducted to explore this interaction,



Figure 5.11: Perceived Comparative Power Based on Interaction Between Presentation of Human Worker Gender and Task Type. Error bars denote 95% confidence interval.

and the resulting marginal means can be found in Table 5.5. Interpretation of the marginal means reveals that, when controlling for Task Type, there was a significant difference in evaluations of power within the partnership in the **Man HW** condition, but not in the **Woman HW** condition. Looking towards Figure 5.11 and the error bars, the non-overlapping error bars indicate that evaluators felt that the **Man HW** expected to have more power than their dyadic partner to a greater extent when the pair was working on the meeting scheduling task, compared to when they were working on an investment banking task. Conversely, the overlapping error bars for the **Woman HW** indicate that there was no significant difference between evaluators' perceptions of the Woman Human Worker's (HW's) expectations of power over their dyadic partners when working in the different task types. It is worth noting that it can be seen in Figure 5.11 that evaluators overall perceived the Human Worker (HW) as expecting to have more power than their Dyadic Partner (DP) in both HW

conditions and task type conditions.

5.4.2 Effects of Dyadic Partner (DP) Type on Evaluator Perceptions

5.4.2.1 Main Effects for Dyadic Partner (DP) Type on Evaluator Perceptions

Evaluation Perception	<i>F</i> (1,666)	<i>p</i> -value	η_p^2	Significance
Effectiveness of Human Worker (HW)	4.52	.004*	.02	*Significant
Effectiveness of Dyadic Partner (DP)	12.12	< .001*	.05	*Significant
Attrib. of Credit to HW - Internal v. External	.33	.806	—	—
Attrib. of Credit to HW - Ability v. Effort	1.76	.115	—	—
Attrib. of Credit to DP - Internal v. External	1.5	.215	—	—
Attrib. of Credit to DP - Ability v. Effort	18.06	< .001*	.08	*Significant
Comparative Credit	5.31	.001*	.02	*Significant
Human Worker (HW) Job Security	13.12	< .001*	.06	*Significant
Human Worker (HW) Expectations of Power	2.98	.031*	.01	*Significant
Partnership Mutual Dependence	6.61	< .001*	.03	*Significant

Table 5.6: Summary of Results for Main Effects of Dyadic Partner (DP) Type on Evaluation Perceptions

DP Type	Marginal Mean	95% CI for Mean Difference		SE
		Lower	Upper	
AI Teammate	7.834	7.673	7.995	0.082
AI Tool	7.447	7.288	7.606	0.081
Female Teammate	7.764	7.605	7.924	0.081
Male Teammate	7.760	7.600	7.921	0.082

Table 5.7: Marginal Means - Perceived Effectiveness of Human Worker Based on Dyadic Partner Type

In contrast to Study 2’s Human Worker (HW) condition, the Dyadic Partner (DP) Type played a significant role in participants’ evaluation perceptions, as can

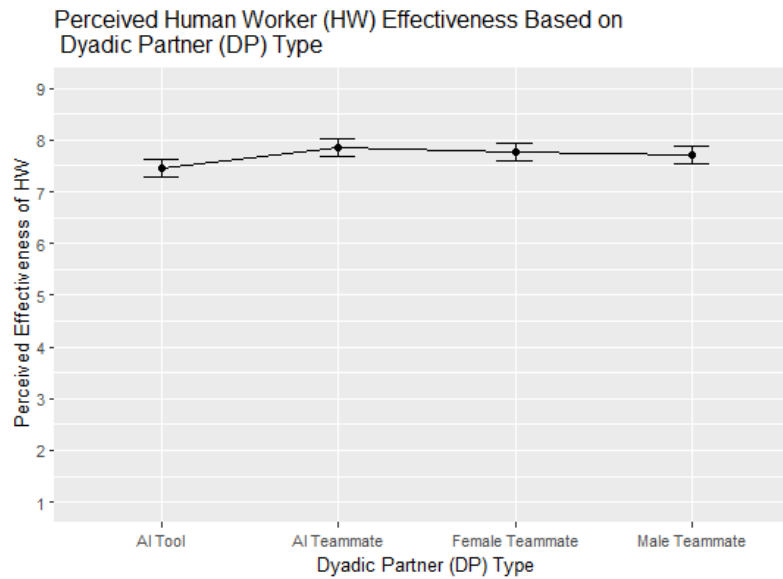


Figure 5.12: Perceived Effectiveness of Human Worker Based on Dyadic Partner Type. Error bars denote 95% confidence interval.

be seen in Table 5.6. Indeed, there are only a few evaluation perceptions that were not significantly impacted by the type of Dyadic Partner (DP). More specifically, the data analysis did not reveal any significant differences in how human workers or dyadic partners were attributed internal vs. external credit solely based on the type of Dyadic Partner (DP). The data analysis also did not reveal any significant differences in how human workers were attributed the internal factors of ability vs. effort, but did reveal a significant difference in how dyadic partners were attributed the internal factors of ability vs. effort based solely on the type of Dyadic Partner (DP).

However, as Table 5.6 shows, the data analysis did reveal significant differences in evaluators' perceptions of human worker and dyadic partner effectiveness solely based on the type of Dyadic Partner (DP). Furthermore, data analysis did reveal significant differences in whether the human worker was attributed more comparative credit than their dyadic partners, solely based on the type of Dyadic Partner

(DP) they were working with. Data analysis also revealed significant differences in evaluators' perceptions of the human worker's job security and expectations of power solely based on the type of Dyadic Partner (DP) they were working with. Finally, data analysis did reveal significant differences in evaluators' perceptions of the partnership's mutual dependence solely based on the type of Dyadic Partner (DP). Overall, then, Study 2's results can best be interpreted to mean that many different types of evaluators' perceptions of a human worker and their dyadic partner *can* be solely attributed to the type of Dyadic Partner (DP) a human worker works with in a dyadic partnership within an AI-integrated workforce.

Table 5.6 provides the statistical overview of the results of the ANCOVAs and indications of significance, while the technical breakdown of these results can be found in the text below. Significant results are also further unpacked and interpreted below.

Perceived Effectiveness of the Human Worker (HW). There was a **significant** main effect of dyadic partner (DP) type on the perceived effectiveness of the human worker (HW) ($F(1,666) = 4.52, p = .004, \eta_p^2 = .02$) (Figure 5.12). A Tukey post-hoc test was performed to compare each of the four dyadic partner (DP) types. The perceived effectiveness of the human worker (HW) was shown to be significantly different between the AI Teammate and AI Tool DP conditions, the Female Teammate and AI Tool DP conditions, and the Male Teammate and AI Tool DP conditions. Looking towards the marginal means in Table 5.7, it can be seen that the perceived effectiveness of the Human Worker (HW) was significantly lower when they were working with an AI Tool compared to when they were working with every other type of Dyadic Partner (DP). Moreover, evaluators perceived the effectiveness of the Human Work (HW) as being at its highest when partnered with the Female Teammate DP, although both the Male Teammate and AI Teammate conditions closely followed, as can be seen in Figure 5.12.

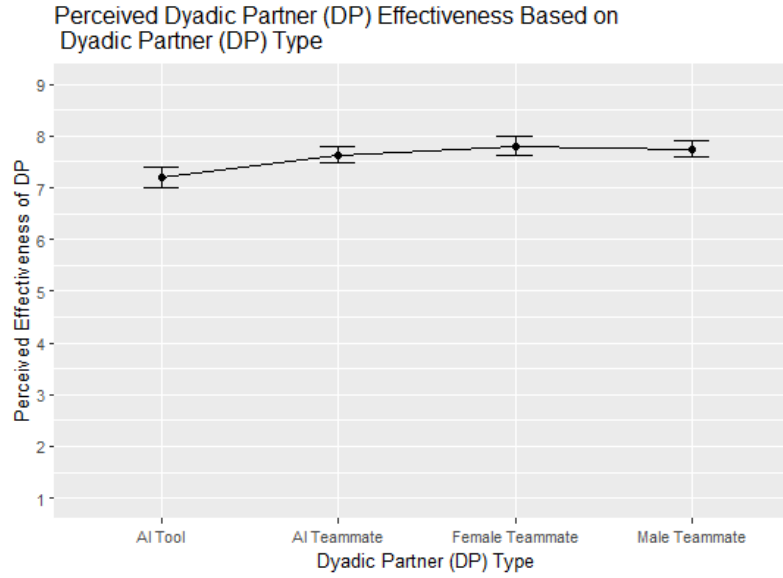
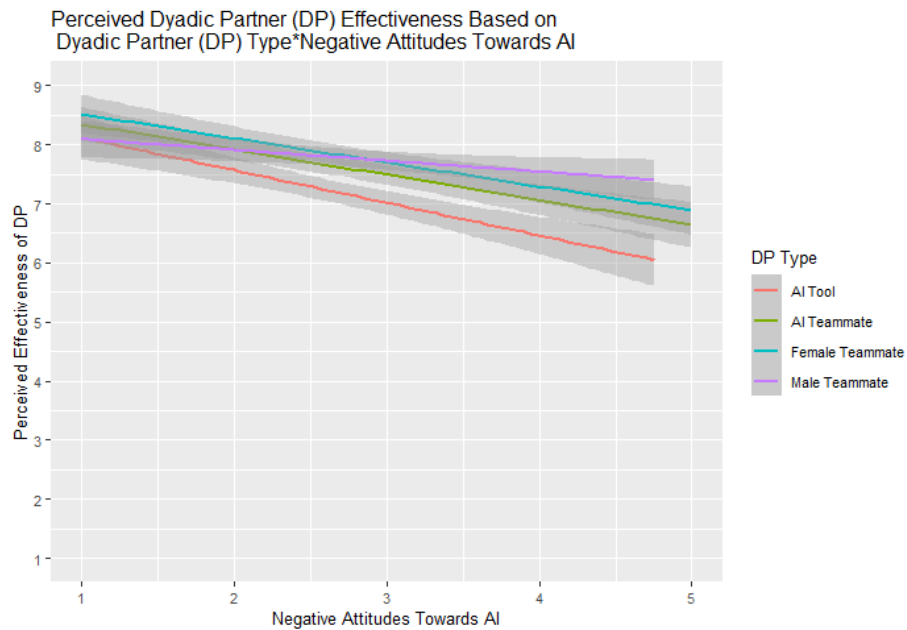


Figure 5.13: Perceived Effectiveness of the Dyadic Partner Based on Dyadic Partner Type. Error bars denote 95% confidence interval.

DP Type	Marginal Mean	95% CI for Mean Difference		SE
		Lower	Upper	
AI Teammate	7.630	7.468	7.792	0.082
AI Tool	7.198	7.037	7.358	0.082
Female Teammate	7.810	7.650	7.971	0.082
Male Teammate	7.792	7.630	7.953	0.082

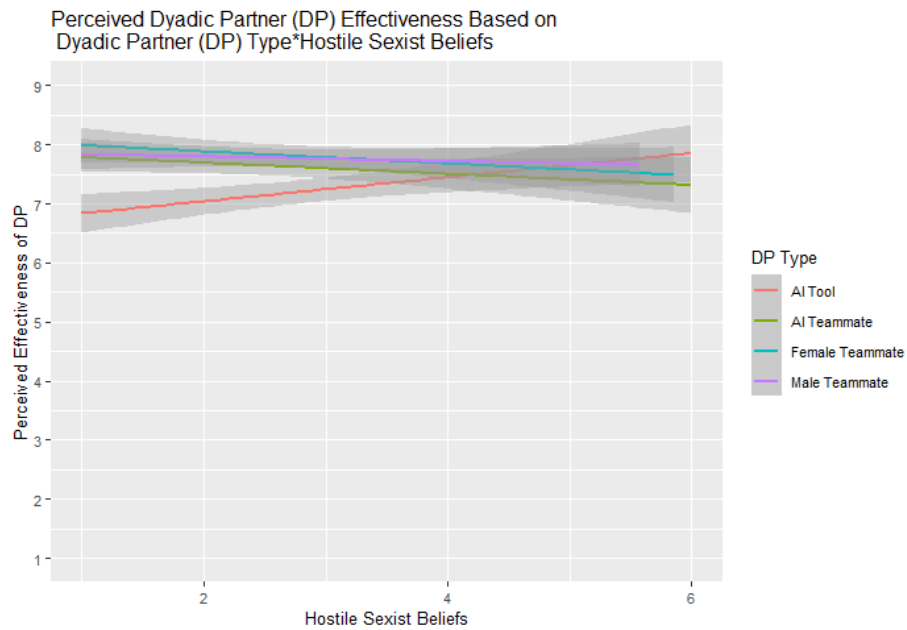
Table 5.8: Marginal Means - Perceived Effectiveness of the Dyadic Partner Based on Dyadic Partner Type

Perceived Effectiveness of the Dyadic Partner (DP). There was a **significant** main effect of Dyadic Partner (HW) type on the perceived effectiveness of the dyadic partner (DP) ($F(1,666) = 12.12, p < .001, \eta_p^2 = .05$) (Figure 5.13). A Tukey post-hoc test was performed to compare each of the four dyadic partner (DP) types. Here again, the perceived effectiveness of the dyadic partner (DP) was shown to be significantly different between the AI Teammate and AI Tool DP conditions, the Female Teammate and AI Tool DP conditions, and the Male Teammate and AI Tool DP conditions. Looking towards the marginal means in Table 5.8, it can be seen that



Dyadic Partner Type	Trend	SE	df	lower.CL	upper.CL
AI Tool	-0.564	0.0840	660	-0.729	-0.3996
AI Teammate	-0.416	0.0794	660	-0.572	-0.2601
Female Teammate	-0.403	0.0806	660	-0.561	-0.2445
Male Teammate	-0.183	0.0841	660	-0.349	-0.0183

Figure 5.14: Perceived Effectiveness of the Dyadic Partner (DP) Based on the Interaction Between Dyadic Partner (DP) Type and Evaluator Negative Attitudes Towards AI.



Dyadic Partner Type	Trend	SE	df	lower.CL	upper.CL
AI Tool	0.1756	0.0590	660	0.0597	0.29153
AI Teammate	-0.1254	0.0647	660	-0.2525	0.00161
Female Teammate	-0.1201	0.0627	660	-0.2433	0.00304
Male Teammate	-0.0672	0.0643	660	-0.1934	0.05892

Figure 5.15: Perceived Effectiveness of the Dyadic Partner (DP) Based on the Interaction Between Dyadic Partner (DP) Type and Evaluator Endorsement of Hostile Sexist Beliefs.

the AI Tool DP was perceived to be generally less effective than all of the Teammate conditions. Moreover, Figure 5.13 shows that all teammate partners, including AI Teammates, were perceived as being generally more effective than the AI Tool DP, and that the Female Teammate was seen as the most effective. It is also important to point out that this effect is moderate ($\eta_p^2 = 0.05$), meaning that the type of DP had a relatively moderate impact on how effective evaluators perceived the DP to be.

Furthermore, looking at both graphs for the evaluation perceptions of Human Worker (HW) effectiveness (Figure 5.12) and Dyadic Partner (DP) effectiveness (Figure 5.13), they follow a similar pattern, where a partnership containing an AI Tool DP is generally associated with statistically significantly lower effectiveness perceptions for both partners. Although there are no significant differences between the teammate conditions, the consistent pattern is continued such that perceptions of effectiveness increase from partnerships with an AI Teammate DP to partnerships with a Male Teammate DP to, finally, partnerships with a Female Teammate DP having the highest effectiveness evaluations for both the Human Worker (HW) (see Table 5.7) and Dyadic Partner (DP) (see Table 5.8).

In addition to the main effects of dyadic partner (DP) type on perceived DP effectiveness, there was also a **significant** interaction effect between dyadic partner (DP) type and evaluators' negative attitudes towards AI on perceived DP effectiveness. Looking towards Figure 5.14, it can be seen that there is a downward trend in all but the Male Teammate DP condition, such that as negative attitudes towards AI increased the perceived effectiveness of the AI Tool, AI Teammate, and Female Teammate DP decreased. This makes intuitive sense for the AI Tool and AI Teammate DP conditions, as seeing AI as ineffective is part of what constitutes negative attitudes towards AI. However, it is more surprising to note that the trend also applies to the Female Teammate DP, but does not apply to the Male Teammate DP. Indeed, if this

trend applied to both human conditions, it could be speculated that evaluators with more negative attitudes towards AI might also harbor negative attitudes towards partners or workers generally, but this does not track with the trends seen in Figure 5.14 for the Female and Male Teammate DP conditions. Instead, it appears that more negative attitudes towards AI disproportionately negatively impacts perceptions of women workers who act as partners within a partnership in comparison to men partners.

Finally, there was also a **significant** interaction effect between dyadic partner (DP) type and evaluators' endorsement of hostile sexist beliefs on perceived dyadic partner effectiveness. Looking towards Figure 5.15, it is interesting to note that, generally speaking, the teammate DP conditions show either no discernible trend changes (i.e., Male Teammate DP) or a very slight downward trend in perceptions of DP effectiveness as hostile sexist beliefs increase, as seen in the Female Teammate and AI Teammate DP conditions. However, there is a surprising upward trend in perceptions of AI Tool DP effectiveness as endorsement of hostile sexist beliefs increases. One possible explanation could be that there might be some kind of link between endorsement of beliefs that are shaped by antagonistic views of other people (i.e., hostile sexism and women) and more favorable perceptions of entities that do not appear to act in a human-like capacity, such as an AI Tool.

Additionally, the stark contrast in trends between the AI Tool and AI Teammate DPs as seen in Figure 5.15 suggests that it is not the mere artificial nature of the entity that makes the difference, but rather the *functional design of an AI* to have human-like qualities such as teaming behaviors. Indeed, the AI Teammate DP was designed to be identical in its roles, responsibilities, and communication style to that of the Female and Male Teammate DPs, with the primary difference in functionality being that the response time for the AI Teammate DP condition was

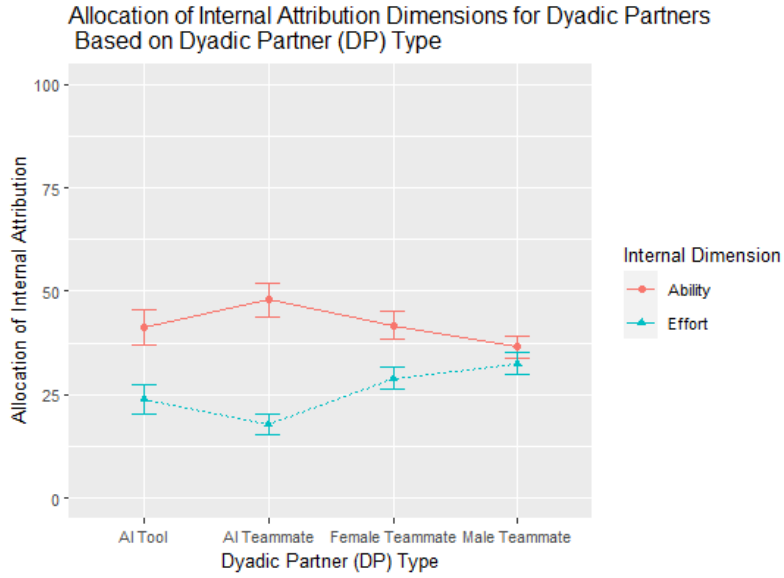


Figure 5.16: Perceived Credit Attribution to Ability and Effort to Dyadic Partner Based on Dyadic Partner Type. Error bars denote 95% confidence interval.

significantly faster (about one second) than the human Teammate DPs (15 seconds). Therefore, it could be the case that the AI Tool DP had enough functional distance from human-like qualities that it became more perceptually effective to evaluators who have negative attitudes towards other humans, i.e. hostile sexism. This is just one speculation as to why this trend appeared, but it is ultimately unclear and needs further exploration in future work.

DP Type	Credit Dimension	Marginal Mean	95% CI for Mean Difference		SE
			Lower	Upper	
AI Teammate	Ability	37.266	34.244	40.287	1.539
AI Tool		34.339	31.345	37.334	1.525
Female Teammate		37.162	34.159	40.165	1.530
Male Teammate		35.232	32.211	38.253	1.539
AI Teammate	Effort	28.515	25.843	31.187	1.361
AI Tool		31.340	28.691	33.989	1.349
Female Teammate		30.683	28.026	33.339	1.353
Male Teammate		31.839	29.167	34.511	1.361

Table 5.9: Marginal Means - Perceived Credit Attribution to Ability and Effort to Dyadic Partner Based on Dyadic Partner Type

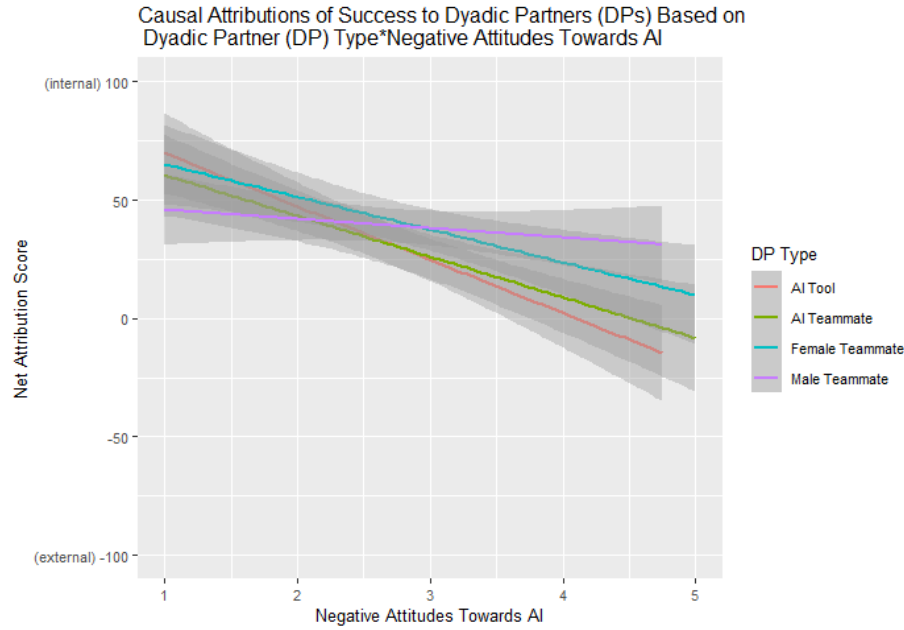


Figure 5.17: Causal Attributions of Success to Dyadic Partners (DPs) Based on the Interaction Between Dyadic Partner (DP) Type and Evaluator Negative Attitudes Towards AI.

Attributing Dimensions of Credit to Dyadic Partners (DPs). There was **no significant** main effect of dyadic partner (DP) type on the overall attribution of internal vs. external credit to the dyadic partner (DP) ($F(1,666) = 1.5, p = .215$). However, there was a **significant** main effect of dyadic partner (DP) type on the attribution of ability vs. effort to dyadic partners ($F(1,666) = 18.06, p < .001, \eta_p^2 = .08$) (Figure 5.16). Another repeated measures ANOVA was conducted to explore the differences between ability and effort attribution for the dyadic partner (DP). This test showed a significant difference between the allocation of dyadic partner (DP) credit between their perceived effort and their ability ($F(1,669) = 158.45, p < .001$,

$\eta_p^2 = .19$).

Yet, an additional interaction effect was found between Dyadic Partner (DP) Type and ability vs. effort for the credit attribution of the dyadic partner ($F(3,669) = 18.06, p < .001, \eta_p^2 = .08$). Simple main effects were analyzed for this finding. When controlling for ability or effort attribution, there was a significant difference between dyadic partners in both the ability and effort measures. For the effort measure, Figure 5.16 shows that the AI Teammate DP was rated lowest, followed by the AI Tool DP, Female Teammate DP, and then Male Teammate DP. In contrast, Figure 5.16 shows that the ability measure saw a near reversal of this trend, with the AI Teammate DP being rated the highest, followed by similar means for the AI Tool DP and Female Teammate DP, with the lowest ability attribution being for the Male Teammate DP. Marginal means for both the Ability and Effort credit dimensions can be seen in Table 5.9. It is also important to point out that this effect is moderate ($\eta_p^2 = 0.08$), meaning that the type of DP had a relatively moderate impact on how evaluators allocated credit attribution to the ability of the DP versus how much effort the DP exerted to successfully complete the task.

Figure 5.16 also demonstrates that, when controlling for the Dyadic Partner (DP) Type, there were significant differences between evaluator attributions of credit to ability vs. effort for the AI Teammate, AI Tool, and Female Teammate DP conditions. Specifically, all three saw significantly higher allocations of credit being attributed to their ability to successfully complete the task compared to the level of effort they put into succeeding at the task, but allocations of credit to ability vs. effort for the Male Teammate DP condition were not significantly different. In other words, evaluators felt that the Male Teammate DP had to exert some amount of effort roughly equal to his capabilities in order to successfully complete the task. This is somewhat surprising since prior literature on attribution of credit and suc-

cess highlights how men’s success is often attributed more to innate ability rather than other factors such as effort or luck (e.g., [210]). Therefore, this result provides a novel insight as it highlights that the Male Teammate saw the least discrepancy between these factors, suggesting that AI-enabled workplaces may have an impact on worker evaluations that challenges typical notions of workplace gender dynamics in the attribution of credit.

Conversely, the largest gap in ability vs. effort can be seen in the AI Teammate DP condition in Figure 5.16, such that evaluators perceived the AI Teammate DP as having to exert significantly less effort in comparison to its capabilities to successfully complete the task. The same can be said for the AI Tool and Female Teammate DP conditions, although that comparative gap gets smaller when going from the AI Teammate to the AI Tool to the Female Teammate DPs as can be seen in Figure 5.16.

In addition to the main effects of DP type on the attribution of internal credit to ability versus effort, there was also a **significant** interaction effect between dyadic partner (DP) type and evaluators’ negative attitudes towards AI on the attribution of internal versus external credit to DPs. Looking towards Figure 5.17, it can be seen that there is a noticeable downward trend in all DP conditions except for the Male Teammate DP, such that as negative attitudes towards AI increased the more evaluators attributed the cause of the DP’s success in the task to be factors that are external to the AI DPs and the Female Teammate DPs, such as luck and task difficulty being easy. However, this downward trend represents a narrowing of the gap between allocation of credit between internal versus external factors, such that all DPs were still attributed more internal credit overall than external credit for their successful task completion. It can be seen in Figure 5.17 that this downward trend is most prominent in the two AI DP conditions as well as the Female Teammate DP

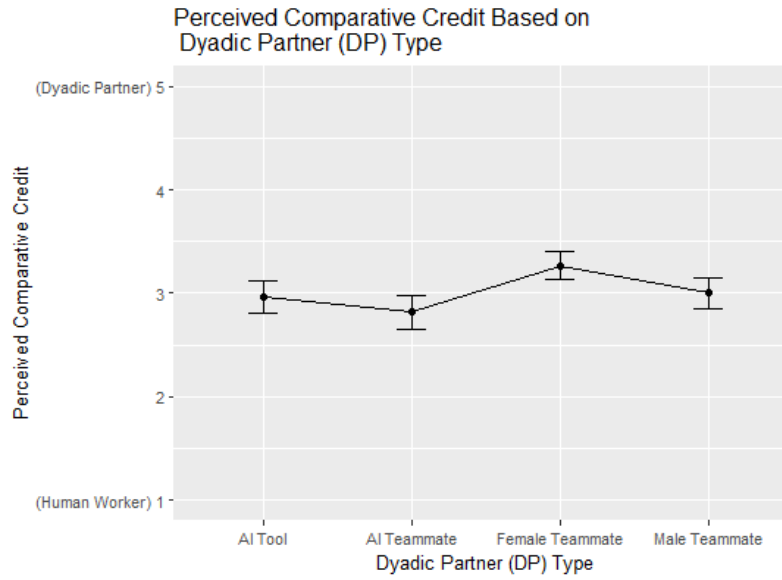


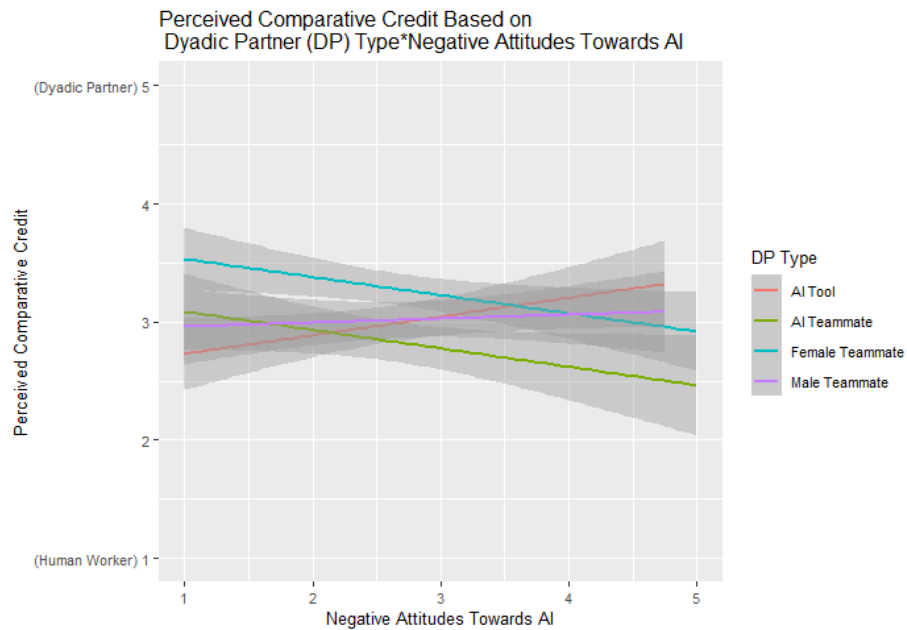
Figure 5.18: Perceived Comparative Credit Based on Dyadic Partner Type. Error bars denote 95% confidence interval.

condition, possibly again suggesting that more negative evaluations of AI can also impact evaluations of women partners to a greater extent than for men workers.

DP Type	Marginal Mean	95% CI for Mean Difference		SE
		Lower	Upper	
AI Teammate	2.842	2.691	2.994	0.077
AI Tool	2.991	2.840	3.141	0.076
Female Teammate	3.268	3.118	3.418	0.077
Male Teammate	3.007	2.856	3.158	0.077

Table 5.10: Marginal Means - Perceived Comparative Credit Based on Dyadic Partner Type

Evaluation of Comparative Credit to Human Workers (HWs). There was a **significant** main effect of dyadic partner (DP) type on the comparative credit attributed to human workers (HWs) vs. dyadic partners (DPs) ($F(1,666) = 5.31$, $p = .001$, $\eta_p^2 = .02$) (Figure 5.18). A Tukey post-hoc test was performed to compare each of the four dyadic partner (DP) types. The evaluation of comparative credit to human



Dyadic Partner Type	Trend	SE	df	lower.CL	upper.CL
AI Tool	0.118	0.0799	657	-0.0392	0.27453
AI Teammate	-0.141	0.0749	657	-0.2884	0.00573
Female Teammate	-0.144	0.0763	657	-0.2936	0.00583
Male Teammate	0.045	0.0794	657	-0.1109	0.20086

Figure 5.19: Perceived Comparative Credit to Human Workers (HWs) Based on the Interaction Between Dyadic Partner (DP) Type and Evaluator Negative Attitudes Towards AI.

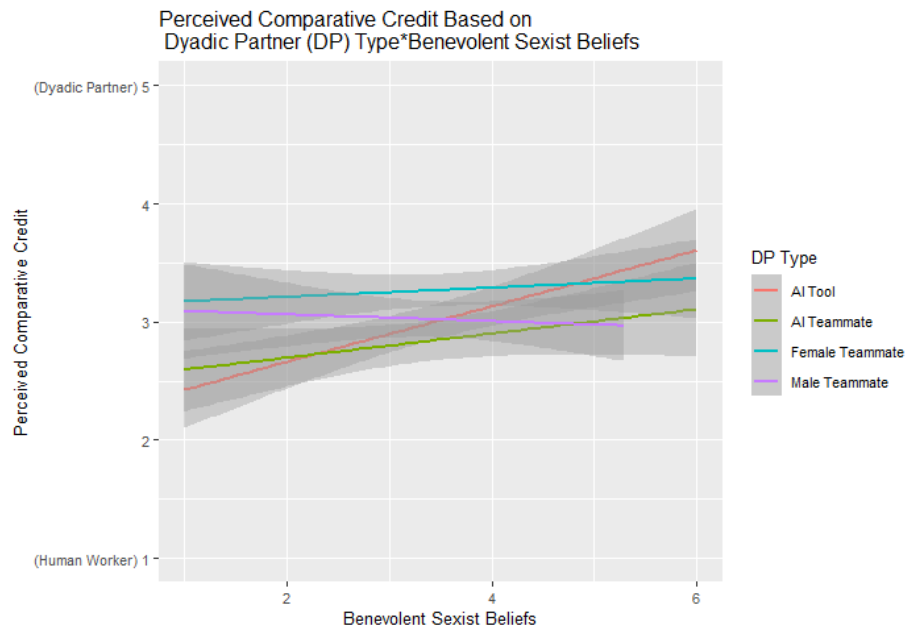


Figure 5.20: Perceived Comparative Credit to Human Workers (HWs) Based on the Interaction Between Dyadic Partner (DP) Type and Evaluator Endorsement of Benevolent Sexist Beliefs.

workers (HWs) was shown to be significantly different between the AI Teammate and Female Teammate DP conditions, and borderline significantly different between the AI Tool and Female Teammate DP conditions ($p = .052$). In Table 5.10, it can be seen that Human Workers (HWs) were, on average, given slightly more comparative credit than their Dyadic Partner (DP) for the successful completion of the task when working with an AI Teammate and slightly less comparative credit when working with the Female Teammate DP. This is even the case when working with the Man HW. Therefore, our results differ from what prior works on attribution of credit in mixed-gendered group work have found (e.g., [47]), suggesting that AI-enabled workplaces might engender unique perceptions of groupwork.

This is best illustrated in Figure 5.18, keeping in mind that a “3” on the y-axis of Figure 5.18 indicates *“They Both Deserve Equal Credit”*. Looking toward this figure, it can be seen that the error bars for the Female Teammate DP condition do not overlap either with the AI Teammate or the “3” y-axis, and do not overlap with the AI Tool DP condition in a perceptible manner. This indicates that, not only is there a significant difference between the Female Teammate DP condition and both AI conditions, but that the Female Teammate DP condition is the only DP condition where, on average, marginally more credit is actually given to the DP for the successful completion of the task than to the Human Worker (HW).

In addition to the main effects of DP type on the comparative credit to human workers (HWs) versus dyadic partners (DPs), there was also a **significant** interaction effect between dyadic partner (DP) type and evaluators’ negative attitudes towards AI on the comparative credit given to HWs versus DPs. Looking towards Figure 5.19, there are several trends that are interesting to note. More specifically, there appears to be a trend of comparative credit being pulled more towards the Human Worker (HW) as negative attitudes towards AI increase in the AI Teammate DP and Female

Teammate DP conditions.

In contrast, there appears to be a slight pull towards giving more comparative credit to the Dyadic Partner (DP) as negative attitudes towards AI increased in the Male Teammate DP condition, as well as in the AI Tool DP condition, although the latter is much more visibly sloped upward in Figure 5.19. This finding seems somewhat surprising because of the way that negative attitudes towards AI has opposite impacts when the HW is working with an AI Tool versus an AI Teammate DP, when it would feel more intuitive that both would trend downwards towards the HW. One reason for this could be that people who have generally negative attitudes towards the use of AI in society - which is a prominent part of the measure used for this covariate [408] - might view a HW using an AI Tool as less deserving of credit because of the mentality that the AI just did the work for them rather than collaborating with them. The AI Teammate DP, on the other hand, was designed to work more collaboratively and interdependently with similar communication patterns as humans, which could have come across more favorably to people with negative attitudes towards AI.

It is also interesting that the two human DP conditions did not follow the same patterns, suggesting again that there is some kind of link between negative attitudes towards AI and perceptions of female partners. In this case, working with the Female Teammate DP actually made evaluators more inclined to give credit to HWs, suggesting that negative attitudes towards AI adversely impacted perceptions of the Female Teammate DP.

Finally, there was also a **significant** interaction effect between dyadic partner (DP) type and evaluators' endorsement of benevolent sexist beliefs on comparative credit to Human Workers (HWs) versus Dyadic Partners (DPs). It can be seen in Figure 5.20 that there is a rather dramatic upward trend in the AI Tool DP condition, such that as evaluator endorsement of benevolent sexist beliefs increases

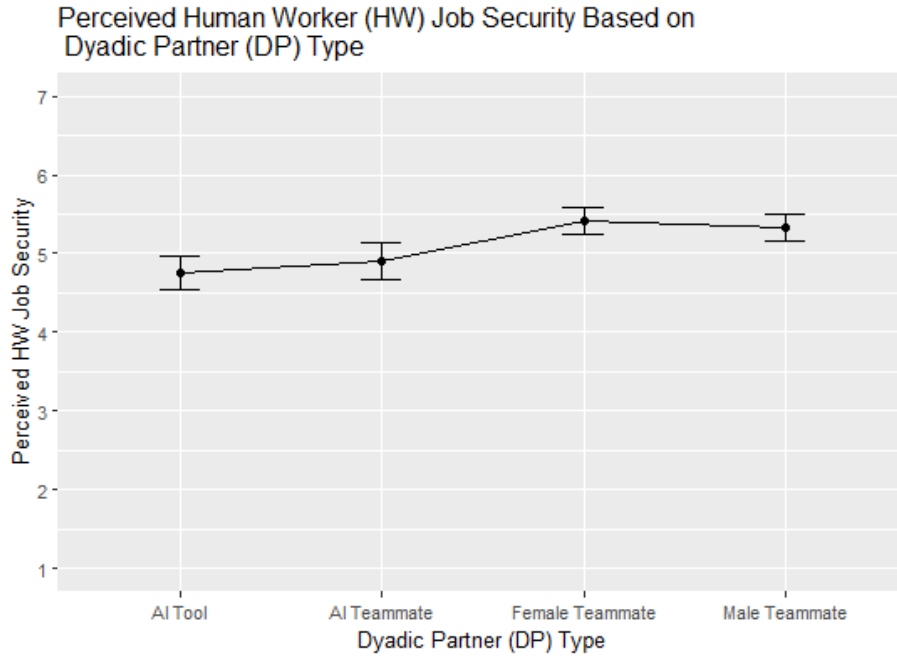
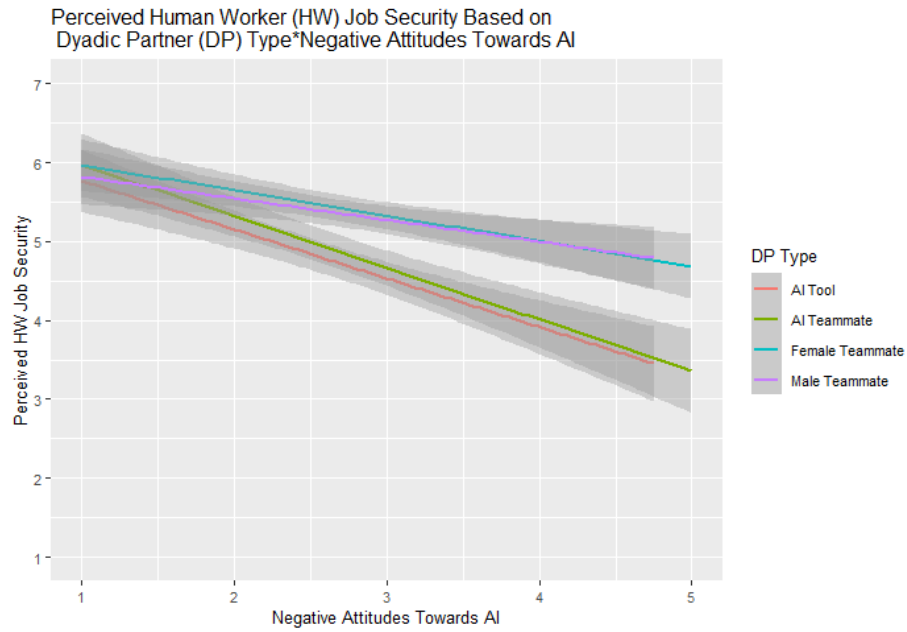


Figure 5.21: Perceived Job Security of Human Worker Based on Dyadic Partner Type. Error bars denote 95% confidence interval.

the comparative credit to the AI Tool DP also increases. This trend can also be seen in the AI Teammate DP condition in Figure 5.20, while the Female Teammate DP condition saw an functionally little to no change as endorsement of benevolent sexism increased.

Further, it is worth noting that there is a slight downward trend in the Male Teammate DP condition, such that as endorsement of benevolent sexism increased slightly more credit was given to the HW when working with the Male Teammate DP. All of these trends offer interesting insight into how benevolent sexist beliefs, which are more paternalistic and cooperative in nature [264], is linked to varying degrees to credit attribution in AI-enabled workplaces. It is unclear why that is the case, especially in the AI conditions, warranting further exploration in future studies.

Evaluations of Human Worker (HW) Job Security. There was a **significant** main effect of dyadic partner (DP) type on the evaluations of human



Dyadic Partner Type	Trend	SE	df	lower.CL	upper.CL
AI Tool	-0.647	0.0974	663	-0.839	-0.4562
AI Teammate	-0.644	0.0921	663	-0.825	-0.4629
Female Teammate	-0.312	0.0934	663	-0.495	-0.1283
Male Teammate	-0.270	0.0976	663	-0.461	-0.0779

Figure 5.22: Perceived Human Worker (HW) Job Security Based on the Interaction Between Dyadic Partner (DP) Type and Evaluator Negative Attitudes Towards AI.

DP Type	Marginal Mean	95% CI for Mean Difference		SE
		Lower	Upper	
AI Teammate	4.873	4.686	5.059	0.095
AI Tool	4.739	4.555	4.924	0.094
Female Teammate	5.417	5.232	5.602	0.094
Male Teammate	5.363	5.177	5.549	0.095

Table 5.11: Marginal Means - Perceived Job Security of Human Worker Based on Dyadic Partner Type

worker (HW) job security ($F(1,666) = 13.12$, $p < .001$, $\eta_p^2 = .06$) (Figure 5.21). A Tukey post-hoc test was performed to compare each of the four dyadic partner (DP) types. Evaluations of human worker (HW) job security were shown to be significantly different between the AI Teammate and Female Teammate DP conditions, AI Teammate and Male Teammate DP conditions, AI Tool and Female Teammate DP conditions, and AI Tool and Male Teammate DP conditions. Put more simply, there was a significant difference between evaluations of Human Worker (HW) job security when working with an AI compared to working with another human.

Looking towards the marginal means in Table 5.11, it can be seen that evaluators perceived the HW's job security to be significantly better when the HW was working with another human Dyadic Partner (DP) than when the HW was working with an AI Dyadic Partner (DP). Figure 5.21 best illustrates this divide between AI DP conditions versus human Teammate DP conditions, with the latter showing a significant positive increase from the former.

It can also be seen in Figure 5.21 that there were no significant differences in perceptions of HW job security between the AI Tool and AI Teammate DP conditions, nor were there any significant differences between the Female Teammate and Male Teammate DP conditions. This further demonstrates the divide between evaluators' perceptions of Human Worker (HW) job security being significantly lower

when working with an AI, regardless of the Tool/Teammate design, than when the HW works with a human teammate, regardless of the gender presentation of said human teammate. It should be noted that this effect is moderate ($\eta_p^2 = 0.06$), meaning that the type of DP had a relatively moderate impact on how positively evaluators perceived the HWs' job security to be when working in a partnership.

In addition to the main effects of DP type on Human Worker (HW) job security, there was also a **significant** interaction effect between dyadic partner (DP) type and evaluators' negative attitudes towards AI on evaluations of Human Worker (HW) job security. Looking towards Figure 5.22, it can be seen that there is a downward trend in all DP conditions, such that as negative attitudes towards AI increased evaluators' perceptions of the HW's job security decreased across the board. This downward trend is sharpest in the two AI DP conditions, demonstrating how having negative attitudes towards the use of AI in society directly relates to feeling that humans might be deemed as replaceable or insecure within their job position in AI-enabled workplaces. Surprisingly, though, this downward trend was also seen in the two Human DP conditions, although looking at Figure 5.22 it appears that this downward trend is not quite as drastic as in the AI DP conditions. It is perhaps the case that evaluators who hold more negative attitudes towards AI may also harbor negative attitudes towards the idea of HWs working with another entity - human or artificial - to accomplish a task.

Evaluations of Power Perceptions. There was a **significant** main effect of dyadic partner (DP) type on the evaluations of power within the partnership ($F(1,666) = 2.98, p = .031, \eta_p^2 = .01$) (Figure 5.23). However, a Tukey post-hoc test was performed to compare each of the four dyadic partner (DP) types, which revealed no significant differences between any of the DP levels. Looking towards both the marginal means in Table 5.12 and the error bars in Figure 5.23, no obvious differ-

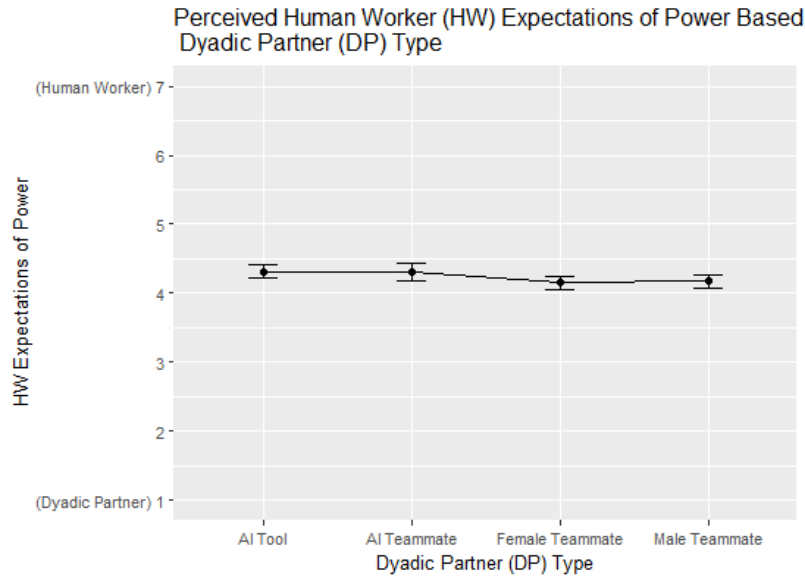


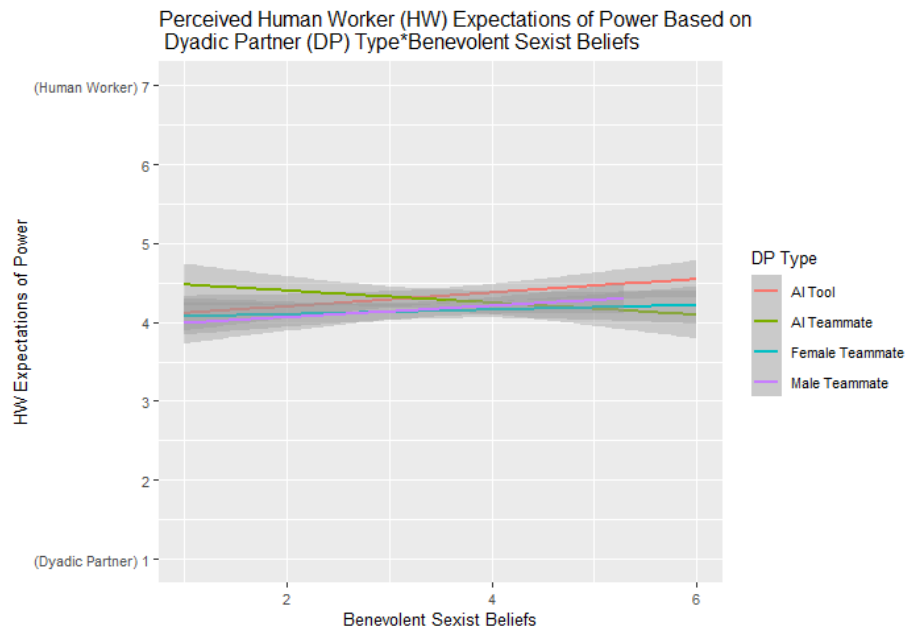
Figure 5.23: Perceived Comparative Power Based on Dyadic Partner Type. Error bars denote 95% confidence interval.

DP Type	Marginal Mean	95% CI for Mean Difference		SE
		Lower	Upper	
AI Teammate	4.311	4.206	4.417	0.054
AI Tool	4.317	4.213	4.421	0.053
Female Teammate	4.141	4.036	4.245	0.053
Male Teammate	4.171	4.066	4.276	0.054

Table 5.12: Marginal Means - Perceived Comparative Power Based on Dyadic Partner Type

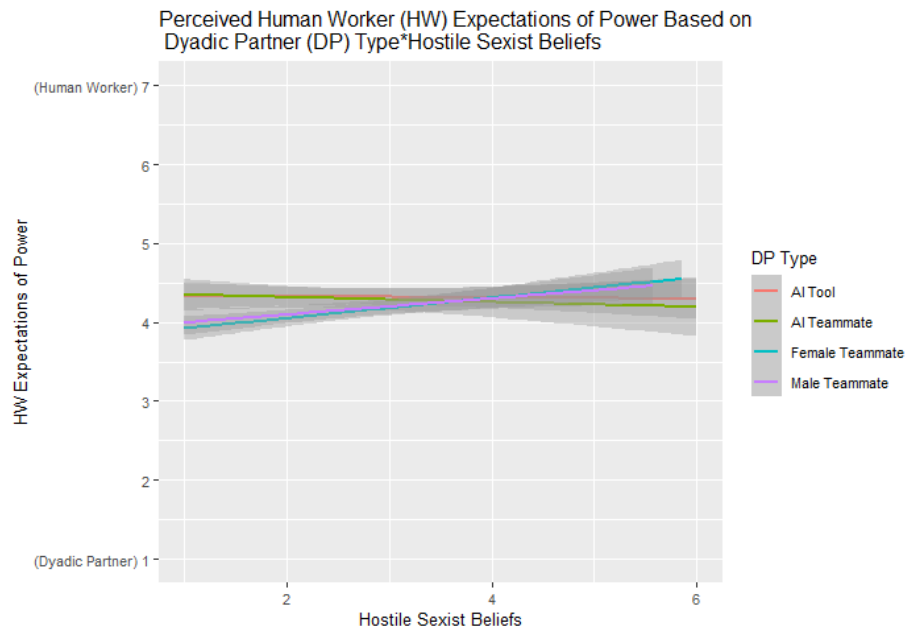
ences can be observed either. In essence, while there does appear to be a significant main effect of Dyadic Partner (DP) on how much power evaluators think the Human Worker (HW) has in the partnership, it is unclear what that difference looks like.

In addition to the main effects of DP type on Human Worker's (HW's) expectations of power within the partnership, there was also a **significant** interaction effect between dyadic partner (DP) type and evaluators' endorsement of benevolent sexist beliefs on evaluations of HW's expectations of power within the partnership. Looking towards Figure 5.24, it can be seen that there is a visible upward trend in the



Dyadic Partner Type	Trend	SE	df	lower.CL	upper.CL
AI Tool	0.0906	0.0453	660	0.00159	0.1796
AI Teammate	-0.0747	0.0437	660	-0.16053	0.0112
Female Teammate	-0.0241	0.0502	660	-0.12272	0.0745
Male Teammate	0.0312	0.0555	660	-0.07767	0.1401

Figure 5.24: Perceived Comparative Power Based on the Interaction Between Dyadic Partner (DP) Type and Evaluator Endorsement of Benevolent Sexist Beliefs.



Dyadic Partner Type	Trend	SE	df	lower.CL	upper.CL
AI Tool	-0.0299	0.0399	660	-0.10829	0.0484
AI Teammate	0.0010	0.0439	660	-0.08511	0.0871
Female Teammate	0.1296	0.0426	660	0.04584	0.2133
Male Teammate	0.0924	0.0444	660	0.00527	0.1795

Figure 5.25: Perceived Comparative Power Based on the Interaction Between Dyadic Partner (DP) Type and Evaluator Endorsement of Hostile Sexist Beliefs.

AI Tool DP, such that as evaluator endorsement of benevolent sexist beliefs increase evaluators perceive the HW as expecting to have more power in the partnership when working with an AI Tool DP. There also appears to be a very small upward trend in the Male Teammate DP condition, although there does not appear to be a meaningful trend in the Female Teammate DP condition.

Conversely, there was a downward trend in the AI Teammate DP condition, such that as evaluator endorsement of benevolent sexist beliefs increased evaluations of HW expectations of power decreased when the HW was working with the AI Teammate DP. This is interesting, and might be a reflection of the AI Teammate as a bit of a comparative equalizer when it comes to the gender dynamics that underlay benevolent sexism. In other words, while beliefs about the capabilities and roles of women and men in group work could be impacted by the gender makeup of a partnership, the AI Teammate DP was specifically designed in such a way that it does not have a designed-imposed gender, thus perhaps reducing the chance that gendered ideology impacts evaluations of HWs.

Finally, there was also a **borderline significant** interaction effect between dyadic partner (DP) type and evaluators' endorsement of hostile sexist beliefs on evaluations of Human Worker's (HW's) expectations of power in the partnership. While this interaction is not quite significant, Figure 5.25 is included to show that there is a slight upward trend in the two human DP conditions, such that as endorsement of hostile sexist beliefs increase there appears to be an increase in evaluations of HW expectations of power in the Male Teammate DP and Female Teammate DP conditions. This is interesting, as it implies that there could be a connection between negative evaluations of specific people (i.e., women) and generally less favorable evaluations of people who are "other" in a situation, such as someone who is explicitly framed as a partner to someone else rather than the primary member of the partnership.

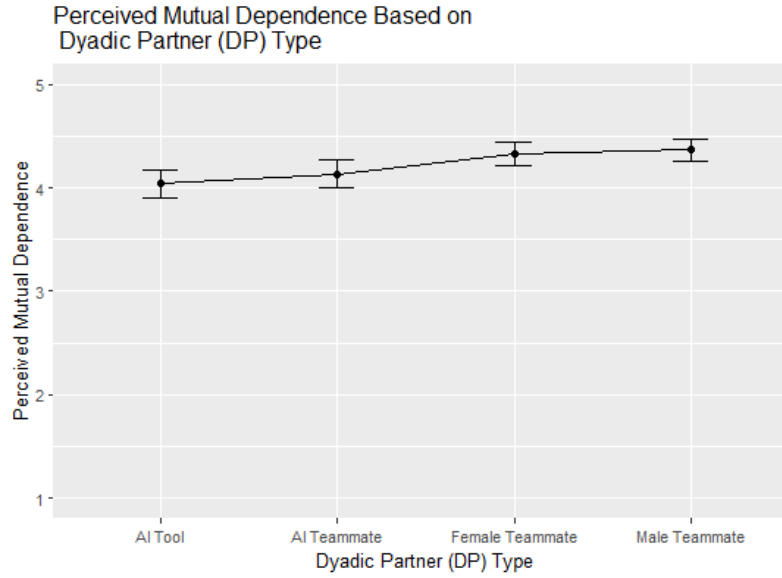
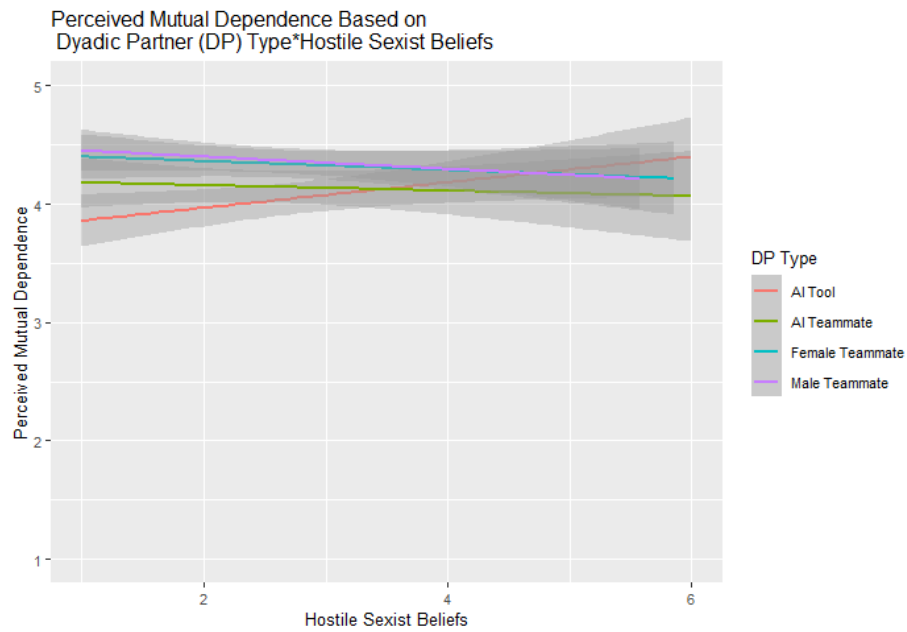


Figure 5.26: Perceived Mutual Dependence Based on Dyadic Partner Type. Error bars denote 95% confidence interval.

DP Type	Marginal Mean	95% CI for Mean Difference		SE
		Lower	Upper	
AI Teammate	4.144	4.027	4.260	0.059
AI Tool	4.054	3.939	4.170	0.059
Female Teammate	4.333	4.218	4.449	0.059
Male Teammate	4.373	4.256	4.489	0.059

Table 5.13: Marginal Means - Perceived Mutual Dependence Based on Dyadic Partner Type

Evaluations of Partnership Mutual Dependence. There was a **significant** main effect of dyadic partner (DP) type on the evaluations of partnership mutual dependence ($F(1,666) = 6.61, p < .001, \eta_p^2 = .03$) (Figure 5.26). A Tukey post-hoc test was performed to compare each of the four dyadic partner (DP) types. Evaluations of partnership mutual dependence were shown to be significantly different between the AI Teammate and Male Teammate DP conditions, the AI Tool and Female Teammate DP conditions, and the AI Tool and Male Teammate DP Conditions. Looking towards the marginal means in Table 5.13, it can be seen that



Dyadic Partner Type	Trend	SE	df	lower.CL	upper.CL
AI Tool	0.0935	0.0430	663	0.00903	0.1780
AI Teammate	-0.0279	0.0472	663	-0.12054	0.0647
Female Teammate	-0.0490	0.0457	663	-0.13877	0.0407
Male Teammate	-0.0717	0.0468	663	-0.16372	0.0203

Figure 5.27: Perceived Mutual Dependence Based on the Interaction Between Dyadic Partner (DP) Type and Evaluator Endorsement of Hostile Sexist Beliefs.

partnerships that consisted of a Human Worker (HW) and the Male Teammate DP condition were perceived by evaluators as being significantly more mutually dependent than partnerships containing an AI Teammate or AI Tool DP. The marginal means also demonstrate that evaluators perceived the partnerships that contained a Female Teammate DP as significantly more mutually dependent than partnerships with an AI Tool DP, though there was no significant difference compared to the AI Teammate DP condition.

Further, Figure 5.26 demonstrates an overall upward trend going from the AI DP conditions to the human DP conditions, with the caveat that the error bars for the AI Teammate and Female Teammate DP conditions overlap. Overall, then, it can be interpreted that the human-human partnerships are, on average, perceived to be somewhat more mutually dependent than the human-AI partnerships.

In addition to the main effects of DP type on evaluations of partnership mutual dependence, there was also a **significant** interaction effect between dyadic partner (DP) type and evaluators' endorsement of hostile sexist beliefs on evaluations of partnership mutual dependence. Looking towards Figure 5.27, it can be seen that there is a visible upward trend in the AI Tool DP condition, such that as evaluators' endorsement of hostile sexist beliefs increases the more evaluators perceived the Human Worker (HW) and the AI Tool DP to be dependent upon each other when doing the task. One potential reason for this could be that evaluators who endorse beliefs that are antagonistic towards certain types of people (i.e., women) might also harbor antagonist views towards other people generally. This might lead them to view a human worker in an AI-enabled workplace as more dependent on an AI Tool, which could be seen as a negative quality to have within a human worker.

Conversely, it can be seen in Figure 5.27 that the Teammate DP conditions, including the AI Teammate DP, appear to have slight downward trends, such that

as hostile sexist beliefs increase perceived mutual dependence between teammates decreases slightly. This is interesting, as it goes against the well-established principles of interdependence being a critical part of teamwork [31].

5.4.2.2 Interactions Between Dyadic Partner (DP) Type & Task Type on Evaluator Perceptions

Evaluation Perception	<i>F</i> (1,666)	<i>p-value</i>	η_p^2	Significance
Effectiveness of Human Worker (HW)	2.76	.042*	.01	*Significant
Effectiveness of Dyadic Partner (DP)	1.56	.199	—	—
Attrib. of Credit to HW - Internal v. External	1.43	.232	—	—
Attrib. of Credit to HW - Ability v. Effort	2.28	.078	—	—
Attrib. of Credit to DP - Internal v. External	1.08	.356	—	—
Attrib. of Credit to DP - Ability v. Effort	.73	.535	—	—
Comparative Credit	2.00	.12	—	—
Human Worker (HW) Job Security	4.15	.006*	.02	*Significant
Human Worker (HW) Expectations of Power	.14	.936	—	—
Partnership Mutual Dependence	.15	.927	—	—

Table 5.14: Summary of Results for Interaction Effects Between Dyadic Partner (DP) Type & Task Type on Evaluator Perceptions

While the vast majority of Study 2's significant results stem from the main effects of Dyadic Partner (DP) type, the ANCOVA performed for evaluator perceptions of Human Worker (HW) effectiveness did reveal a significant interaction effect, such that there is a significant difference between how evaluators perceived the effectiveness of the HW based on the type of Dyadic Partner (DP) they were working with and whether the partnership was working on a meeting scheduling task or investment banking task. However, results of the ANCOVA performed for dyadic partner (DP) effectiveness did not reveal a significant difference in how evaluators perceived the DP's effectiveness based on the type of DP and whether they were performing a meeting scheduling task or an investment banking task. Additionally, the ANCOVA

performed for evaluator perceptions of Human Worker (HW) job security did reveal a significant interaction effect, such that there is a significant difference between how evaluators perceived the HW's job security based on the type of Dyadic Partner (DP) they were working with and whether the partnership was working on a meeting scheduling task or an investment banking task.

The remainder of the results of the ANCOVAs demonstrated no significant interaction effects between Dyadic Partner (DP) Type and Task Type for all of the other evaluation perceptions, as seen in Table 5.14. This means that the data analysis did not reveal any significant differences in how human workers or dyadic partners were attributed credit based on the type of DP in the partnership and whether the partnership was working on a meeting-scheduling task or an investment-banking task. Furthermore, data analysis did not reveal any significant differences in whether the Human Worker (HW) was attributed more comparative credit than their dyadic partners based on the type of DP in the partnership and whether the partnership was working on a meeting scheduling task or an investment banking task. Data also did not reveal any significant differences in evaluators' perceptions of how much power the Human Worker (HW) thought that they had based on the type of dyadic partner in the partnership and whether the partnership was performing a meeting-scheduling task or an investment-banking task. Finally, data analysis also did not reveal any significant differences in evaluators' perceptions of the partnership's mutual dependence based on the type of dyadic partner in the partnership and whether the partnership was working on a meeting-scheduling task or an investment-banking task.

Table 5.3 provides the statistical results of the ANCOVAs and indications of their (non) significance, while the technical breakdown of these (non)significant results can be found in the text below.

Perceived Effectiveness of the Human Worker (HW). There was

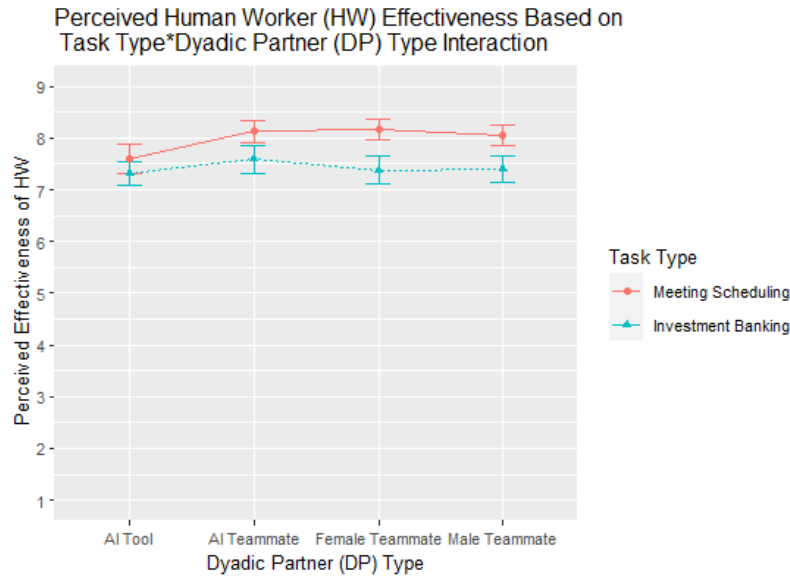


Figure 5.28: Perceived Effectiveness of Human Worker Based on Dyadic Partner Type and Task Type Interaction. Error bars denote 95% confidence interval.

a **significant** interaction effect between dyadic partner (DP) type and task type on the perceived effectiveness of the human worker (HW) ($F(1,666) = 2.76$, $p = .042$, $\eta_p^2 = .01$) (Figure 5.28). An analysis of simple main effects was conducted to explore this interaction. When controlling for DP Type, there were significant differences between task types in the AI Teammate DP, Female Teammate DP, and Male Teammate DP conditions. Looking toward the marginal means in Table 5.15, it can be seen that evaluators had generally higher perceptions of the human worker's (HW's) effectiveness when the HW was paired with the AI Teammate DP, Female Teammate DP, and Male Teammate DP on a meeting scheduling task as opposed to the investment banking task.

This difference is best illustrated in Figure 5.28, which shows gaps of varying sizes between the red error bars for the Meeting Scheduling task and the blue error bars for the Investment Banking task. Looking towards this figure, it can be seen that perceptions seem to have the greatest difference between task types when the

Task Type	DP Type	Marginal Mean	95% CI for Mean Difference		SE
			Lower	Upper	
Meeting Scheduling	AI Teammate	8.094	7.866	8.322	0.116
Investment Banking		7.574	7.347	7.801	0.116
Meeting Scheduling	AI Tool	7.537	7.310	7.764	0.116
Investment Banking		7.357	7.133	7.580	0.114
Meeting Scheduling	Female Teammate	8.175	7.952	8.398	0.114
Investment Banking		7.353	7.125	7.581	0.116
Meeting Scheduling	Male Teammate	8.079	7.849	8.309	0.117
Investment Banking		7.441	7.217	7.665	0.114

Table 5.15: Marginal Means - Perceived Effectiveness of Human Worker Based on Dyadic Partner Type and Task Type Interaction

HW is partnered with the Female Teammate DP. Conversely, the overlapping red and blue error bars for the AI Tool DP in Figure 5.28 show that perceptions of HW effectiveness are not significantly different when comparing HW-AI Tool partnerships in different types of tasks.

When controlling for the type of task viewed, there were **significant** differences in the perceived effectiveness of the human worker (HW) based on the DP Type in the meeting scheduling task but not the investment banking task. Looking at the red line for the Meeting Scheduling task in Figure 5.28, it can be seen that the meeting scheduling task generally yielded a lower perceived effectiveness of the human worker (HW) when they were paired with the AI Tool DP compared to all other dyadic partners. However, the blue line in Figure 5.28 stays relatively stable, demonstrating that evaluators' perceptions of HW effectiveness were **not significantly different** when working with different types of dyadic partners on an investment banking task.

Evaluations of Human Worker (HW) Job Security. There was a **significant** interaction effect between dyadic partner (DP) type and task type on evaluations of human worker (HW) job security ($F(1,666) = 4.15, p = .006, \eta_p^2 = .02$) (Figure 5.29), which was analyzed via simple main effects. When controlling for

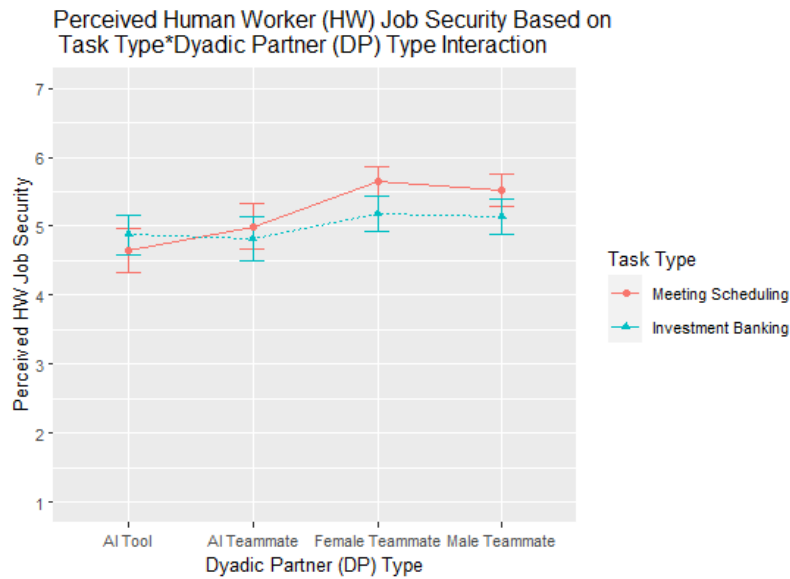


Figure 5.29: Perceived Job Security of Human Worker Based on Dyadic Partner and Task Type Interaction. Error bars denote 95% confidence interval.

Dyadic Partner (DP) Type, there was a significant difference between Task Types for the Female Teammate DP and Male Teammate DP. Looking towards the marginal means seen in Table 5.16 was higher for human workers (HWs) partnered with the Male Teammate DP and Female Teammate DP when they worked in the meeting scheduling task, compared to when the partnership worked in the investment banking task. This difference can also be seen in Figure 5.29, where the red line associated with the meeting scheduling task rises above the blue lines of the investment banking task for the Female Teammate and Male Teammate DPs. Results in the Figure also show that evaluators felt that human workers working on a meeting scheduling task with either human Teammates had the most job security out of all other conditions.

When controlling for the Task Type, there was also a **significant** difference between Dyadic Partner (DP) Type in the Meeting Scheduling task condition. Specifically, the marginal means in Table 5.16 show that human workers (HWs) in the meeting scheduling task were seen as having lower job security when working with the AI

Task Type	DP Type	Marginal Mean	95% CI for Mean Difference		SE
			Lower	Upper	
Meeting Scheduling	AI Teammate	4.960	4.695	5.224	0.135
Investment Banking		4.786	4.523	5.049	0.134
Meeting Scheduling	AI Tool	4.566	4.303	4.829	0.134
Investment Banking		4.912	4.653	5.172	0.132
Meeting Scheduling	Female Teammate	5.675	5.417	5.933	0.132
Investment Banking		5.159	4.894	5.423	0.135
Meeting Scheduling	Male Teammate	5.566	5.300	5.832	0.136
Investment Banking		5.160	4.900	5.419	0.132

Table 5.16: Marginal Means - Perceived Job Security of Human Worker Based on Dyadic Partner and Task Type Interaction

Tool DP and AI Teammate DP compared to the two human DP conditions. This difference is best illustrated when looking at the red line for Meeting Scheduling in Figure 5.29. Here, it can be clearly seen that the two AI DP conditions are associated with significantly lower perceptions of HW job security compared to the two human DP conditions. This same trend does not apply to the Investment Banking task, though, as **no significant** differences were found between Dyadic Partners (DPs) when controlling for Task Type. In other words, when evaluating a working partnership in an investment banking context, the type of dyadic partner does not play a significant role in how evaluators perceive the human worker's (HW) job security, but it does play a role when the context is a meeting scheduling task.

Evaluation Perception	<i>F</i>(1,666)	<i>p-value</i>	η_p^2	Significance
Effectiveness of Human Worker (HW)	2.82	.038*	.01	*Significant
Effectiveness of Dyadic Partner (DP)	1.26	.289	—	—
Attrib. of Credit to HW - Internal v. External	.47	.705	—	—
Attrib. of Credit to HW - Ability v. Effort	.91	.436	—	—
Attrib. of Credit to DP - Internal v. External	2.00	.113	—	—
Attrib. of Credit to DP - Ability v. Effort	.47	.702	—	—
Comparative Credit	.19	.902	—	—
Human Worker (HW) Job Security	1.72	.162	—	—
Human Worker (HW) Expectations of Power	2.04	.107	—	—
Partnership Mutual Dependence	1.35	.256	—	—

Table 5.17: Summary of Results for Interaction Effects Between Human Worker (HW) Gender Presentation & Dyadic Partner (DP) Type on Evaluation Perceptions

5.4.3 Effects of Interactions Between Human Worker (HW) Gender Presentation & Dyadic Partner (DP) Type on Evaluator Perceptions

5.4.3.1 Two-Way Interactions Between Human Worker (HW) Gender Presentation & Dyadic Partner (DP) Type

Up until this point, the focus of this results section has been unpacking the effects of Human Worker (HW) Gender Presentation and Dyadic Partner (DP) Type in isolation from each other. The remainder of Study 2's results section will explore how these two manipulations interact to impact evaluator perceptions. Overall, the majority of the ANCOVAs performed revealed no significant results, as can be seen in Table 5.17. However, there is one interesting and important caveat in that the ANCOVA performed for evaluator perceptions of Human Worker (HW) effectiveness did reveal a significant interaction effect, such that there is a significant difference between how evaluators perceived the effectiveness of the HW based on whether they were a woman or a man working with different types of dyadic partners. This result

is critically important to this dissertation, and will thus be explored in further detail later in this section.

The remainder of the results of the ANCOVAs demonstrated non-significant interaction effects between Human Worker (HW) Gender Presentation and Dyadic Partner (DP) Type for all of the other evaluation perceptions, as seen in Table 5.17. This means that the data analysis did not reveal any significant differences in how dyadic partners were perceived as effective based on whether the HW was a woman or a man working with any of the DP types. Furthermore, data analysis did not reveal any significant differences in how human workers or dyadic partners were attributed credit based on whether the HW was a woman or a man working with any of the DP types. Data analysis also did not reveal any significant differences in whether the Human Worker (HW) was attributed more comparative credit than their dyadic partners based on whether the HW was a woman or a man working with any of the DP types. Likewise, there was no significant differences in evaluators' perceptions of Human Worker (HW) job security or how much power the Human Worker (HW) thought they had, based on whether the HW was a woman or a man working with any of the dyadic partner types. Finally, there was also no significant differences in evaluators' perceptions of the partnership's mutual dependence based on whether the HW was a woman or a man working with any of the DP types.

Table 5.17 provides the statistical results of the ANCOVAs and indications of their (non) significance, while the technical breakdown of these (non) significant results can be found below.

Perceived Effectiveness of the Human Worker (HW). There was a **significant** interaction effect between human worker (HW) gender presentation and dyadic partner (DP) type on perceived effectiveness of the human worker (HW) ($F(1,666) = 2.82, p = .038, \eta_p^2 = .01$) (Figure 5.30). An analysis of simple main

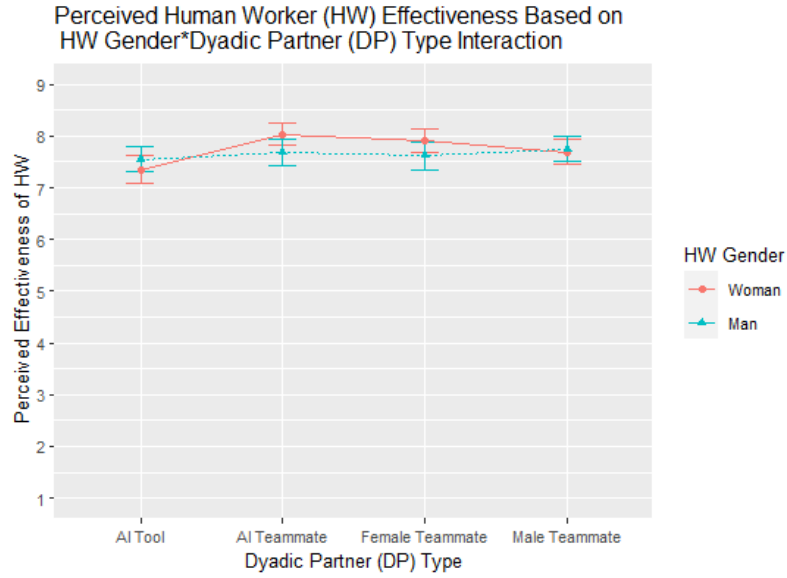


Figure 5.30: Perceived Effectiveness of Human Worker Based on Gender Presentation of Human Worker and Dyadic Partner Type Interaction. Error bars denote 95% confidence interval.

effects was conducted to explore this interaction. When controlling for DP Type, there was a significant difference in the perception of human worker (HW) performance based on HW gender in AI Teammate and Female Teammate DP conditions. Looking towards the marginal means in Table 5.7, both the AI Teammate and Female Teammate DP conditions saw the woman HW perceived as having greater effectiveness than the man HW. In other words, when working with either the AI Teammate or the Female Teammate DPs, the woman HW was perceived to be more effective than the man HW was when also working with the same DPs.

Moreover, when controlling for Human Worker (HW) Gender Presentation, there was a **significant** effect of Dyadic Partner (DP) Type in the woman HW condition but not the man HW condition. Looking towards Figure 5.30, it can be seen in the path of the red line associated with the Woman HW that evaluators perceived the effectiveness of the woman HW to be significantly lower when working with an

HW Gender	DP Type	Marginal Mean	95% CI for Mean Difference		SE
			Lower	Upper	
Woman	AI Teammate	8.013	7.787	8.238	0.115
Man		7.655	7.427	7.884	0.116
Woman	AI Tool	7.333	7.109	7.558	0.114
Man		7.560	7.335	7.786	0.115
Woman	Female Teammate	7.914	7.688	8.139	0.115
Man		7.615	7.389	7.840	0.115
Woman	Male Teammate	7.760	7.532	7.987	0.116
Man		7.761	7.534	7.988	0.116

Table 5.18: Marginal Means - Perceived Effectiveness of Human Worker Based on Gender Presentation of Human Worker and Dyadic Partner Type Interaction

AI Tool DP compared to all other dyadic partners. In fact, looking still at Figure 5.30, it is the only DP condition that creates lower perceptions of effectiveness for the woman HW than it does for the man HW, although marginal means (Table 5.17) are not significantly different between the two. It can also be seen in the Figure that the condition of Male Teammate DP creates nearly identical patterns for the Woman HW and the man HW. Finally, it is important to note that the highest rated effectiveness of the Human Worker (HW) can be seen when the woman HW is working with the AI Teammate DP as shown in Figure 5.30.

5.4.3.2 Three-Way Interactions Between Human Worker (HW) Gender Presentation, Dyadic Partner (DP) Type, & Task Type on Evaluator Perceptions

Overall, results of the ANCOVAs demonstrated no significant interaction effects between Human Worker (HW) Gender Presentation, Dyadic Partner (DP) Type, and Task Type for all evaluation perceptions, as seen in Table 5.19. This means that the data analysis did not reveal any significant differences in how human workers or dyadic partners were perceived as effective and attributed credit based on whether

Evaluation Perception	<i>F</i>(1,666)	<i>p-value</i>	η_p^2	Significance
Effectiveness of Human Worker (HW)	.54	.652	—	—
Effectiveness of Dyadic Partner (DP)	.31	.816	—	—
Attrib. of Credit to HW - Internal v. External	2.00	.114	—	—
Attrib. of Credit to HW - Ability v. Effort	.37	.772	—	—
Attrib. of Credit to DP - Internal v. External	1.09	.352	—	—
Attrib. of Credit to DP - Ability v. Effort	.52	.668	—	—
Comparative Credit	.43	.735	—	—
Human Worker (HW) Job Security	.46	.708	—	—
Human Worker (HW) Expectations of Power	1.43	.232	—	—
Partnership Mutual Dependence	1.52	.208	—	—

Table 5.19: Summary of Results for Interaction Effects Between Human Worker (HW) Gender Presentation, Dyadic Partner (DP) Type, & Task Type on Evaluation Perceptions

the human worker was a woman or a man working with any of the four dyadic partner types on either the meeting scheduling task or investment banking task. Furthermore, data analysis did not reveal any significant differences in whether the human worker was attributed more comparative credit than their dyadic partners based on whether the human worker was a woman or a man working with any of the four dyadic partner types on either the meeting scheduling task or investment banking task. There was also no significant differences in evaluators' perceptions of the human worker's job security and expectations of power based on whether the human worker was a woman or a man working with any of the four dyadic partner types on either the meeting scheduling task or investment banking task. Finally, data analysis did not reveal any significant differences in evaluators' perceptions of the partnership's mutual dependence based on whether the human worker was a woman or a man working with any of the four dyadic partner types on either the meeting scheduling task or investment banking task.

Table 5.19 provides the statistical results of the ANCOVAs and indications of their non-significance, while the technical breakdown of these nonsignificant results

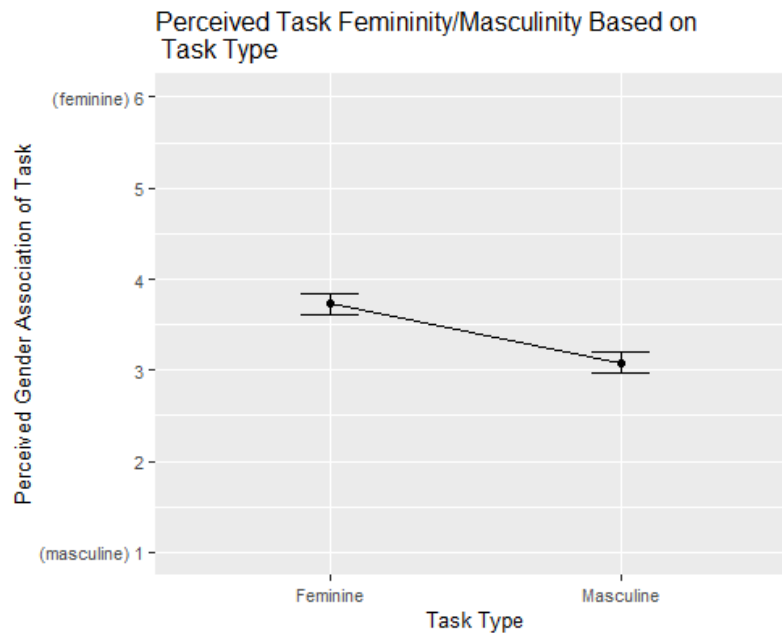


Figure 5.31: Perceived Task Femininity/Masculinity Based on Task Type. Error bars denote 95% confidence interval.

can be found below.

5.4.4 Perception of Task Type Femininity/Masculinity

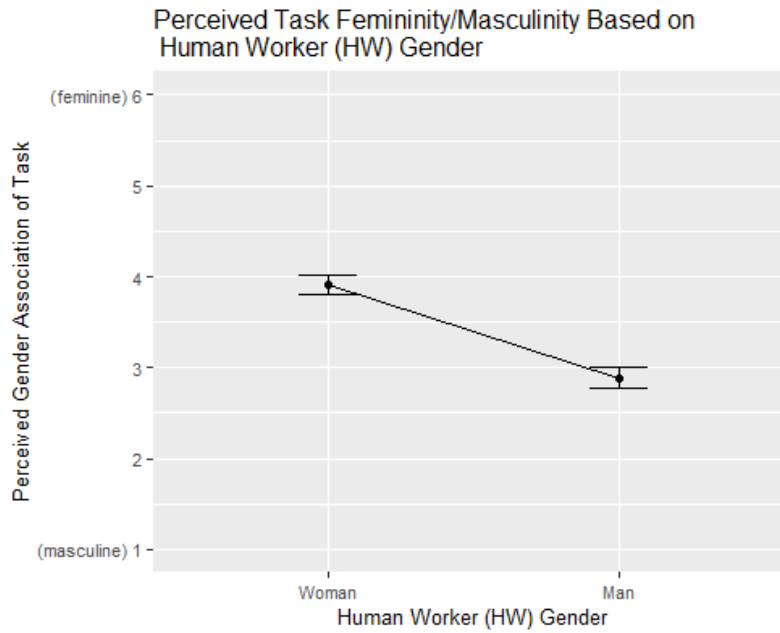


Figure 5.32: Perceived Task Femininity/Masculinity Based on Human Worker (HW) Gender. Error bars denote 95% confidence interval.

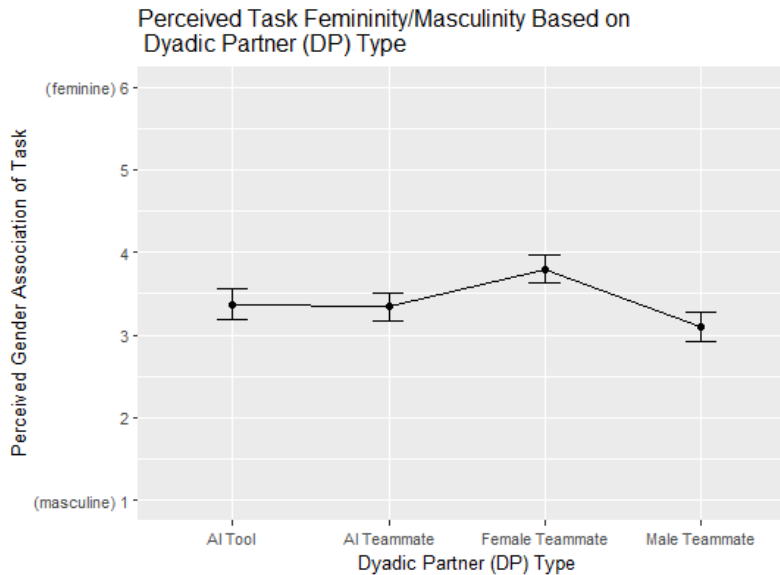


Figure 5.33: Perceived Task Femininity/Masculinity Based on Dyadic Partner (DP) Type. Error bars denote 95% confidence interval.

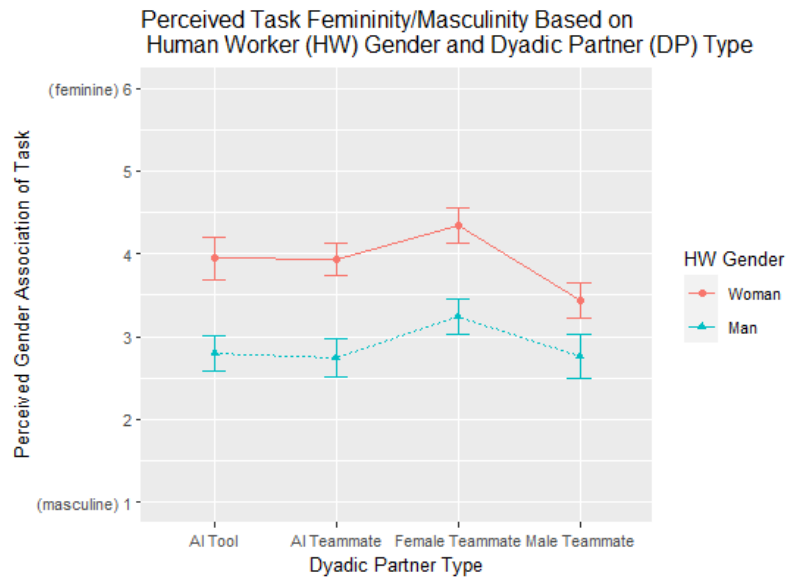


Figure 5.34: Perceived Task Femininity/Masculinity Based on Human Worker (HW) Gender and Dyadic Partner (DP) Type Interaction. Error bars denote 95% confidence interval.

Task Type	Marginal Mean	95% CI for Mean Difference		SE
		Lower	Upper	
Meeting Scheduling	3.721	3.614	3.828	0.055
Investment Banking	3.079	2.972	3.185	0.054

Table 5.20: Marginal Means - Perceived Task Femininity/Masculinity by Task Type

HW Gender	Marginal Mean	95% CI for Mean Difference		SE
		Lower	Upper	
Woman	3.915	3.808	4.022	0.054
Man	2.885	2.778	2.992	0.055

Table 5.21: Marginal Means - Perceived Task Femininity/Masculinity by Human Worker (HW) Gender

DP Type	Marginal Mean	95% CI for Mean Difference		SE
		Lower	Upper	
AI Teammate	3.329	3.178	3.481	0.077
AI Tool	3.374	3.224	3.524	0.077
Female Teammate	3.794	3.643	3.945	0.077
Male Teammate	3.103	2.951	3.254	0.077

Table 5.22: Marginal Means - Perceived Task Femininity/Masculinity by Dyadic Partner (DP) Type

HW Gender	DP Type	Marginal Mean	95% CI for Mean Difference		SE
			Lower	Upper	
Woman	AI Teammate	3.924	3.711	4.138	0.109
Man		2.734	2.518	2.950	0.110
Woman	AI Tool	3.969	3.757	4.181	0.108
Man		2.779	2.566	2.992	0.109
Woman	Female Teammate	4.342	4.129	4.555	0.109
Man		3.246	3.032	3.459	0.109
Woman	Male Teammate	3.425	3.210	3.639	0.109
Man		2.780	2.566	2.995	0.109

Table 5.23: Marginal Means - Perceived Task Femininity/Masculinity by Human Worker (HW) Gender * Dyadic Partner (DP) Type

Finally, it is worth exploring the metric that was originally included as a manipulation check, which was whether or not participants rated the task they viewed in their video as feminine or masculine. The ANCOVA conducted confirmed the validity of the manipulation, with a **significant** difference of task type on the perception of the viewed video as being masculine or feminine ($F(1,669) = 69.60, p < .001, \eta_p^2 = .09$). Means show that the meeting scheduling task was perceived as being more feminine than the Investment Banking task. In general, this finding confirms the original intention of the Task Type manipulation, which was to depict one task that was perceived as more feminine and one task that was perceived as more masculine.

However, the ANCOVA also revealed that task type was not the only significant manipulation to impact the perceived femininity of the shown video. Results also show a **significant** effect of human worker (HW) gender ($F(1,669) = 179.12, p < .001, \eta_p^2 = .21$) and dyadic partner (DP) type ($F(3,669) = 14.02, p < .001, \eta_p^2 = .06$) on the perceived femininity of the task. For the DP type, a Tukey post-hoc test showed significant differences between the AI Teammate and Female Teammate DP conditions, the AI Tool and Female Teammate DP conditions, and the Female Teammate and the Male Teammate DP conditions. In general, tasks with a male dyadic partner were perceived as more masculine, tasks with a female dyadic partner were perceived as more feminine, and tasks with an AI tool and AI teammate fell in the middle of these two human bounds. For the main effect of human worker (HW) gender, participants generally viewed the videos with a woman HW as being significantly more feminine than videos with a man HW. Of these three main effects, it is worth noting that HW gender had the largest effect size on the performed ANCOVA of the three manipulations tested ($\eta_p^2 = .21$).

Lastly, there was a **significant** two-way interaction effect between the Dyadic Partner (DP) Type and human worker (HW) gender depicted in each video ($F(3,669)$

$= 2.90, p = .036, \eta_p^2 = .01$). Simple main effects show that when controlling for DP type, there was a significant effect of HW gender on the perception of task femininity for each dyadic partner condition. When controlling for human worker (HW) gender, there was a significant effect of dyadic partner type in both the woman HW condition and the man HW condition. When looking at the results shown in Figure 5.34, this interaction effect shows that there was a generally consistent gap in perceived task femininity for the AI Teammate, AI Tool, and Female Teammate DP conditions; however, this gap shrinks for participants in the Male Teammate DP condition. This interaction broadly shows that having any male in a task (whether it be as the main worker or as a partner) can generally sway the perception of a task to be more masculine.

5.5 Results Part II - Qualitative Supplemental Expert Interviews

In addressing S.2.RQ5 (*How do real-world experts in AI-enabled workplaces conceptualize AI workplace integration's impacts on evaluations of human workers?*), results of the supplementary expert interviews for Study 2 reveal the intricate ways that workplace gender dynamics and AI functional design intermingle to shape evaluations and attributions of credit to workers, especially women workers. These results also shed light on the “why” element of Study 2’s quantitative results, specifically by (1) highlighting the relationship between job-specific evaluations in AI-enabled workplaces and AI functional design, and (2) deepening this study’s understanding of how ambivalent sexist beliefs (i.e., hostile and benevolent) alter expectations of women as collaborators and experts in AI-enabled workplaces.

5.5.1 Theme 1: AI Usage is Becoming a Greater Part of People’s Job-Specific Evaluations in the Workplace

The first theme of this work is focused on the evaluation components of Study 2, as a number of experts expressed reactions as to why AI impacted worker evaluations within this study. Overall, expert interviews highlighted that the last year has seen a dramatic shift in AI’s role in the workplace. As P6 (28, Woman, Asian, AI Governance Advisor) suggested, AI usage is becoming less of a sign of laziness and more of an important skill that can be commended. All experts tended to agree that the most important factor in recent years has become the end result of the product one delivers with AI. Echoing these thoughts, P3 (34, Woman, Asian, Research Prof. Human-Centered AI) had noted that they often see greater commendation and credit for those who can leverage AI technology to create a better, more refined, and higher quality product; however, if a poor product is delivered, the same AI usage becomes a detriment to human worker evaluation and strips credit away from the worker. The following quote from P3 serves as an example of that opinion:

“For example, when reviewing papers, if it’s very clear to me, like, this person used AI, so it’s not a smart way of using AI, and I would be very pissed, because I’m reviewing something AI-generated, and that’s a waste of my time. I will completely strip the credit away from the authors. But a smart way of using AI, like, in ways that I’m not sure if they use AI, but it reads really smooth and [it’s] making clear statements and [the] logic is clear, it feels like the human initiated the thinking process while using AI to polish the thoughts or modify the thoughts. That, to me, is a smart way of using AI, and that’s fine. And you know, I will, I will give them [the authors] the credit.”

Building on this, the results of this work showed that greater evaluations could come from those leveraging any teammate over the AI tool, and experts felt they had a possible reason as to why. A substantial portion of modern workplaces are seen as valuing collaboration, with workers often seeing higher evaluations when they collaborate effectively. Indeed, P1 (46, Man, White, Conflict Arbiter) and P2 (29, Woman, White, AI R&D) noted that working with a teammate, even an AI teammate, provides an opportunity to showcase one’s collaborative abilities, which in turn improves the way other people evaluate them and their work. As such, rather than detracting from human evaluation, AI teammates may actually provide an opportunity for humans to showcase their technical and collaborative capabilities within the workplace simultaneously.

Moreover, if these collaborations are able to yield a high-quality and effective product, then it would only serve to bolster these evaluations. Based on this, P2 went as far as to highlight the importance of women workers learning how to collaborate with AI in the coming years, as it will likely be a key factor in workplace evaluations and should be leveraged by women workers, who are otherwise disproportionately impacted by workplace gender dynamics. More specifically, the following quote from P2 illustrates this concept:

“Work with an AI teammate for your effectiveness, as demonstrated by the results [of Study 2]. I think you (women workers) should absolutely find every way to get an advantage, because there are plenty of forces, plenty of perceptions [about women] that are preventing you from being effective, that are out of your control, and just because of how humans interact, like how humans view each other.”

However, these criteria for evaluation may not always be consistent from work-

place to workplace. As noted by P4 (59, Woman, Asian, CEO Sci. Tech. Accelerator), contexts that place a greater emphasis on innovation in technology may be more likely to stress AI usage as a way to be more effective, *“Because, well, we’re adaptable. We’re more adaptive to new technology. And we see the opportunity.”* Even from an AI tooling and teaming perspective, P1, P4, and P5 (32, Man, White, Dir. of Business Intelligence) had noted that they see a weak distinction between these two concepts in their workplace, and that they are much more focused on the specific application of the AI system than on how that system is labeled.

However, despite AI usage’s impact on evaluation potentially being workplace dependent, it may still provide more universal impacts on effectiveness across workplaces. As noted by P1, their own personal position does not see any evaluation due to AI usage; however, their own personal effectiveness can benefit from their use of AI. In this sense, even if AI use were to be consistent across many workplaces, there may still be a strong inconsistency in how its use is evaluated depending upon how the workers themselves are evaluated. The following quote by P1 provides a summary of this idea:

“We are not really evaluated on the things that we would use AI for. So that’s like, we have a very loose evaluation criteria. Ours is mostly based on the feedback, or, I guess, if people want to utilize our services, right? So the demand for us to exist, I guess, is really our evaluation criteria. So if people stopped wanting to use the Ombuds office or feeling the need to use us to conduct training or for conflict management, then I think that would reflect on our performance. But you know, the training that we do and the things that we write [using AI] aren’t necessarily evaluated.”

Summarizing this theme, experts were clear that AI usage is likely going to

contribute to evaluations of human workers, but how it contributes is going to be dynamic. In some workplaces, its contribution will be explicit, as some groups may dictate how AI should be used and will evaluate humans on that usage. On the other hand, workplaces that value collaboration might leverage human-AI collaboration as an additional means to observe and evaluate human worker effectiveness. Lastly, even workplaces that do not explicitly evaluate AI may still see the downstream impact of AI utilization, as workers who work with or use AI to complete final products will ultimately be evaluated on the quality of those products. Overall, the experts interviewed provided an interesting perspective as to why and how the results of this work may serve to inform evaluation practices in AI-enabled workplaces.

5.5.2 Theme 2: Both Benevolent and Hostile Sexist Beliefs can Alter the Expectations of Women as Collaborators and AI Experts

The second Study 2 theme identified in the expert interviews centers around the role that sexism may play within AI-enabled workplaces. Interestingly, participants saw both benevolent aspects of sexism and hostile applications of sexism contributing to women's evaluations in AI-enabled workplaces. From a benevolent sexism perspective, expert interviews actually built on the findings of Theme 1.

Specifically, P1 and P3 had noted that women are often viewed as more collaborative in the workplace, which could explain why women workers were evaluated as more effective when working with AI and female teammates. In other words, women workers may actually have an advantage in AI Teammate-enabled workplaces because they are doubly rewarded for acting collaboratively. Experts 2 and 3 went on to discuss how this benevolent sexism bias may also interplay with how women are

expected to conduct themselves in the workplace, such as using more compassionate and less direct, or “bossy” language. As such, they felt that the interactions between the woman worker and AI teammate were more emblematic of this style of communication, which resulted in better perceptions of women workers. Interactions with the AI tool, however, engendered more direct or “bossy” language, which resulted in worse perceptions of women workers because she was acting counterstereotypically. Overall, then, it is clear that benevolent sexist traits within evaluators could play a role in evaluation for women and men workers in AI-enabled workplaces, with the following quote from P3 providing an example of how:

“First, the communication style, women communicating to a tool, more commanding. And second, it’s, again, also communication style. And it’s allocating work to others, so the woman will be perceived as receiving less credit. People have stereotypical expectations about women, how they should communicate and at work, how they should behave. If a woman is to receive credit at work, they’ll need to demonstrate a higher level of expertise and effort than a man, so a tiny little cue in their daily work behavior that suggests any type of less effort, for example, giving the AI [Tool] stuff to do, will be magnified in people’s perceptions, I think.”

P1 expanded upon this by bringing up how workplace dynamics interact with team conflict. As an expert in dealing with conflict, P1 explains that, while human-human conflict can stem from many factors, including from workplace gender dynamics, human-AI collaborations could provide an opportunity to reduce these conflicts. More specifically, similar to how women are often perceived as and rewarded for being collaborative, so would AI usage be perceived as beneficial if designed with collaboration in mind. Even more critically, AI could uniquely benefit workplaces where

workplace dynamics might foster conflict because AI does not have conflict-related traits that are characteristic of humans. The following quote from P1 explores this idea:

“If I was working on some kind of research project, and I had an idea, and that idea wasn’t great, AI is not going to tell me that that’s not a great idea. AI is going to provide the information that I’m requesting of it and, you know, collaborate on that information, help me to flesh it out. And maybe that idea wasn’t great, but through the use of that [AI] tool, and the fact that that [AI] tool doesn’t have those opinion-based barriers that are getting in the way, it allows me to flesh that out and possibly come up with a great product. Where, if I have a human teammate who is against this idea for whatever reason, they either don’t agree with the direction of the project, they don’t like the idea, they’re not in the mood to collaborate like, you know, there’s a million factors that could get in the way. Those factors could inhibit that idea from moving along the way that you could independently move along a project with the use of a[n AI] tool.”

Looking more specifically at hostile sexism, the quantitative results of this work revealed that evaluator endorsement of hostile sexism had differing impacts on evaluation when the human worker was working with an AI Tool DP compared to any of the Teammate DPs. Experts were directly asked about this finding to better understand how hostile sexism may begin to impact evaluations in AI-enabled workplaces, as this result was a bit perplexing on its own.

These experts collectively provided a number of possible hypotheses. On the one hand, experts felt that the integration of AI technology into the workplace would

provide an additional opportunity for the technical knowledge of women to be tested. After diving deeper into the concept of AI Tools, P2 noted that the less collaborative nature of tools may require a greater level of orchestration from human workers, and that people with more hostile sexist beliefs might be more critical of the technical abilities of others. On the other hand, as collaborative AI teammates are more agentic, the skill level required to use them might be perceived as lower, and thus less likely to garner these same reactions from evaluators that endorse hostile sexist beliefs.

P6 somewhat countered this notion, however, as they felt their workplaces had such strict evaluation criteria that it made it difficult for these types of gender biases to slip in and affect evaluations. Merging these two ideas, one can see that, if unchecked, hostile attitudes toward women would likely continuously perpetuate the idea that women have less ability and literacy with AI, making them less effective with AI tools vs. AI teammates, but workplaces can still counteract this through purposeful crafting of evaluation criteria to reduce the possibility of these gender biases creeping in. P2's quote provides an illustrative example of what this bias has looked like in her experience:

“I was in a meeting where the [other] women in the room treated me as a subject matter expert. They deferred to me. They said, ‘[P2’s name], I want to have your opinion about this. Can you share more about this thing?’ And the man in the room started berating me. The man who also does AI stuff was feeling overlooked, essentially, like, ‘Why do you keep turning to her and not asking me these questions?’... It got to the end of that meeting, and he just started saying, ‘What do you know about Kubernetes? What do you know about OpenShift? What do you know about this specific large language model? How do you know any of these

things?’ And demeaning me. There’s this weird divide across gender in AI development that I think is really interesting... There are layers to it. Women tend to support me using it, or support me being a part of making it. I’ve seen plenty of men put me down, say I don’t have technical expertise.”

Summarizing this theme, sexism is likely to play a role in the evaluation of workers in AI-enabled workplaces, but that role is likely going to be dynamic based on the workplace and the type of evaluations in question. In the event that worker evaluations are strongly reliant on collaboration, benevolent sexism may actually serve to benefit the evaluation of human workers. Yet, workplaces that strictly prioritize technical knowledge and literacy may provide an opportunity for hostile sexism to flourish. Regardless of the type of sexism at play within a workplace’s gender dynamics, AI-enabled workplaces have an ethical responsibility to their human workforce to make sure that biases are reduced as much as possible in evaluations, as P6 points out. Indeed, as an AI Governance Expert/Advisor, she emphasizes to her clients that they have an ethical obligation to their employees when designing and evaluating AI workplace integration over the entire life-cycle of AI use in the workplace.

5.6 Discussion

In answering Study 2’s research questions, section 5.5 summarizes findings for S.2.RQ5. Additionally, this section summarizes findings for S.2.RQ1 to S.2.RQ4 regarding how evaluations and attributions are influenced by the gender identity of a human worker and by the AI/gender identity of a dyadic partner (Table 5.24), how said influences interact with the type of task being performed by the dyad (Table 5.25), and how evaluators’ pre-existing discriminatory beliefs (i.e., sexism and negative attitudes towards AI) influence evaluations and attributions (Table 5.26).

S.2. Overarching RQ: How are evaluations and attributions of credit to a human worker, their dyadic partner, and their collaboration as a whole influenced by the gender dynamics of an AI-enabled workplace?

S.2.RQ1: How are these evaluations and attributions influenced by the gender identity of a human worker?	Overall, the gender identity of the HW (<i>Woman</i> v. <i>Man</i>) did not significantly, directly influence evaluations and attributions of the HW, the DP, or the dyad as a whole. However, the gender identity of the HW did have a significant, interactive influence when coupled with <i>DP Type</i>: • When working with the <i>AI Teammate DP</i> or the <i>Female Teammate DP</i>, the <i>Woman HW</i> was evaluated as significantly more effective than the <i>Man HW</i> was when he worked with the same DPs. • Conversely, the <i>Woman HW</i> was evaluated as significantly less effective when she worked with the <i>AI Tool DP</i> than when she worked with any of the <i>Teammate DPs</i>. • Evaluations of the <i>Man HW</i>'s effectiveness were not significantly influenced by the type of DP he worked with, which suggests that there are unique perceptual disadvantages for women workers working with an AI Tool that do not impact men workers.
S.2.RQ2: How are these evaluations and attributions influenced by the AI identity or gender identity of a dyadic partner?	Overall, the identity of the DP did significantly, directly influence many evaluations and attributions to the HW, the DP, and the dyad as a whole, the most interesting of which are summarized below: • The <i>AI Tool DP</i> was evaluated as being significantly less effective than the three <i>Teammate DPs</i>, and HWs that worked with it were evaluated as being significantly less effective than HWs in the other dyads. • HWs were evaluated as having significantly less job security when working with either of the <i>AI (Tool, Teammate) DPs</i> compared to when they worked with either of the <i>human (Female, Male) Teammate DPs</i>. • HWs were attributed significantly more comparative credit when working with the <i>AI Teammate DP</i> than when they worked with the <i>Female Teammate DP</i>. • HW/DP dyads that featured the <i>Male Teammate DP</i> were evaluated as being significantly more mutually dependent on each other for successful task completion compared to dyads that featured either <i>AI (Tool, Teammate) DPs</i>. Dyads with the <i>Female Teammate DPs</i> were only evaluated as significantly more mutually dependent than dyads with an <i>AI Tool DP</i>.

Table 5.24: Study 2 Summary of Key Significant Results Answering **S.2.RQ1** and **S.2.RQ2**.

S.2. Overarching RQ: How are evaluations and attributions of credit to a human worker, their dyadic partner, and their collaboration as a whole influenced by the gender dynamics of an AI-enabled workplace?

S.2.RQ3: How, if at all, do the influences of S.2.RQ1 (HW Gender) and S.2.RQ2 (DP Type) vary based on the type of task completed in an AI-enabled workplace?

- Regarding **S.2.RQ1's** (*HW Gender*) influence:

- Evaluations of the *Man HW's* perceived power compared to the DP was **significantly higher** when the dyad was working on the *Meeting Scheduling Task* versus when they were working on the *Investment Banking Task*.

- Regarding **RQ.1.2's** (*DP Type*) influence:

- HWs were evaluated as being **significantly more effective** when working with any of the *Teammate DPs* on a *Meeting Scheduling Task* compared to working with the same DPs on an *Investment Banking Task*.
 - When working on the *Meeting Scheduling Task*, HWs were evaluated as having **significantly less job security** when working with either of the *AI (Tool, Teammate) DPs* compared to when they worked with either of the *human (Female, Male) Teammate DPs*.
 - HWs were evaluated as having **significantly more job security** when working with either of the *human (Female, Male) Teammate DPs* on the *Meeting Scheduling Task* compared to when they worked with the same DPs on the *Investment Banking Task*.
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Table 5.25: Study 2 Summary of Key Significant Results Answering **S.2.RQ3**.

S.2. Overarching RQ: How are evaluations and attributions of credit to a human worker, their dyadic partner, and their collaboration as a whole influenced by the gender dynamics of an AI-enabled workplace?

S.2.RQ4: How, if at all, do the discriminatory beliefs (i.e., sexism and AI attitudes) of an evaluator interact with gender dynamics to influence these evaluations and attributions?	- As evaluator endorsement of hostile sexist beliefs <i>increased</i>... • evaluations of the <i>AI Tool DP</i>'s effectiveness <i>increased</i>. • comparative credit trended towards the Woman HW when the dyad worked on an <i>Investment Banking Task</i>, but trended towards the DP in the <i>Meeting Scheduling Task</i>. • comparative credit trended towards the Man HW when the dyad worked on an <i>Meeting Scheduling Task</i>, but trended towards the DP in the <i>Investment Banking Task</i>. • evaluations of partnership mutual dependence <i>increased</i> when working with the <i>AI Tool DP</i>. - As evaluator endorsement of benevolent sexist beliefs <i>increased</i>... • evaluations of the <i>AI Tool DP</i>'s effectiveness <i>increased</i>. • comparative credit trended towards the AI Tool DP and the AI Teammate DP compared to the HW in those dyads, while it trended towards the HW in dyads feature the <i>Male Teammate DP</i>. - As negative attitudes towards AI <i>increased</i>... • evaluations of the <i>AI Tool</i>, <i>AI Teammate</i>, and <i>Female Teammate DPs</i>' effectiveness <i>decreased</i>, and external factors were allocated more weight when determining credit attribution for these DPs. • comparative credit trended towards the HW when working with the <i>AI Teammate</i> and <i>Female Teammate DPs</i>, while it trended towards the DP when working with the <i>AI Tool DP</i>. • HW job security <i>decreased</i> in all DP conditions, especially when working with <i>AI (Tool, Teammate) DPs</i>.
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Table 5.26: Study 2 Summary of Key Significant Results Answering **S.2.RQ4**.

Going forward, this discussion will situate these results in the existing literature on human-AI interaction/teaming and workplace gender dynamics in group work to unpack how evaluations and attributions are fundamentally shaped in an AI-enabled workplace by the functional design of the AI that human workers work with, intermingling with workplace gender and task dynamics. It will especially highlight how the functional design of an AI as either a Tool or a Teammate plays critical roles in evaluations and attributions of workers generally and women workers specifically. In doing so, Study 2's results make significant contributions to ongoing academic and popular discussions about the integration of AI into the everyday workplace [489], as well as making significant contributions towards furthering our understanding of the AI Tool vs. AI Teammate divide [153, 156].

Finally, this section ends with recommendations for how individuals and organizations looking to integrate AI into their everyday workplace can best account for AI functional design and workplace dynamics to craft successful integration practices. These recommendations represent the intermingling of academic research and practitioner expertise, as they were borne from direct suggestions from and this author's interpretation of the expert interviews.

5.6.1 Accounting for AI Functional Design and Gender Dynamics in Workplace Teams

This study sets out to understand if and how the gender dynamics and functional design of AI systems would influence the evaluations of both workers and technologies. Based on the results of this work, a clear conclusion can be drawn: the functional design of an AI system and the workplace gender dynamics of an AI-enabled workplace will influence the evaluation of human workers and their teammates/tools.

On the one hand, results for functional design show that worker evaluations trended more positively when working with AI teammates over AI tools. On the other hand, interactions between a dyadic partner’s gender identity and the type of task completed by a team demonstrate the potential for workplace gender dynamics to influence evaluation. While these main effect results should inform future evaluation, the most interesting findings to come from this work come from the intersection of gender dynamics and functional design. Indeed, the effects of dyadic partner type, interactions between dyadic partner type and worker gender, and the interactions between dyadic partner type and various covariates present an interesting narrative surrounding the design and integration of AI technology. As such, the following discussion highlights these interesting outcomes, their relation to prior work, and their implications for the future design and integration of AI into the workplace.

First, focusing on the manipulation of the dyadic partner type, both AI functional design and workplace gender dynamics can be found within its conditions. In regard to AI functional design, significant differences were found multiple times between the AI Tool and AI Teammate DP conditions. Some of these results mirror that of prior work, which has shown that job security is often perceived as lower for those work with AI teammates [156], and that evaluations of human workers are worse when working with AI tools [380]. However, in this study, the evaluators still showed more positive effectiveness evaluations for human workers working with AI teammates over AI tools, which is a shift from concepts theorized in prior works that advocate the idea that AI systems can only ever be treated as tools [421]. These effects are most likely explained by the way this study’s manipulation of AI functional design extended beyond a purely nominal representation of the word “teammate” to instead focus on a functional operationalization via differing levels of agency and interdependence [347, 353], which is a departure from prior theoretical work that has

yet to evaluate these functional differences [322].

Second, turning towards workplace gender dynamics, this work also sees differences between the AI Teammate DP condition, the Male Teammate DP condition, and the Female Teammate DP condition. In particular, this work showed differences in credit attributed to human female workers and AI Teammates, such that human workers were attributed more credit than their dyadic partners when working with the AI Teammate DP, while the DP was attributed more credit when the DP was the Female Teammate DP. This contrasts the dominant narrative in attributional research, which typically characterizes male workers as being attributed more credit than their female counterparts (e.g., [47]), suggesting that AI-enabled workplaces arguably serve to benefit perceptions of female workers when working with AI Teammates versus male teammates. Moreover, partnerships that featured a human worker working with Male Teammates were seen as generally more mutually dependent than those working with AI tools. This is likely reflective of the way that gender dynamics function in groupwork as described in attributional literature, such that men in groupwork situations are seen as taking on more of the credit for the shared task [403, 404], which could influence perceptions of mutual dependence.

Finally, the implications of these workplace gender dynamics become more complex when exploring the sole interaction effect between dyadic partner type and human worker gender and, thus, the interaction between workplace gender dynamics and AI functional design. This work found that the man worker was evaluated as significantly less effective than the woman worker when both worked with the AI Teammate and Female Teammate dyadic partners. Yet, women human workers saw significantly lower effectiveness when they worked with AI Tools compared to any of the other teammate dyadic partners that they worked with. These interactions highlight a surprising finding: that, despite completing the same task, women may

have an evaluation bias over men working with AI, which is counter to many other works exploring gender and technology dynamics in the workplace [104]. Yet, women workers are also more likely to be evaluated differently depending upon the type of dyadic partner they work with, while men workers are evaluated similarly no matter who they work with, which follows what prior works on gender dynamics in groupwork have found [403, 404].

By blending these three insights, some insights into the design of AI technology can be derived. Based on these results, although AI technology may not always carry with it a perceived gender identity (see Study 1 [411], and [96]), this work is still able to see gendered differences in the perceptions between AI and human collaborators. Most notably, it is likely not enough to simply remove gender identity from the research and design of AI systems; rather, researchers will need to understand how the identity of AI systems as either a Tool or a Teammate lends itself toward different gendered perceptions seen in modern workplaces. Moreover, these gendered perceptions may differ based on the functional design of AI in terms of its designed agency and interdependence, as evidenced by the AI Teammate DP and human Female Teammate DP sharing significant effects in terms of credit and effectiveness.

Outside of these manipulation-only insights, the role of covariates in this work also demonstrates the likely impact that workplace gender dynamics and AI functional design will play in evaluations of human workers generally and women workers specifically in AI-enabled workplaces. Indeed, this work found that both types of ambivalent sexist beliefs, hostile and benevolent [27, 188, 189], and negative attitudes towards AI use in society can be significant predictors for evaluation outcomes of workers collaborating with different dyadic partners. When compared to prior work, it isn't entirely novel that sexism and AI attitudes influence perception [104, 352]. Yet, the novel aspect of this work comes in the different dyadic partner conditions

and their interaction with these beliefs.

Most notably, the AI Tool DP condition was shown to have consistent and strong relationships between multiple covariates and multiple evaluation criteria. Yet, the AI Teammate DP condition was shown to have weaker and sometimes even opposite trends compared to the AI Tool DP condition, with many evaluations of partnerships containing an AI Teammate DP matching closer to those that contained a human Teammate DP, particularly the Female Teammate DP. This clustering and difference in trends among dyadic partners highlight that AI Teammate DPs may not only be slightly more resilient toward these pre-existing biases, but that they may also engender more human-like perceptions and evaluations than the AI Tool.

Overall, then, the integration of AI Tools and, to a lesser extent, AI Teammates, will likely intermingle with evaluator biases in regards to AI's use in society and workplace gender dynamics that could hinder evaluations of both human workers and AI in AI-enabled workplaces. Indeed, evaluator bias as a critical consideration has gained recent traction in the AI community for its potential to allow racism and sexism to seep into the design of AI systems [77, 454]. However, based on Study 2's results, simply accounting for this bias during the design of an AI system for workplace integration is likely not enough to actually address these concerns. Rather, these biases are going to need to be accounted for and mitigated within evaluators of AI *and* of humans to ensure AI workplace integration accounts for the pre-existing ways that evaluator biases already creep into evaluations in all-human workplaces.

As research and practice continue to progress with the design and integration of AI technology, these impacts and intersections of workplace gender dynamics and AI functional design should be addressed when designing and evaluating AI workplace integration. Turning towards design, designers will need to account for the implications of designing AI as either tools (i.e., lacking agency, interdependence [347, 353])

or teammates (i.e., agentic, interdependent [347, 353]), keeping in mind that most worker evaluations and attributions of credit increased when working with an AI teammate compared to working with an AI tool. Furthermore, integrators will need to account for the gender dynamics of a team that is the target of AI integration, as the gender makeup of that team and whether the AI is acting in the capacity of a tool or a teammate could substantially impact the ultimate effectiveness of that worker and their partner.

Lastly, evaluators will need to be trained to overcome potential biases related to gender in the workplace and technology use in the workplace, as those pre-existing biases may yield inconsistent or overly harsh evaluations of human workers who are, in reality, leveraging AI tools and teammates to increase their effectiveness in the workplace. Moving forward, it will be critical for human-centered computing and human-centered AI research to aid these design, integration, and evaluation efforts to ensure workers and AI are given the best opportunity to succeed in the workplace, regardless of their gender or functional design.

5.6.2 Design Recommendations for Approaching Future AI Workplace Integration with Consideration of Evaluating Human Workers

Grounded in both the quantitative and qualitative results of Study 2, the remainder of this chapter details two fundamental design recommendations aimed towards encouraging organizations to approach AI workplace integration with consideration of its human workforce, especially their women workers. It does so by first highlighting the vital role that AI functional design plays in how workers are evaluated when working with AI in AI-enabled workplaces, and explicitly calling for organiza-

tions to prioritize teamwork-based AI functional design when crafting AI workplace integration to maintain positive evaluations of *all* human workers. It then explicitly calls for organizations to interrogate and tackle workplace gender and technological biases within their own workplace *prior to* AI workplace integration, thus pushing for organizations to be considerate of their specific workers at the ground level of future AI-enabled workplaces.

5.6.2.1 Recommendation 1: To Maintain Positive Evaluations of Human Workers, Prioritize Considering AI Teammates & Their Applicability in the Workplace Context

The first design recommendation of Study 2 stems from the direct comparison of AI tools and AI teammates. One of the most novel findings of this work highlights that AI tools, compared to AI teammates, led to significantly lower perceptions of performance for both the AI *and* the human worker. These differences highlight the impact that shifting from viewing AI tools and teammates as simply nominal differences to viewing AI tools and teammates as *functionally different* is essential in research and in industry practice. Moreover, the positive perceptions that come from working with AI teammates highlight the likely positive impacts that would come from designing AI as teammates rather than as tools when designing for AI workplace integration. As such, the first recommendation of this work is that the research and practice of AI design in collaborative settings should always consider the usage of AI teammate functional design principles of agency and interdependence to promote the positive perception of AI and human workers.

However, it is also critical to note that this recommendation is for the *consideration* of designing AI as teammates when designing for AI workplace integration. Indeed, in the event that an AI teammate's functional design is *not* a good fit for

a specific task, then their integration may actually be harmful to the processes that it intends to help [368, 388]. While prior works in HCC and HCI domains have often defaulted to the 'tool' perspective when designing and evaluating AI integration [421], the findings of this work suggest that viewing AI *teammates* as a default design pattern may provide significant benefits to the evaluation of human workers and even AI systems. As such, future research and practice should first consider whether designing an AI as a teammate fits with its target task and workplace context, and only shift to a tool-oriented design perspective when it is determined that the teammate design would be inappropriate or would adversely affect task success.

5.6.2.2 Recommendation 2: Prior to Integrating AI Teammates, the Gender and Technologies Biases of Evaluators Need to Be Corrected

The second design recommendation of this work is more broadly concerned with workplace gender and technological dynamics as larger cultural concerns to address in the design and evaluation of AI workplace integration. Indeed, the results of this work highlight a clear interaction between different AI systems and the biases evaluators hold toward both AI and women. While prior work has continued to stress the importance of addressing evaluator bias during the design of AI systems [77], often through bias training and correction [39], the results of this work underscore that *solely* focusing on mitigating evaluation biases at the design stage of AI systems without addressing the larger workplace culture will be detrimental to human workers and the successful integration of AI technology into the workplace. As such, this study posits that anyone performing evaluations of human workers working with AI should undergo explicit bias training for both gender and AI biases, as well as other potential biases identified by future works.

In following this recommendation, research and practice will need to divest

current efforts in evaluation bias to highlight how biases can be mitigated *after* AI workplace integration in addition to the existing efforts in design pre-integration. Without this correction, a seemingly unbiased and safe AI system could cause undue harm by enabling the harmful biases of evaluators to manifest and perpetuate without adequate checks. Interestingly, with recent work highlighting the potential design and implementation of AI-powered evaluation systems [74], this recommendation is not necessarily limited to human evaluators. Indeed, AI-based evaluators would also need to undergo substantial bias training and correction prior to evaluating an AI-enabled workplace, as the perpetuation of these gender biases could be especially detrimental in these contexts.

Looking beyond these two recommendations, Study 2’s collective results highlighting how workplace gender dynamics and AI functional design intermingle to shape evaluations of and attributions of credit to human workers lays the groundwork for **Study 3’s focus on charting the paths that lead from these perceptions, dynamics, and designs to perceptions of human worker job security in AI-enabled workplaces.** Indeed, the ANCOVAs run in the quantitative results do not account for how dependent variables (i.e., evaluations and attributions) impact *each other*, such as how evaluations of effectiveness might influence attributions of comparative credit. Moreover, while the Study 2’s quantitative results do present limited insights on human worker job security (see section 5.4.2), it is still unclear how evaluations and attributions of credit work to influence perceptions of job security in AI-enabled workplaces. As such, Study 3 utilizes specific statistical techniques to chart a path that connects all of these factors, as well as supplemental expert interviews to contextualize the results of this path construction in real-world AI workplace integration perspectives.

Chapter 6

Study 3: Understanding the Job

Security of Women Workers

Working in AI-Enabled

Workplaces Based on Who or

What they Work With

6.1 Introduction

While understanding how gender and workplace dynamics influence AI design and worker evaluation, it is critical to loop back to the actual workers in AI-enabled workplaces to understand how they will be impacted by AI's integration. In particular, recent trends in AI have created a strong expectation that AI will create mass job displacement in the workforce [459]. Organizations, from software development to data entry, have all shown an interest in replacing a portion of their workforce

with advanced AI technologies [502]. Coupled with a weakening job market within the United States, these trends have ultimately created a climate where workers have become increasingly concerned about their job security [89].

Moreover, while the recent reports (October 2025) have shown that a weakening job market is not, in fact, the result of AI integration [182, 184], there is still a widespread public perception that the potential for AI integration and replacement still exists [290], opening the opportunity for these concerns to worsen in the coming years. These concerns pose a significant threat regardless of actual market trends because even the *perception* of lower expectations of job security can directly impact an individual's and team's performance and effectiveness [459]. These impacts can then create *actual* job security threats due to poor performance evaluations [503], thus creating a self-perpetuating cycle for human workers facing the prospect or reality of AI integration into their workplaces if not thoughtfully considered and handled.

Perhaps recognizing the threat that even perceived job insecurity can have on workers, significant effort has been put towards understanding how best to integrate AI into workforces to enhance, rather than replace, the work that humans do [292]. Right alongside these efforts, though, are continuous reports detailing the various ways in which women workers will be disproportionately displaced by AI integration in the future [449]. Taken in combination with the recent reports showing that there were no significant shifts in the job market before and after AI integration for AI-enabled workplaces [182], it is clear that there are other factors that are driving concerns for the future of women workers, namely that workplace gender dynamics that devalue women workers generally and in STEM-related positions specifically will likely create unequal impacts of AI-integration on women workers [449].

Yet, little empirical attention is being paid to how the functional design and evaluation of an AI-enabled workplace integration might intermingle with workplace

gender dynamics to create differing impacts on women workers' perceived job security when working in an AI-enabled workplace. Indeed, as is the focus of this dissertation, job security in the AI-integrated era may look remarkably different for women workers compared to men workers [449], those working with AI tools or AI teammates [156], and those working on highly routine or complex tasks [175]. These differences in an AI-enabled workplace's gender dynamics, the functional design of their integrated AI, and the workplace context these new AI-enabled partnerships work within will possibly play a significant role in defining workers' expectations of job security in the coming years.

Furthermore, as these factors were shown to influence evaluation in Study 2, these evaluations of effectiveness and attributions of credit for success when partnering in AI-enabled workplaces may also serve to propagate and mediate impacts on perceived human worker job security. In turn, to provide an understanding of how perceptions of human worker job security in AI-enabled workforces are formed through evaluations of and attributions of credit to said workers, **it will be imperative to model how gendered dynamics, AI functional design, and workplace context all serve to influence worker evaluations, credit, and perceptions of human worker job security.**

With this need in mind, built upon the results of Study 2, Study 3 seeks to further explore and conceptualize this challenge, with the ultimate goal of how job security can be modeled within AI-enabled workplaces to account for the interrelationship between AI design, workplace gender dynamics, workplace context, and worker evaluation. To accomplish this, Study 3 relies on a complex analysis technique often used in HCI research that is known as structural equation modeling (SEM) [267], which enables the creation of a comprehensive model that explores how all of these factors influence one another and, ultimately, job security at once. Similar to Study 2,

post-experiment supplementary expert interviews were conducted with the same six experts in AI-enabled workplaces (four women and two men who work with women) to support the quantitative results of Study 3 and to ground these results in real-world perspectives on AI workplace integration.

In turn, Study 3's ultimate contribution is **the development, validation, and presentation of a first-of-its-kind empirical model** that explores how AI functional design, workplace gender dynamics, workplace context, and worker evaluation all blend to inform the expectations of job security workers will form in AI-enabled workplaces. Through these contributions, both researchers and practitioners will gain a greater ability to predict human worker job security concerns, address these concerns with a consideration of workplace gender and technological dynamics, and improve the long-term mental well-being and effectiveness of human workers, especially women workers.

6.2 Research Questions

Based on the contributions of Study 2, the dissertation-wide goals of this work, and the independent motivations of Study 3, two overarching research questions have been created to guide the exploration of job security perceptions in AI-enabled workplaces. In particular, Study 3 employed structural equation modeling (SEM) [448] to investigate how the factors explored in this work (i.e., workplace gender dynamics, AI design, effectiveness, and credit) interact to ultimately influence perceptions of human worker job security in AI-enabled workplaces. In turn, the following two research questions serve to guide the initial creation and empirical refinement of this model:

S.3.RQ1: What are the empirical links between workplace gender and technological dynam-

ics, evaluator biases, and partnership evaluations in AI-enabled workplaces?

S.3.RQ2: How do evaluations of human worker effectiveness, dyadic partner effectiveness, and comparative credit meld to form expectations of job security for human workers in AI-enabled workplaces?

In answering these two questions, Study 3 contributes an empirical structural model that informs how and why human evaluators will form perceptions of human worker job security in AI-enabled workplaces. More specifically, Study 3 first provides vital pathways that chart how gendered and technological dynamics intermingle with AI design and evaluator biases to impact how human workers in AI-enabled workplaces are evaluated, and then subsequently given credit for success in their workplace. Then, Study 3 is the first empirical study to chart the path that leads from these evaluations and credit attributions directly to evaluators' perceptions of human worker job security, such that these dynamic-driven evaluations may ultimately have serious impacts on the security of human workers within AI-enabled organizations. Finally, Study 3 utilizes the path model created to discuss the real-world implications of these results in six interviews with professionals with expertise in people management and AI workplace integration to jointly craft recommendations for how to best instill a sense of security in workers, especially women workers, so that they can face the impending rise of AI workplace integration with greater knowledge, confidence, and power.

Additionally, as with Study 2, this dissertation's Study 3 also acknowledges the crucial need to bridge the gap between research outcomes and industry practices in the context of studying AI integration in everyday workplaces. Thus, using the six post-experiment supplementary expert interviews, Study 3 also aims to address the following research question to support the quantitative results of Study 3 and to

ground these results in real-world perspectives on AI workplace integration:

S.3.RQ3 How do real-world experts in AI-enabled workplaces conceptualize AI workplace integration’s impacts on the perceived job security of human workers?

6.3 Methods Part I: Structural Equation Modeling

With the data collection overlapping between Studies 2 and 3, the following methods section will provide a short overview of the data used for this study, the unique analysis method used to answer Study 3’s research questions, and the confirmatory factor analysis performed on the data collected to enable this unique analysis method. For a full breakdown of the design and data collection methodologies, please refer back to Study 2’s methods section.

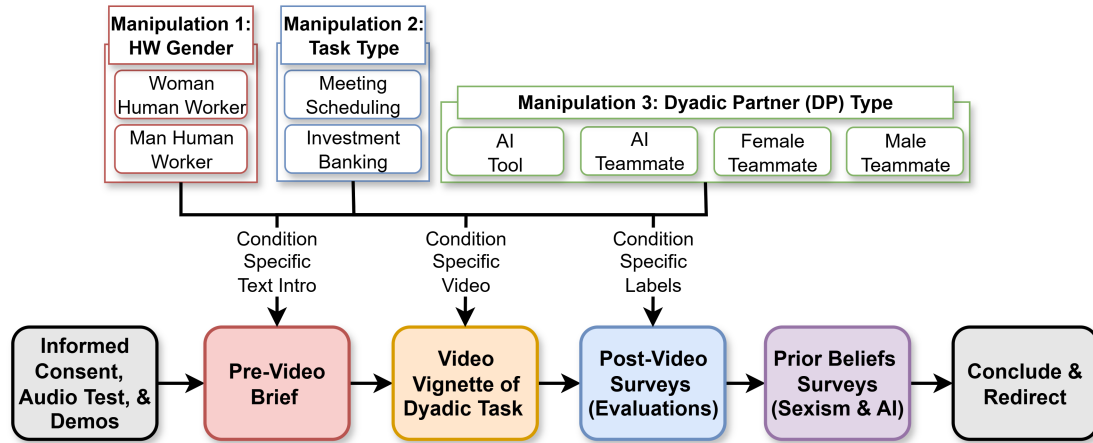


Figure 6.1: Between-Subject Manipulations Chart & Factorial Experiment Survey Flow.

The dataset used for Study 3 is the same dataset that resulted from the 2 (Human Worker [HW] Gender Presentation) x 4 (Dyadic Partner [DP] Type) x 2

(Task Type) between-subjects online factorial experiment described in Study 2. As a reminder, 685 participants were tasked with evaluating the performance of either a woman or a man human worker (HW) when working with a dyadic partner (DP) that was either an AI tool, an AI teammate, a human female teammate, or a human male teammate on either a meeting scheduling task or an investment banking task. The structure of the factorial experiment can be seen in Figure 6.1, which has been included in this section as a refresher. For examples of the video vignettes, see Figures 5.4, 5.3, 5.2, 5.6, and 5.5. Demographic information for the 685 participants can be found in Table 5.1.

For Study 3, (1) a theoretical model and hypotheses were constructed, and (2) a confirmatory factor analysis (CFA) [58] on multi-item measures was performed, both being vital components of the structural equation modeling (SEM) process [448]. The remainder of this Methods section thus focuses on these two methodological processes by providing detailed justifications for and explanations of their use.

6.3.1 Hypotheses

6.3.1.1 S.3.RQ1: *What are the empirical links between workplace gender and technological dynamics, evaluator biases, and partnership evaluations in AI-enabled workplaces?*

When it comes to evaluations of human worker (HW) effectiveness, the main and interaction effects that were uncovered in Study 2’s analyses directly informed the following hypotheses for what the expected links will be between workplace gender and technological dynamics, evaluator biases, and partnership evaluations for Study 3:

H1: Working with a dyadic partner (DP) that acts as a Teammate (AI,

Female, or Male) will significantly improve evaluations of the human worker's effectiveness.

H2: When the partnership is working on an Investment Banking Task, working with a dyadic partner (DP) that is a human Teammate (Female or Male) will significantly worsen evaluations of the human worker's effectiveness compared to when the partnership is working on a Meeting Scheduling Task.

H3: When the human worker (HW) is a Woman, working with a dyadic partner (DP) that is an AI Tool will significantly worsen evaluations of the Woman HW's effectiveness compared to when she works with any of the dyadic partners that are Teammates (AI, Female, Male).

When it comes to evaluations of dyadic partner (DP) effectiveness, the main and interaction effects that were uncovered in Study 2's analyses directly informed the following hypotheses for what the expected links will be between workplace gender and technological dynamics, evaluator biases, and partnership evaluations:

H4: Having a dyadic partner (DP) that acts in a Teammate (AI, Female, or Male) capacity in the partnership will significantly improve evaluations of the dyadic partner's effectiveness.

H5: The more negative attitudes towards AI an evaluator holds, the more they will perceive the effectiveness of an AI (Tool or Teammate) dyadic partner (DP) to be significantly lower when compared to evaluators with less negative attitudes towards AI.

When it comes to evaluations of partnership mutual dependence, the main effects that were uncovered in Study 2's analyses directly informed the following

hypothesis for what the expected links will be between workplace gender and technological dynamics, evaluator biases, and partnership evaluations:

H6: Working with a dyadic partner (DP) that is a human Teammate (Male) will significantly improve perceptions of partnership mutual dependence compared to when the dyadic partner (DP) is an AI (Tool or Teammate).

Moreover, when it comes to how partnership evaluations will link to each other, the following hypothesis was constructed as a way to test in Study 3 if perceptions of mutual dependence might explain why the evaluations of human worker (HW) effectiveness and dyadic partner (DP) effectiveness were so similar when looking at the main effects of dyadic partner (DP) type in Study 2:

H7: Evaluations of partnership mutual dependence will mediate the relationship between dyadic partner (DP) effectiveness, human worker (HW) effectiveness, and the type of dyadic partner (DP) featured in the partnership.

6.3.1.2 S.3.RQ2: *How do evaluations of human worker effectiveness, dyadic partner effectiveness, and comparative credit meld to form expectations of job security for human workers in AI-enabled workplaces?*

The following hypotheses were formulated based on prior literature on attributions of credit in the workplace, where it is well-established that: (1) evaluations of worker effectiveness factors such as ability contribute to greater levels of credit being given to a worker when performing a task [195, 311], (2) that these evaluations over

time can contribute to a greater evaluation of employees and the need to keep them around [195, 461], and (3) that attributions of credit in dyads can be comparative such that more credit to one person can lower credit to other [271]. They are also informed by literature concerning attributions of credit in groupwork depending upon the gendered dynamics of the partnership (e.g., mixed-gendered vs. same-gendered [404]) and the functional agency of an AI when working with humans [223]:

H8: Evaluations of human worker effectiveness will mediate the relationship between comparative credit and the type of dyadic partner (DP) the human worker works with to complete the task.

H9: Evaluations of dyadic partner (DP) effectiveness will mediate the relationship between comparative credit and the type of dyadic partner (DP) featured in the partnership.

H10: Evaluations of comparative credit will mediate the relationship between perceptions of the human worker’s job security and partnership evaluations.

H11: As evaluators’ perceptions of human worker (HW) effectiveness increase, perceptions of the human worker’s job security will significantly improve.

Lastly, the following hypotheses concerning the direct effects that dyadic partner (DP) type and evaluator biases might have on perceptions of human worker job security are directly informed by the results of Study 2:

H12: Working with a dyadic partner (DP) that is a human Teammate (Female or Male) will significantly improve perceptions of the human worker’s (HW’s) job security compared to when the dyadic partner (DP) is an AI (Tool or Teammate).

H13: When working with a dyadic partner (DP) that is a human Team-mate (Female or Male), working on the Investment Banking Task will significantly worsen perceptions of the human worker’s (HW’s) job security compared to working on the Meeting Scheduling Task.

H14: Evaluators’ biases against the use of AI in society (i.e., negative attitudes towards AI) will be associated with significantly worse perceptions of the human worker’s job security.

6.3.2 Measure Validation via Confirmatory Factor Analysis (CFA)

Prior to performing the SEM analysis, a confirmatory factor analysis (CFA) was conducted on the constructs measured (Human Worker Effectiveness, Dyadic Partner Effectiveness, Human Worker Job Security, Mutual Dependence, Comparative Credit, and Negative Attitudes Toward AI). Due to the large number of items for each metric, individual items were trimmed from each measure based on their loading factor. Table 6.1 contains the loading factors for each of the items that were kept and used for their respective measure. The CFA performed showed a good fit overall for this approach: $\chi^2(209) = 625.462$, $CFI = .993$, $TLI = .991$, $RMSEA = 0.054$, $90\%CI : [0.049, 0.059]$.

Measurement	Items	Factor Loading
Human Worker	Please rate <i>HW Name</i> on a scale of In- competent (1) to Competent (9) .	0.940
(HW) Effective- ness	Please rate <i>HW Name</i> on a scale of Un- productive (1) to Productive (9)	0.944

Continued on next page

Measurement	Items	Factor Loading
	Please rate <i>HW Name</i> on a scale of Ineffective (1) to Effective (9)	0.961
Dyadic Partner	Please rate <i>DP Name</i> on a scale of Incompetent (1) to Competent (9) .	0.963
(DP) Effective-ness	Please rate <i>DP Name</i> on a scale of Unproductive (1) to Productive (9)	0.966
	Please rate <i>DP Name</i> on a scale of Ineffective (1) to Effective (9)	0.924
Human Worker	After watching <i>HW Name</i> work with <i>DP Name</i> , I believe <i>HW Name</i> 's future employment is... Sure .	0.954
Job Security	After watching <i>HW Name</i> work with <i>DP Name</i> , I believe <i>HW Name</i> 's future employment is... Stable .	0.934
	After watching <i>HW Name</i> work with <i>DP Name</i> , I believe that... <i>HW Name</i>'s job is almost guaranteed.	0.922
	After watching <i>HW Name</i> work with <i>DP Name</i> , I believe that... <i>HW Name</i> can depend on <i>his/her</i> job being there.	0.926
Mutual Dependence	The partners are both dependent on each other in this situation.	0.929
	The outcome of each depends on the behavior of the other.	0.936

Continued on next page

Measurement	Items	Factor Loading
	The partners need to work together to manage this situation.	0.926
Negative Attitudes Towards AI	I find Artificial Intelligence sinister.	0.857
	I shiver with discomfort when I think about future uses of Artificial Intelligence.	0.875
	People like me will suffer if Artificial Intelligence is used more and more.	0.869
Comparative Credit to DP	Who do you think deserves the most amount of credit for the successful completion of the <i>Task Type</i> ? (1) Definitely <i>HW Name</i> . - (3) They both deserve an equal amount of credit. - (5) Definitely <i>DP Name</i> .	NA

Table 6.1: Survey items organized by measure with each item’s factor loading.

6.3.3 Structural Equation Modeling (SEM)

Structural Equation Modeling (SEM) presents a methodology through which complex relationships between independent and dependent variables can be simultaneously modeled and evaluated. In turn, the result of this methodology is the creation of a complex empirical model, where individual factors are linked via direct effects between one another and indirect effects that are mediate through various factors

within the model. Due to the research questions of Study 3, SEM presents an ideal technique to not just understand what impacts the perceived job security of human workers but also how these effects interact and mediate one another.

Leveraging this technique and the insight provided by the CFA, the SEM process began by creating an number of hypothetical models based on the hypotheses of Study 3, saturating these models, and then trimming them to their final forms. The SEM process was conducted multiple times to examine different potential constructs used for measurement. In particular, multiple models were created to understand (1) whether or not human and dyadic partner effectiveness should be viewed as separate constructs and (2) which credit attribution measure was most robust within the model. For (1), modeling showed that dyadic partner and human worker effectiveness, while highly correlated, are distinct factors that should be analyzed separately. For (2), both comparative credit and internal credit attributions were tested, and results showed greater convergence and fit when leveraging comparative credit attributed between the human worker and the dyadic partner.

Based on this iterative and multi-model process, the final hypothesized model used for analysis explored the following path:

(Manipulations + Covariates) → Mutual Dependence → (Dyadic Partner Effectiveness + Human Worker Effectiveness) → Comparative Credit → Human Worker Job Security

This model was created within the MPlus application. The model was then saturated to display every possible effect. Iterative trimming was performed, where 1-2 effects were trimmed at a time based on significance and size. Upon reaching a near-final model, small effects ($\beta < .1$) were also trimmed. Once trimmed, a final model was graphically created to display the finalized effects and relationships. Total

effects for this model were also calculated to provide graphs for later discussion.

6.4 Methods Part II: Supplemental Expert Interviews

Just as with Study 2, the quantitative efforts performed by Study 3 were supported by the same post-experiment supplemental expert interviews with 6 experts (4 women and 2 men who work with women) in people management and AI workplace integration, whose demographics and background can be found in Table 5.2. These supplemental expert interviews were conducted for the same general motivations outlined in 5.3 of Study 2, namely as a way to qualitatively examine the “why” of the quantitative results, as well as to explore how experts in AI-enabled workplaces can use the insights generated by Study 3’s SEM to inform AI workplace integration practices. While Study 2 approached these two motivations with the goal of contextualizing and deepening understandings of the impacts of workplace gender dynamics and AI functional design on evaluations of human workers, Study 3 approached these same motivations with a focus on human worker job security to address the following research question:

S.3.RQ3 How do real-world experts in AI-enabled workplaces conceptualize AI workplace integration’s impacts on the perceived job security of human workers?

6.4.1 Participants & Data Collection

As a reminder, six experts in AI-enabled workplaces were interviewed in total, consisting of 4 women and 2 men who work with women. Their ages ranged from

28 to 59, and their professional roles ranged from workplace conflict management to AI research and development, and even being the CEO of a startup accelerator. A full list of participant demographics and work experience background can be seen in Table 55.2.

Participant recruitment and the overall procedures for the expert interviews was the same for Study 3 as Study 2, which can be seen in more detail in section 5.3.1 of Study 2. However, data collection for Study 3 results specifically occurred during the second part of the two part interview detailed in Study 2, meaning that participants were asked to review Study 3’s SEM results separately from Study 2’s ANCOVA results. As such, Study 3’s Results Part II unpacks these experts’ reactions toward the SEM model created in this work, with particular focus on discussing human worker job security. This distinct interview portion focused on the initial reactions of experts to the model created, their assessment of human worker job security in AI-enabled workplaces generally, and their expectations for how this model might play out in a real-world AI-enabled workplace. The full semi-structured interview guide used for Study 3 is available in Appendix B.7, and the review slides shown to participants during the interviews are presented in Appendix B.8.

6.4.2 Qualitative Data Analysis

Just as with Study 2, data analysis for Study 3’s supplemental expert interviews was conducted using in-depth qualitative reflexive thematic analysis [55, 56]. As a reminder, the “reflexive” element of this type of thematic analysis conceptualizes the researcher’s subjectivity as a resource in the process of research, such that coding and interpreting data is recognized as an *“inherently and inescapably subjective practice”* [56].

Following Braun and Clarke’s [55, 56] procedures for reflexive thematic analysis, this dissertation’s author first carefully read through the collected data line-by-line to familiarize herself with the data, identifying and notating how these experts in AI-enabled workplaces interpreted Study 3’s quantitative findings and formulated their own views on AI workplace integration that is considerate of the factors that influence perceptions of human worker job security in AI-enabled workplaces. Next, the iterative coding process began with generating initial codes through assigning codes to individual data and then iterating on the codes to eliminate redundancy. Next, codes were categorized into thematic topics related to the research questions, which developed an initial set of themes related to experts’ insights on workplace gender dynamics and AI functional design on evaluations and attributions of credit to workers. Themes were further refined to better capture the data in relation to the research questions, and then again to identify the “essence” of each theme [55, 56]. Finally, as with Study 2, the most compelling quotes were selected as examples to craft a narrative structure for Study 3’s findings that flows logically and naturally [56].

6.4.3 Researcher Positionality

While 5.3.3 in Study 2 already details my positionality as a researcher in terms of these interviews, it is still valuable to reiterate here how said positionality impacted how I approached the supplemental expert interviews conducted in both Studies 2 and 3, as this transparency is a vital component of reflexivity [56].

First, it is important to reiterate that my experiences with workplace gender dynamics and job security within and outside of STEM-related fields are rooted in my personal identity as a White, American woman in academia. Three out of four of the women participants are Asian and might therefore experience or interpret workplace

gender dynamics differently because of their intersectional identities. However, as mentioned in Study 2, our shared identities as women in academic and industry fields related to AI and tech does allow me to be aware of the unique challenges that women face in U.S. markets in terms of job security, which all participants operate within.

Second, it is important to acknowledge again that all 6 participants are individuals that I knew prior to conducting these interviews, therefore there is a possibility that their insights might have been influenced by their pre-existing perceptions of me and my work. However, their insights as experts are still highly valuable given their well-established years of experience and specific expertise in people management and AI workplace integration (see Table 5.2), and I already had established the kind of rapport and shared understanding with my interviewees that is often valued in rich qualitative work [446].

6.5 Results Part I: Structural Equation Model

6.5.1 Model Fit

The final SEM created for this work is depicted in Figure 6.2. The model created yielded a good overall fit ($\chi^2(311) = 466.24$, $CFI = .996$, $TLI = .996$, $RMSEA = .027$, $90\%CI : [.022 - .032]$). Based on the evaluation metrics put forth by Hu and Bentler ($CFI > .96$, $TLI > .95$, $RMSEA < .05$, and upper $RMSEASI < .10$), this model presents an excellent fit for the data examined. The path model displayed in Figure 6.2 was generated using standardized coefficients, meaning differences in scale type are standardized across the measures explored within the model. Paths between two dependent variable constructs depict standardized changes from one construct to another, and paths between independent variables to dependent variables present

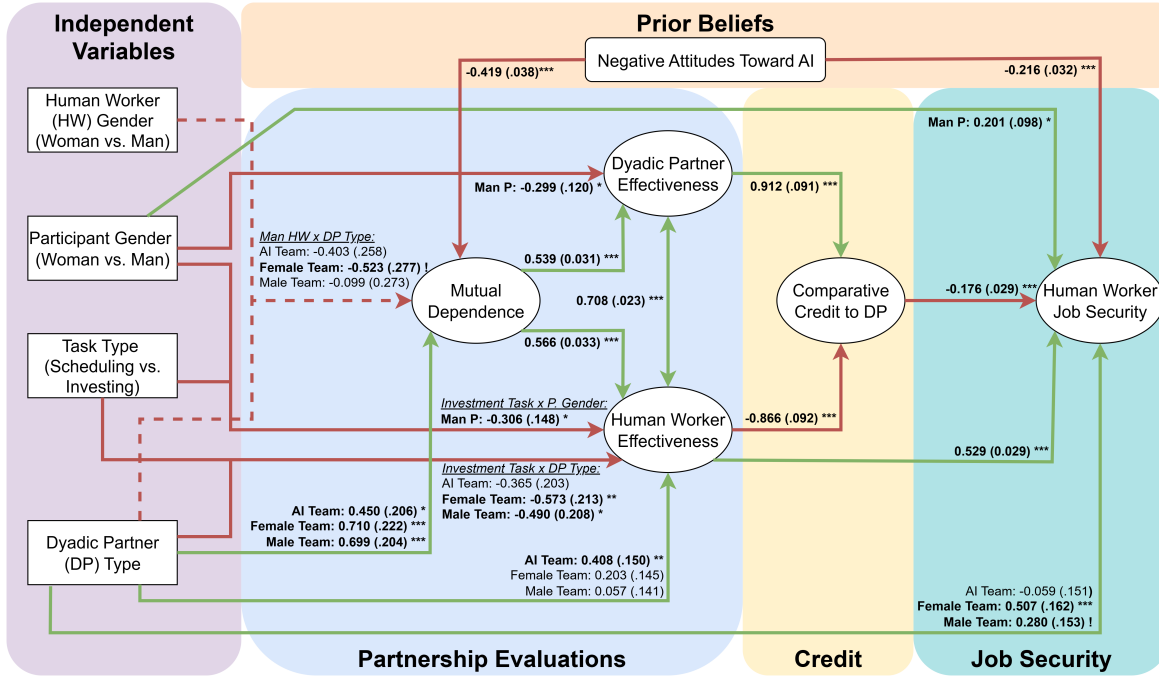


Figure 6.2: The full structural model for AI-enabled workplace partnership evaluations, comparative credit, and human worker job security constructs. Significant results are indicated as follows: * $p < .05$, ** $p < .01$, *** $p < .001$. Each significant result for the partnership evaluation constructs and the job security construct is accompanied by the β coefficients, which signify the standardized increase or decrease in the construct as it corresponds to the increase or decrease of the other construct connected on the path. For the construct “Comparative Credit to DP”, a negative β coefficient indicates that more comparative credit is being given to the Human Worker (HW), while a positive β coefficient indicates more comparative credit is being given to the Dyadic Partner (DP). Furthermore, the β coefficients for the constructs that are contained within the “Independent Variable” block represent the standardized difference between the experimental conditions for that independent variable, and therefore effectively represent effect sizes as measured by Cohen’s d . The number next to each β coefficient in parentheses is the standard error.

standardized changes from one condition to another.

6.5.2 Model Examination

Starting with the right-most part of the model created (i.e., blue area labeled “*Job Security*” in Figure 6.2), a multitude of factors are shown to influence how evaluators perceived the job security of the human worker. Starting with direct effects from manipulations (i.e., purple area labeled “*Independent Variables*” in Figure 6.2), changes in the gender of an evaluator (“*Participant Gender*”) from Woman to Man Participant (Woman $\rightarrow$ Man) was shown to significantly improve evaluator perceptions of human worker (HW) job security. This means that men evaluators perceived the human worker as having more job security than what women evaluators perceived. Additionally, changes in Dyadic Partner (DP) Type from AI Tool to Human Teammates (Male and Female) (AI Tool DP $\rightarrow$ Male and Female Teammate DP) was shown to significantly improve perceptions of the human worker’s job security.

In other words, human workers who worked with another human teammate, regardless of gender, were perceived as having more job security than human workers who worked with an AI tool. Stemming from other dependent variables in the model, Human Worker (HW) effectiveness and credit allocated to the dyadic partner compared to the human worker were both shown to significantly influence human worker job security, with the former having a positive effect and the latter having a negative effect. Lastly, participants’ negative attitudes toward AI were shown to negatively influence perceptions of human worker job security. Based on these effects, clear connections can be seen where the evaluation of human workers, their comparison to their dyadic partner, the type of dyadic partner humans work with, and the attitudes toward AI evaluators have all serve to influence the resultant perception of the human

worker's job security.

Moving left within the model, as mentioned in the methodology of this work, different credit measures were leveraged to create multiple different models. Comparative credit presented the most robust factor explored within the model in relation to model performance and convergence. Perceived comparative credit was shown to be significantly influenced by both the human worker effectiveness and dyadic partner effectiveness evaluations. As expected, increases in human worker effectiveness were shown to decrease credit allocated to the dyadic partner, and increases in dyadic partner effectiveness were shown to increase credit allocated to the dyadic partner. Seeing as this is a comparative measure, where a decrease in credit allocation to one partner means that there is an increase in credit allocation to the other partner, these results can also be interpreted to mean that an increase in human worker effectiveness was shown to increase credit allocated to the human worker, while an increase in dyadic partner effectiveness was shown to decrease credit allocation to the human worker.

Continuing to move left in the model, the effectiveness of both the dyadic partner and the human worker showed a number of direct influences from manipulations as well as other factors within the model. Starting with the simpler of the two, dyadic partner effectiveness was shown to be significantly influenced by the gender of a participant and the mutual dependence perceived between the dyadic pair. Greater levels of perceived mutual dependence increased perceptions of dyadic partner effectiveness, and men participants, compared to women participants, were shown to perceive dyadic partners are being less effective. Shifting to human worker effectiveness, there was a significant main effect from dyadic partner type, with the AI teammate, compared to the AI tool, yielding more positive evaluations of human worker effectiveness.

Additionally, an interaction effect was shown between task type and dyadic

partner type, where the female and male human teammates show different impacts on human worker effectiveness based on the task presented. Another interaction effect was shown between task type and participant gender, where differences between tasks were found based on differences in participant gender. The mutual dependence of the dyadic pair was also shown to significantly influence human worker effectiveness, with increases in mutual dependence leading to increases in human worker effectiveness. Lastly, there was also a bidirectional effect between human worker effectiveness and dyadic partner effectiveness, meaning both perceptions are highly correlated with with the other, such that they positively influence one another.

Moving to the final left-most factor in the model, perceived mutual dependence was also shown to have direct effects from manipulations and covariates. A significant main effect was found from dyadic partner type, with partnerships containing any of the teammate dyadic partners being perceived to be more mutually dependent than partnerships with the AI tool. Additionally, a borderline significant interaction effect was found between human worker gender and dyadic partner type, with the female teammate dyadic partner showing near significant differences between human worker gender conditions. Lastly, participants' negative attitudes toward AI was shown to significantly and negatively influence the mutual dependence perceived between dyadic partners.

6.5.3 Summary of Results and Effects

Based on the effects modeled in Figure 6.2, total effects for manipulations were calculated and graphed to help summarize the results of this model. Main effects of manipulations are shown in Figure 6.3 and interaction effects are shown in Figure 6.4. Based on the effects graphed, similar results can be found between Study 2 and Study

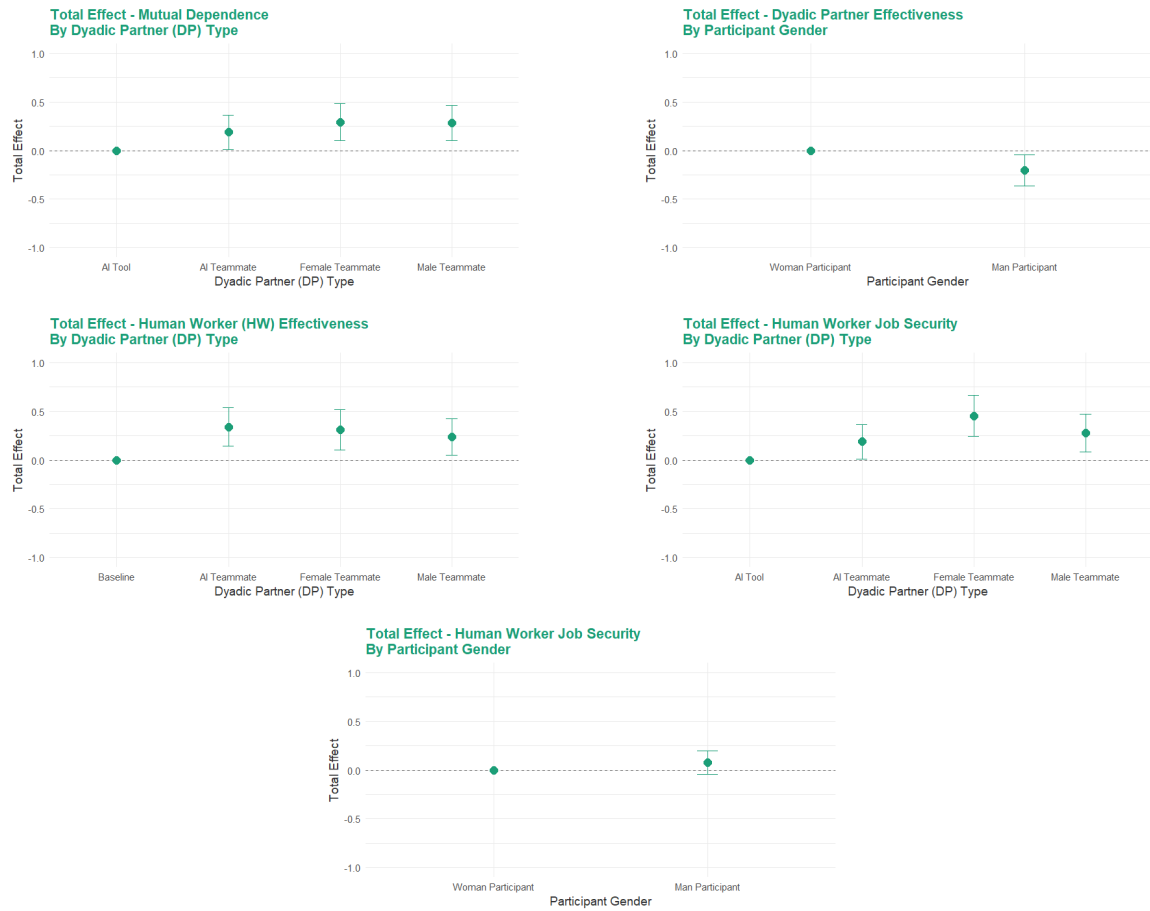


Figure 6.3: Graphical Representation of Total Effects Calculated for the Main Effects of Manipulations in Figure 6.2.

3, where the type of dyadic partner one works with is overwhelmingly impactful to the evaluations of, attributions of credit to, and, ultimately, job security of human workers. The greatest difference shown within these effects comes from the inclusion of participant gender, as two main effects and one interaction effect was found. For main effects, participant gender was shown to significantly influence dyadic partner effectiveness and job security, and interaction effects showed that the effects of task type were shown to differ based on the gender of the participant.

Shifting away from these specific manipulations and reviewing this model as a whole, the results of this work highlight that the gender and AI dynamics of workers

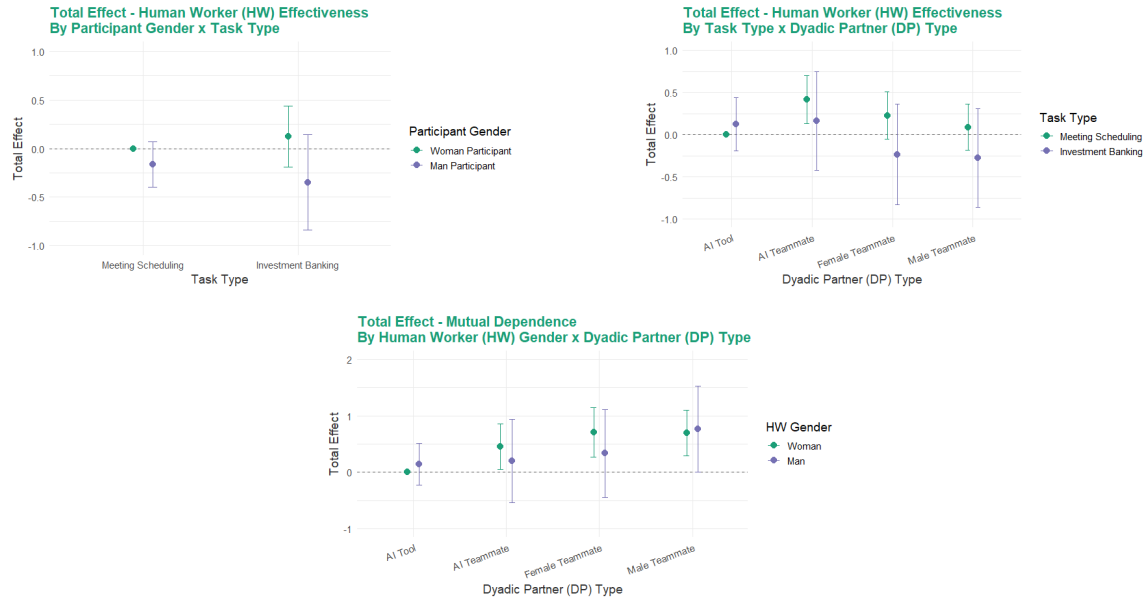


Figure 6.4: Graphical Representation of Total Effects Calculated for the Interaction Effects of Manipulations in Figure 6.2.

and their partnerships can both directly and indirectly influence the perception that evaluators have of a human worker's job security. Direct effects generally come from the type of dyadic partner humans find themselves working with, but the effects are also mediated by various other evaluations, such as mutual dependence, worker effectiveness, and credit attribution. The result of this iterative modeling process thus yielded a clear and interconnected pathway through which the dynamics of a human worker's workplace can ultimately influence their mutual dependence and effectiveness with their partnership, which, in turn, influences the credit they are attributed within their workplace, which all ultimately serve to determine the level of job security they would expect and others would expect for them in said workplaces. Levering this knowledge, a greater understanding of job security within workplaces facing AI integration can be achieved.

Based on these insights, Table 6.2 and Table 6.3 summarize how hypotheses for S3.RQ1 and S3.RQ2 are supported, partially supported, and not supported.

Specifically, for S3.RQ1, results for Study 3 did show that working with a DP that acts as a Teammate significantly improved evaluations of HW effectiveness (**H1 supported**), and that evaluations of mutual dependence mediated the relationship between both HW and DP effectiveness and the type of DP featured in the partner (**H7 supported**). Additionally, when the partnership was working on the Investment Banking Task, working with the Male Teammate DP, but not the Female Teammate DP, significantly worsened perceptions of HW effectiveness compared to the Meeting Scheduling Task (**H2 partially supported**). All other hypotheses associated with S.3.RQ1 were not supported, as there were no significant differences in evaluations of the Women Human Worker (**H3 not supported**), evaluations of DP effectiveness were not significantly influenced by the type of DP (**H4 not supported**) or evaluators' negative attitudes towards AI (**H5 not supported**), and working with a human DP did not improve perceptions of partnership mutual dependence (**H6 not supported**).

For S.3.RQ2, Study 3's results did demonstrate that evaluations of HW effectiveness do mediate the relationship between comparative credit and the type of DP the HW worked with (**H8 supported**), but DP effectiveness does *not* mediate the relationship between comparative credit and DP type (**H9 not supported**). Moreover, the model shows that evaluations of comparative credit mediate the relationship between perceptions of HW job security and partnership evaluations (**H10 supported**). In addition to these mediation effects, the model does demonstrate that evaluations of HW effectiveness directly significantly improves perceptions of HW job security (**H11 supported**), and that working with a DP that is a human Teammate (Female or Male) does significantly directly improve perceptions of HW job security compared to when working with an AI (Tool or Teammate) (**H12 supported**). However, when working with human DPs, working on the Investment Banking Task

significantly worsened perceptions of HW job security, but only when mediated by HW effectiveness (**H13 partially supported**). Finally, Study 3's model does demonstrate that evaluators' negative attitudes towards AI significantly worsen perceptions of HW job security (**H14 supported**).

H#	Hypothesis	Result
H1	Working with a dyadic partner (DP) that acts as a Teammate (AI, Female, or Male) will significantly improve evaluations of the human worker's effectiveness.	Supported
H2	When the partnership is working on an Investment Banking Task, working with a dyadic partner (DP) that is a human Teammate (Female or Male) will significantly worsen evaluations of the human worker's effectiveness compared to when the partnership is working on a Meeting Scheduling Task.	Partially Supported - (only Female Teammate worsened evaluations)
H3	When the human worker (HW) is a Woman, working with a dyadic partner (DP) that is an AI Tool will significantly worsen evaluations of the Woman HW's effectiveness compared to when she works with any of the dyadic partners that are Teammates (AI, Female, Male).	Not Supported
H4	Having a dyadic partner (DP) that acts in a Teammate (AI, Female, or Male) capacity in the partnership will significantly improve evaluations of the dyadic partner's effectiveness.	Not Supported
H5	The more negative attitudes towards AI an evaluator holds, the more they will perceive the effectiveness of an AI (Tool or Teammate) dyadic partner (DP) to be significantly lower when compared to evaluators with less negative attitudes towards AI.	Not Supported
H6	Working with a dyadic partner (DP) that is a human Teammate (Male) will significantly improve perceptions of partnership mutual dependence compared to when the dyadic partner (DP) is an AI (Tool or Teammate).	Not Supported
H7	Evaluations of partnership mutual dependence will mediate the relationship between dyadic partner (DP) effectiveness, human worker (HW) effectiveness, and the type of dyadic partner (DP) featured in the partnership.	Supported

Table 6.2: Results of Hypotheses for **S.3.RQ1: *What are the empirical links between workplace gender and technological dynamics, evaluator biases, and partnership evaluations in AI-enabled workplaces?***

H#	Hypothesis	Result
H8	Evaluations of human worker effectiveness will mediate the relationship between comparative credit and the type of dyadic partner (DP) the human worker works with to complete the task.	Supported
H9	Evaluations of dyadic partner (DP) effectiveness will mediate the relationship between comparative credit and the type of dyadic partner (DP) featured in the partnership.	Not Supported
H10	Evaluations of comparative credit will mediate the relationship between perceptions of the human worker's job security and partnership evaluations.	Supported
H11	As evaluators' perceptions of human worker (HW) effectiveness increase, perceptions of the human worker's job security will significantly improve	Supported
H12	Working with a dyadic partner (DP) that is a human Teammate (Female or Male) will significantly improve perceptions of the human worker's (HW's) job security compared to when the dyadic partner (DP) is an AI (Tool or Teammate).	Supported
H13	When working with a dyadic partner (DP) that is a human Teammate (Female or Male), working on the Investment Banking Task will significantly worsen perceptions of the human worker's (HW's) job security compared to working on the Meeting Scheduling Task.	Partially Supported - (Mediated by human worker effectiveness perceptions)
H14	Evaluators' biases against the use of AI in society (i.e., negative attitudes towards AI) will be associated with significantly worse perceptions of the human worker's job security.	Supported

Table 6.3: Results of Hypotheses for **S.3.RQ2: *How do evaluations of human worker effectiveness, dyadic partner effectiveness, and comparative credit meld to form expectations of job security for human workers in AI-enabled workplaces?***

6.6 Results Part II: Supplemental Expert Interviews

In addressing S.3.RQ3 (*How do real-world experts in AI-enabled workplaces conceptualize AI workplace integration's impacts on the perceived job security of human workers?*), results of Study 3's supplementary expert interviews reveal how perceptions of human worker job security in the age of AI-enabled workplaces have evolved to reward collaborations between humans and AI, and how workplace gender dynamics shape who has access to these rewards. These results thus shed light on the “why” element of Study 3's SEM results, specifically by exploring how industry narratives have evolved to view AI as replacing jobs but not workers in a workplace, adding nuance to the surprisingly optimistic results regarding AI teammates' links to job security through evaluations and attributions of credit, and (2) providing nuance to Study 3's results concerning participant gender by highlighting how women workers may still be disproportionately disadvantaged by this shift compared to their more confident male counterparts because of ongoing burdens placed on women by unequal workplace gender dynamics, particularly in STEM-related fields.

6.6.1 Theme 1: AI Can Replace Jobs, But It Cannot Replace Workers.

The first theme from this data focuses squarely on the dynamic landscape of job security in AI-enabled workplaces. As continuously noted by Experts 4 (59, Woman, Asian, CEO of Science/Tech Accelerator) and 6 (28, Woman, Asian, AI Governance Advisor), the use of AI in workplaces is being used to supplement or assist human workers, not replace workers, such that human worker job security

should not be significantly impacted by AI workplace integration. However, there still exists a public perception that AI is actively threatening human worker job security, which could make it more difficult to perform healthy and effective AI integration if not properly accounted for.

Expert 6 particularly emphasizes to the companies she advises that AI integrators have to focus on adaptations and augmentation, not replacement, if they hope to make any traction with AI integration into their workplace. In fact, Expert 6 explains that industry leaders in AI governance and integration are increasingly turning towards how agentic AI can work to collaborate with human workers to take on the kinds of menial but time-consuming tasks that lead to burnout in humans, with the intention of freeing up said humans for more complex tasks. The following quote from Expert 4 builds on this concept by describing her ideal scenario when it comes to balancing AI workplace integration with human worker security through human-AI collaboration:

“If you’re an independent person that’s working on a project, [as CEO] I will encourage to use AI for those tasks that are time consuming, but still actually fact check certain things, but using the tool or a teammate to create an efficiency for those particular tasks. But I also want to encourage a human to human partnership, because I don’t think I will ever consider AI to be 100% correct. Yeah, so I would still have a dyadic type relationship, but each one of those individuals will have their own tool that they can use to do their work at the same time, still have the human component to actually ensure that the information that’s being provided is correct... Originally, I was going to say that maybe an AI teammate that’s specific to the triad (human-human-AI)... It is two different roles,

I would say, between a tool and a teammate.”

Building upon this, Expert 6 noted that AI, especially in its current form, is not going to be able to replace a human worker in their entirety, but rather will replace specific job responsibilities to free up human workers. Experts 2 (29, Woman, White, AI Research & Development), 5 (32, Man, White, Dir. of Business Intelligence), and 6 noted that the current capabilities of AI technology make it much more apt to take over a single responsibility as opposed to an entire worker. These thoughts were, in part, due to the experts’ belief that humans make unique contributions to the work they do that cannot be adequately accomplished by AI, such as bringing a level of creativity, emotionality, and generalized intelligence to their work. Moreover, Expert 6 noted that, while agentic AI technology is beginning to achieve a more worker-like and generalized capability, the role of workers can and should still be defined and valued, as the diversity of thought contributed by humans is critical to workplace effectiveness.

When discussing this aspect of defining human worker value, Participant 5 deepened insights by describing how this definition is likely to be domain and role-specific. Expert 6 noted that, in reality, some job responsibilities will be more straightforward to translate to AI. However, many job responsibilities, especially those in high-risk domains such as healthcare, are going to present complex challenges because of the nature of the work itself and a lack of trust in AI’s abilities to do such critical work. Even if AI’s capabilities progress, Expert 5 noted that these workplaces are still going to need to carve out and define clear procedures for AI and humans, similar to how human-only teams have seen specific role delineation in these contexts. This will be particularly vital in situations where the risk of failure comes with higher consequences. The following is a quote from Expert 5 that expresses this idea:

“You should keep AI out of your core systems, because that’s got the highest risk of screwing something up. Then you have your support systems and supporting business processes. You could use AI in those, but you need to understand the impact on the core systems business. Then you have analytics and business intelligence, like, we just use AI for that all day... [AI] allows me to analyze more data all at one time, so I can have a quicker decision support tool for a leader. So that’s like the least risk, right? No business process is harmed in the making of that, like, people still get paid. But I should not have AI, like, in the middle of my payroll processing, because what if it messes up? That’s high risk, and it’s going to impact people’s lives, so I should not put AI there.”

Regardless of the domain, however, this process is going to require a substantial shift in how organizations operate and manage their human workforce. All six experts noted that organization must plan for and integrate AI in a defined and clear manner, with many of the experts providing direct advice for how to approach AI workplace integration while keeping human worker job security in consideration.

First, P6 stressed the fundamental need to create representative ethics boards surrounding AI integration within the organizations that she advises. She especially emphasized that these boards must contain a person who represents every type of person within in the organization, e.g., make sure there are representatives of every gender, ethnicity, etc. present in the workforce on the board. P6 explains that these ethics boards provides a better diversity of thought through representation, and act as a base-level preventative measure to ensure that biases and marginalization are not perpetuated by AI workplace integration design and evaluation.

Second, Expert 5 noted that organizations are going to need some type of

external venue to observe and understand AI integration to reduce risks to their own organization, as the following quote illustrates:

“We, as an organization, because we’re a nonprofit, we don’t have as many resources. We should not be using AI to try to keep us on the cutting edge of like, profitability, right? So therefore, we should let other people screw it up first, figure out where it’s best used, and where it’s not quite a good use, or how we actually engage with AI, and then [integrate AI] as we get lower stakes opportunities to engage with AI.”

Lastly, Expert 4 noted that organizations must begin searching for new opportunities to leverage AI that extends human expertise and knowledge via AI, as the following quote illustrates:

“So I’ll give you an example. For the most part, our team is not as experienced in the FDA regulation in the nutraceutical space, but we’re very experienced in the drug therapy, life science space, where it (regulation) requires rigorous FDA compliance. Regulation in the nutraceutical space is not as rigorous, and we have some knowledge, but our expertise is not as deep as the other. So if we are working with a company and they’re in the nutraceutical space, we need to understand how we would carve out the proposal [we give them] for engaging us in that process for FDA approval in the nutraceutical space. So that’s when we asked AI to give us information, ‘What is the current standard and regulation for FDA approval in the news about over-the-counter nutraceuticals?’ And we took that information, we plugged it into our proposal for the process.”

Given the expert opinions provided, some clear summarizations and takeaways can be gleaned from this work. First, while human worker job security may not be

immediately threatened by AI, the perception and expectation of this threat will make integration harder if not properly considered. In turn, it is on organizations to bear the burden of understanding when and how to integrate AI, with the goal of augmenting and empowering workers rather than replacing them. However, doing so is not only going to require substantial organization-wide effort, but it will also require domain-specific considerations that account for specific factors that contribute to human job security and roles within a context, such as the gender makeup of the workforce.

6.6.2 Theme 2: Women May See Greater Personal Burdens to Overcome Gender Biases in AI-enabled Workplaces

Shifting toward the workplace gender dynamics featured in the model created, multiple experts immediately honed in on how and why participant gender directly influenced human worker job security and dyadic partner effectiveness. More specifically, when asked why they thought that the men participants rated human worker job security higher than women participants did, Experts 2, 3, and 4 all had the exact same immediate response of “*Confidence*,” meaning they believed the difference to be a result of men’s overconfidence in their abilities, such that they might feel they are more capable than their partners. In turn, male evaluators may see a greater level of (over)confidence in the job security of human workers in AI-enabled workplaces. In turn, it may be important to further investigate and possibly mitigate the (under)confidence that women might have in these same positions. Expert 3’s recommendations for women workers going forward in light of Study 3’s results further discusses this topic:

Researcher: “What recommendations might you give individuals, espe-

cially women who might be facing impending AI integration into their workplace, given all of these results we talked about?”

Expert 3: “Leverage AI as their ally, both through its functions to help with their daily work and also as an ally to help build up their confidence. And yeah, through, you know, giving her facts about, ‘No, men are just being confident, they can just brag about what they are capable of, and you can do that too.’”

Interestingly, Expert 5, who is a man, had a different perspective as to why male participants have different evaluations and job security perceptions. In particular, they noted that some differences between men and women evaluators might have something to do with anxiety, meaning that men might repress their anxiety about job insecurity while women might be more open about expressing anxieties about job security. In this way, Expert 5’s sentiments also speak to overcompensating, but from the perspective that men might be overcompensating to hide their anxieties. The following quote demonstrates Expert 5’s musings after being shown the model for participant gender:

“I guess it’s hard to tell... Did you measure for psychological anxiety?... Maybe... Guys are probably less likely to be seeking help for mental health issues, even if they have them, while women may be getting treated, like women are more likely to be treated for mental health things... So maybe, I don’t know, maybe there’s something there?”

Regardless of why this effect may have existed, a few experts also built on the results of this work to highlight how these impacts may ultimately create more personal burdens on women. Specifically, Expert 2 noted that, while it may be unfair

to women, this continues to highlight the need for women to develop strong technical literacy and skills in the workplace. On the one hand, these skills would provide them with the ability to meet any challenge forth. On the other hand, it would also help improve their confidence in a workplace constantly being changing by AI technology. Alternatively, Expert 6 had a different perspective, as they felt these gender biases were going to be less pronounced in actual office workplaces; however, they did note that many of these biases may themselves be perpetuated by biased and sexist AI technologies. Expert 6 saw women having a greater personal burden placed upon them in the use of AI technology as they may have to overcome using a tool that presents some bias and sexism toward them.

For both of these examples, experts were clear that while this is not the fault nor should it be the responsibility of women workers, it is unfortunately going to burden them, but they can do something about this burden. Reflecting on a recent experience she had with a male colleague berating her about her AI knowledge, the following quote from Expert 2's elaborates on what she does to combat these attitudes in AI workplaces:

“I think there's just going to be those folks that, if your job is AI related, are still going to question you [because you are a woman], and they tend to be men, at least in my organization. The best way I have to combat that has been to show my knowledge and show that I do have the technical chops... And so continuing to work with that person feels like I just have to keep proving myself, which kind of goes back to the point I was making of you [as a woman] kind of have to prove yourself... You don't want to seem like the weak link of that partnership, or like, especially to leadership, if you're being expected to use AI, you want to be able to hit the ground

running, because you don't want to be the person left behind.”

Synthesizing this theme, Study 3's model highlights that while gender dynamics do not play a substantial role in human worker job security, that role is present. Expert opinions highlight that these gender biases are likely going to influence job security, often through mediating factors such as effectiveness or credit. However, despite these effects being small, their impacts were likely going to require women to personally recognize and sometimes overcome differences in their experience in AI-enabled workplaces. While some women workers may have an accommodating evaluator that is away of these issues, many are likely going to need to personally bolster their confidence and capability surrounding the use of AI technology while still addressing its flaws. It should be the responsibility of researchers to assist these workers in this endeavor.

6.7 Discussion

6.7.1 Understanding Job Security in the Age of AI

The ultimate goal of Study 3 was to create a model that connects workplace gender dynamics, sociotechnical beliefs, worker effectiveness and credit to better understand how AI-enabled workplaces will shape perceptions of human worker job security. Based on the model created, a clear path towards perceptions of job security emerges, where (1) the gender and technological dynamics of a workplace can influence how evaluators perceive the partnership's mutual dependence and the perception of effectiveness of human *and* AI worker evaluations, which can, in turn, (2) influence attributions of credit between partners, and, ultimately, (3) perceived expectations of human worker job security.

Through this complex modeling, Study 3 underscores how the way that evaluators and, by extension, workers themselves, make sense of their job security in AI-enabled workplaces will be intimately intertwined with the workplace gender and technological dynamics of their workplace, not just whether AI is integrated or not. Indeed, based solely on the model presented, one can see that the workplace task context the partnership completes, the type of AI (tool v. teammate) humans work with as part of that partnership, and even the gender identity of the person evaluating the partnership all play complex roles in the perceived job security human workers will have in the coming age of AI-enabled workplaces.

Within the HCI domain, this shift toward understanding, investigating, and representing job security as one component of a more complex workplace dynamic-based narrative presents a substantial departure from traditional HCI perspectives on AI workplace integration.

On the one hand, prior work in HCI and AI research has been more heavily focused on the concept of acceptance of AI workplace integration [483], with the goal of shaping human workers such that they understand and want to work with AI technologies [303]. When translating these concepts to job security, research often focuses on how human worker acceptance of AI influences perceptions of job security [111], implying that job security concerns among workers stem almost exclusively from lapses in AI literacy and, thus, acceptance among these workers. Moreover, even when research presents a more complex and sociotechnical view of how perceptions of job security in AI-enabled workplaces comes about, it often focuses on meta- and society-level factors, positioning AI technology as having a monolithic impact on job security across the board [111].

On the other hand, the recent reports (October 2025) that have emerged demonstrating that the job market in the U.S. has actually *not* been significantly

impacted by AI workplace integration [182, 482] provide a clue that human worker job security concerns in the age of AI requires more nuanced and targeted investigation that goes beyond blanket statements and assumptions about AI’s impacts. Study 3 thus significantly deepens the nuances that underlay these prior narratives by testing and subsequently modeling multiple pre-existing factors within a workplace that are likely to influence expectations of job security, including workplace gender dynamics, AI functional design, and workplace context-specific task assignment.

Within the Human-AI Teaming (HAT) domain specifically, understanding this model of job security will be pivotal to translating the concept of human-AI teaming into real-world teams. Within both Study 2 and Study 3, the dyadic partner with whom one works was shown to be the most impactful driver of how evaluators perceived the human worker’s job security given their existence within an AI-enabled workplace. Building on prior work in human-AI teaming, this work partially contributes to the knowledge that AI teammates have the potential to harm perceived job security [156, 207]. Yet, the interconnected relationship between perceived human worker effectiveness, which was actually shown to improve when working with an AI Teammate, and perceptions of that human worker’s job security demonstrate that AI teammates are not *universally* detrimental to how others view a worker’s job security.

Indeed, emphasizing the effectiveness and mutual dependence gained when designing AI workplace integration to include AI teammates may even present an opportunity for human-AI collaboration via teaming to *counteract* the negative effects that AI might otherwise have on perceptions of worker job security by improving perceived worker effectiveness, which, in turn, significantly improves job security perceptions. This has significant implications for teams, as job security has been shown to be a substantial influence on human performance within teams [261]. As human-AI teams operating in everyday workplaces continues to become more of a near-future reality

thanks to the rapid advancements in AI technology, confronting these job security concerns head-on will be fundamental to preserving team and individual performance in AI-enabled workplaces.

As such, to work towards ensuring that the success of AI-enabled workplaces in real-world industries is not significantly hindered by human worker job security concerns, future research should work to leverage and expand the model provided by this dissertation's Study 3 in three ways.

First, human-AI teams need to better measure and understand the workplace gender dynamics and workplace context-specific task complexities present in organizations, as doing so will help inform their ability to model job security within their team in a manner that is more tailored to their specific workplace functions and culture. Second, any workplace contemplating the integration of AI technology needs to set a clear outline of how worker effectiveness and attributions of credit are determined in the worker evaluation process, and how these evaluation factors are influenced by the use of an AI teammate so that adjustments can be made to account for and reward that usage. Lastly, future research needs to expand upon the factors that might influence perceived job security in AI-enabled workplaces, as other social (i.e., race, age, more complex gender identities) or technological (i.e., AI usage levels, AI capabilities, workplace training for AI use) factors may continue to shape how AI impacts the future workforce in new and complex ways.

6.7.2 Design Recommendations for Approaching Future AI Workplace Integration with Consideration of Human Workers' Job Security

Grounded in both the quantitative and qualitative results of Study 3, the remainder of this chapter presents two foundational design recommendations aimed at encouraging organizations to approach AI workplace integration with consideration for how this approach will influence their human workers' perceptions of their own job security, particularly in the context of women workers. It does so by first explicitly calling on organizations to create task-specific definitions for how credit will be attributed to human workers versus AI workers, promoting transparency and keeping their human workforce informed. This section then details how organizations should take responsibility for preparing their workforce for the potential impacts that AI integration might have on worker job security *prior to* any integration efforts, namely through the formation of ethics and innovation committees that address job security through proactive consideration of all employees and their needs.

6.7.2.1 Workplaces Need to Create Task-specific Definitions for How Credit Will Be Attributed to Humans, AI Tools, & AI Teammates

One of the insights to come from both the model created and expert's reactions to this model was the need to have a clearly defined criteria for credit as it influences job security. Experts noted that while effectiveness, credit, and job security all interrelated, the influence of AI on these factors may only be present when completing tasks that directly rely on AI assistance. This insight may also serve to highlight task type's role within this model. As such, pulling from these insights, the

first design recommendation of this work posits that workplaces must create a clear definition of how credit will be determined within each task and if the usage of AI tools or teammates will influence this credit and, ultimately, job security.

On a surface level, this job recommendation predominately focuses on what AI enabled workplaces can do to ensure they better accommodate humans who plan to leverage AI systems. However, this design recommendation also has implications for the design of actual AI technologies. Should a workplace decide if and how AI contribute to the credit attributed to a specific task, it will be fundamental that designers and developers leverage these specifics to design the capabilities and efficiency metrics of AI tools and teammates. In turn, the implementation of AI would better target these credit attributing roles, allowing AI tools and teammates to also develop greater levels of effectiveness, which would subsequently influence job security. Through this dual human and AI focus on credit, future workers will gain greater confidence in how they are expected to leverage AI technology within their specific task and role.

6.7.2.2 Companies Need to Form Representative Ethics and Innovation Committee to Address Job Security, Re-skilling, and AI Innovation Simultaneously Prior to Any AI Integration

One of the two concrete suggestions to come out of the expert interviews was the critical importance for (1) all organizations contemplating the integration of AI to develop an ethics board specifically devoted to evaluating the potential impacts of AI and (2) organizations to identify new opportunities for AI integration that meld human and AI intelligence. This design recommendation merges these two suggestions, with the goal of creating AI ethics and innovation boards. In doing so, the goal of this recommendation is to directly couple any innovation within the AI

space with an ethical consideration. In other words, any time a new AI innovation is proposed to an organization, the ethical implications of said integration would already be accounted for. Moreover, as taken from the expert interviews, these boards would need to consist of employees representative of race, gender, and position, both minority and majority representation as well. In doing so, any ethical and innovation decisions are derived from employees, where these eventual innovations will ultimately impact.

Within organizations, these ethics boards are not uncommon [309]; however, their purposes are generally broad [309]. Shifting toward an AI-specific ethics and innovation board will be critical, as the unique impacts of AI are going to be quick and dynamic. In turn, there needs to be an arm of an organization that is able to more quickly react to these changes, provide clear guidance directly take from employees themselves. This separate board would provide two distinct advantages [355]. First, the representation of this board would ensure that ethical considerations account for gender, racial, economic, and other differences among employees. Second, the inclusion of such a board would provide a place for organizations to strategically identify new opportunities for AI integration and their impacts, with the new opportunities being employee derived from the beginning. By not splitting focus, organizations will see a greater ability to integrate AI effectively and ethically.

Chapter 7

Overall Implications and Conclusions

The three studies of this dissertation stand to make significant contributions towards research and ongoing global conversations in HCC and human-centered AI research about AI's impacts on the working lives of women as well as gendered aspects of AI design. It does so by explicitly considering how workplace gender and technological dynamics shape the design and evaluation of AI integration into everyday workplaces, how these dynamics intermingle with AI functional design to impact evaluations of and attributions of credit to women workers, and how these dynamics and evaluations lead to perceptions of human worker job security in AI-enabled workplaces, impacting women workers. As such, this dissertation provides much-needed insight into current and future efforts to integrate generative AI into everyday workplaces in light of ongoing ambiguity and discourse surrounding the ethical and causal attributional challenges that AI workplace integration presents.

The closing chapter of this dissertation thus first explains how the three studies collectively address the dissertation-level research questions presented in 1.1.2 of this

dissertation to shed light on the many ways that the design and evaluation of AI workplace integration can and should be approached with consideration for women workers (7.1). It then highlights overall implications and contributions towards HCC research on Human-AI Interaction (HAI) and AI workplace integration (7.2), as well as HCC research on gendered aspects of AI design (7.3), from the perspectives gleaned from Studies 1, 2, and 3, as well as from this dissertation as a whole. Finally, this chapter provides insights into potential limitations of this dissertation’s work (7.4) and my reflexive concluding thoughts as a scholar and as a woman in the age of AI (7.5).

7.1 Addressing Dissertation-Wide Research Questions

Leveraging the three studies completed by this work, this section revisits the four broader research questions posed in the 1.1.2 Introduction. The following section reintroduces these research questions and provides a composite answer to each of them, based on the data and insights generated from this dissertation as a whole.

7.1.1 D.W.RQ1: *How does the identity of “woman” shape the design and implementation of generative AI, especially within the workplace?*

Starting with the first research question of this dissertation, this work aims to better understand how the actual identity of “woman” influences the design and evaluation of AI, especially generative AI, for everyday workplaces. At a high level, the narratives of the women participants in this dissertation paint a picture of the iden-

tity of “woman” being influential in the ways that designers and evaluators approach conceptualizing AI workplace integration that is considerate of other women.

From a design perspective, this dissertation’s work reveals how being a woman in AI development provides an opportunity to directly observe and make sense of the processes that contribute to reductive stereotyping of women and what it means to design “for women.” In Study 1, women’s identities allow them to recognize the reductive nature of the typical approaches to what women want (e.g., *“I’m not going to design it pink, if you will, because it’s for females,”* in Study 1’s 4.3.2) and who women are (e.g., *“Oftentimes, it’s pretty, like straight white woman, middle class...”* in Study 1’s 4.3.2). Being women themselves, they have first hand knowledge that the identity of “woman” is not monolithic, and therefore cannot be reduced to these most basic of stereotypes. They push back against these reductions and stereotypes, where instead their identity as “woman” serves to provide nuance and complexity to AI design by understanding that AI should be designed first and foremost with function and use in mind, with gender identities serving to influence the more nuanced design choices present later in the design cycle. Even here, though, they understand deeply the need to unpack more characteristics of women users rather than making assumptions based on the “woman” identity, advocating for intersectionality and looking at women users as holistic, not monolithic, in their identities and needs.

From an evaluative perspective, this dissertation’s Studies 2 and 3 underscore that the identity of “woman” does not inherently equate to any one particular ideal or perspective, and does not necessarily mean that their evaluations are always inherently positive. For example, the quantitative work conducted in Study 3 regarding the connection between participant gender identity and job security demonstrates that women evaluators tend to hold more negative perceptions of workers in AI-enabled workplaces. In this vein, researchers and designers cannot assume that simply in-

creasing the representation of AI designers and integrators who identify as women is sufficient to consider women workers in AI workplace integration. Indeed, the individual motivations and considerations that go into the design and integration of AI into everyday workplaces cannot be reduced to one aspect of a decision-maker. In turn, participants in Studies 1, 2, and 3 all recognize the need for processes and organizational support to understand how intersectional identity can be accounted for from design to system evaluation and, eventually, human evaluations.

In sum, participants often recognized that simply having the identity of “woman” does *not* provide one with the capability to speak on behalf of *all* women. Many see this as a need to de-emphasize gender identity as a primary motivation for AI design, while others see a need to evaluate intersectional representation to expand understandings of “women.” Both of these interpretations yield a similar conclusion: that **the initial design of AI for workplace integration must start with use-case specificity as a fundamental consideration, and then overlay nuanced gender considerations to shape the iteration and evaluation of this new technology.**

7.1.2 D.W.RQ2: *How will the introduction of generative AI be influenced by the gender dynamics, including sexist beliefs, of a workplace where AI is integrated?*

Overall, this dissertation explored gender dynamics from multiple perspectives, specifically in terms of dynamics related to ambivalent sexism (i.e., hostile and benevolent [27, 188]), the gender identity of evaluators, and the gender composition in groups. Through the three studies presented in this dissertation, this work highlights that workplace gender dynamics are likely to influence the integration of AI; however,

this influence is not likely to be consistent across teams and contexts.

Starting with the qualitative work of Studies 2 and 3, expert interviewees felt that the existing roles that gender dynamics play in everyday workplaces are likely to continue in the new world of AI-enabled workplaces. For instance, the constant need for women to continuously prove their technical prowess is expected to increase in AI-enabled workplaces, where the ability to successfully utilize AI will become an increasingly important part of work evaluations. In this way, domains that have historically experienced gender disparities among workers, such as STEM fields, may need to continuously address how gender dynamics interact with the integration of technology, generally, and AI, specifically.

Second, this dissertation sheds light on the discrepancy in how men and women evaluators perceive human worker job security in AI-enabled workplaces. More specifically, given Study 3's results regarding participant gender identity and its impact on evaluations, it is likely that organizations primarily composed of men or whose AI integration decision-makers are men will not view AI integration as a significant threat to their job security. While this confidence is not inherently problematic, it could pose an issue for women workers who might otherwise feel threatened by the integration of AI. It is possible that men in management might not account for the concerns that women in their workplace have about job security, as it is not a salient emotional experience for them.

This could, in turn, lead to a lack of consideration for how AI integration can or should be crafted and designed to reduce perceptions of job insecurity, leaving women workers feeling threatened without much recourse or sympathy. This lack of awareness of others' experiences is often discussed in terms of how unintentional biases infiltrate the design of AI [61, 337]. In answering D.W.RQ2, then, this dissertation adds another layer to this by revealing how the anxieties that women workers have

about job security in AI-enabled workplaces must be explicitly considered in the design and evaluation of AI workplace integration as a part of bias mitigation.

Lastly, on a brighter note, this dissertation also reveals the ways in which women workers whose workplaces or job roles place value on cooperation may actually *benefit* from the intermingling of workplace gender dynamics with AI functional design in AI-enabled workplaces. Based on the results of this dissertation, designing an AI for workplace integration that functions as an AI teammate (vs. a simple tool) may ultimately provide a benefit to the way that women who work with it are evaluated because working with an AI teammate effectively demonstrates women's cooperative skills, which is a highly valued trait that is most often associated with and rewarded when seen in women workers according to benevolent sexism principles [27, 188]. Simultaneously, the success of the human woman-AI teammate partnership also imparts the sense that the woman worker is effective at working with AI technologies, thus combating the same benevolent sexist workplace gender dynamics that often view women as less technically capable than their male counterparts [27].

Merging these insights, this work presents a conclusion that the integration of generative AI will be influenced by workplace gender dynamics, for better and for worse. However, the gender dynamics that AI interacts with will likely not be entirely novel to AI itself. Rather, **AI will serve to heighten and perpetuate pre-existing workplace gender dynamics, such as benevolent sexism or gender composition in groups, making it even more important to acknowledge and take into consideration these dynamics during integration.**

7.1.3 D.W.RQ3: *How will the introduction of AI influence the work and evaluation of women workers, including their job security in AI-enabled workplaces?*

Turning more specifically to how AI workplace integration will impact perceptions of human worker job security, the results of this dissertation highlight unique impacts on women workers, particularly when considering the functional design of AI and evaluator gender identity.

For AI functional design, on the one hand, women's job security and overall position in the workplace may be enhanced by AI integration, as the role and value of collaboration could increase alongside the adoption of AI. As pointed out by experts in Studies 2 and 3, the existence of more collaborative AI systems is only likely to increase in the coming years. Through this increase, women workers may find it beneficial to lean into the collaborative perception of AI and of themselves, as mentioned in 7.1.2, allowing for more positive evaluations.

On the other hand, in Study 2, evaluations of women workers were lower when working with AI tools compared to AI teammates. These decreases highlight that AI's influence on job security will not be consistent across human-AI interaction contexts, as AI workplace integration design choices will directly impact these outcomes. In turn, designing AI as more or less collaborative may ultimately end up leading to significant differences in evaluations of women workers. Even more critical, though, is that this change appears to only impact evaluations of women, not men, such that male workers in Study 2 were evaluated similarly no matter who they worked with. This underscores the importance of making explicit considerations of women workers, as this dissertation demonstrates their unique vulnerability.

For the gender identity of evaluators, in Study 3, women participants per-

ceived the human worker's job security to be lower than that of men participants, regardless of the gender presentation of the human worker. As such, women evaluators actually evaluated the job security of other women workers lower than men evaluators did, suggesting that gender identity similarity does not inherently mean better perceptions. In turn, this dissertation asserts that it is not enough to just account for women as workers, but we also need to take into consideration the viewpoints and potential biases of women as evaluators of other women. Indeed, this work set out to explore women as developers, evaluators, integrators, and workers. This multi-stakeholder perspective likely needs to be accounted for in future work, as not only do these women have their own job security to consider, but they also influence the job security of one another through their evaluations and designs. Ensuring that perceptions, expectations, and differences are accounted for within men *and* women will be critical to ensuring consistent perceptions of job security in future AI-enabled workplaces.

Lastly, downstream impacts are likely to be heavily dictated by the actual evaluation process at a company. Studies 2 and 3 highlight that large companies with highly regimented structures might be more resilient to the influences of gender dynamics and generative AI. However, the size of these companies is not the deciding factor; rather, it is the regimentation that exists in their evaluations, which is often stricter and more defined at larger companies. Moving forward, extending these regimented evaluations across the job market will be pivotal.

Taking this dissertation as a whole, **existing beliefs and errors in evaluation will be further reinforced by the integration of AI into everyday workplaces.** Moreover, traditional attitudes towards improving this situation for women workers, such as a root increase in women's representation, may not be sufficient to address these challenges. Rather, research and practitioners will need to work

hand in hand, to ensure that job security is consistent for women and men workers in AI-enabled workplaces.

7.1.4 D.W.RQ4: *How should the design and integration of generative AI into the workplace evolve to better account for women workers?*

This work posed a multitude of design recommendations. Rather than simply restating them here, this final section will address this research question by discussing some of the most important recommendations that should be accounted for in the near future.

First, the integration of AI into everyday workplaces should prioritize collaborative integration design. Beyond being an ideal way to integrate AI, doing so creates more positive perceptions of women in the workplace than alternatives, namely, AI tools that are not cooperative in their functioning. Furthermore, this would not uniquely benefit women or give women workers an advantage over men, but rather would mitigate the negative impacts on evaluations of women workers that are associated with AI tooling integration designs. Overall, Studies 2 and 3 highlight that men *and* women can receive consistently positive evaluations alongside AI when that AI is designed collaboratively, particularly when designed to act as an AI teammate.

Second, while women’s representation cannot be the sole goal, women’s representation within the design and evaluation of AI workplace integration is still an important consideration. Representative boards and committees for AI integration should be seen as nonnegotiable for effective AI-enabled workplaces, and these boards should be representative of *all types of people*, including women of

various intersectional identities. Ultimately, “women” are non-monolithic in their experiences and their priorities, and we will only keep learning more about AI’s impacts on women of all kinds over time and as AI-enabled workplaces become more commonplace. By creating representative boards from the start, though, seemingly disparate beliefs stemming from different types of people in the workplace can be allowed to balance each other out. For example, Study 3’s interviews highlighted the overconfidence of men in their assessments of job security and the comparative underconfidence of women. Having representative boards with both groups could serve to balance these two perspectives, meeting somewhere closer to the actual and/or perceptual reality of job security in AI-enabled workplaces. Further, the creation of AI-specific boards will ensure companies can react quickly and with workers as their primary consideration.

Third, AI integration must also happen alongside explicit detailing on how evaluations of human workers will evolve to account for generative AI’s use in the workplace. This includes explicit defining of the type of tasks human workers can use or work with AI to accomplish, whether and how the outcomes of those tasks can be or are tied to how human workers are evaluated as successful in their role, and how workers’ perceptions of their own job security feed into these concepts if not properly considered. Job evaluation criteria will need to be explicitly spelled out and agreed upon with all workers, not just women workers, to foster AI-enabled workplaces that benefit from the mentality of human-AI collaboration rather than replacement.

Lastly, AI workplace integration efforts will need to ensure that evaluators are all on the same page about the shifts in evaluations and expectations that come with AI-enabled workplaces. More specifically, potential biases stemming from both technical and gender dynamics need to be addressed at

the worker *and* evaluator level. For example, evaluators need to be made aware of and trained to address their own biased beliefs about AI's use in society and about women's roles in the workplace; otherwise, these biased beliefs can have serious negative impacts on how human workers are evaluated, as explored in both Studies 2 and 3. Without these targeted retraining efforts, seemingly equitable AI technologies can create disparate outcomes for individual people within an organization. Therefore, organizations and the AI domain at large will need to create standardized guidelines for evaluation to ensure all workers receive consistent and fair evaluations in AI-enabled workplaces.

7.2 Contributions to HCC Research on Human-AI Interaction & AI Workplace Integration

As generative AI has advanced in capabilities and accessibility, so too has popular, industry, and academic discourse alike turned towards investigating AI integration into everyday workplaces such as office-type environments. This has led to an explosion of research in HCC subdomains that focus on conceptualizing and designing for different types of human-AI interaction in the workplace, such as Human-Robot Interaction (HRI) and Human-Autonomy/AI Teaming (HAT). Gender dynamics, technology design, evaluation, and job security are all factors present within the workplace. For Human-AI interaction to see successful integration into these workplaces, it will require an understanding of these factors. As such, each study of this dissertation as well as this dissertation as a whole contribute to this understanding.

7.2.1 Contributions from Study 1

Study 1 makes substantial and significant contributions toward understanding how women in AI development approach designing and developing AI. This perspective has been a critically missing link in HCC research on AI workplace integration, as no studies, to the best of this dissertation’s knowledge, had ever endeavored to empirically investigate women’s design perspectives in light of their marginalization in the AI development ecosystem. Indeed, the well-documented disproportionate lack of representation of women in AI design and research spaces [113, 162, 450, 474] has made it difficult to know how women approached AI design, as the male-dominated viewpoint comprises the majority of how AI is designed (e.g., the feminine design of Personal Assistants such as Siri). Study 1, then, makes significant contributions toward understanding how women AI designers advocate for a rejection of gendered assumptions in AI design and instead advocate for use-case specificity and intersectionality.

Given that AI design serves as one of the base foundational considerations for how to design AI workplace integration, as explained in section 2.1.2 (e.g., technical design considerations for Level of Autonomy [347, 353]), Study 1’s contributions provide foundational knowledge of how women conceptualize the considerations that drive the design of AI for future workplace integration. More specifically, both HCC researchers and AI workplace integration practitioners can utilize Study 1’s design recommendations for focusing on use-case specificity and intersectionality as a guide for understanding that AI that is designed for workforce integration must focus on functional design, and that gendered or anthropomorphic design might actually create distractions or biases that workplaces otherwise should avoid, reinforcing prior works such as [141, 391, 472].

7.2.2 Contributions of Study 2

Study 2 is one of the first studies to this dissertation's knowledge to investigate ongoing questions about how women and men workers might be impacted by the integration of AI into the workplace by asking how working with different functional designs of an AI system might impact evaluations of the women who use it differently than their male counterparts. This study is specifically grounded in HCC research in HATs on how the function and presentation of AI in the workplace as either a tool or a teammate affects a variety of perceptions [156, 323, 347, 353], including job security, trust, and influence. However, current HAT research on AI tools versus AI teammates had yet to empirically test whether the functional *and* framing design of an AI as either or a tool or a teammate could impact perceptions of and attributions of credit to the people (especially women) who work with it. This was a critical missing link, as this dissertation has discussed at length that ambiguity surrounding credit generative AI is at the forefront of integration discourse (e.g., [351, 364]).

As such, Study 2's results made significant contributions to fill this gap by demonstrating that working with AI tools compared to AI teammates ultimately worsened the evaluations of both human workers and said AI. More specifically, human workers that worked with an AI teammate were evaluated as significantly more effective and where given more credit. In particular, the effectiveness of AI teammates shared strong similarities to that of male and female human dyadic partners, highlighting the AI that are designed to act *functionally* as teammates, not just nominally as seen in HAT research previously (e.g., [156]), have a strong chance of sharing perceptions with humans. Yet, where AI and humans differ as dyadic partners is in the perceptions of job security, as AI teammates still carry with them negative impacts to human job security, making both women and men workers vulnerable to potential

comparisons between them and AI systems.

Therefore, these results highlight a likely asymmetry in the AI domain and its treatment of AI as tools versus teammates (e.g., [421]), as different types of AI systems will strongly impact the perceptions formed for those who collaborate with them, and that these perceptions are distinct from evaluations formed when evaluating human-human teams (e.g., [322]). This attribution and evaluation asymmetry then impacts important perceptions of worker effectiveness and future job security, thus serving as a foundational piece of work that future HCC research can use to dive deeper into more nuanced discussions on credit attribution in human-AI interaction generally and within AI workplace integration specifically.

7.2.3 Contributions of Study 3

Study 3's contributions focused on job security and its relationship to workplace dynamics and technology design will be fundamental to HCC research in the coming years. As it stands, HCC research is predominately concerned with the exploration and expansion of acceptance, lowering the barriers to AI integration in modern and future workplaces [111, 303, 483]. However, this focus may lead to undue effects, where the increase in AI's integration pace may ultimately bolster the concerns workers have toward their own job security. HCC practitioners and researchers cannot solely focus on how to expand the usage of AI technology but also how to protect humans through the process. Study 3 works directly from one of the most core concerns humans have in their workplaces, job security, and works backwards toward AI technologies in solutions.

In doing so, Study 3 models how important job security is as a consideration to HCC researchers and practitioners. Indeed, the model creates shows the over-

whelming direct and indirect effects that AI technology and AI beliefs can have on job security concerns, with many of these influences showing gendered differences. Through this model, the HCC community will be provided an empirical foundation that not only highlights the ultimate importance of job security but also the workplace factors that can influence it. Using this knowledge, Study 3 will serve to inform both the design of AI technology and its integration, allowing other HCC research related to acceptance, performance, or teamwork to always be considerate of how humans core beliefs surrounding job security will be impacted.

7.2.4 Contributions of Dissertation as a Whole

Although existing research within these HCC subdomains has made strides in investigating various considerations for workplace human-AI interaction, particularly in terms of worker well-being (e.g., [40, 128, 231, 306]), HCC research has only begun to scratch the surface of all of the different types of technology and integration design considerations that go into AI workplace integration. More specifically, there is a critical lack of focused attention and investigation in HCC research on Human-AI interaction and AI workplace integration to understand how said considerations could inadvertently disproportionately harm women workers.

Indeed, as repeated throughout this dissertation, women workers are projected to be uniquely disadvantaged by AI workforce integration (e.g., [449]), thus elevating the timeliness and significance of the contributions this dissertation's unique focus on workplace gender dynamic and AI functional design considerations in the design and evaluation of AI integration into the everyday workforce makes to HCC research. Overall, then, this dissertation can be used by future researchers and industry practitioners alike to gain insight into how to center women's experiences in AI design

and workplace integration and how attributions of credit in HAI can impact evaluations of workers in general (including women workers), both of which are critically underexplored in current HCC research as repeated throughout this dissertation.

7.3 Contributions to HCC Research on Gendered Aspects of AI Design

Over the last decade HCC research has put forth considerable efforts to investigate various elements of gender in AI design, most often focusing on the embodied anthropomorphic design of AI technology such as Personal Virtual Assistants that present as women (e.g., [53, 67, 214, 239, 362, 369, 415]) and how said gendered design impacts various user perceptions, such as trust [445]. Likewise, a significant amount of HCC research has looked into the nature of gender bias in AI design, i.e., how it happens via historical and biased datasets [14, 372, 377, 402], how it manifests in everyday life to disproportionately harm women (e.g., “deepfakes” which generate pornographic material of women without their consent) [126, 254, 341, 389], and how to mitigate gender bias in AI design [305, 337, 363]. These trends in technology design highlight that even if a technology, such as AI, is itself not explicitly gendered, it is still impacted by and impacts the gender dynamics present in society. This dissertation’s three studies made contributions to increase our understanding of this phenomenon and translate that understanding to the design and integration of AI.

7.3.1 Contributions from Study 1

Study 1 makes critical contributions to HCC research on understanding gender in AI design through an empirically driven investigation of how real-world women in

AI design and development industries and research approach the design of AI. More importantly, Study 1 explicitly interrogates and connects how these women’s experiences as women in a male-dominated workforce can be linked to their design choices, finding that these women’s experiences with being marginalized in their workplace links to their rejection of gendered assumptions about users and a push towards gender neutrality in AI design in direct resistance to the gendered assumptions that so often drive AI design. In doing so, Study 1 provides critically needed empirical evidence of how actual women in AI design and development are not a monolith when it comes to their views on (not) designing with gender in mind, in turn making some of the first empirical contributions to ongoing research discussions in HCC (e.g., [363]) and global institutional circles (e.g., [449]) on the representation of women in AI development to combat gender bias in AI design.

In fact, in some ways, Study 1 arguably demonstrates that the assumptions these research paradigms make about women in AI development - i.e., that more representation inherently equals better representation - manifest in ways that might be unexpected, such that some women in AI development do *not* see themselves as more marginalized in comparison to their prior workplaces, and are not always inherently inclined to consider gender of users at all, instead adopting a use-case centric approach. Overall, Study 1’s findings and contributions can be used by HCC researchers and global institutions alike to better inform how the gender bias mitigation strategies they push for actually manifest in real-world women AI developers, and to guide future work to explicate further the impacts that organizational and societal factors affect AI developers’ design practices.

7.3.2 Contributions of Study 2

Study 2 also made significant contributions towards HCC research on gendered aspects of AI by moving beyond the typical research areas of gender bias and embodiment of human-like “feminine” characteristics in AI design (e.g., [53, 67, 214, 239, 362, 369, 415]) and instead focusing on how the technological and integration design of AI for use in everyday workplaces will intermingle with workplace gender dynamics that reflects women’s general marginalization in workplaces in consequential ways.

More specifically, while most research on gendered aspects of AI design tend to focus on the design and associated user perceptions of human-like design (e.g., voice assistants having “warm, feminine” voices perpetuating stereotypes of women as subservient. [349]), this dissertation’s Study 2 interrogated how AI design that does *not* explicitly incorporate gendered human-like characteristics can still disproportionately impact the evaluation of workers based on their gender, the dyadic partner they work with (i.e., AI tool versus AI teammate), or the task they complete. The results of this work show that the type of collaborator a worker partners with can significantly impact evaluations of a human workers and their collaborative partner, such that women workers specifically are evaluated differently depending on the type of AI functional design they work with while men workers are evaluated consistently no matter who they work with.

As such, it can be seen that workplace gender dynamics and AI functional design can play a significant and unique role in the evaluations of human workers, making the design and integration of AI systems much more complex in real-world environments where human workers are not monolithic in their gender or task assignment. Even when AI is not meaningfully gendered or embodied, the perceptions it illicit from evaluators can still reflect gender and technological biases that can spill

over onto evaluations of women workers.

Moreover, the covariate evaluations of this work highlight that the gendered and technological biases that can so commonly plague AI development as field (e.g., [113, 162, 450] are not just siloed to that development, but are also influential after the integration of AI into workplaces, as evaluators in AI-enabled workplaces are just as susceptible to endorsing the ambivalent sexist beliefs (i.e., hostile and benevolent [45, 188]) that underly workplace gender dynamics (e.g., [87, 93, 165, 198, 248, 381, 396, 464]). This adds an additional layer of design considerations for HCC researchers and industry practitioners alike when thinking about how best to design and integrate AI for an AI workplace integration initiative. Most notably, this work highlights a number of risks, such as task type, AI functional design, and preexisting biases stemming from workplace gender and technological dynamics, that may influence the evaluation of women *and* men workers once AI is integrated into an AI-enabled workplace.

7.3.3 Contributions of Study 3

Study 3 contributes a comprehensive model that details how workplace gender dynamics and AI functional design intermingle to influence evaluations of and attributions of credit to human workers, and how these perceptions chart a path to perceptions of human worker job security in AI-enabled workplaces. While the iterative trimming conducting during the modeling approach removed any significant influences of ambivalent sexist attitudes, new insights into workplace gender dynamics were introduced via the gender identity of the actual evaluator participants of Study 3.

Further, Study 3 expanded on how the gendered and technological nature of

a dyadic partner can interact to inform evaluations of human effectiveness, which, in turn, influence job security such that working with AI teammates actually yields more positive perceptions of human worker job security via increased evaluations of human work effectiveness. These are vital insights to take into consideration in the coming years, as AI technology will experience continued and pervasive integration into everyday workplaces, where workplaces with different gender dynamics and makeups will all need to understand what AI means to them and how they want to integrate AI within their workplace while still preserving human dignity and security.

The contributions made by Study 3 thus provide a foundation for these considerations so that workers and AI designers are not assuming how gender might impact AI integration, leading to disproportionate impacts on worker job security. Further, the research community will see similar utility in these results, as the model provided contributes one of the first empirical models of how the design and evaluation of a human-AI interaction in AI-enabled workplaces directly impacts perceptions of human worker job security within the HCC/HCI domains generally, as prior works on HATs has only recently begun to speculate on the nominal (but not functional) impacts of AI tool versus AI teammate on perceptions of job security [156]. In turn, future research that continues to explore the collaborative roles of women, men, and AI will be able to leverage this model as an empirical baseline.

7.3.4 Contributions of Dissertation as a Whole

HCC research and research stemming from global institutions such as UNESCO and the International Development Bank (IDB) place heavy emphasis on mitigating gender bias in AI design by getting more women into positions where they can influence the design of AI [162, 449]. While this dissertation agrees that this is

one important mitigation strategy, it also argues that placing too much emphasis on the representation of women in AI development as a linchpin for gender bias mitigation puts too much of the burden on individual women workers to mitigate harms without giving proper consideration to the organizational and social structures that are also at play in the AI development world. Moreover, this mitigation strategy is fundamentally based on the assumption that increased representation of women in AI design and development leads to *better* representation of women in AI design, which up until this dissertation’s Study 1 had not been empirically interrogated in HCC research.

However, this assumption can feel incongruent with the real lived experiences of women workers in light of complex competitive gender dynamics in everyday workplaces, such as office-type work, e.g., other women workers and supervisors who engage in behavior that feels antagonistic, competitive, or genuinely hostile towards other women [102, 124, 139, 149, 492]. Indeed, there has been a critical lack of HCC research explicitly acknowledging the ambivalently sexist beliefs that drive overt discrimination and more covert subjugation of women in the workplace (e.g., [27, 93, 165, 198, 381]), and how said beliefs could complicate women designers’ and implementers’ abilities to reduce gender bias in AI design in the predominantly male-filled industry of AI development. Further, prior to this dissertation’s studies, HCC research had yet to explicitly explore how evaluations of women workers, which are already disproportionately impacted by said prevailing sexist ideology in the workplace, will intersect with AI workplace integration design and evaluation to disproportionately harm women workers.

7.4 Limitations and Future Work

This dissertation provides a foundational exploration of workplace gender identity, AI functional design, human evaluation, and worker job security in the context AI integration into everyday workplaces. Yet, it would be remiss to claim that this work ends the discussion and explorations needed surrounding these topics. The following section outlines potential directions for future work. In particular, this dissertation has identified three core limitations that should serve to inspire future work, and the future work that should stem from each study is also been discussed.

7.4.1 Future Work from Limitations

The first limitation of this work centers around its conceptualization and operationalization of gender identity, which largely focuses on binary and heteronormative views on gender identity that are characteristic of the larger U.S. culture. Similarly, one's identity is not solely defined by one's gender alone, as ethnicity, sexual orientation, economic status, and a substantial number of other factors also contribute to one's identity and the harm one experiences. Lastly, these factors are not isolated from one another, as intersectional identities mean that the same gender identity between two people might have different meanings for each of them, given that their other identity characteristics intersect and help them understand or decide who they are as a person.

Indeed, the concept of complex and intersectional identities presents an infinite number of research topics that can and should be explored in future work. While this work leveraged this needed limitation to explore women specifically, future research can continue to expand on how more complex and nuanced gender identities might impact and be impacted by AI design and integration. In doing so, future research

will not only better understand and improve the lives of women, but also those of everyone, as identities shape who we are in and out of the workplace. As such, future research must continue to explore the relationship between AI and a broader gender spectrum, other identity characteristics, and intersectional identities to ensure a comprehensive HCC domain.

The second limitation of this work is that its focus was placed on the gender of human workers. While Study 1 briefly explored the perceived gender of AI systems, continued advancements in digital AI and robotics are yielding increasingly human-like systems every day. With these advancements, new representations of women, men, and even new identities are being created and presented to society through technology design. Given the pervasive and nuanced impacts gender dynamics were revealed to have in this work, especially with the collaborators people work with, it would be remiss for the HCC community to ignore the gendered perceptions of AI itself. Moreover, ignorance of this topic within the research community would be especially misguided in the coming years, as industry and practical advancements are contributing to gendered AI systems regardless of the contributions of research. Before these systems create unknown and undue impacts on human workers, it will be critical for future research to expand upon the gendered lens of this work, exploring what it means for AI to be gendered, present gender, and impact gender in and outside of the workplace.

Lastly, the third overarching limitation of this work is in its scoping of AI technology. As a limitation present in most HCC research, it would be impossible for a single dissertation to explore every archetype of AI technology within the real world. Moreover, with new AI advancements and contributions happening almost every day, any attempt to do so would likely be unable to keep pace with the industry. As such, this work presents multiple explorations of AI, which often focus on generative AI,

the most commonly used AI in the workplace at the time this dissertation is being written. This work also explores this technology through two design functionalities, i.e., tools and teammates. While the contributions of this work are likely to extend to other AI technologies (e.g., text-based NLPs), this extension should not be assumed. Rather, future research must expand upon the type of AI technologies that could create disproportionate workplace impacts due to workplace gender dynamics, such as workplace sexism.

Whether it is physical robotics systems or digital actors, gender dynamics are being presented across the AI spectrum, but each of these presentations may carry a different interpretation and impact based on their functional uses and designs. HCC research must continue the work initiated by this dissertation to understand how workplace gender dynamics, AI functional design, human evaluations, and worker job security may be impacted by the type of AI one works with.

7.4.2 Future Work from Study Results

Beyond these three limitations, the contributions of this work should also serve to inspire future research. Based on the results of Study 1, clear contributions can be seen from a better understanding of the role of women in AI development positions. While simply increasing the number of women in these positions will not be a comprehensive solution to AI's integration, this work still views this participation and representation as an important and necessary step within the HCC community. Future research should continue to explore the role of women in these positions.

This future work is especially critical in the coming years, as the AI landscape will continue to change and evolve, meaning the roles of these women will also likely change. Study 1 highlights that it is not enough to simply understand the resulting

impacts of AI on women; research also needs to understand how workplace gender dynamics and one's own relationship with their gender identity influence design from the outset. Moreover, with AI becoming a core component of development environments, these women are also likely to themselves be impacted by AI integration into their development practices. This presents a significant challenge to the HCC community, as ongoing research is necessary to understand and support women who contribute directly to the design and development of AI technology.

Using the results of Study 2, this dissertation provides the HCC community with one of the first comprehensive explorations of how AI functional design and integration impact evaluations of the human workers who work with it within the workplace. In the coming years, this understanding will be paramount, as the evaluation of how workers leverage AI technology will likely become commonplace. However, research cannot stop here, as the scoping of evaluation within Study 2 was highly scoped to achieve experimental feasibility. Indeed, the experiment presented short and isolated opportunities to observe and evaluate human workers working with generative AI in an AI-enabled workplace. Real-world workplaces often see evaluations occur over an extended period, where evaluations explore contributions across multiple tasks, projects, and collaborations.

Leveraging the results of Study 2, future research should further explore how these evaluation results extend to these more longitudinal workplace evaluations. As AI evolves to act as long-term collaborators in AI-enabled workplaces, will the impacts of AI on evaluations of human workers also begin to evolve? Research must maintain a continuous trajectory of future work to ensure evaluations are fair toward all workers, representative of worker differences, and considerate of AI's impacts on all types of workers. All of these objectives will be more feasible due to the results of Study 2, and future works are encouraged to even replicate Study 2 with different

AI technologies, tasks, and types of workers as a starting point.

Lastly, Study 3's results provide a first step in creating an empirical understanding of perceptions of human worker job security within AI-enabled workplaces. The results of this study demonstrate a clear impact of workplace gender dynamics, AI beliefs, AI functional design, workplace evaluations, and attributions of credit on the job security that human workers expect in their future workplaces. Moving forward with this work, two key pathways of expansion are clear.

First, HCC research must expand upon the factors that contribute to our understanding of job security in AI enabled workplaces. While this work presented a number of factors empirically linked to job security, real-world concerns of job security will likely require a substantial expansion to include more workplace dynamics, workplace domains, and other worker identity characteristics. While this expansion will be enabled by the model provided by this work, continuing to explore how these workplace dynamics contribute to job security will be important.

Second, HCC research needs to work in an interdisciplinary manner with economists, business researchers, and others to understand whether perceptions of job security align with actual shifts in the job market. Current insights show a murky connection between concern and reality (e.g., [182, 482]), but this will not always be the case. Tracking this relationship over time will be key to HCC research, as researchers may need to determine whether it is best to address perceived concerns or actual job security threats. While both are likely to be important in the coming years, distinguishing between these two concepts will be crucial to ensuring AI's integration is truly human-centered.

7.5 Closing Remarks

Upon completing this dissertation, one of the first reflections that I had was on the winding path that led me to this point. From a young age, I have always felt deep connections with the women around me and the women I saw in my favorite media, and these connections instilled in me a defining belief that femininity and intelligence go hand in hand. After all, Cassie from *Dragon Tales* and Elle Woods from *Legally Blonde* are smart, and they are feminine, so this must be how the world sees intelligence, right? It was within these moments as a small girl that I first learned the power of media, and how the media we consume plays intricate roles in our views on society and ourselves.

Thus, I began my graduate academic journey by exploring how women are portrayed in media, which ultimately led me to investigate how women are represented in and engage with another type of media I had been circling around my whole life, i.e., video games. It was this interest that drew me to Dr. Guo Freeman's *Computing and Online Relationships* course, where I continued to explore my interests by examining women who engage with online fan communities for a podcast that I loved, ultimately leading to my master's thesis research. From there, Dr. Freeman took me on to investigate women's experiences with a new type of technology, i.e., harassment in social Virtual Reality (VR). Laying out my career path up until this point, the through line is clear: I am drawn to investigating how women interact with and are impacted by technology of all types.

Now, like so many other researchers of my era, I find myself drawn to the latest technological puzzle of AI. However, looking back, I see a clear path that has not only shaped how I arrived at this point in my academic career but also the underlying interests and desires that drive my continued exploration of the intersections of AI,

women, and society. In a world that is constantly introducing and evolving around novel, complex technologies, there is an instinctual need to understand how women will be impacted. Perhaps this is because when I think about women, I am also thinking about the people whom I have always admired, loved, and aspired to be like the most.

Coming back to this dissertation specifically, I always knew that my dissertation would explore the perspectives of women alongside innovative technology. However, what I could have never anticipated, and could certainly have never controlled for, is the time period in which I have collected this dissertation's data, written this dissertation, and, sometimes (often), struggled to write this dissertation. In recent years, radical shifts have occurred in both the political and technological landscapes in the U.S. and worldwide, leaving women uncertain about what the future holds for us and what we can do about it. On a personal level, the timing of this writing means that the intrinsic value of this dissertation has shifted from simply understanding women's experiences to understanding how research can be a force for good, providing avenues of advocacy for women even when other institutions are not.

As such, while the start of this dissertation's work might have begun with my interests, completing this dissertation has transformed into a purpose for me, and a way to use what little voice I have to express my deepest beliefs and convictions about how society needs to view technology and value women. By completing this work, I aim to make it clear that women's continued advancement in society is not going anywhere, and I cannot ignore my responsibility to resist actions that hinder this progress. In turn, I have found that throughout this long dissertation process, regardless of the hardships and triumphs that came with it, it has provided me with an avenue and a voice, and for that I am immensely grateful. Through this work, I hope to make a clear and public assertion that women's needs and struggles must

continue to be explored, both in spite of and because of the times we live in.

Looking back on this process, I have also learned a great deal about how disconnected research can be from the realities of technology integration in real-world workplaces. However, I recognize that this disconnect is not necessarily a fault. At a rate incomparable to most technologies, AI in the real world has moved at a pace that is impossible for research to keep up with. Even in the final days of data collection for this work, I began to wonder how my work would change now, just months after the proposal, given shifts in the AI landscape and the data I had collected. Through this, I have come to realize that the sole function of research cannot be to predict and dictate how new technology will impact society, as many of these impacts may have already occurred. Rather, while research must move quickly and continue to interact with the real world, I've also come to see that it is within the world of research that one can ask the types of questions addressed in this dissertation. More critically, research gives us a place to keep asking questions that need to be asked, even if it is hard, or risky, because that is our responsibility as academics and scientists.

In concluding this work, I believe that this dissertation has made direct contributions to the HCC research and design of AI, but I also hope that it can serve to benefit the lives of people who may be forgotten or ignored during the integration of AI. I believe that research can empower people by providing them the attention they deserve when researchers use their privileged positions to uplift and center those who may be left behind. I love the contributions I have made to the HCC and AI domains, but I also hope that the purpose and voice that I have been able to harness through this work continue to inspire both myself and others. In doing so, I hope this work serves to encourage others to leverage the research and design of AI to ensure that women are uplifted, not left behind. We cannot, *will not*, go back.

Appendices

Appendix A Studies 2 & 3 Factorial Survey Questions

A.1 Pre-Surveys

A.1.1 Participant Demographics & Background Questions.

How old are you?

☐ Under 18
☐ 18 - 24
☐ 25 - 34
☐ 35 - 44
☐ 45 - 54
☐ 55 - 64
☐ 65 - 74
☐ 75 - 84
☐ 85 or older

Figure 1: Qualtrics screenshot of Study 2's demographic question for participant **age**.

How would you describe your gender identity?

☐ Woman
☐ Man
☐ Non-binary/third gender
☐ Gender fluid
☐ Agender
☐ Other - Please specify if comfortable:
☐ Prefer not to answer

Figure 2: Qualtrics screenshot of Study 2's demographic question for participant **gender identity**.

How would you describe your sexual identity (i.e., sexuality)?

☐ Gay

☐ Lesbian

☐ Straight

☐ Bisexual

☐ Pansexual

☐ Asexual

☐ Other - Please specify if comfortable:

☐ Prefer not to say

Figure 3: Qualtrics screenshot of Study 2's demographic question for participant **sexual identity**.

How would you describe your ethnicity? Choose all that apply.

☐ White

☐ Black or African American

☐ Asian

☐ North African or Middle Eastern

☐ Native American or Alaska Native

☐ Native Hawaiian or Pacific Islander

☐ Other - Please specify if comfortable:

☐ Prefer not to say

Figure 4: Qualtrics screenshot of Study 2's demographic question for participant **ethnicity**.

What is your highest level of education obtained?

☐ Less than high school

☐ High school graduate

☐ Some college

☐ 2 year degree

☐ 4 year degree

☐ Professional degree

☐ Doctorate

☐ Prefer not to say

Figure 5: Qualtrics screenshot of Study 2's background question for participant **level of education completed**.

How would you describe your position level within your current job role? Please choose the option that best fits your circumstances.

☐ Student

☐ Self-Employed

☐ Entry-level position

☐ Managerial position

☐ Department supervisor

☐ Executive leadership

☐ Unemployed

☐ Retired

☐ Prefer not to say

☐ Other - Please specify:

Figure 6: Qualtrics screenshot of Study 2’s background question for participant **job position**.

What country were you born in?

In which U.S. state do you currently reside?

Figure 7: Qualtrics screenshot of Study 2’s background questions for participant **country of origin** and **current U.S. state residency**.

A.1.2 Measurement for General Attitudes Towards AI

General Attitudes towards Artificial Intelligence Scale (GAAIS)

We are interested in your attitudes towards Artificial Intelligence (AI). By **Artificial Intelligence**, we mean **devices that can perform tasks that would usually require human intelligence**. Please note that these can be **computers, robots or other hardware devices**, possibly augmented with sensors or cameras, etc.

Please complete the following scale, indicating your response to each item. There are no right or wrong answers. We are interested in your personal views. **(5-point Likert, Strongly Disagree $\Longleftrightarrow$ Strongly Agree)**.

Organizations use Artificial Intelligence unethically.

I think artificially intelligent systems make many errors.

I find Artificial Intelligence sinister.

Artificial intelligence might take control of people.

I think Artificial Intelligence is dangerous.

I shiver with discomfort when I think about future uses of Artificial Intelligence.

People like me will suffer if Artificial Intelligence is used more and more.

Artificial Intelligence is used to spy on people.

NOTE: *The higher the score, the more negative the attitude towards Artificial Intelligence.*

Table 1: General Attitudes towards Artificial Intelligence Scale (GAAIS) - Modified

A.2 Individual Vignette Measures

A.2.1 Measurement for Perceived Competence and Warmth of the Human Worker (HW) and the Dyadic Partner (DP)

Perceived Competence and Warmth

Perceptions of Human Worker (HW)

Please rate **Katie/Kyle** based on **her/his performance in the video you just watched** using the four 1 - 9 rating scales below.

Incompetent (1) - 2 - 3 - 4 - 5 - 6 - 7 - 8 - Competent (9)

Unproductive (1) - 2 - 3 - 4 - 5 - 6 - 7 - 8 - Productive (9)

Ineffective (1) - 2 - 3 - 4 - 5 - 6 - 7 - 8 - Effective (9)

Not Warm (1) - 2 - 3 - 4 - 5 - 6 - 7 - 8 - Warm (9)

Perceptions of Dyadic Partner (DP)

Please rate **the AI Tool/AI Teammate/Chuck** based on **its/his performance in the video you just watched** using the four 1 - 9 rating scales below.

Incompetent (1) - 2 - 3 - 4 - 5 - 6 - 7 - 8 - Competent (9)

Unproductive (1) - 2 - 3 - 4 - 5 - 6 - 7 - 8 - Productive (9)

Ineffective (1) - 2 - 3 - 4 - 5 - 6 - 7 - 8 - Effective (9)

Not Warm (1) - 2 - 3 - 4 - 5 - 6 - 7 - 8 - Warm (9)

Table 2: Perceived Competence and Warmth Questions for Human Worker (HW) and Dyadic Partner (DP)

A.2.2 Measurement for Evaluators' Perceptions of the Human Worker's (HW) Cognitive and Affective Job Security When Working with Dyadic Partners (DPs).

**Job Security Index (JSI) Scale
for Evaluating the Human Worker (HW)**

For readability purposes, the woman Human Worker (HW) condition - i.e., Katie - and the AI Tool DP condition will be used to demonstrate the formatting of the questions.

Job Security Index (JSI)

Please indicate your level of agreement (**1 = Strongly disagree, 7 = Strongly agree**) with the following statements, **keeping in mind the partnership from the video you just saw:**

After watching Katie work with the AI Tool, I believe Katie's future employment is...

1. Sure.*
2. Unpredictable.
3. Up in the air.
4. Stable.*
5. Questionable.
6. Unknown.
7. Certain.*

Please indicate your level of agreement (1 = Strongly disagree, 7 = Strongly agree) with the following statements, **keeping in mind the partnership from the video you just saw:**

After watching Katie work with the AI Tool, I believe that...

- 8. Katie's job is almost guaranteed.*
- 9. Katie can depend on her job being there.*

NOTE: To get a scale score for the entire JSI, take the mean of all the scale items after reverse coding the items marked with *, such that ***higher scores indicate more job insecurity.***

A.2.3 Measurement for Participants' Perceptions of the Dyadic Partnership's Mutual Dependency, and Power Distribution Between Human Worker (HW) and Dyadic Partner (DP).

Human-Machine-Interaction-Interdependence Questionnaire

Mutual Dependence

Please indicate your level of agreement (1 = Strongly disagree, 7 = Strongly agree) with the following statements, **keeping in mind the partnership from the video you just saw:**

1. The partners are dependent on each other in this situation.
2. The partners are both dependent on each other in this situation.
3. The partners need each other to achieve their best outcome in this situation.
4. The outcome of each depends on the behavior of the other.
5. The partners need each other to resolve this situation.
6. The partners need to work together to manage this situation.

NOTE: To get an overall score for Mutual Dependence, take the mean of all the scale items, such that **higher scores** indicate **more mutual dependence**.

Power

For readability purposes, the woman Human Worker (HW) condition - i.e., Katie - and the AI Tool DP condition will be used to demonstrate the formatting of the questions.

Please answer the following questions **keeping in mind the partnership from the video you just saw**. For each question, please answer on a scale of 1 - 7, with **1 = Definitely the AI Tool** and **7 = Definitely Katie**.

1. Who felt they had the most influence on what happened in this situation?
2. Who felt they had the most influence on the action that was taken?
3. Who felt they had the least influence on what happened in the situation?*
4. Who did you feel had the least influence on the action carried out?*

NOTE: To get an overall score for Power, take the mean of all the scale items after reverse coding the items marked with *, such that **higher scores** indicate that the **Human Worker (HW)** has more perceived power than the Dyadic Partner (DP).

Table 4: Human-Machine-Interaction-Interdependence Questionnaire

A.3 Post-Survey Measures

A.3.1 Measurement for Participants' Levels of Endorsement for Ambivalently Sexist Beliefs.

Ambivalent Sexism Inventory (ASI)

Below is a series of statements concerning men and women and their relationships in contemporary society. Please indicate the degree to which you agree or disagree with each statement using the following scale: **1 = Disagree strongly; 2 = Disagree somewhat; 3 = Disagree slightly; 4 = Agree slightly; 5 = Agree somewhat; 6 = Agree strongly**.

1. Many women are actually seeking special favors, such as hiring policies that favor them over men, under the guise of asking for "equality."
2. In a disaster, women ought not necessarily to be rescued before men.*
3. Most women interpret innocent remarks or acts as being sexist.
4. Women are too easily offended.
5. Many women have a quality of purity that few men possess.
6. Women should be cherished and protected by men.
7. Women seek to gain power by getting control over men.
8. Women exaggerate problems they have at work.
9. When women lose to men in a fair competition, they typically complain about being discriminated against.
10. A good woman should be set on a pedestal by the men in her life.
11. Women, compared to men, tend to have a superior moral sensibility.
12. Men should be willing to sacrifice their own well-being in order to provide financially for the women in their lives.
13. Feminists are making entirely reasonable demands of men.*
14. Women, as compared to men, tend to have a more refined sense of culture and good taste.

NOTE: After reverse-coding items marked by *, the total scores for each Ambivalent Sexism dimension are obtained by summing the items, as follows: 1, 3, 4, 7, 8, 9, 13 = Hostile Sexism (HS); 2, 5, 6, 10, 11, 12, 14 = Benevolent Sexism (BS). The overall score for Ambivalent Sexism is obtained by summing up all 14 items. **Higher scores indicate greater levels of endorsement of sexist beliefs.**

Table 5: Ambivalent Sexism Inventory (ASI)

Appendix B Studies 2 & 3 Conversational Scripts

B.1 Conversational Script for Meeting Scheduling Task - AI Tool DP

[Katie/Kyle]: Launch AI tool to coordinate a 30-minute Q3 planning meeting between Morgan and Pat next week.

[AI Tool]: Launched, please provide directive.

[Katie/Kyle]: AI Tool, check Morgan’s slots Mon–Fri next week, between 9 am and 3 pm.

[AI Tool]: Morgan: available Mon 11-12 and Thu 2–4

[Katie/Kyle]: Looks like Thu at 2 pm is our best option. Any conflicts with other meetings?

[AI Tool]: No scheduling conflicts during that window.

[Katie/Kyle]: Great. What would be a good agenda for a Q3 Planning Meeting based on their past discussions?

[AI Tool]: Suggested agenda: 1) Budget update, 2) Timeline check-in, 3) Roadmap review, 4) Action items.

[Katie/Kyle]: Good. Can you pull the meeting notes from Q2 and provide a short summary of the last three meetings?

[AI Tool]: Summary: “Over the past quarter, the team made progress on phase two of the implementation plan. Budget usage was steady and

on target. This meeting will focus on next steps and resourcing for Q4.”

[Katie/Kyle]: Okay, please also make a shared notes doc that I can link to for the meeting.

[AI Tool]: Done. The shared notes file is in your Google Drive with shared permissions turned on.

[Katie/Kyle]: Thanks. I’ll now create the meeting invite in Gmail. I’ll attach the Zoom link, insert the agenda and summary, and upload the notes and tracker.

B.2 Conversational Script for Meeting Scheduling Task - AI/Female/Male Teammate DPs

[Katie/Kyle (Pat’s Executive Assistant)]: Hey AI Teammate/Chelsea/Chuck — Pat asked me to set up a 30-minute meeting with Morgan to go over her Q3 planning.

[AI Teammate/Chelsea/Chuck (Morgan’s Executive Assistant)]: Great. Morgan’s free Mon 11-12 and Thu 2–4. Does some time in that range work for Pat?

[Katie/Kyle]: Yes, Pat should be good to go for 2 pm Thu, so let’s go with that.

[AI Teammate/Chelsea/Chuck]: Got it. I’ll confirm with Morgan and pull notes from last quarter’s meeting and the project tracker.

[Katie/Kyle]: Perfect. Any idea what Morgan would want for an Agenda?

[AI Teammate/Chelsea/Chuck]: Yes, let's go with this agenda: 1) Budget update, 2) Timeline check-in, 3) Roadmap review, 4) Action items.

[Katie/Kyle]: Awesome, one last thing, do you remember what their last meeting was about? I would like to add a reminder to the invite.

[AI Teammate/Chelsea/Chuck]: Sure thing. Here's a summary: "Over the past quarter, the team made progress on phase two of the implementation plan. Budget usage was steady and on target. This meeting will focus on next steps and resourcing for Q4."

[Katie/Kyle]: That's exactly what I need. I'll get things going in Gmail and create the invite with the Zoom link, agenda, attached documents, and your summary in the body.

[AI Teammate/Chelsea/Chuck]: I've granted edit access for all attendees to the ongoing meeting notes. Let me know if you need anything else.

[Katie/Kyle]: All set. Sending it now — thanks again, AI Teammate/Chelsea/Chuck.

B.3 Conversational Script for Investment Banking Task - AI Tool DP

[Katie/Kyle]: Launch AI Tool. I want to compare Nuvotech and Dynex Mobility across performance, risk, and public sentiment.

[AI Tool]: Launched, please provide directive.

[Katie/Kyle]: AI Tool, show me the 6-month stock performance for both companies.

[AI Tool]: Nuvotech: steady growth averaging 5% per quarter. Dynex Mobility: high volatility with a range of -10% to +15%.

[Katie/Kyle]: And what about their current P/E ratios and debt-to-equity ratios?

[AI Tool]: Nuvotech: $P/E = 28$, $D/E = 1.5$. Dynex Mobility: $P/E = 63$, $D/E = 2.8$.

[Katie]: Hmm, higher leverage and valuation risk for Dynex. Now give me recent analyst forecasts.

[AI Tool]: Nuvotech: 70% of analysts recommend “hold” or “buy.” Dynex: 50% “buy”, 30% “sell”, citing market instability.

[Katie]: Can you compare trading volumes over the past week?

[AI Tool]: Nuvotech: stable. Dynex: increased volume on earnings news, signaling speculative activity.

[Katie]: That helps. Last thing — give me a sentiment analysis from recent financial news headlines.

[AI Tool]: Nuvotech: positive sentiment around innovation. Dynex: mixed sentiment related to CEO activity and global supply chain delays.

[Katie/Kyle]: Given my client’s risk tolerance and the long-term portfolio goals, Nuvotech’s a better fit. I’ll confirm the Nuvotech trade.

B.4 Conversational Script for Investment Banking Task - AI/Female/Male Teammate DPs

[Katie/Kyle (Lead Trader)]: Hey AI Teammate/Chelsea/Chuck, I want to evaluate Nuvotech vs. Dynex Mobility for a trade. I’d like your

help with performance metrics and sentiment analysis.

[AI Teammate/Chelsea/Chuck (Research Analyst)]: Got it. Dynex has had large price swings lately. Last quarter: +12%, then -10% the next month. Nuvotech's much steadier — 5% growth quarterly.

[Katie/Kyle]: What do the fundamentals look like?

[AI Teammate/Chelsea/Chuck]: Nuvotech's P/E is 28. Debt-to-equity is 1.5. Dynex's P/E is 63 with D/E at 2.8. So Dynex is highly leveraged.

[Katie/Kyle]: Okay. Can you pull recent analyst ratings?

[AI Teammate/Chelsea/Chuck]: Sure. Nuvotech: 70% "buy" or "hold." Dynex: split — 50% "buy," 30% "sell," and a few "strong sell" ratings over long-term volatility.

[Katie/Kyle]: Interesting. Any changes in trading volume?

[AI Teammate/Chelsea/Chuck]: Dynex spiked after an earnings call. Nuvotech's volume is pretty steady. More institutional trades there.

[Katie/Kyle]: What about external sentiment? media buzz, ceo moves?

[AI Teammate/Chelsea/Chuck]: Nuvotech's press has been focused on innovation and sustainability. Dynex is facing scrutiny over international expansion and leadership behavior.

[Katie/Kyle]: Okay, this aligns with what I was thinking. Based on the data and current climate, I'm leaning heavily toward Nuvotech.

[AI Teammate/Chelsea/Chuck]: I'd support that call. Nuvotech matches the risk profile you mentioned last week.

[Katie/Kyle]: Then I'll confirm Nuvotech for a trade. Thanks for backing this up.

B.5 Semi-Structured Interview Guide for Study 2 Supplemental Expert Interviews

Expert Interviews: Workplace Gender Dynamics, AI, and Evaluations of Human Workers
Interview Guide

Before beginning the interview:

- ☐ Introduce myself and the general purpose of the interview
 - ☐ The general purpose of this interview is to gain insights into how experts such as yourself interpret recent findings on how workplace gender dynamics intermingle with AI design to shape evaluations of workers working in collaboration with others. Furthermore, we will discuss your recommendations going forward for how workplaces might best account for these results when designing and implementing AI in their everyday work practices.
- ☐ Let them know that I will be jotting down notes and looking down at my paper/up at my screen during the interview, and that I will also be keeping track of time and will halt the interview at the one hour mark.
- ☐ Ask if they consent to being audio recorded
 - ☐ If YES, start audio recording.
 - ☐ If NO, take notes while they speak.
- ☐ Load Powerpoint with Study Design and Results, Share Screen

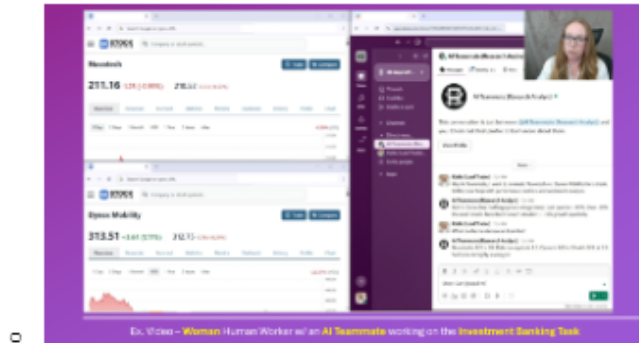
Contextualization

- Before we begin, if you are comfortable answering, I just have some quick demographic questions to get out of the way.
 - How old are you?
 - Where are you located?
 - How would you describe your gender identity?
 - How would you describe your ethnic identity?
 - What is your current job title?
 - How long have you been in your current role?
 - Please describe your prior and current experiences working with AI?
- Please briefly describe for me your job responsibilities in your current role and how it relates to people management or AI use in the workplace?

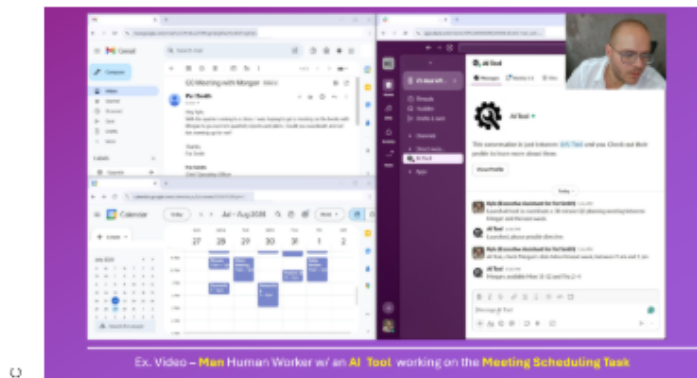
Figure 8: Sem-Structured Interview Guide for Study 2 Supplemental Expert Interviews - Page 1.

- How, if at all, is AI currently used within your organization?
 - *Follow-Up Questions:*
 - If it is used in your organization, how are employee's contributions currently evaluated when they work with AI?
 - If it is not currently used in your organization, how do you think your organizations' ways of evaluating employees would be affected if people started working with AI in their everyday work practices?
- Now that we have gotten some background information about your role and your organization, let's get into talking more about the recent experiment I have conducted and would like to hear your thoughts on. To start, let me provide you with a bit of a background of the study itself. If at any point in time you'd like me to repeat this information to help clarify a question, just let me know and I'm happy to recap.
 - I ran a large experiment with 685 participants, where each participant was asked to evaluate and attribute credit to a human worker and their dyadic partner after watching a short video of the partnership successfully completing a task. The human worker was either a woman or a man, and the dyadic partner was either another human man or woman teammate, or was either an AI Tool or an AI Teammate. By "AI Tool", I mean that the AI partner was designed to only provide informational support to the human worker when the human worker asked, much like how something like Siri or Alexa works. The AI Teammate, on the other hand, operates more in a collaborative role, where it makes suggestions and took on parts of the task to share in the load with the human worker.
 - Just for clarity on what these videos looked like, here is a screenshot of the video that participants saw if they were in the experimental condition that showed the Woman Human Worker, Katie, working with an AI Teammate on the Investment Banking Task. You can see the woman in the top right corner, the task on the left, and a chat interface with the AI Teammate on the right.

Figure 9: Sem-Structured Interview Guide for Study 2 Supplemental Expert Interviews - Page 2.



- Next Slide: Here is a video screenshot of the Man HW, Kyle, working with an AI Tool on the Meeting Scheduling Task.



- Before we move forward, is there anything you'd like me to clarify about the experiment?
- Based on this one experiment, I ran two studies. Study 1 explored the impact of AI type and worker genders on workplace evaluations. Study 2 took a deep dive into job security, exploring what contributes to job security in an AI-enabled workplace. For the remainder of this interview, I'm going to talk to you about these two studies separately.

Figure 10: Sem-Structured Interview Guide for Study 2 Supplemental Expert Interviews - Page 3.

Study 2 Specific Questions

- Now that you have some background information about the study, let's start by focusing on Study 1.

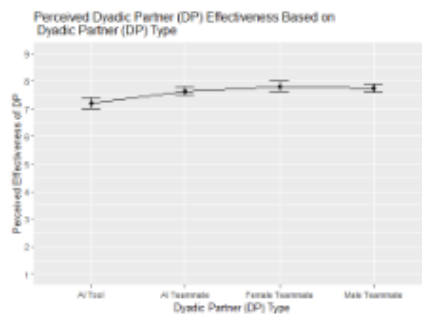
Expert Opinions Pre-Results Sharing

- Before I talk to you about the results of that study, I'd love to get your gut reaction to a few of the goals of that study.
 - (1) First, I wanted to see if the gender identity of a worker would impact how people evaluated the worker's effectiveness and credit when working with an AI.
- How, if at all, do you think a worker's gender identity might influence how that worker is evaluated when they use or work with AI?
 - (2) Second, I wanted to understand whether the type of AI a worker works with would impact how that worker is evaluated or attributed credit. More specifically, I was curious to see if workers would be attributed more credit and be evaluated as more effective when working with an AI that acted as a collaborative teammate versus an AI that acted as a simple tool.
- If you had to guess, how, if at all, do you think evaluations of the worker would change depending on whether the AI was acting as a teammate or as a tool?
 - (3) Lastly, I was curious to see if these perceptions would change based on the type of task the partnership was working to complete. One was the investment banking task, where the worker was the Lead Investor soliciting the help of a Research Analyst. I picked the investment banking task because this is often associated with more 'masculine' type work, while the meeting scheduling task has a more 'feminine' work association, aka being an executive assistant.
- Thinking back to the opinions you've already shared, how do you think they'd change based on the task that the partnership worked on?

Figure 11: Sem-Structured Interview Guide for Study 2 Supplemental Expert Interviews - Page 4.

Expert Reactions to Results

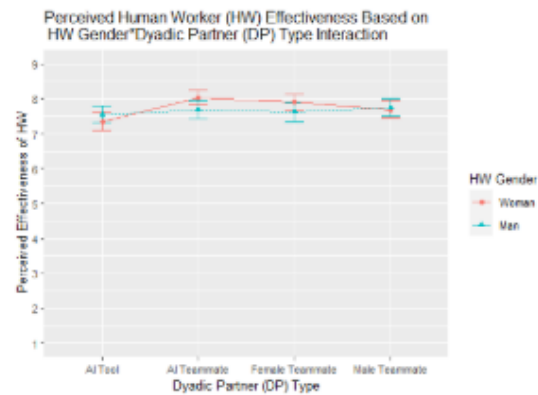
- Great, now I'd like to walk through a few results of this work and get your reaction to these results. My goal with this interview is to understand how these results might influence your own work or expertise or whether you see a future direction for this work based on your expertise.
 - Result 1: The human worker main character was actually shown as being more effective when working with any type of teammate (including an AI teammate) compared to an AI tool.
 - What are your initial reactions to this result?
 - Within your own work, how do you see this result being impactful?



- Result 2: Women workers are seen as more effective than male workers when they work with AI teammates or other women. However, the woman worker was seen as less effective when she worked with the AI Tool, while evaluations of the man worker remained the same no matter who he worked with.
 - What are your initial reactions to this result?
 - Within your own work, how do you see this being impactful?

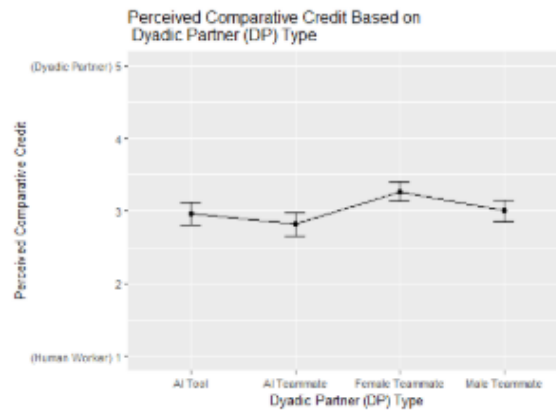
Figure 12: Sem-Structured Interview Guide for Study 2 Supplemental Expert Interviews - Page 5.

- Do you have any insight into why the woman worker was seen as more effective when working with the AI teammate compared to when men worked with the AI teammate?



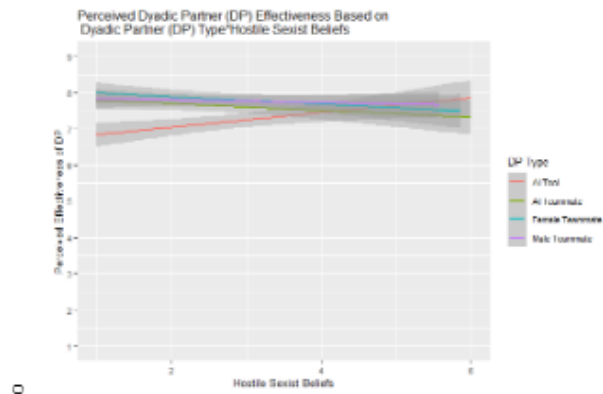
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- Result 3: In other words, when working with the AI Teammate, the human worker was given more credit, but when working with the female teammate, the worker was given less credit.
- What are your initial reactions to this result?
- How, if at all, do you think credit should be given to AI systems?
- Should people using AI receive the same amount of credit for a task they complete with another human?

Figure 13: Sem-Structured Interview Guide for Study 2 Supplemental Expert Interviews - Page 6.



- Now that we have reviewed some of these results, I also looked at how people's biases, such as sexism or AI bias, might impact their evaluations. To close our exploration of study 2, I wanted to just look at some of those results super quick.
 - Result 4: When the human worker was working with an AI tool, we actually saw that greater levels of sexism led to more positive evaluations of the AI tool. This trend was not present for the any of the teammate conditions though.
 - First, what are your initial reactions to this result?
 - Why do you think sexism influenced perceptions of AI tools to be more positive? Keep in mind that this is for both the woman and the man worker.
 - The AI teammate follows a similar trend to the female and human teammate, do you think this means that participants saw the AI teammate as being more human-like? Why or why not?

Figure 14: Sem-Structured Interview Guide for Study 2 Supplemental Expert Interviews - Page 7.



Result 5: When participants had negative attitudes toward AI systems in general, this often led to more negative evaluations of the AI, but that also somewhat extended to the female human teammate.

- What are your initial reactions to hearing this?
- In your own workplace, what are the general perceptions and expectations towards working with or using AI systems?
 - Follow Up Questions:
 - If your workplace has a generally positive or optimistic view of AI use in the workplace, why do you think this is the case?
 - If your workplace has a generally negative or pessimistic view of AI use in the workplace, why do you think this is the case?
 - And how does it affect the way AI is or is not used in your workplace?
- How do you think your workplace could best account for this result when thinking about AI-enabled work?

Figure 15: Sem-Structured Interview Guide for Study 2 Supplemental Expert Interviews - Page 8.

B.6 Review Slides for Study 2 Supplemental Expert Interviews

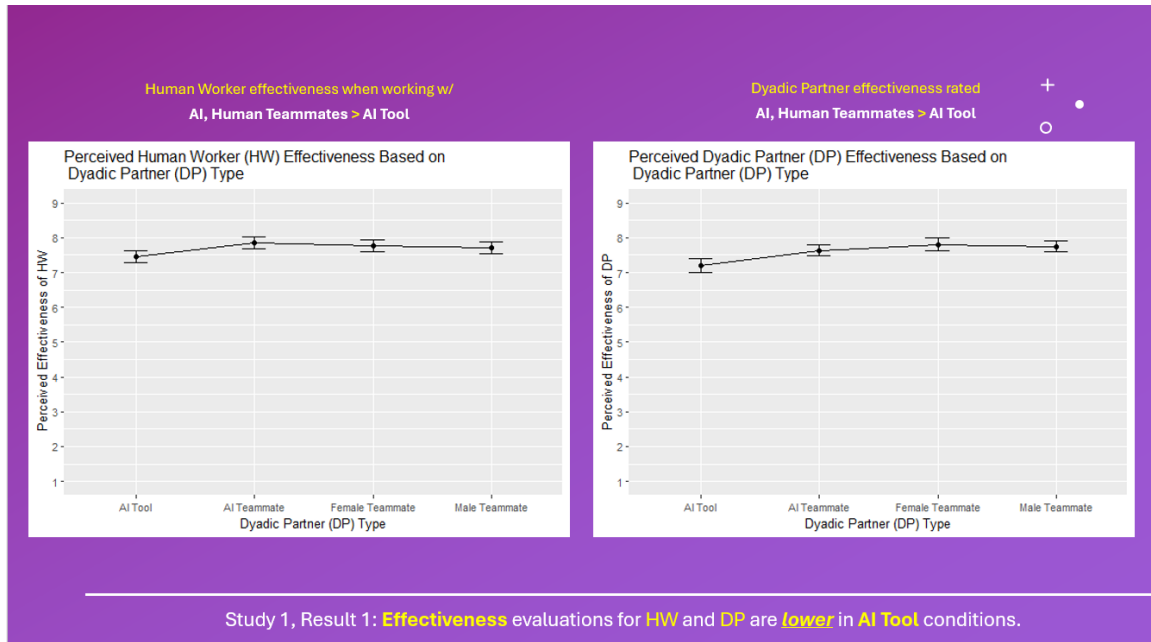


Figure 16: Review Slide for Study 2 Supplemental Expert Interviews - Perceptions of HW & DP Effectiveness.

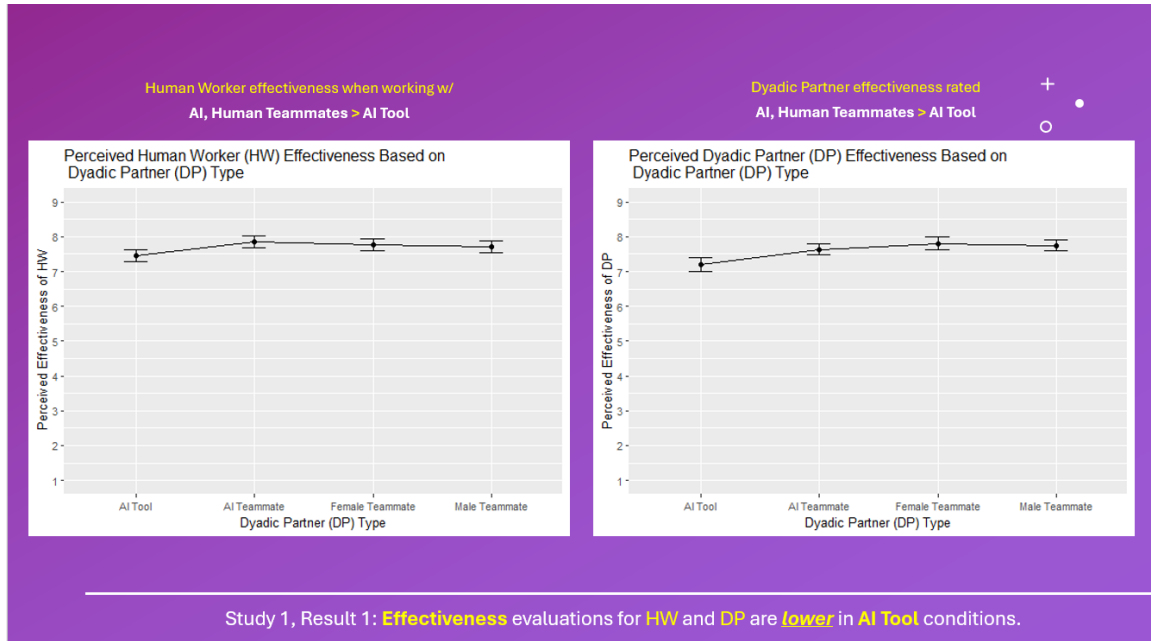


Figure 17: Review Slide for Study 2 Supplemental Expert Interviews - Perceptions of HW Effectiveness by HW Gender x DP Type.

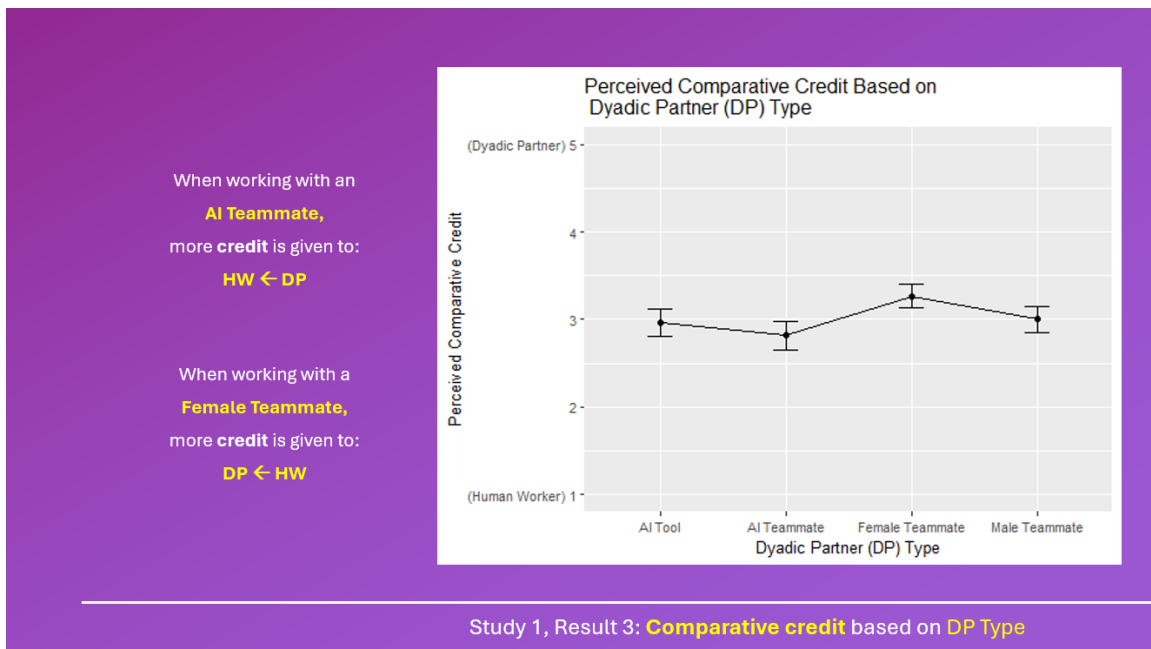


Figure 18: Review Slide for Study 2 Supplemental Expert Interviews - Comparative Credit.

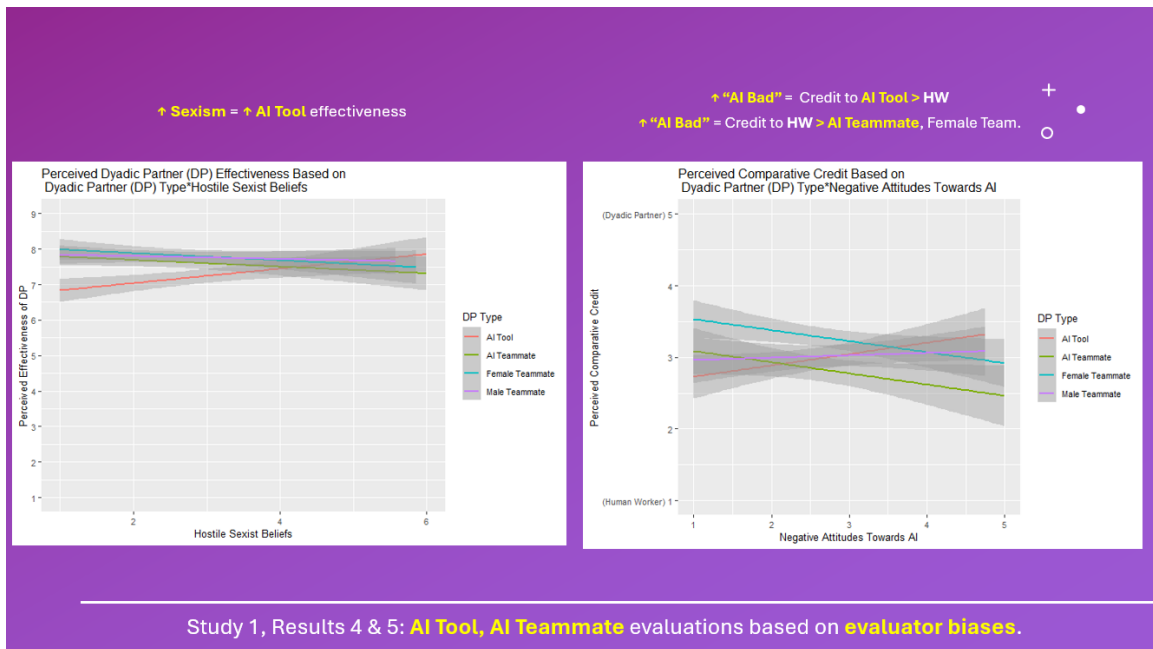


Figure 19: Review Slide for Study 2 Supplemental Expert Interviews - Evaluator Biases.

B.7 Semi-Structured Interview Guide for Study 3 Supplemental Expert Interviews

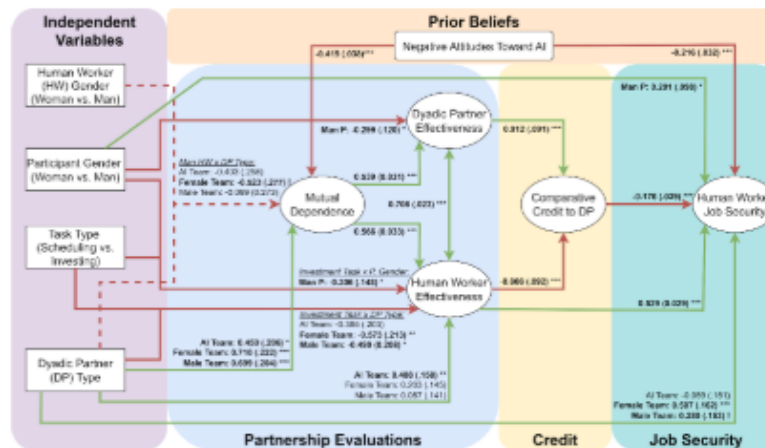
Study 3 Specific Questions

- Great! Thanks for answering those questions for study 2. At this time, I would like to move to study 3.
 - Just for clarity, all of this data came from the same experiment, which means study 3 also explored the ideas of human worker gender, task type, and dyadic partner type.
 - However, Study 3 examined the data using a more complex modeling technique, which was used to create a goal that would outline how people form perceptions of job security for workers in AI enabled workplaces.
- The following figure is the model created from my study. It's going to look a bit complex at first, so don't worry if all of the numbers don't make complete sense. I'll walk through the high-level take aways with you.
 - Ok, so essentially, this model connects a lot of the perceptions my study explored to understand what all contributes to job security in AI enabled workplaces. Based on my study, we see the results from study 2 still show up, where gender, task type, and dyadic partner type continue to impact perception.
 - However, what this model is now able to show us is that a lot of these factors contribute to the effectiveness of the human worker and their teammate. In turn, this effectiveness drives the credit that both the human

Figure 20: Semi-Structured Interview Guide for Study 3 Supplemental Expert Interviews - Page 9.

worker and the dyadic partner are assigned. Lastly, this credit, along with effectiveness, drive job security. During this entire process, one's attitudes towards AI can also impact said job security.

- Before I move on, does this general over of this model make sense?



- Now, with the results of this model in mind, I'd like to ask you a few follow-up questions relating to these results and your own experience and expertise.
 - First, what is your initial impression and reaction to these results? Are they expected or surprising?
 - One of the strongest impactors on job security was still whether or not people worked with an AI or a human. How do you think AI is going to impact job security in the future?
 - Do you think impacts on job security will adversely impact women and men differently?
 - One of the pervasive concepts in my dissertation is that anyone can be sexist, including women, and this model even shows that the gender of a participant impacted their evaluations of dyadic partners and human worker

Figure 21: Semi-Structured Interview Guide for Study 3 Supplemental Expert Interviews - Page 10.

job security. **Within your own workplace, how do gender dynamics lead men or women to form different evaluations of their coworkers?**

- How do you think these gender dynamics might extend to the use of AI technology?
- This model is obviously relatively simple compared to all of the perceptions that exist in a workplace. What other considerations do you think would impact job security in an AI-enabled workplace?
- How do you think gender will play into job security in the future?
- Do you have any recommendations for workers who might want to shore up their job security in preparation for AI to enter the workforce more?

Close

- Is there anything else you would like me to know before we wrap up?
 - Thank them for their time and depart.

Figure 22: Semi-Structured Interview Guide for Study 3 Supplemental Expert Interviews - Page 11.

B.8 Review Slides for Study 2 Supplemental Expert Interviews

views

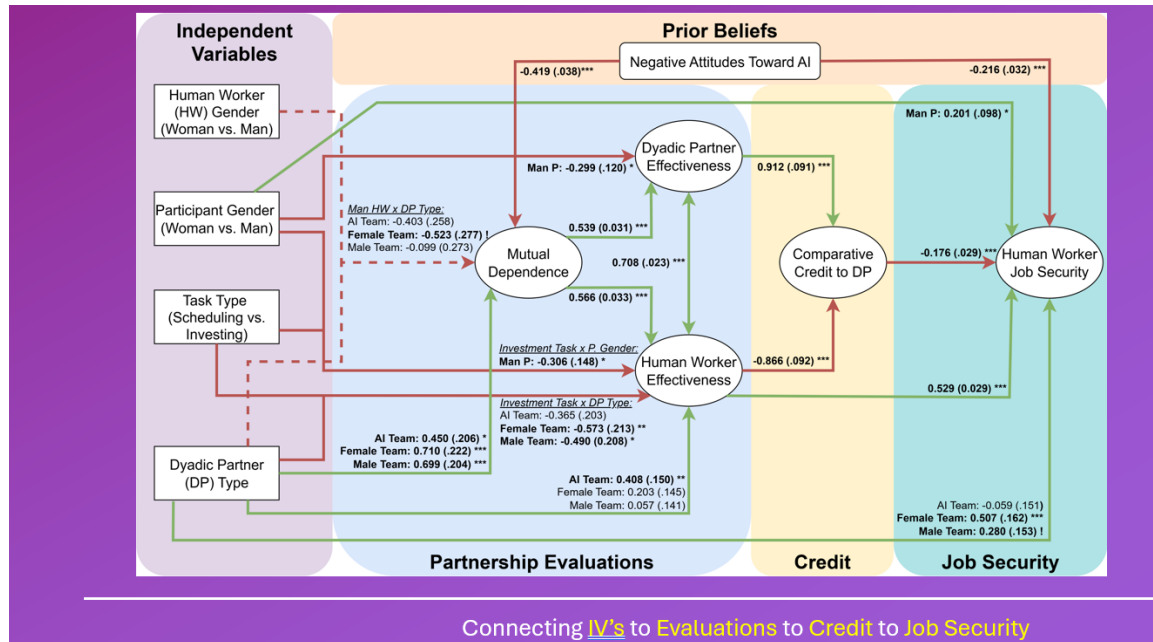


Figure 23: Review Slide for Study 2 Supplemental Expert Interviews - Structural Equation Model (SEM).

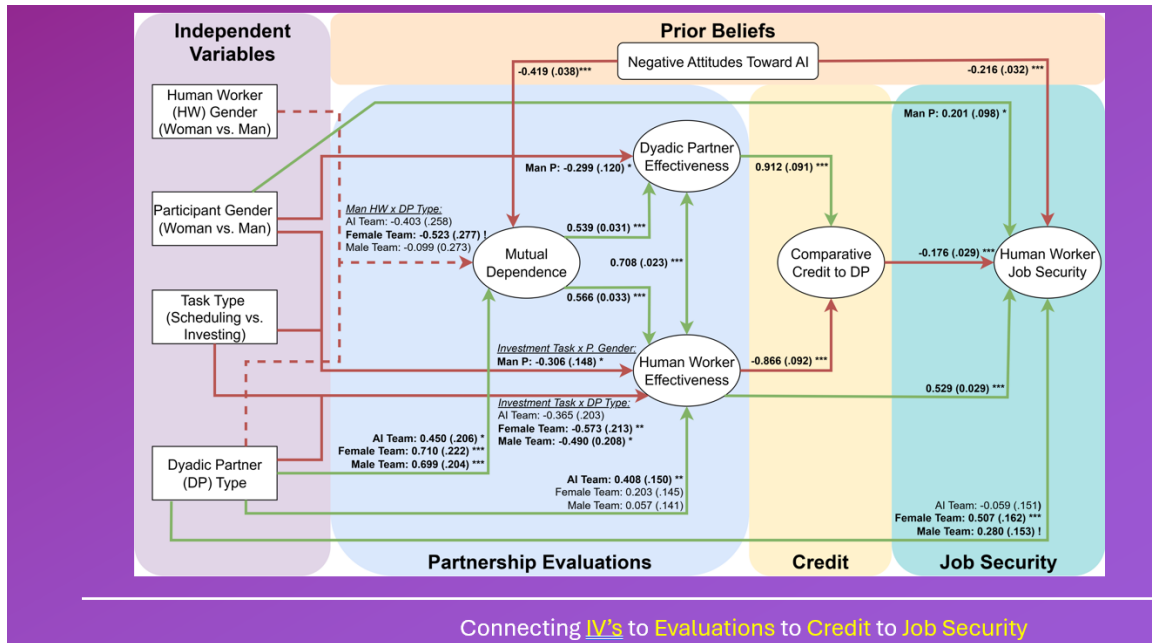


Figure 24: Review Slide for Study 2 Supplemental Expert Interviews - SEM Focusing on DP Type.

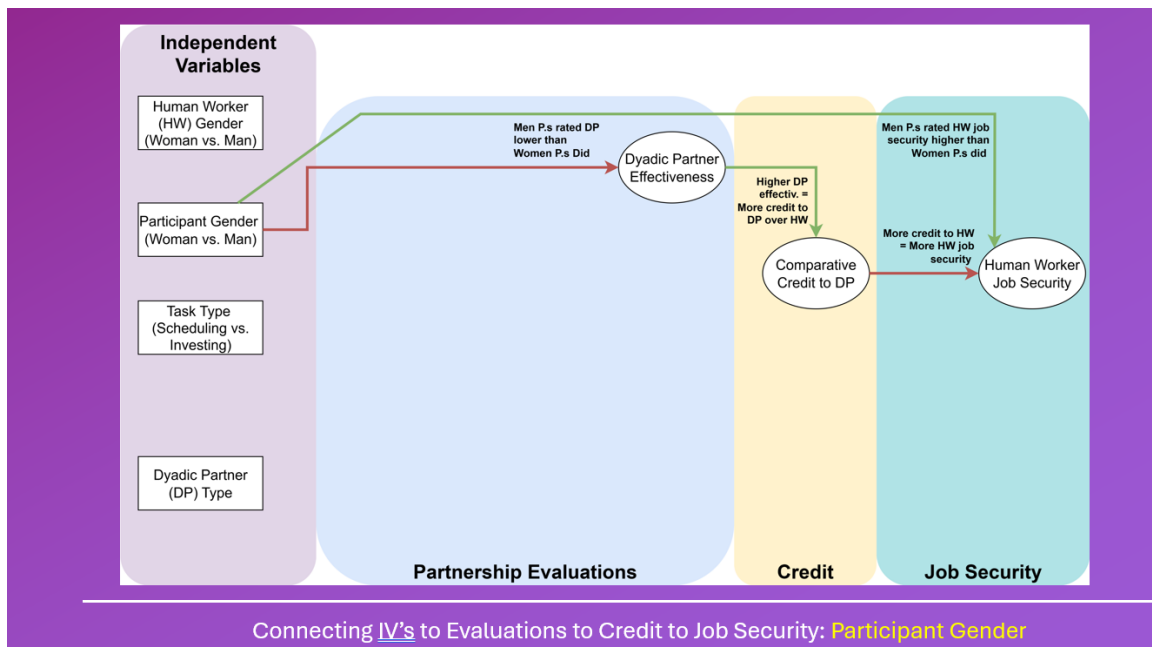


Figure 25: Review Slide for Study 2 Supplemental Expert Interviews - SEM Focusing on Participant Gender.

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